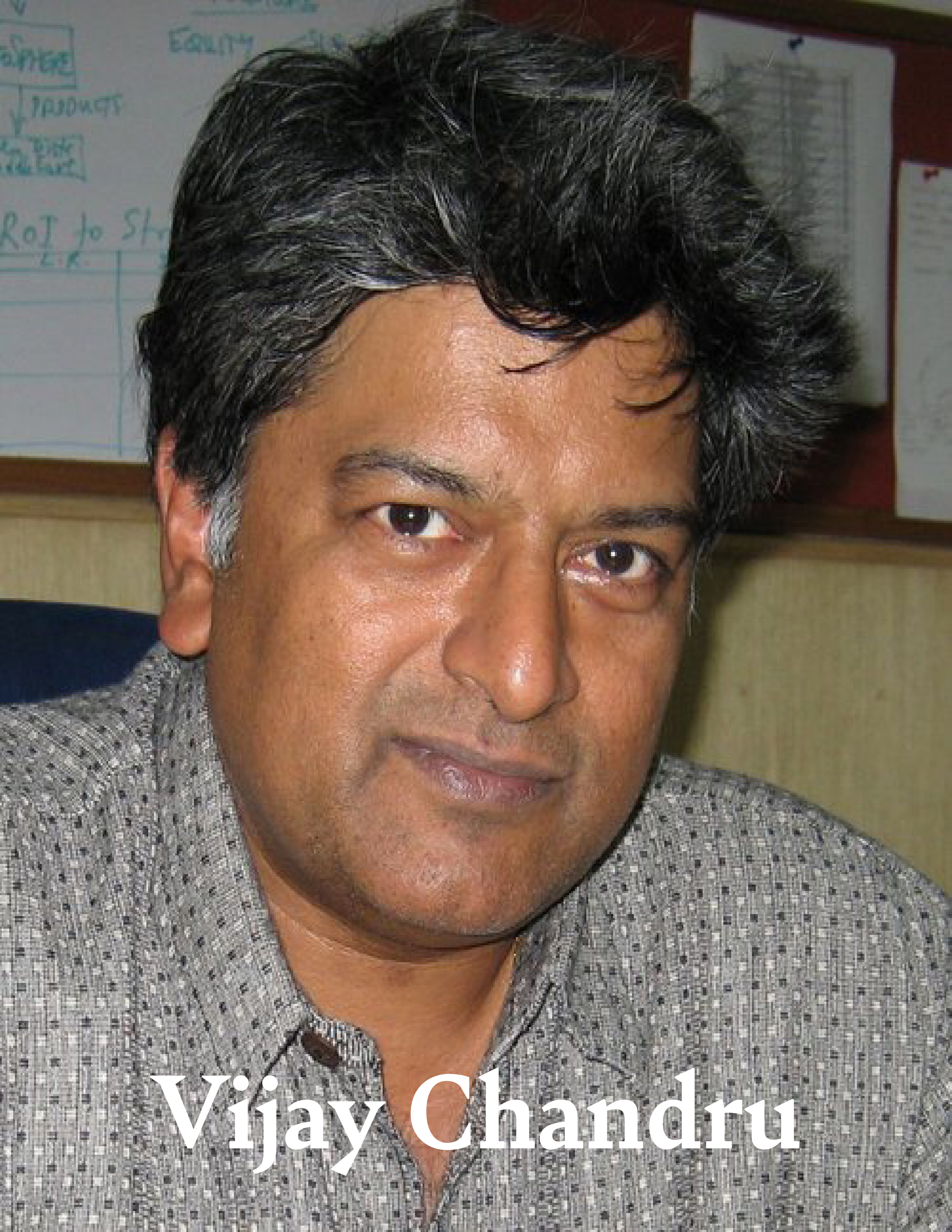


Vijay Chandru

ABLE Association of Biotechnology Led Enterprises

The collective face of the Indian Biotech Industry

PRICEWATERHOUSECOOPERS

Leadership in Affordable Therapeutic Products

A BIO PHARMA STRATEGY FOR INDIA

VISION 2020

An **ABLE - PwC** Report for

सत्यमेव जयते

Department of Pharmaceuticals
Ministry of Chemicals & Fertilizers
Government of India

Evolution of new drugs continues to be an increasingly daunting task. The en-block patents expiry of older small molecule drugs with little to add by way of discovery of new ones is at once a challenge and an opportunity.

This challenge is on one hand leading to hunt of new molecules from the depths of the ocean to prospecting new chemical complexes through nanotechnologies while opening up larger market opportunities in the generics space. However the generics market will see a lot of consolidation due to entry of large global players whose energies would be diverted from the hitherto patents of small molecules, to now, in the changed context, generics, which offers the traditional low hanging fruit to profit from.

It is in this context that the development of biopharma drugs becomes very important especially in the context of India. With the advent of the product patent regime, there is an air of expectancy to develop new ideas. There can be no better area to do this than Biopharma. Biopharma as we know encompasses the range from recombinant technology based biosimilars to development of large antibodies structures – the mABs. While in the year 2009 the global biopharma market was said to be about $ 137 billion strong, India has but a miniscule share of 1.4% in it. However Indian biopharma is growing at a scorching pace and registered a 17% growth over the last year, which is really the bright silver lining.

The Department of Pharmaceuticals in the Government of India has taken up the task of addressing this opportunity. In partnership with the premier biotechnology industry body ABLE and pre-eminent consultant leader in the life sciences sector – PwC, the Department of Pharmaceuticals has sought to prepare a detailed document – The Vision 2020 BioPharma Strategy. This would for the first time bring out the multifaceted challenge and the break-through possibilities in the biopharma sector for the Indian industry.

I am sure that the industry would benefit from it and so would the government for finding new ways to address the growth and development of the biopharma industry in India for making it the future leading global hub. Thank you.

Ashok Kumar
Secretary, Department of Pharmaceuticals
Ministry of Chemicals & Fertilizers
Government of India
New Delhi
12th July 2010

The Vision 2020 BioPharma Strategy: This would for the first time bring out the multifaceted challenge and the break-through possibilities in the biopharma sector for the Indian industry.

The Biopharma Challenge

The biopharma drugs sector is the next big block-buster awaiting right techno-scientific entrepreneurship. India cannot afford to miss this opportunity. The present growth rates of over 30% but scratch the surface.

The challenge is how to do it given the promises on one hand and constraints of world class human resources, infrastructure, discovery funding and the appropriate policy mix on the other. The Vision 2020 BioPharma Strategy document would be path-breaking in pointing out direction to accomplish this difficult task. ABLE and PwC have come together through painstaking research and wide experience including first hand interviews to prepare this document.

I hope the industry, the academia and all the stake-holders find it useful. We encourage them to contribute to it and join us in the task of building India as the next big destination for global biopharma.
Thank you.

Devendra Chaudhry
Joint Secretary
Department of Pharmaceuticals
Ministry of Chemicals & Fertilisers
Government of India
12th July 2010

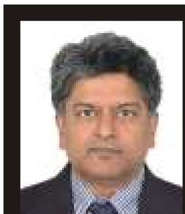
The Vision 2020 BioPharma Strategy document would be path-breaking in pointing out direction...

The Indian Pharmaceutical Industry has been a global leader in the cause of providing high quality affordable medicines to the world. With a gradual shift from small molecules to biologics, the vision, the mission and the tactical planning for the Indian Biopharmaceutical Industry has yet to be evinced. This was the task given to us by the Department of Pharmaceuticals and this report is the result. At the outset we would like to thank the Department for reposing faith in ABLE & PwC to deliver on this important task.

This report analyses the global environment with reference to markets, regulations, government initiatives and identifies key areas for the Government to play a major role in building the required capacities and capabilities. An enabling ecosystem - infrastructure, regulatory framework and a skilled workforce are some of the factors that will ensure that India becomes a leading producer of affordable biopharmaceutical drugs in the next decade.

Recognising that the strategic way forward is through Innovation and the ability of Indian companies to create Intellectual Property, the report identifies long term initiatives that the Government can implement to foster innovation.

We hope this report presents an understanding of the opportunity for Indian Biopharma and a direction for all stakeholders to realize this opportunity.



Vijay Chandru
President - ABLE

Sujay Shetty
Leader – Pharma Life Sciences
PricewaterhouseCoopers

Recognising that the strategic way forward is through Innovation and the ability of Indian companies to create Intellectual Property...

In 2010, the Department of Pharmaceuticals (DoP) of the Government of India (GOI) set the nation's biopharmaceutical industry (BioPharma) a lofty goal: to become a leading global producer of affordable "biopharmaceutical" products by 2020. The market is currently worth about US $137 billion, but it is growing very rapidly. Indeed, industry experts estimate that it could be worth US $319 billion by 2020. Moreover, at least 48 products with combined sales of nearly US $73 billion in 2009 are due to come off patent over the next decade. So the potential is huge.

However, biopharmaceutical products are very much more difficult to develop and manufacture than traditional pharmaceuticals. The competition – both from manufacturers of branded products and from other emerging countries keen to make their mark in the biopharmaceutical space – is also likely to be intense.

If India is to achieve its aim, it will therefore have to act fast – and the GOI will have to play a major supporting role by creating a suitable physical, financial, legislative and regulatory infrastructure. The private sector will invest in building the necessary manufacturing capacity, if that enabling infrastructure is in place. But it cannot provide the roads and ports, fiscal incentives, laws, regulations and other such features that will also be needed to put India at the forefront of biopharmaceutical production.

This report focuses on the changes that will be required to provide such an infrastructure. It explores the key differences between biopharmaceutical therapies and conventional therapies, together with the implications for development and manufacturing; analyses the competitive landscape; and identifies the measures the GOI will need to implement within the next five and 10 years, respectively. It does not attempt to quantify the precise amount of manufacturing capacity that will be required; it focuses, rather, on the big picture.

We believe that India should aim to capture 10% of the global market for biosimilars – i.e., follow-on versions of original biopharmaceutical products – by 2020, and become one of the top five producers in the world. We estimate that the GOI will need to invest at least US$1 billion over the next five years to implement the measures we have identified. Doing so could yield rich returns; if India's Biopharma industry succeeds in realising this aspiration, it will bring in additional revenues of US $4.3 billion a year.

The GOI will have to play a major supporting role by creating a suitable physical, financial, legislative and regulatory infrastructure.

The Vision for 2020

Chapter 1

India should aim to become one of the world’s five leading producers of affordable biopharmaceutical drugs by 2020. Biopharmaceutical products include: vaccines, blood and blood components, somatic cells, gene therapies and recombinant therapeutic proteins.

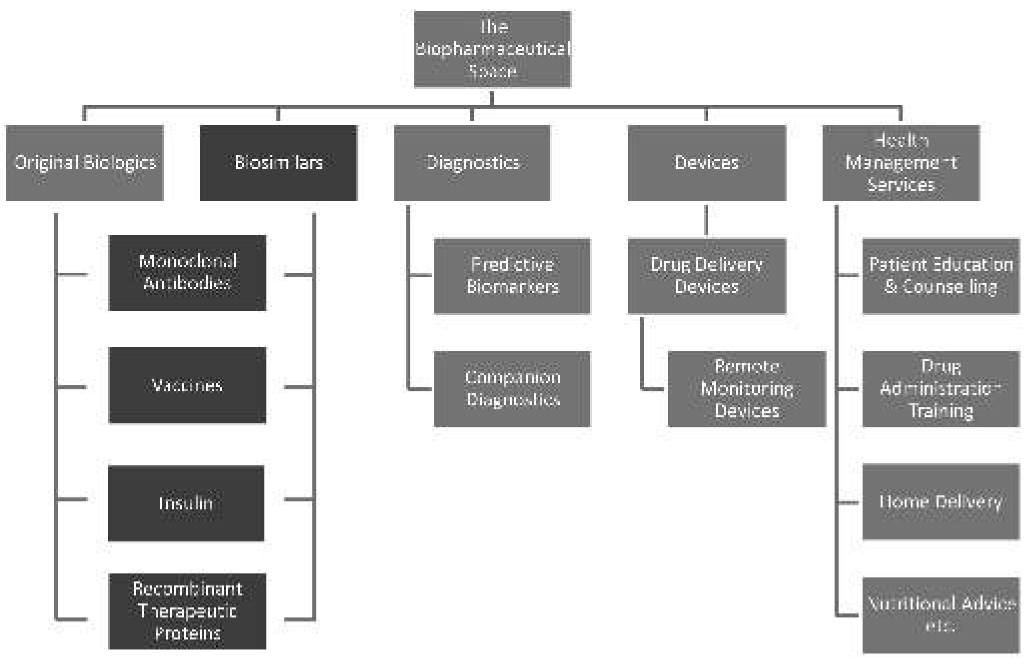
1.1 WHAT ARE BIOPHARMACEUTICAL PRODUCTS?

The word “biopharmaceutical” is a composite of the words “biotechnology” and “pharmaceuticals”, and reflects the convergence of what were once two distinct industries. So it is probably helpful to begin by discussing precisely what biopharmaceutical products comprise. They are medicines typically derived from living systems and produced using biotechnology – e.g., vaccines, blood and blood components, somatic cells, gene therapies and recombinant therapeutic proteins.

Over the past decade, the BioPharma industry has developed many such “biologics”, as they are also called. And follow-on versions of some of the earliest biologics made via recombinant DNA technology, including biosynthetic “human” insulin and human growth hormone, are now available. These are known as “biosimilars”.

However, the biopharmaceuticals market extends well beyond medicines. It also includes diagnostics, sophisticated drug delivery and remote monitoring devices, and health management services. Only a few biopharmaceutical companies – e.g., Fresenius and Baxter Healthcare – currently offer such services, but other companies are likely to follow suit as the spotlight switches to the secondary-care sector (see Figure 1). Diagnostics, devices and health management services are major subjects in their own right. This report therefore focuses on the burgeoning market for biosimilars – and, more specifically, on how India can best prepare to capture a share of that business.

Figure 1: The Biopharmaceutical Space



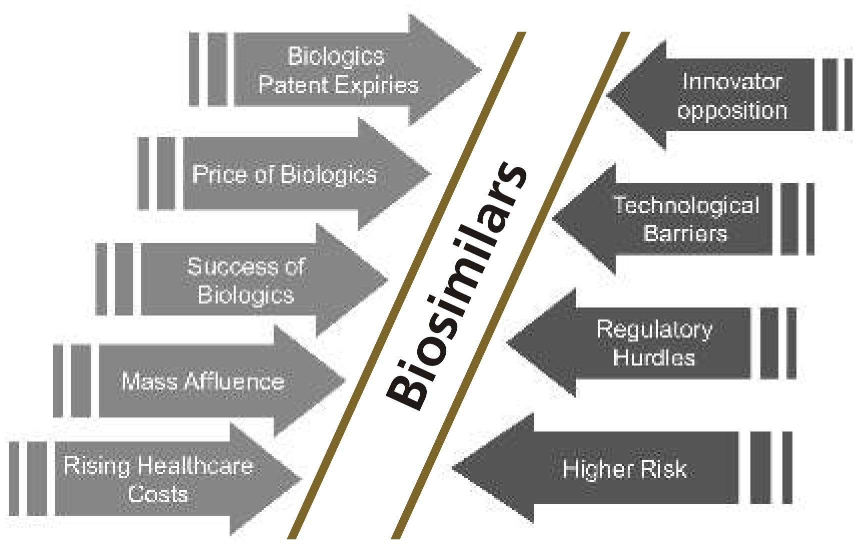
Source: PricewaterhouseCoopers

1.2 THE SIZE OF THE PRIZE

The biosimilars market is still in its infancy, and is largely concentrated in less heavily regulated regions such as Latin America and Asia at present. But industry analysts estimate that it could be worth more than US $43 billion by 2020. (All subsequent references are to US dollars, and all figures have been converted into dollars for ease of comparison, using the exchange rates detailed in Appendix 1.)

Aging populations and higher expectations are boosting demand for good medicines, and biologics now have a very successful track record. However, many biologics cost thousands – or even hundreds of thousands – of dollars. Biosimilars, by contrast, typically sell for between about 15% and 75% of the price of the original versions. So they are more affordable – in both mature economies with increasingly cash-strapped healthcare systems and emerging economies with increasingly affluent inhabitants (see Figure 2).

Figure 2: The Forces Shaping the Biosimilars Market



Source: PricewaterhouseCoopers

Biosimilars have considerable commercial potential, but exploiting that potential will not be easy. For one thing, biosimilars are much more difficult to develop and manufacture than traditional generics. The regulatory pathways for getting them approved are also less well established, and the competition – both from innovator companies and from rival biosimilars producers – is likely to be intense. We shall discuss these challenges, and how India should respond to them in more detail in the following pages.

Overview of the Global Biologics Sector

Chapter 2

With global sales of biologics reaching nearly $137 billion in 2009 and the patents on at least 48 biologics due to expire over the next decade, industry experts predict that the global biosimilars market could be worth more than $43 billion by 2020. But biologics differ from conventional pharmaceuticals in some fundamental ways.

2.1 THE SIZE AND COMPOSITION OF THE GLOBAL BIOLOGICS MARKET

In 2009, global sales of biologics totalled $136.6 billion (see Table 1). Avastin (bevacizumab) headed the list of best sellers, with sales of $5.74 billion, while Rituxan (rituzimab) and Humira (adalimumab) came second and third, respectively (see Table 2).

Table 1: The Global Market for Biologics in 2009

Country	2009 Sales ($ bn)
US	69.02
Europe	41.68
Japan	10.29
Asia/Africa/Australasia	14.40
Latin America	1.20
Total Biologic Drugs Market	136.59

Source: visiongain & PricewaterhouseCoopers analysis

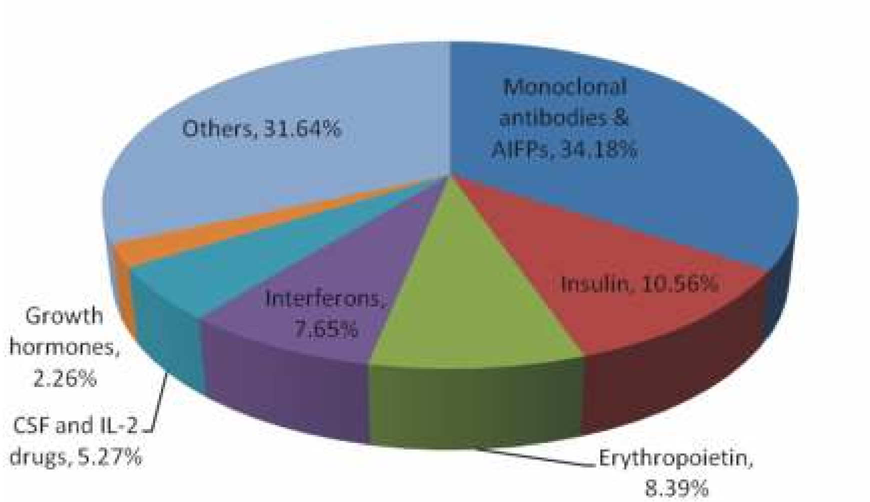
Table 2: The 10 Top Selling Biologics in 2009

Brand	Drug Name	2009 Sales ($bn)
Avastin	bevacizumab	5.74
Rituxan	rituximab	5.62
Humira	adalimumab	5.48
Herceptin	trastuzumab	4.86
Lantus	insulin glargine	4.29
Enbrele	tanercept	3.87
Remicade	infliximab	3.51
Neulasta	pegfilgrastim	3.35
Epogen	epoetin alfa	2.56
Avonex	interferon beta-1a	2.32

Source: EvaluatePharma

Measured by drug type, monoclonal antibodies (MAbs) led the market. They accounted for more than one-third of all sales of biologics in 2009, followed by insulin and erythropoietin (see Figure 3).

Figure 3: Share of the Total Biologics Market by Drug Type in 2009 (%)



Source: visiongain

2.2 THE SIZE AND COMPOSITION OF THE GLOBAL BIOSIMILARS MARKET

Biosimilars accounted for sales of just $1.23 billion – less than 1% of the total biologics market – in 2009 (see Table 3). However, there is considerable potential for growth.

Table 3: The Global Market for Biosimilars in 2009

Country	2009 Sales ($ bn)	Market Share of Biosimilars (%)
US	0.06	4.9
Europe	0.14	11.4
Other Countries (incl. China and India)	1.03	83.7
Total Biosimilars Market	1.23	100.0

Sources: IMS Health & visiongain

Some 30-odd biosimilars are currently in development or close to securing regulatory approval. The patents on at least 48 biologics with combined sales of nearly $73 billion in 2009 are also due to expire within the next 10 years, paving the way for the development of additional follow-on products. (Please see Appendices 2 and 3 for further information on the biosimilars pipeline and biologics with patents expiring between 2010 and 2020.) Furthermore, healthcare payers around the world are becoming increasingly interested in biosimilars, as they struggle to contain healthcare bills that are soaring as a result of aging populations – and, given that treating a single patient with a MAb can cost as much as $100,000 a year, it is easy to see why more economic alternatives might appeal. Lastly, the US, which is the world's largest biopharmaceuticals market, has just created a new regulatory pathway for approving complex biosimilars. For all these reasons, industry experts predict that the global biosimilars market could be worth more than $43 billion by 2020.

2.3 THE KEY DIFFERENCES BETWEEN BIOLOGICS AND CONVENTIONAL PHARMACEUTICALS

However, biologics differ from conventional pharmaceuticals in some fundamental ways. In the next section, we shall discuss these differences, together with their implications for the development, manufacturing and marketing of biosimilars.

2.3.1 Differences in Therapeutic Targets

Biologics typically address diseases conventional drugs cannot treat very effectively – such as cancer and genetic disorders. They can therefore command premium prices in most markets when they are first launched, because there are no effective therapeutic alternatives.

2.3.2 Differences in Product Attributes

But biologics are very complex products. Conventional drugs are derived from chemicals, consist of relatively few molecular ingredients, are quite small and can easily be characterised through their chemical structures, using established analytical techniques such as mass spectrometry. Biologics, by contrast, are derived from genetically modified microorganisms or animal cell lines. Most biologics also consist of many molecular ingredients, are much larger than conventional drugs (between 100 and 1,000 times larger) and have complex structures that cannot be completely characterised by the methods used for conventional drugs.

2.3.3 Differences in Manufacturing Processes and Infrastructure Required

Biologics likewise differ crucially from conventional drugs in that the methods by which they are manufactured greatly influence their therapeutic characteristics. The same starter ingredients may deliver quite different results, depending on the system that is used. Similarly, a slight variation in the starter ingredients or external manufacturing conditions may yield a different product, even if the living system from which it is derived is the same. These differences can render a biologic unsafe or ineffective. Hence the fact that the patents protecting biopharmaceuticals often include the processes used to manufacture them as well as their chemical composition.

Moreover, the process flows typically used to manufacture proteins or antibodies, and the unit operations involved in those process flows, have very little overlap with the process flows and unit operations typically involved in the production of small molecules (see Tables 4 and 5). These differences in the manufacturing processes and infrastructure required to produce small molecules and proteins or MAbs mean that the latter are more expensive to manufacture in terms of both capital costs [per square feet (sq. ft.)] and operating costs.

Biologics likewise differ crucially from conventional drugs in that the methods by which they are manufactured greatly influence their therapeutic characteristics.

Table 4: Typical Process Flows	
Small Molecules	Proteins/MAbs
Chemical reaction	Cell banking
Solvent extraction	Seed fermentation (or inoculum development)
Crystallization	Production fermentation/cell culture
Vacuum or air drying	Harvesting (cell separation)
Milling	Concentration and purification (multiple steps depending on product and host organism)
Blending	Bulk formulation, bulk lyophilisation
	Aseptic fill-finish (vials, cartridges, syringes, etc.)

Source: PricewaterhouseCoopers

Table 5: Key Differences in Manufacturing of Conventional Drugs and Biologics	
Small Molecules	Proteins/MAbs
Chemical synthesis	Expressed in micro-organisms (bacteria,yeast, fungi, mammalian cells)
May involve harsh conditions like extremes of pH, temperature and pressure, flammable organic solvents	Generally aqueous processing using mild conditions
Bulk active pharmaceutical ingredient available as a stable solid at room temperature	Requires formulated bulk solution with cold storage
Formulated for oral delivery (tablets, capsules, syrup) or topical application (ointment, spray)	Formulated as a sterile vial, pre-filled syringe or cartridge for injection or infusion without terminal steam sterilization
Manufacturing facility is generally not designed for aseptic processes	Manufacturing facility is designed for aseptic processes
Supply chain may include different manufacturers for drug intermediate, bulk drug and formulated drug product	Vertically integrated up to formulated bulk; at most fill-finish can be decentralized. No concept of drug intermediate

Source: PricewaterhouseCoopers

More specifically, whereas most of the equipment needed to manufacture small molecules can be sourced locally, most of the equipment needed to manufacture biologics must be imported, although some subsidiaries of European and US vendors have now set up shop in India (e.g., Millipore Corporation, Sartorius and Clestra). The initial capital investment is therefore much greater. The skills required to manufacture biologics – e.g., a knowledge of cell line engineering, fermentation, microbiology, protein purification (chromatography, membrane separations, lyophilisation, protein folding, etc.) and aseptic processing – are also much more demanding.

2.3.4 Differences between Biosimilars and Generics

Biosimilars are thus more difficult to develop and manufacture than traditional generics for a number of reasons. First, it is almost impossible to produce an exact replica of a biologic because changes to the structure of the molecule can take place during the production process. Biosimilars, unlike generics, must therefore be subjected to additional clinical testing to ensure that they have similar pharmacokinetic profiles to those of the original products, and do not cause unexpected adverse responses or immune reactions. Second, the processes used to manufacture biosimilars are inherently much more complex than those used to manufacture generics. And, lastly, since biosimilars are injectable products which cannot be terminally steam sterilised, all processing must be performed under aseptic (sterile) conditions.

These challenges mean that it typically costs $10-40 million to develop a biosimilar, compared with just $1-2 million for a traditional generic. The gestation period for clinical development, regulatory approval and scale-up to commercial production is also much longer (typically, about seven years versus two or three years), so the risk to capital is much higher. The cost differential between a biosimilar and the original product is thus much smaller – a fact that some countries, such as Japan, have recognised by introducing differentiated pricing regulations for biosimilars and generics.

Moreover, since biosimilars are not easily shown to be bioequivalent to the original products (as traditional generics are), they are not normally interchangeable (i.e., a pharmacist cannot substitute a biosimilar for the original version). So any company that manufactures a biosimilar will need to employ a specialised sales force (or enter into a marketing agreement with a third party) to encourage physicians to prescribe it.

2.3.5 Differences in Regulation

The many differences between biologics and conventional drugs – and hence between biosimilars and generics – have resulted in very different criteria for regulatory approval. Small molecule bulk is approved based on analytical characterisation of the active moiety and identification of any impurities above a specified threshold. The formulated product is approved based on dissolution tests and/or bioequivalence/bioavailability studies in healthy volunteers. These are generally quick, relatively inexpensive and do not require a sophisticated clinical infrastructure.

However, all biosimilars currently require pre-clinical testing in animals (rodents, dogs or other relevant species) to Good Laboratory Practice (GLP) standards and clinical testing to Good Clinical Practice (GCP) standards to establish their pharmacokinetics, immunogenicity and comparative efficacy against the original product. Some countries (or groups of countries, like the European Union) also insist on clinical trials in their local populations or sourcing of the original reference product from the country or region concerned.

Fulfilling these stipulations is very time-consuming and expensive. It also requires clinical and bioanalytical expertise and infrastructure, and the use of clinical research organisations (CROs) with specific domain and country knowledge. Furthermore, the original product may not be one that is routinely prescribed in India. So only a few investigators and trial sites may have the clinical experience to comply with internationally accepted protocols.

There are other obstacles, too. It is relatively easy to implement most process changes, scale changes and site changes for small molecules by showing in-vitro characterisation data through high-performance liquid chromatography (HPLC), low molecular weight mass spectrometry and other such techniques. But there are not yet any guidelines for biosimilars in terms of defining which changes can be permitted through in-vitro testing and whether in-vitro testing can be permitted for "well characterised proteins" (which are mainly smaller proteins without complex glycosylation).

Sophisticated protein characterisation tools may be needed to justify use of in-vitro testing to establish comparability (e.g., mass spectrometry and high frequency nuclear magnetic resonance (NMR) spectroscopy for macromolecules, cell-based assays, affinity measurements, specificity to receptor binding, enzyme-linked immunosorbent assays (ELISA) and gel electrophoresis). The specifications for recombinant proteins and MAbs generally also include establishing limits for host cell proteins, host cell DNA and endotoxins, while products made using mammalian cell cultures have specifications on viral clearance. Performing the analysis and characterisation tests required to meet such specifications requires different skills from those involved in the analysis and characterisation of small molecules.

2.4 THE REGULATORY ENVIRONMENT

Regulatory pathways for approving original biologics have been created around the world. However, the member states of the European Union, Japan and Australia (which adopted the European system) are currently the only countries with established biosimilar market authorisation pathways. The US, which is the world's largest biologics market, has just passed legislation to create such a pathway, but some of the details still have to be finalised.

2.4.1 The Regulatory Environment in the US

In the US, original biologics gain access to the market through two regulatory pathways:

- The Public Health Service Act (PHS, 1944) – which covers the majority of biologics and is enforced by the FDA via two departments: the Center for Biologics Evaluation and Research (CBER), which mostly regulates blood products, cellular products and vaccines; and the Center for Drug Evaluation and Research (CDER), which mostly regulates biologics produced by biotechnological methods (e.g., MAbs and therapeutic proteins). Biologics falling under the PHS are approved through the Biological License Applications (BLA) procedure.

- The Food Drug & Cosmetic Act (FD&C, 1938) – which covers conventional pharmaceuticals and certain natural proteins (e.g., insulins and growth hormones) and is also enforced by the FDA. Biologics falling under the FD&C are approved through the New Drug Application (NDA) procedure.

At present, biosimilars can only be approved for originals authorised under the FD&C, which means that it is only possible to market a very limited range of biosimilars (the simpler proteins) in the US. Applications must be submitted to the CDER through the Abbreviated NDA (ANDA) pathway. Application data from the already approved reference product can be used in support of the biosimilar, although the FDA can request further documentation of the new product's safety and efficacy relative to the reference product (e.g., through clinical trials).

However, in June 2010, the US Federal Government passed the "Biologics Price Competition and Innovation Act" to create a regulatory approval pathway for biosimilar versions of more complex biologics approved under the PHS. The Act distinguishes between biosimilars and "interchangeable" biosimilars, and establishes a different burden of proof for each category; a biosimilar must possess no clinically meaningful differences in safety, purity and potency from the original product, whereas an "interchangeable" biosimilar must produce the same clinical result as the original product in any given patient and present no additional risk if a patient is switched from the original product to the biosimilar. The Act also includes a 12-year period of data exclusivity for all original products (with a six-month extension for products supported by paediatric studies) and introduces a new "patent information exchange" process, in which the biosimilar applicant is required to provide information about its manufacturing process to the innovator company and both parties are then required to identify the key patents they believe either need to expire or can be successfully challenged.

The FDA must still provide guidance on the clinical data it requires, although the nature of that guidance has not yet been specified. And the process is unlikely to provide a fast route to market; there is considerable scope for disagreement during the patent information exchange period, for example. Nevertheless, industry commentators widely regard the Act as welcome progress.

2.4.2 The Regulatory Environment in the European Union

Manufacturers of all pharmaceuticals, including biologics, can apply to the European Medicines Agency (EMA) for marketing authorisation in the 27 European Union member states. Approval is regulated through the 2001 Code for Human Medicines Directive (CHMD) and its subsequent amendments. The CHMD provides an abbreviated pathway for the approval of traditional generics and, following two amendments in 2003 and 2004, also covers biosimilars.

The EMA has issued various guidelines on the data biosimilar manufacturers need to supply with their applications, including specific guidelines for the approval of growth hormones, insulins, epoetin products and granulocyte-colony stimulating factors (G-CSF). It is still preparing guidelines for the approval of interferons and MAbs, but has already approved a number of biosynthetic growth hormones as well as biosimilar versions of epoetin and filgrastim.

The EMA assesses a biosimilar based on whether its safety, quality and efficacy are comparable to that of the original drug. The extent to which clinical trials have to be conducted to provide this information is at the agency's discretion. The regulations further stipulate that original drugs approved before 2005 enjoy marketing exclusivity for 10 years, if they were approved through the centralised European Union procedure. If they were approved in individual member states, varying marketing exclusivity periods apply. However, for original drugs approved after 2005, a data exclusivity period of eight years applies. An additional two years of marketing exclusivity – or three years, if the product is approved for a new indication with significant clinical advantages – can be granted.

2.4.3 The Regulatory Environment in Japan

The Japanese Ministry of Health, Labour and Welfare (MHLW) issued guidelines for the approval of biosimilars in March 2009. All manufacturers of biosimilars are required to support their applications with data from clinical trials, information on the manufacturing methods used and evidence of the product's long-term stability, as well as supporting data from use of the product in other countries.

The MHLW has now approved biosimilar versions of somatropin and erythropoietin. It has also introduced a reimbursement pricing regime specifically for biosimilars. Whereas all generics must be priced at no more than 70% of the price of the original drugs in order to be admitted to the National Health Insurance (NHI) list of reimbursable drugs, biosimilars command a 10% premium on this ceiling – and can thus be priced at up to 77% of the price of the original products.

Overview of the Indian Biologics Sector

Chapter 3

The Indian biopharmaceuticals market is currently worth nearly $2 billion a year and growing rapidly. Moreover, approximately 20 companies are already producing biosimilars, and about 50 such products (including imports and multiple brands of the same products) are already available on the domestic market. So India has made a good start.

3. 1 THE SIZE AND COMPOSITION OF THE INDIAN BIOLOGICS MARKET

The Indian biologics market consists primarily of vaccines, monoclonal antibodies, recombinant proteins and diagnostics. In the 2009/10 financial year, it was worth $1.9 billion – 62% of the $3 billion generated by the biotechnology industry as a whole (i.e., including bioagricultural and bioindustrial products, bioinformatics and bioservices). The top 15 biopharmaceutical companies accounted for nearly $1.2 billion of this sum (see Table 6).

Table 6: The Top 15 Biopharmaceutical Companies Operating in the Indian Market

Rank in 2010	Company	2009-10 Revenues1 ($ millions)	2008-09 Revenues1 ($ millions)	% Change
1	Biocon	257.00	198.71	29.34
2	Serum Institute of India[2]	185.13	242.63	-23.7
3	Panacea Biotec	153.15	130.06	17.76
4	Reliance Life Sciences[2]	98.01		
5	Novo Nordisk[2]	74.49	71.87	3.64
6	Shantha Biotech	73.00	53.80	35.32
7	Indian Immunologicals	59.43	50.41	17.89
8	Bharat Biotech	59.17	52.50	12.70
9	Eli Lilly	40.73	35.72	13.85
10	Bharat Serums	38.12	30.50	25.00
11	Hafkine Biopharma	36.80		
12	Cadila Healthcare	32.12	20.41	57.40
13	GlaxoSmithKline	26.86	18.18	47.75
14	Intervet India	26.48		
15	Intas Biopharma	25.05	19.44	28.82

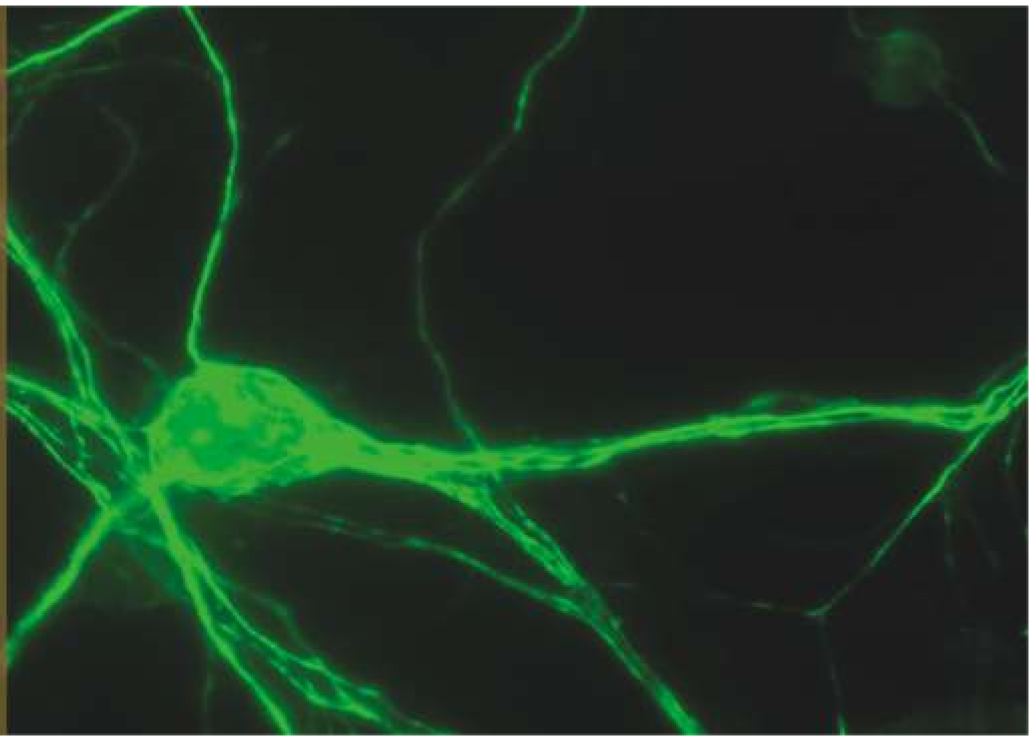
Source: BioSpectrum
Notes: (1) Revenues are for the Indian fiscal year, which runs from 1 April to 31 March; (2) BioSpectrum estimates.

3.1.1 Vaccines

The vaccines sector (including human and animal vaccines) represented the largest slice of the pie, with estimated sales of $475 million in 2009/10, up from $436 million the previous year. Human vaccines generated about 80% of this revenue, with domestic sales reaching $218 million and exports reaching $163 million. Sales of human vaccines are forecast to grow by 10-13% a year over the next five years, as better education and awareness about disease prevention, rising disposable incomes and government participation boost demand.

However, the market is clearly shifting from traditional "whole-cell" pertussis vaccines to combination vaccines and "acellular" preparations. Domestic players such as Bharat Biotech and Shantha Biotech (which was bought by Sanofi-Aventis for $783 million in July 2009) have already received massive orders for pentavalent vaccines from the GOI for immunisation programmes in the states of Himachal Pradesh, Kerala, Tamil Nadu, Jammu and Kashmir and Karnataka. Demand for newer products like the pneumococcal conjugate, meningococcal conjugate and human papillomavirus vaccines is also stimulating the paediatric and adolescent segment of the market, while flu vaccines will continue to play a big role in expanding the adult segment. And breakthrough products like Shanchol – the bivalent oral cholera vaccine jointly developed by Shantha Biotech and the International Vaccine Institute – will boost demand in the market as a whole.

Oncology products are a very profitable line of business for many Indian biopharmaceuticals manufacturers because they address an area of high unmet need and thus command premium prices.



3.1.2 Diagnostics and Targeted Therapeutics

The diagnostics and therapeutics sectors have also expanded in recent years. The diagnostics market is currently worth about $436 million, with molecular diagnostics accounting for sales of about $300 million in 2009/10. The market is growing at 15-20% annually, with revenues split equally between the multinationals – e.g., Roche, Siemens (which has acquired Bayer Diagnostics) and Abbott – and domestic players – e.g., Tulip Group, Transasia Biomedicals, RFCL (Diagnova), Span Diagnostics and Trivitron. Gradual acceptance of the concept of personalised medicine is driving much of this growth.

Meanwhile, the therapeutics sector accounted for 15% of India's biologics market in 2009/10, with cancer therapies clocking up sales of $68 million. Oncology products are a very profitable line of business for many Indian biopharmaceuticals manufacturers because they address an area of high unmet need and thus command premium prices. Uptake of such medicines is also increasing, as domestic producers make less expensive versions than those made by the multinationals and a growing number of Indian patients get medical insurance.

3.1.3 Oral Diabetes Drugs and Insulins

The oral diabetes market is currently worth about $338 million, while the insulin (and insulin analogues) market is worth about $133 million. Novo Nordisk dominates the latter, with more than 50% of all sales, followed by Eli Lilly with 22%. However, both markets are growing rapidly, as India becomes the "diabetes capital" of the world. Between 1995 and 2005, the number of patients with diabetes doubled from 20 million to 40 million, and it is projected to increase by another 70 million over the next 25 years. Demand for insulin analogues is growing especially rapidly; the market increased at a compound annual growth rate of 32%, measured in terms of value, in 2007-09. Novel delivery devices will also contribute to the expansion of the market in the future.

3.1.4 Biosimilars

About 20 Indian companies are already producing biosimilars. Dr. Reddy's Laboratories, Ranbaxy, Biocon, Shantha Biotech, Reliance Life Sciences, Panacea Biotec and Intas Biopharmaceuticals are among those that lead the way. But several other well-known companies have recently entered the field, including Glenmark, Cipla and Lupin Pharma. In June 2010, for example, Cipla announced that it was spending $65 million on stakes in two biotechnology companies – MabPharm and BioMab, based in India and Hong Kong, respectively – to bolster its presence in the global biosimilars space.

About 50 biosimilars have already reached the Indian market, and they are typically sold at discounts of as much as 85%, putting them within reach of the masses. In 2009/10, domestic sales of erythropoietin rose to $22 million while sales of c-GCSF rose to $11 million, sales of interferons rose to $22 million and sales of streptokinase rose to $15 million. Moreover, demand is likely to grow considerably, as India becomes more affluent. US investment bank Goldman Sachs estimates that the number of Indians with annual incomes of between $6,000 and $30,000 (measured in terms of purchasing power parity) will increase by 250-300 million during the next decade alone.

The global biosimilars market has even more potential for the most efficient Indian biosimilars manufacturers, since the market will be characterised by price competition, even when there are only a very limited number of rival products. That said, the manufacturers of branded products are likely to use second-generation products with more convenient administration schedules as a means of defending their territory. Some of these manufacturers may also try to crowd out the competition by producing their own biosimilars. So the competition is likely to be intense.

Competitive Analysis

Chapter 4

India's main national competitors in the biopharmaceutical space include China, Israel, South Korea, Singapore and Taiwan. These countries have initiated a number of infrastructure support and policy initiatives that provide an environment conducive to the growth of strong domestic industries.

Big BioPharma is not the only threat to India's biosimilars manufacturers. The industry is still in its infancy; however, a number of emerging economies have been actively building up their biopharmaceutical expertise, and some countries are clearly positioning themselves to capitalise on increasing demand for biosimilars. The following section provides an overview of India's key competitors. (We have treated Central and Eastern Europe, and Latin America collectively.)

4.1 AUSTRALIA

Australia is one of the world's leading established centres of biopharmaceutical expertise (together with the US, Europe and Canada), thanks to a first-class research base and robust patent regime. The country is home to some 450 biotechnology companies and 600 medical technology companies. In March 2010, more than 150 biotechnology and healthcare companies with a combined market capitalisation of $47 billion were listed on the Australian Stock Exchange – by far the biggest being CSL, which specialises in pharmaceuticals, vaccines and plasma products.

4.1.1 Government Initiatives

The Australian Government is keen to maintain this lead. It allocated about $7 billion for investment in science and biotechnology in 2009/10, although the two sectors fared less favourably in this year's budget. The Commonwealth Scientific and Industrial Research Organisation (CSIRO) is the national body responsible for promoting scientific research in Australia. It oversees various initiatives, including the Cooperative Research Centres programme, which provides funding and other forms of support for long-term public-private collaborations aimed at commercialising scientific innovations.

In 2009, the federal government also launched "Commercialisation Australia", a new initiative to commercialise Australian research and ideas. The programme offers various forms of help, including up to $43,667 to pay for specialist advice and services; up to $174,670 (payable over two years) to assist with the recruitment of experienced executives; proof-of-concept grants of $43,668 to $218,338 to test the commercial viability of a product, process or service; and repayable early-stage commercialisation grants of $218,338 to $1.7 million to develop a new product, process or service to the stage where it can be taken to market.

Individual states have supplemented these initiatives with their own programmes. Queensland is particularly notable for its efforts; it has invested about $3 billion in the sector, and now has 90 core biotechnology companies employing 1,900 scientists as well as some 66 biopharmaceutical research institutes employing 5,700 researchers. Queensland will also host Australia's first major contract manufacturing facility for biologics, which is due to be completed in 2012.

4.1.2. Research Base

Australia has numerous world-class medical research organisations, including the Garvan Institute, Institute for Molecular BioScience, Menzies Research Institute, John Curtin School of Medical Research, and Australian Institute of Bioengineering and Nanotechnology.

4.1.3 Regulation

The Australian Therapeutic Goods Administration (TGA) is responsible for approving and regulating the marketing of medicines in Australia. The TGA has adopted the EMA's guidelines for regulating biosimilars.

4.1.4 Investment Incentives

The Australian Government offers tax concessions for biotechnology-related R&D. The existing system will be replaced in July 2010 with a new 45% refundable tax credit for companies with an annual turnover of less than $17 million. Companies with a turnover of more than $17 million can claim 40%.

4.2 CENTRAL & EASTERN EUROPE

The biotechnology sector in Central and Eastern Europe is smaller and less developed than it is in Western Europe. There are currently about 260 biotechnology companies in the region; 29 are engaged in developing medicines, 55 operate in other areas such as veterinary therapeutics and industrial biotechnology, and the remaining 176 provide biotechnology services like contract research, diagnostics, manufacturing and analytical services.

Hungary, Poland and the Czech Republic lead the way. Hungary has 67 companies engaged in developing human medicines or providing bioservices. Poland has 38 such companies, and the Czech Republic 29. We have therefore focused on these three countries in the details below.

4.2.1 Government Initiatives and Investment Incentives

In 2004, the Hungarian Government established a Research and Technology Innovation Fund to support the country's biotechnology efforts. It contributes half the fund's income; the rest comes from Hungary's 26,000 private companies, which are expected to pay at least 0.25% of their turnover into the fund as an "innovation contribution".

In 2005, the Hungarian Government also launched a five-year plan to develop the biotechnology sector, with corporate tax breaks for foreign direct investments and new companies. Tax credits are offered on R&D investments, and R&D corporate tax allowances are generous, particularly if a company locates its laboratory at a university or public research institute. Tailor-made incentive packages are available for biotechnology investments of more than $8 million that result in the creation of at least 10 new jobs, and highly educated students can be employed tax-free in educational and research activities.

The Polish Government likewise provides grants for companies that conduct biopharmaceutical R&D, via the biotechnology division of the State Committee for Scientific Research. Grants typically range from $50,000 to $100,000. And the Czech Government supports biotechnology companies through CzechInvest, its investment and business development agency. CzechInvest has identified nine key areas of investment, including life sciences, medical devices and R&D – with particular emphasis on molecular biology, biomedicine and biotechnologies, and the development of new materials designed to advance the life sciences.

4.2.2 Research Base

Most of the biotechnology-related R&D conducted in Central and Eastern Europe is performed at universities or institutes, and collaboration with industry is limited. Hungary has eight research institutes specialising in biomedical research, including a genetics and immunobiology department at Semmelweis University, Budapest. Poland has about 40 universities, hospitals and research institutes engaged in biotechnology-related R&D, as well as five technology parks. In December 2009, it also set up a new incubator, called Science2Business, to support the commercialisation of scientific discoveries and inventions. The incubator will cover five sectors: biotechnology, telemedicine, renewable energy, informatics and mobile technologies. The Czech Republic has 300 biotechnology institutes and 10 established technology parks and clusters. The government is currently building a new biotechnology park at Masaryk University, in Brno, and an international clinical research centre that it hopes will be able to compete with the top research hubs around the world.

4.2.3 Regulation

All the new European Union member states were required to bring their regulatory frameworks into line with the rules prevailing in the rest of the bloc during the accession process. They were also required to harmonise their patent protection laws with those in Western Europe, although full harmonisation will not occur until 2011-2019, depending on the country and product.

4.2.4 Access to Capital

There is very little private funding from venture capital sources and angel investors in Hungary, Poland and the Czech Republic. The majority of the funding for biotechnology-related research comes from state or European Union framework programmes and structural funds.

4.2.5 Access to Talent

All three countries have a highly educated workforce and labour costs are very much lower than they are in Western Europe. However, Hungary has clearly stated that it does not want to compete on the basis of low wages and is actively trying to woo talented Hungarian scientists working overseas back to their homeland.

4.3 CHINA

China's biopharmaceuticals sector is entering what has been called a "golden age", thanks to substantial investment from the central government, which wants to see it become one of the country's leading industries by 2020. A number of Chinese biopharmaceutical enterprises have entered into alliances with global biopharmaceutical research institutions and multinationals, primarily to develop therapeutic vaccines, MAbs and recombinant proteins. The central government is also promoting the development of stronger regulations, better enforcement of the patent protection laws and various other measures to support the industry.

4.3.1 Government Initiatives

China expressly stated in its National Medium- and Long-Term Science and Technology Development Plan Outline (2006–2020) that it aims to acquire expertise in various cutting-edge technologies in the biopharmaceutical space over the next 15 years. These include target discovery, drug molecule design, gene operation and protein engineering, stem cell engineering and a new generation of industrial biotechnology tools. The central government also issued a new regulation on the registration of drug technology transfers in August 2009, which should promote technological innovation, while the Chinese National Development and Reform Commission (NDRC) has secured an additional $65 million of funding to support advances in biological medicine, bio-breeding and biomedical engineering.

However, the government has already implemented 19 price-cutting initiatives for medicines and plans to make more cuts in the future. This may affect the country's biopharmaceutical revenue growth, but will benefit Chinese patients – and the overall increase in volume may be sufficient to offset the reduction in per-unit sales.

4.3.2 Protection of Intellectual Property

China is making significant efforts to improve the protection of intellectual property rights in compliance with the requirements of the World Trade Organisation. Several multinationals have now established R&D centres in China, indicating that there is growing global confidence in the country's patent-protection regime.

4.3.3 Regulatory and Funding Reforms

The Chinese State Food and Drug Administration (SFDA) is introducing new policies to encourage innovation, restrict imitation and encourage biopharmaceutical outsourcing. China is also in the process of establishing an effective venture capital system to attract foreign venture capital into the domestic BioPharma industry.

4.3.4 Biotechnology Clusters

China's central and local governments have built more than 100 biotechnology parks. Some of these parks have been very successful – e.g., Shanghai Zhangjiang High-tech Park, which has attracted numerous multinational and domestic companies, including Roche, Novartis Biomedical Research, Kirin Kunpeng Biopharma, Wuxi PharmaTech and Shanghai Lead Discovery. However, many parks still have high vacancy rates.

4.3.5 Enterprise Development

A number of Chinese biopharmaceutical companies have joined forces to enhance their overall strength. The biopharmaceutical giant CNBG is one such instance; it was formed by merging six major biological product institutes in Beijing, Shanghai, Changchun, Wuhan, Lanzhou and Chengdu with two biopharmaceuticals manufacturers (Beijing Tiantan and Chengdu Rongsheng). CNBG now employs 10,000 people across China.

4.3.6 Increasing Focus on R&D

Chinese biopharmaceutical companies are increasingly aware of the need to invest in R&D. Shenyang Sunshine Pharma, which invests 10% of its revenues in R&D, has developed several successful products, for example, and various other companies are now investing in technology platforms for upstream R&D and clinical studies. China's support bioservices subsector is also expanding rapidly. Several hundred CROs currently operate in China, offering integrated or specialised services, and some of these CROs are evolving from one-off study vendors into viable, strategic long-term players.

4.3.7 Greater International Collaboration

A growing number of Chinese nationals who have resided overseas are returning to their home country and taking important positions in the domestic BioPharma industry. They are using their connections to forge powerful international alliances.

4.4 ISRAEL

The Israeli biopharmaceuticals sector has grown impressively over the last decade, as a result of its strong track record in research. The country devotes 35% of its research activities to the life sciences and has the largest number of scientists per capita in the world. Its expertise in neurological disorders, cancer and autoimmune diseases is particularly noteworthy, but it is also a global leader in regenerative medicine and cell therapy. Moreover, with a high percentage of graduates in mathematics, physics and computer sciences, it is well placed to make an impact in interdisciplinary technologies such as bioinformatics and proteomics.

4.4.1 Government Initiatives

The Israeli Government declared the life sciences industry – and biotechnology in particular – a "preferred sector" in 2005. The Office of the Chief Scientist (OCS) of the Ministry of Industry, Trade and Labour is responsible for administering government policy regarding the support and encouragement of industrial R&D. It operates various support programmes, including grants of up to 50% of the approved budget for market-driven competitive R&D projects, up to 66% for start-up companies and advanced generic technology, and up to 85% for technology incubators. The OCS also participates in bi-national funding initiatives with the US, UK, Australia, Singapore, Canada and Korea; and recently launched a new funding route to bridge the gap between basic and applied research, with grants of up to $95,000.

4.4.2 Research Base

Biopharmaceutical research is carried out at seven universities, five colleges and 10 specialised institutes as well as the major hospitals. The leading universities also have commercial offices of technology transfer to facilitate the commercialisation of their research.

4.4.3 Technology Incubators

Israel has 23 high-technology incubators, one of which is dedicated to biotechnology and two to industry-related technologies. Each incubator houses up to 15 companies and provides funding of about $500,000 per company for the first two or three years of its life, when the risk is highest and private funding is scarce. An estimated 1,000 high-tech companies have “graduated” so far, but a report submitted to the Knesset finance subcommittee for promoting and assisting high-tech technology suggests that the success rate has fallen in recent years. Between July 2006 and June 2009, only 100 companies managed to raise $500,000 or more. The subcommittee has now convened to consider a number of options, including extending the incubation period.

4.4.4 Regulation

Israel has aligned its regulatory framework with those in the US and European Union, and medications approved by the FDA or EMA are approved much more quickly than those that have not received such approval. The Israeli Ministry of Health and FDA have also signed a mutual recognition agreement to enable full acceptance and recognition by the FDA of clinical trials conducted in Israel for new drugs and medical devices. Israel’s IP laws concerning technology-based products and services, pharmaceuticals and other emerging sectors likewise closely resemble those of the US and European Union.

4.4.5 Investment Incentives

The Israel Investment Centre (IIC) of the Ministry of Industry and Trade offers companies in knowledge-based industries and their investors various tax incentives and grants. Investors may apply for one of two incentives: government participation of up to 32% in capital investments for creating jobs in export industries; or accelerated depreciation and a tax holiday of up to 10 years on undistributed profits, based on the capital that has been committed to job creation in export industries. (The precise amount of the benefit depends on the location.)

4.4.6 Access to Capital

Israel has a robust financial infrastructure and well-developed venture capital industry. However, fledgling biopharmaceutical companies have had a hard time raising capital following the global downturn in the capital markets last year. In the first quarter of 2010, Israeli high-tech companies raised only $234 million from venture investors (both local and foreign), less than in any preceding quarter for the past five years. The life sciences sector attracted over a third of this money, but much of it was directed towards more developed businesses.

The Knesset finance subcommittee for promoting and assisting high-tech technology is now reviewing a number of proposals for supporting the sector. They include creating a private fund with state leveraging, offering incentives for venture capital investment in seed companies and establishing “second-stage incubators” to promote “industrialisation”.

4.5 LATIN AMERICA

Latin America’s biotechnology industry is growing at an impressive rate. Bioagricultural and bioindustrial applications (biofuels) account for most of this growth, but the BioPharma sector is also getting stronger. The most active countries in the region are Argentina, Brazil and Chile. Brazil has more than 100 biopharmaceutical companies, while Argentina and Chile have about 23 and 15 such companies, respectively. All three countries have also become centres for clinical research. In 2009, there were more than 425 trials in Chile alone.

4.5.1 Government Initiatives

The Argentinian Government has implemented several programmes to promote basic research and technological investment. In 2007, it also established a Ministry of Science and Technology specifically to encourage scientific and technological innovation. The Ministry administers two government-backed funds: the Fund for Scientific and Technological Research and the Argentine Technology Fund.

Brazil introduced a national biotechnology policy in the same year, and undertook to invest $5.8 billion over a 10-year period, with approximately 60% of the funds coming from public sources. Chile has likewise taken significant steps to incorporate biotechnology into its economy. The National Innovation Board for Competitiveness, which reports to the President, is responsible for coordinating the efforts of the government agencies concerned. These include the Chilean National Commission for Scientific and Technological Research (CONICYT), which supports basic research and the development of resources, principally by financing academic research and collaborations between universities and private firms; and the Chilean Economic Development Agency (CORFO), which promotes the commercialisation of new technologies and products.

4.5.2 Research Base

Argentina has more researchers per active person than any other country in Latin America, and over 115 Argentinian research institutes and universities are engaged in biotechnology-related research. But most of this work is in the field of bioagriculture. Brazil, by contrast, has a strong reputation in fundamental biopharmaceutical research, although it has been less successful in commercialising that research. It can call on several strong public research institutions specialising in health biotechnology, including the Oswaldo Cruz Foundation in Rio de Janeiro and the Institute Butantan in SãoPaulo.

Meanwhile, Chile has 61 university research centres, 15 incubators and 10 technology transfer centres. It is also building a number of research nuclei as part of its Millennium Scientific Initiative, which has been partly funded by the World Bank. Five such nuclei have now been established, including the Millennium Institute for Advanced Studies in Cell Biology and Biotechnology, the Center for Scientific Studies and the Millennium Institute for Fundamental and Applied Biology. Again, however, human health accounts for a relatively small percentage of the country's biotechnology activities.

4.5.3 Regulation

Most Latin American countries (including Argentina, Brazil and Chile) have now aligned their local regulations with the Declaration of Helsinki, Council for International Organisations of Medical Sciences, and International Conference on Harmonisation-Good Clinical Practice (ICH-GCP) guidelines, and all have a regulatory agency responsible for pharmaceuticals, medical devices and clinical research. But there are still significant differences in the level of regulatory sophistication in the region.

4.5.4 Intellectual Property Protection

Argentina, Brazil and Chile are all signatories to the Agreement on Trade Related Aspects of Intellectual Property Rights (TRIPS). But although Chile scores highly for its enforcement of the rules on patent protection, Argentina and Brazil fare less well. The International Property Rights Index (2010) awards the latter two countries scores of 64 and 84, respectively (significantly below India's score of 53).

4.5.5 Investment Incentives

Argentina offers various incentives for biotechnology-related research under Law 26,270, including accelerated amortisation of income tax, early reimbursement of value-added tax (VAT) and tax credit bonds for services purchased from government-owned research institutes. Brazil has also enacted an "Innovation Law", which provides tax incentives for R&D, deductions on expenses related to patent filing and prosecution, and partial payment of the salaries of R&D scientists employed by biotechnology companies. Chile has adopted a similar approach, with tax incentives for R&D, subsidies of up to $30,000 to finance internships and subsidies of up to $730,000 to support innovation in the areas of goods, services, processes and organisational or trading methods.

4.5.6 Access to Capital

Latin America's venture capital industry is still undeveloped. However, experts anticipate that it will grow rapidly in Brazil and Chile (though not in Argentina) over the next few years. Brazil is one of the emerging countries with the greatest appeal for the investment community, thanks to the good economic management of successive democratic governments, the size of its domestic market, the strength of its industrial sector and the significant tax incentives available to venture investors. Chile also has a relatively strong economy, based on commodity production, and a dynamic, modern entrepreneurial class.

4.6 SINGAPORE

Singapore is well-known for being politically stable and business-friendly. It offers clear and consistent government guidelines, a reliable patent regime, a first-rate physical infrastructure and a favourable tax environment; indeed, in 2009, the World Bank dubbed it the world's "easiest place in which to do business". The city-state has also created a powerful biopharmaceutical nexus. More than 30 multinationals have established regional headquarters there, and Novartis chose Singapore as the home for its prestigious Institute for Tropical Diseases.

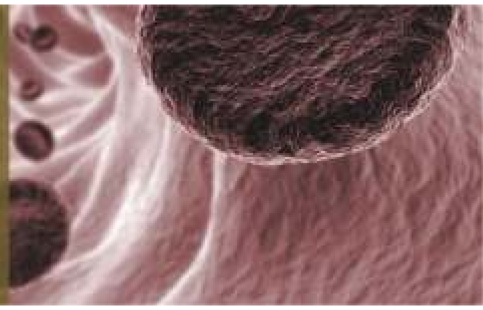
4.6.1 Government Initiatives

In the late 1990s, Singapore identified the biomedical sciences as an area with tremendous growth potential. Between 2000 and 2005, it put in place the key building blocks to establish a core biomedical research base. In the second phase of the initiative (2006-2010), it has been focusing on strengthening its capabilities in translational and clinical research. It has put serious money into these efforts, with a commitment of $9.8 billion for the four years ending December 2010.

4.6.2 Research Base

Singapore has established a strong scientific foundation with seven research institutes and five research consortia covering clinical sciences, genomics, bioengineering, molecular/cell biology, medical biology, bioimaging and immunology. It has also made significant progress in translational and clinical research; dedicated Investigational Medicine Units, many of them co-located with institutes of higher learning, conduct early-phase trials in public hospitals, while the Singapore Clinical Research Institute focuses on later-stage trials.

Singapore's Health Sciences Authority (HSA) is actively involved in defining new regulatory frameworks and pursuing new areas of research in regulatory science.



4.6.3 Biotechnology Clusters

Singapore has built a major biomedical research hub that co-locates public-sector institutes and private-sector corporate laboratories. It is currently expanding the Biopolis (as the hub is known), with a 460,000 square-feet expansion that will bring its total research space to more than three million square feet.

4.6.4 Regulation

Singapore's Health Sciences Authority (HSA) is actively involved in defining new regulatory frameworks and pursuing new areas of research in regulatory science. The HSA has forged strong links with many of the world's leading regulatory agencies. In October 2009, Singapore was also accepted into the OECD's Mutual Acceptance of Data framework, which means that data from GLP-compliant pre-clinical trials conducted in Singapore can be accepted by 30 OECD and non-OECD members, including the US, EU and Japan.

4.6.5 Manufacturing Expertise

Singapore has a strong track record in both small-molecule active pharmaceutical ingredient and secondary manufacturing. It is also building a substantial biologics manufacturing base. Baxter, GlaxoSmithKline Biologicals and Roche have already opened biologics manufacturing plants in Tuas Biomedical Park, and Lonza is currently constructing a cell therapy manufacturing plant which should be operational in 2011.

4.6.6 Access to Talent

Singapore has worked hard to develop its expertise in the biosciences. Some 4,000 researchers are now engaged in biopharmaceutical R&D, including about 1,200 foreign nationals. In 2001, Singapore's Agency for Science, Research and Technology (A*STAR) also launched a national scholarship programme to fund the education of 1,000 postgraduate students at the world's top universities. To date, the agency has awarded more than 500 biomedical sciences scholarships, and more than 100 recipients have returned to work in the city-state after completing their PhDs. A*STAR has likewise introduced various awards to attract bright young researchers and develop a cadre of local clinician scientists.

4.7 SOUTH KOREA

In the late 1990s, the South Korean Government launched a programme to make the country one of the top biotechnology powers by 2010. South Korea's life sciences industry is now the fourth largest in Asia. But the government is not resting on its laurels; in 2006, it announced a new initiative called "BioVision 2016" to invest another $14.3 billion in the industry, with the aim of turning it into a $60-billion market over the next 10 years.

4.7.1 Government Initiatives

In 2008, the government spent some $930 million supporting biotechnology research and commercialisation, as part of its BioVision 2016 programme. It had three key objectives: strengthening the scientific infrastructure to emphasise bioinformatics, nanobiotechnology and synthetic biology; fostering the globalisation of the biotechnology industry; and creating biotechnology clusters.

4.7.2 Regulation

The Korean Food and Drug Administration (KFDA) has established a clear regulatory approval pathway for biosimilars, with an amendment to the "Rules Governing the Approval and Examination of Biological Products" (effective as of July 15, 2009), which establishes the definition of a biosimilar, and the publication of guidelines for the evaluation of biosimilars on July 27, 2009.

4.7.3 Research Base

South Korea is still at the forefront of stem cell research, despite the setback that occurred in 2006 when one of the country's leading researchers was charged with using fraudulent data. It also aims to become a global leader in biosimilars. In July 2009, shortly after the regulatory pathway for biosimilars was established, South Korean electronics giant Samsung announced that it would invest $389 million in biosimilars over the next five years. Other significant developments include the successful completion of a South Korean clinical trial of a biosimilar version of Enbrel produced by Taiwanese drug development company Mycenax Biotech, and an agreement between US-based Hospira and Celltrion to develop and market eight biosimilars.

4.7.4 Biotechnology Clusters

South Korea is currently investing about $5 billion in the construction of two high-tech medical-industrial centres. The bigger of the two – the Osong Bio-Health Science Technopolis, based in Cheongju (about 60 miles south of Seoul) – is scheduled for completion in 2012. It will house the Korean Food and Drug Administration (KFDA), Korean National Institute of Health, Korea Centers for Disease Control and Prevention, and various other government bodies as well as private companies.

4.7.5 Clinical Trials Expertise

South Korea has earned a solid reputation as a place in which to perform clinical trials. It has highly trained physicians and a good IT infrastructure, as well as plenty of urban hospitals (which make it easy to recruit trial patients).

4.7.6 Investment Incentives

The South Korean Government offers foreign investors in high-tech companies a wide range of tax incentives, including tax credits on corporate and personal income tax, acquisition tax, registration tax, property tax and aggregate land tax. It also operates a number of free trade zones and free economic zones.

4.8 TAIWAN

The Taiwanese Government formulated a "Promotion Plan for the Biotechnology Industry" in 1995, under which it enacted various laws and regulations relating to biotechnology; and encouraged the private sector to invest in R&D, technology transfer, and the training and development of personnel. It has stepped up these efforts during the past decade, and a growing number of businesses are now focusing on the sector.

4.8.1 Government Initiatives

Between 2000 and 2008, the Executive Yuan's National Development Fund invested $394 million in the biotechnology industry. Then, in March 2009, the Executive Yuan launched the Taiwan Biotechnology Takeoff Package, aimed at boosting the sector and capturing a share of the global market. This package comprises four key elements: strengthening the country's translational research and commercialisation skills; establishing a biotechnology venture capital fund; founding a Food and Drug Administration; and creating an integrated biotech incubation centre.

4.8.2 Biotechnology Clusters

Taiwan is establishing a number of large science parks focusing on biopharmaceutical research and manufacturing. It aims to build two clusters: one in the Southern Taiwan Science Park (STSP) and the other in northern Taiwan's Hsinchu Biomedical Science Park (HBSP). The STSP is already open for business, with nearly 20 biopharmaceutical companies focusing on vaccines, bioagriculture, biomedical inspection and floriculture, as well as companies that manufacture medical equipment and devices. Construction of the HBSP started in October 2009 and, once the park is complete, it is expected to host about 30 biopharmaceutical manufacturers.

The government also plans to create an integrated incubation centre in a third park, to be called the National Biotechnology Development Park, in the Nangang District of Taipei City. This will contain an incubation centre, animal testing centre, biopharmaceutical R&D service centre and legal information centre.

4.8.3 Regulation

The Taiwanese Department of Health established a new Food and Drug Administration in January 2010 to improve the oversight of food and medicines, and link the standards of the local BioPharma industry more closely with those of the international market. Taiwan also set up an Intellectual Property Office in 1999, but it has relatively little experience in this area.

4.8.4 Investment Incentives

Companies in the biomedical and special chemical industries sectors meeting the criteria of the Incentive Programme for the Manufacturing and Technical Service Aspects of Major Emerging Strategic Industries may choose to take advantage of a five-year tax holiday or shareholder investment credit. The Taiwanese government has also established a series of tax breaks and other incentives to encourage local and overseas biopharmaceutical entities to invest in Taiwan. These include tax breaks for expenditure on R&D, training and automation, and energy-saving equipment; corporate and personal tax credits for investors in companies in newly emerging strategic industries; import tax exemptions for imported machinery and equipment; and tax exemptions for certain kinds of income, where a company sets up its headquarters in Taiwan.

4.8.5 Access to Capital

The government plans to create a $1.9 billion biotechnology venture capital fund to attract private capital to the sector and give financial assistance to start-ups. The Executive Yuan's National Development Fund will contribute 40% of the funding, with the remaining 60% to come from private investors.

4.8.6 Access to Talent

Various Taiwanese universities have established life sciences departments to cultivate the highly educated professionals the BioPharma industry needs. The Academia Sinica (Taiwan's foremost research institute), National Health Research Institutes, Industrial Technology Research Institute, Development Centre for Biotechnology and other related organisations also provide training. The number of university graduates with degrees in the biopharmaceutical and biotechnological sciences has risen accordingly; it was 53,283 in 2008, with 12,660 having obtained masters or Ph.D. degrees. However, the country is still very short of trained personnel, particularly those with international experience.

Key Recommendations for the Medium Term (2015)

Chapter 5

If India's biopharmaceutical players are to compete effectively on the global stage and capture a significant share of the biopharmaceuticals market, they will need to make a substantial investment in building new manufacturing capacity. However, the GOI will also need to play its part by providing the necessary enabling environment.

India already has a strong pharmaceutical manufacturing base. It is thus well positioned to capitalise on the opportunities arising from the "patent cliff" and growing demand for biosimilars. But the domestic market will remain relatively small in the near term, so India's efforts should be directed towards becoming a global manufacturing hub.

If India is to capture 10% of the global biosimilars market by 2020 – the goal for which we believe it should aim – the private sector will have to invest a considerable amount of capital in building the necessary manufacturing capacity and skills base. The GOI will also need to provide the environment required to enable that expansion. This chapter covers the infrastructure improvements, fiscal incentives, regulatory changes and policy initiatives that we believe will be crucial. We estimate that they will require an investment of at least $1 billion over the next five years. We have divided our recommendations into six sections: R&D; manufacturing and commercialisation; human capital; the regulatory framework; innovation; and intellectual property.

5.1 RESEARCH & DEVELOPMENT

5.1.1 Build Protein Characterisation Laboratories and GLP-Certified Animal Study Facilities

Foreign regulators require evidence that the product development process has been performed to the applicable standards as part of the regulatory submission. The product development process includes both the method of manufacture and high-end bioanalytical product characterisation to verify that the method being used delivers a product that is consistently comparable in quality to the original drug. The product that is manufactured through such a "validated" process must then be tested in a "GLP-certified" animal laboratory and the data from these tests must be submitted, together with data on the process and analytical characterisation of the product, to the regulator for permission to conduct a clinical trial.

India currently has very few GLP-certified animal laboratories and only one GLP-certified protein characterisation laboratory at the National Centre for Biological Sciences (NCBS) in Bengaluru. The industry needs at least four more such laboratories, all of which will have to be GLP-certified. The GOI should therefore fund the construction of the necessary facilities in national scientific institutions and laboratories as well as in CROs. This would not only support two of the most crucial stages in the biopharmaceutical product development cycle, it would also help to promote collaboration between academia and industry.

In addition, the GOI should offer duty waivers or other incentives to encourage existing service providers to branch into biologics development, using the Research-as-a-Service (RaaS) model – which would, in turn, attract multinationals wanting to outsource such activities and improve revenue generation. India's public institutions are not equipped to provide such services because they do not have sufficient understanding of the regulatory requirements.

5.1.2 Create a National Animal Breeding Facility

India needs a "National Animal Breeding Facility" to produce high-quality animals for pre-clinical studies and to generate certified data that meet international standards and scrutiny. The Chinese Government has set up two dozen primate breeding facilities, but many Indian firms are currently breeding their own animals. In January 2010, the GOI announced plans to establish a large centre to breed dogs and monkeys for use in clinical research, and the Department of Pharmaceuticals has invited expressions of interest from both public and private institutions with relevant biomedical expertise. However, at least one rodent facility and two large-animal facilities are required to develop MAbs and biosimilars, and it takes at least two years to build a rodent facility, so the GOI should start creating this infrastructure immediately.

It should also set up quarantine centres for imported animals at the places where they are imported and make provision for inspecting these facilities properly. The Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), acting under the aegis of the Department of Forestry, currently controls all animal facilities, including registrations. But it lacks the expertise to conduct proper scientific inspections and is under-staffed. The GOI should therefore equip the CPCSEA to match international standards.

5.1.3 Expand Viral Testing Facilities

India does not have enough facilities for testing and evaluating the viral safety of biologics derived from characterised cell lines of human or animal origin, in compliance with International Conference on Harmonisation guidance ICH Q5A (R1). The GOI should therefore support the construction of more such facilities. The Indian regulatory agencies should likewise start requiring viral clearance as a pre-condition for approval of all domestic biomanufacturing plants.

5.1.4 Provide Financial Assistance for Ensuring Compliance with Global Standards

All facilities for characterising proteins, breeding animals, conducting animal studies and performing viral testing will need to operate to international standards. So the GOI should also provide financial assistance with the cost of hiring consultants to advise on the global regulations.

5.1.5 Promote the Development of Pre-clinical Service Providers

Very few centres for offering services such as outside toxicology and bioanalysis currently exist – and they are too few to cater for the increasing number of early drug candidates being developed by companies that want to outsource these activities. One senior industry analyst estimates that the market for toxicology and bioanalytical services is worth about $12 billion, and that only 15% is outsourced. The GOI should thus provide soft loans to help pre-clinical service providers establish such facilities, as well as ensuring that subsidised infrastructure support is available.

5.1.6 Provide Practical Support for Clinical Trials

Any Indian biosimilars company that wants to sell a biosimilar in a regulated market will be required to conduct clinical trials of that biosimilar against the reference product in the country concerned. This represents a major financial risk, so the GOI should provide assistance in engaging consultants to design and execute world-class trials.

5.1.7 Simplify the Procedures for Importing and Exporting Biologics

The procedures for importing comparator drugs, test materials, Genetically Modified Organisms (GMOs) and Living Modified Organisms (LMOs) into India for research purposes, and for exporting biologics out of India, for clinical studies in other countries, are currently very cumbersome. India also lacks cold storage facilities (and most biologics are heat-sensitive).

The GOI should therefore simplify the process for importing biological samples and participate in negotiating government-to-government treaties for handling biologics (which would make it easier to export biosimilars that are manufactured in India). In addition, it should encourage the construction of cold storage throughout the supply chain and mandate faster clearance times at Customs to avoid loss of material in transit.

5.2 MANUFACTURING & COMMERCIALISATION

The domestic market is too small to sustain a significant amount of manufacturing capacity by itself, and most Indian biopharmaceutical companies are not yet ready to launch their products in developed countries with highly regulated markets. So the best way for India to become a manufacturing hub is to encourage multinationals to set up manufacturing facilities here, both by building the necessary physical infrastructure and by providing a commercially attractive environment.

Various kinds of infrastructure are essential to support the manufacturing of biologics. They include an uninterrupted power supply (since power failures can result in the contamination or death of the production organism or other process deviations); potable water to maintain sterility and yields; common effluent treatment plants so that individual sites need only invest in primary treatment (rather than zero discharge); and good road and rail freight links, especially in semi-urban and rural areas.

5.2.1 Create a Single-Window System for Approvals and Clearances

A biopharmaceutical company that wants to establish a manufacturing site in India must secure various approvals and clearances governing land use, access to power and potable water, and so forth. Different local state authorities may administer these regulations, and different states may have different tariff structures. The GOI should ensure that the format, content and interpretation of all such regulations is common throughout the country and introduce a single-window system for securing all clearance and approvals from the various central and state agencies.

5.2.2 Introduce Flexible Pollution Controls

Under the current rules for controlling pollution, a biopharmaceutical manufacturer must secure approval from the local pollution control board whenever it changes its product mix or yield, even if there is no net increase in the total amount of effluent. However, biopharmaceutical products typically generate easily biodegradable, aqueous effluents. The regulations should therefore be revised to allow for flexibility in terms of changes in product mix, especially when those products use the same process flows and generate the same amount of effluent.

5.2.3 Invest in Better Transport Links and Cold-Chain Facilities

The GOI should also invest in constructing more roads and ports, especially in areas of intensive biopharmaceutical manufacturing such as Mumbai, Pune, Goa, Hyderabad and

Bengaluru, and in creating a distribution network for the temperature-controlled transportation of biologics throughout the supply chain. At present, only a few airports like Chhatrapati Shivaji International Airport in Mumbai and Bengaluru Airport have temperature-controlled areas, and Indian pharmacies frequently suffer from power cuts or shortages.

5.2.4 Provide Fiscal Incentives

Any biologics manufacturer planning to enter the highly regulated markets must have a plant that is capable of meeting foreign regulatory standards, but such factories are much more expensive than normal chemical plants. The GOI should therefore endeavour to attract such companies (as well as manufacturers of ancillary products like Heating, Ventilating and Air Conditioning (HVAC) systems, Water For Injection (WFI) systems, sterilisers, process tanks and purification systems) by providing investment incentives, either on a standalone basis or through dedicated special economic zones and industrial parks. This would help to build India's future manufacturing capacity.

However, financing is not the only challenge for companies investing in more expensive manufacturing facilities to match Western expectations. The additional expenditure means that it is more difficult for such organisations to compete in cost-conscious markets like India and other emerging countries. So the GOI should consider extending the weighted tax deduction of 200% on expenses incurred for in-house R&D to cover expenses incurred from outsourcing to CROs. It should also expand the definition of R&D expenditure to include incidental costs, whether incurred in India or elsewhere (e.g., consultancy fees for patents and product registrations in foreign countries).

In addition, it should permit accelerated depreciation on capital expenditure incurred in constructing R&D laboratories or manufacturing facilities that are either accredited by international certification agencies or approved by foreign regulators like the FDA and EMA. It should also re-set the clock on tax holidays, so that they start at the point when a company moves into profit (as is the case in Malaysia, for example), rather than at the date of commissioning; favour companies that make a material investment in the Indian BioPharma industry (be they indigenous or foreign-owned) in government tenders; and reduce import duties on high-end equipment purchased from international manufacturers to encourage the creation of state-of-the-art bioanalytical facilities.

5.3 HUMAN CAPITAL

India's BioPharma industry faces several workforce challenges, including a shortage of skilled job candidates and difficulties in retaining employees who possess the right skills because there are too few opportunities for growth. More and better-educated graduates and postgraduates, and better training programmes, are essential to overcome these problems – and both will require closer collaboration between academia, the industry and government.

5.3.1 Expand India's Capability in Toxicity Studies

India's strength in toxicological studies is limited; it needs to create a pool of some 1,000 well trained toxicologists. The GOI should therefore borrow from best practice elsewhere and set up training institutes in toxicity studies. (We estimate that the cost of training 100 people per year at CRO sites would be about $500,000 annually.)

5.3.2 Foster a Trilateral Relationship between Industry, Academia and Government

Collaboration between the industry, academia and government has recently improved, but more work is required to create clear lines of communication and coordinate the sharing of resources, thereby reducing the duplication of expenditure and effort. Stakeholders from all three sectors should meet regularly to keep each other informed about their activities in this regard. The GOI should also help Indian companies form partnerships with foreign universities.

5.3.3 Improve and Expand the Workforce Development Pipeline

India's universities should offer interdisciplinary undergraduate programmes covering the physical sciences, information sciences, translational research and other related subjects, as well as a core specialisation, since many of the new skills required to be innovative combine two or more disciplines (e.g., bioinformatics). They should also involve the industry in designing these programmes; place greater emphasis on laboratory skills, problem-solving skills and independent learning; and provide more tutoring and mentoring. The GOI can play a key role here by providing organisations like the National Institute of Pharmaceutical Education and Research (NIPER) with modern equipment. Meanwhile, the industry itself should support academia's efforts by providing internships to give students hands-on experience and prepare them more effectively for work life.

5.3.4 Establish Exchange Programmes, "Finishing Schools" and Scholarships

The GOI should establish exchange programmes with foreign training institutes and professional bodies [like International Society for Pharmacoepidemiology (ISPE) and Drug Information Association (DIA)] for local graduates. It should also create "finishing schools" to impart industry-specific skills – e.g., vocational training in biopharmaceutical plant operations, quality control and cGMP regulations – and devise an accreditation programme for the purposes of quality control, drawing on the industry for guidance. In addition, it should set up scholarships for new recruits to help defray in-house training costs, and absorb half their salary costs during the training period.

5.3.5 Increase Public Awareness about Career Opportunities in the Industry

Academia and the industry should collaborate to increase public awareness about the industry and the career opportunities it offers by holding careers fairs and job fairs, and creating and distributing literature on these opportunities – e.g., brochures, fact sheets and newsletters.

5.3.6 Provide More Training for Existing Employees

The industry is rapidly evolving, with the result that employees must constantly maintain and update their expertise. Training programmes should therefore be developed so that employees can acquire and strengthen the technical and soft skills they need to advance. Tax relief should also be provided on all expenditure on training made by biosimilars manufacturers.

5.4 THE REGULATORY FRAMEWORK

India is becoming an increasingly attractive place in which to conduct clinical trials, thanks to its large patient population, well-trained investigators and relatively low costs. It is also building a regulatory regime based on the International Conference on Harmonisation guidelines for GCP. However, before making any changes in policy, it is important to evaluate the lacunae in the existing policies. Very often, it is not a particular policy that is deficient but, rather, its implementation and enforcement. Patent law reforms and regulatory reforms both require knowledgeable people to implement or enforce them, for example. Hence the importance of building both capability and capacity in the various regulatory agencies.

5.4.1 Simplify the Procedure for Approving Biologics

The Drug Controller General of India (DCGI) and Central and State Drugs Control departments like the Central Drugs Standard Control Organisation (CDSCO) and Drug Regulatory Authorities (DRAs) are responsible for regulating most aspects of most biopharmaceuticals. However, in certain cases – e.g., where a product is developed or manufactured using recombinant-DNA technology – the approval of various other agencies or committees is essential. These include the Genetic Engineering Approval Council (GEAC), Recombinant DNA Advisory Committee (RDAC), Review Committee on Genetic Manipulation (RCGM), Institutional Biosafety Committee (IBSC) and many more such committees, at the state and district levels.

The approvals procedure is also very cumbersome. In most countries, a single central body is responsible for reviewing the data from animal studies and granting permission to start clinical trials. In India, by contrast, the IBSC reviews the data before forwarding it to the RCGM for additional reviewing and approval. The GOI should therefore review and rationalise the roles and responsibilities of the IBSC, RCGM and DCGI in order to streamline and simplify the approvals procedure.

5.4.2 Create an Independent Inspection Facility

The GOI should establish an independent inspection facility to audit the manufacturing and containment facilities of all companies involved in the production of recombinant drugs. This would help to ensure that Indian rDNA products are acceptable in the global marketplace. Establishing strong agencies for providing GLP and GCP accreditation, and monitoring compliance with the rules would likewise help to create a reliable network of service providers, which would in turn help to lower costs and attract foreign investment in domestic product development and manufacturing facilities.

5.4.3 Modify the Regulations on Process Validation

Under the current system, process validation batches must often be supplied prior to (or during) the clinical development phase. Large-scale batches are also needed to supply representative samples for clinical trials and export registration. These large-scale production lots are taken prior to receipt of a local manufacturing licence, which is issued only once a drug has been approved. However, biopharmaceutical products have long manufacturing cycle times. The regulations should therefore be changed to permit the use of product from validated batches for commercial launch in India after receipt of the manufacturing license, provided that the product is within its established shelf life. This would reduce development costs and time-to-market, and make the products more affordable.

5.5 INNOVATION

For every dollar India invests in innovation, China invests as much as 10 times that amount. The Chinese public sector also plays a key role in biopharmaceutical innovation, whereas, in India, the burden falls almost exclusively on the private sector. If India is to compete with China and other such emerging economies, and attract investment from the multinationals, it must adopt a much more proactive stance.

5.5.1 Provide Seed Funding for Innovation

The GOI should invest more money in setting up “venture” funds specifically to support all companies that create innovation in India, be they foreign-owned or locally owned. It should also encourage more innovation by providing seed funding to help indigenous biopharmaceutical companies license new technologies and intellectual property from other countries and build on them. Such funding could take the form of mixed grants and loans, with repayment of the loan element contingent on success.

The existing funding programmes provide support for companies that have products in the later stages of development, but there is a crippling paucity of risk capital for early-stage innovation. Moreover, assuming that such innovation will only be performed at academic institutes is too restrictive an approach. Start-up biotechnology companies can be effectively groomed to conduct directed research that yields useful insights into disease biology and novel targets with validation screens, but this does require substantive grant support from government funding agencies.

5.5.2 Construct Biotechnology Clusters

Setting up biotechnology clusters would also help to improve the sector’s productivity and competitiveness by leveraging the synergies of common areas of research and technology, supporting industries and human capital. Clusters foster innovation through increased ease of experimentation with new activities and asset configurations, and facilitate the commercialisation of research by increasing the visibility of opportunities.

India will probably need about five biotechnology clusters, each containing a molecular sciences facility, analytical laboratories, pilot production facilities and laboratories for pre-clinical work and nanotechnology research.

5.5.3 Promote Translational Research

The GOI should create incentives for academic institutions and companies working on translational research, and allocate funding for providing grants to encourage further work in this field.

5.5.4 Provide Financial Support in Specific Areas of Innovation

The GOI should provide grants or soft loans for expenditure incurred in filing patents in foreign countries.

5.6 INTELLECTUAL PROPERTY

India has made genuine efforts to develop an intellectual property rights regime that meets international criteria. It has adopted an approach which aims to balance the need for a regime that encourages investment in scientific and technological innovation with the performance of its social responsibilities, including the protection of public health, biodiversity and traditional knowledge. The creation of a clear framework is particularly important for the BioPharma industry, since lack of clarity often results in wasteful litigation.

5.6.1 Protect Innovation

The GOI has implemented various measures to ensure that India complies with the TRIPS Agreement. However, from an international perspective, further improvements are needed. These include resolution of the legal situation regarding data exclusivity (i.e., whether data submitted to the regulatory authorities for testing for approval should be protected); and government-funded training to strengthen the Indian Patent Office.
Addressing these issues would encourage the manufacturers of branded products to shift some of their production to India. A number of companies have already set up secondary manufacturing operations outside their home countries. Ireland and Singapore have already become global manufacturing hubs, for example, and India should aim to do likewise.

5.6.2 Approve the Bill Liberalising the Commercialisation of Intellectual Property generated in State-Funded Institutions

A revised draft law on the intellectual property rights of state-funded institutions was re-submitted to parliament for discussion in June 2010. The revised bill introduces relaxed norms for patenting inventions and also gives the GOI more control over the non-exclusive licensing of the intellectual property to third parties, if this is deemed to be in the public interest.

The bill – which was originally introduced as the "Indian Bayh-Dole Act" in 2008 – has gone through considerable changes during the consultation process. However, it is hoped that the bill will have the same effect as the original Bayh-Dole Act did in the US, in encouraging investigators and institutions to commercialise the intellectual property generated from state-funded research with direct financial benefits flowing to both parties. Biotechnology research often emanates from the laboratories of such state-funded institutions in India, so the lowering of the legal barriers to the commercialisation of that research can only be good. It would be particularly useful if the Department of Pharmaceuticals and the Department of Biotechnology could launch awareness programmes alerting the technology transfer officers of state-funded institutions to the new statute, once it is passed, so that they can advise their researchers on the resulting opportunities.

Key Recommendations for the Longer Term (2020)

Chapter 6

India will not only need to build a robust biopharmaceutical infrastructure, it will also need to become a source of innovation. There are at least five initiatives the GOI could implement to help the domestic BioPharma industry leverage its unique advantages.

We have made a number of suggestions for strengthening India's biopharmaceutical resources over the next five years. But will these be sufficient to help the country realise its aspirations? We believe not. If India is indeed to become the world's leading provider of affordable biopharmaceutical products by 2020, it cannot simply count on biosimilars and vaccines; it must also become a source of innovation. More specifically, it should aim to have at least 10 original biologics on the local market and at least two on the global market by 2020.

In other words, India cannot simply learn to play the game better; it must also change the game. We have identified five longer-term initiatives that we think would help it to do this – and we recommend that the GOI make an additional $1 billion available over the next 10 years to provide for them.

6.1 CREATE A FRAMEWORK FOR INTELLIGENT REGULATION

India should be bold and create a completely new framework for regulating biopharmaceutical products. The current regulations do not meet the standards of many foreign regulatory authorities – and the obvious answer would be to harmonise them with the regulations in the US or other developed countries. However, history has played a large part in shaping those regulations and it is sometimes difficult to put history aside, even when conditions have changed dramatically. The current paradigm of clinical development, with its reliance on double-blinded, randomised control tests, is one such instance. It was devised in the wake of the Thalidomide tragedy but has altered very little since then, despite the fact that it is inefficient and, arguably, unethical (in its use of placebos).

Two changes, in particular, will have a major bearing on how biopharmaceutical products should be regulated in the future:

- The development of diagnostic tests for distinguishing between patients with related but different disease subtypes. More than 40 such "companion diagnostics" are already on the market, and there are many more tests in the pipeline.

- The development of sophisticated remote monitoring technologies (e.g., ingestible microchips and implantable devices that can be connected to a wireless network).

These advances will enable healthcare providers to diagnose patients much more accurately and monitor them continuously, eliminating the need for some of the precautions that are currently required. The GOI should therefore establish a special committee to formulate a new framework for regulating biopharmaceutical products intelligently in an intelligent age. That committee should include experts in biostatistics, public health, pharmacovigilance, genomics and molecular diagnostics, and healthcare informatics.

6.2 INVOKE "MARKET PULL"

Various healthcare studies have documented the impact of "market pull" in stimulating the creation of world-class healthcare systems in emerging economies. Affluent patients in these countries typically have health insurance, suffer from many of the same chronic diseases as patients in the mature economies and are abreast of the latest medical advances. Venture capitalists and private equity players are therefore interested in reaching them, as are global vendors with local manufacturing, sales and delivery facilities.

India already has experience of this phenomenon; several high-profile physician-entrepreneurs are currently creating first-rate healthcare infrastructures in urban centres. But the GOI could boost the effect if it started investing much more heavily in the public health system, as China is doing. Providing healthcare for its citizenry should be motivation enough. However, such a move would have a welcome secondary effect in stimulating the market for biopharmaceutical products. (It is worth noting that China plans to spend nearly $500 billion a year on healthcare by 2017, and that the domestic biopharmaceuticals market is forecast to reach $61 billion in 2013, making it the third-largest market in the world.)

Public health in India remains a huge developmental challenge that needs serious attention from the nation's most strategic thinkers, and the support of political giants to ensure implementation. The Ministry of Health's flagship programme, the National Rural Health Mission (NRHM), is a step in the right direction, but it needs to be scaled up massively. The central government allocated $3.86 billion for healthcare in its annual budget in fiscal 2009, of which $2.3 billion was earmarked for the NRHM. Nevertheless, India's citizens still bear most of the financial burden, with public resources covering less than 20% of overall healthcare costs.

The mega-healthcare establishments serving the urban centres have proved that extraordinary economies of scale can be leveraged to make healthcare affordable at levels unimaginable in the West; take the case of Narayana Hrudalaya, which performs paediatric cardiac operations for just $1,400. These models must be taken to the countryside. The spread of India's telecommunications connectivity must also be leveraged to make the practice of telemedicine much more routine. Lastly, the logistics of healthcare delivery must be addressed with the same efficiencies the fast-moving consumer goods (FMCG) sector achieves today. Suitable infrastructure and financial incentives should be provided to those who lead the way.

The mega-healthcare establishments serving the urban centres have proved that extraordinary economies of scale can be leveraged to make healthcare affordable at levels unimaginable in the West...

6.3 SEEK OPPORTUNITIES FOR GROWTH THROUGH ACQUISITIONS

Indian companies in various sectors (e.g., steel manufacturing, automotive manufacturing and telecommunications) have shown great skill in acquiring surplus manufacturing capacity from other organisations, integrating it with their own operations and managing it profitably. The consolidation of the global BioPharma industry may present more such opportunities, as the leading manufacturers dispose off surplus plants – enabling the Indian BioPharma sector to acquire capacity in markets that are attractive.

The DoP could therefore contract an international consultant to scope out the opportunities on behalf of the indigenous industry. It could also offer Indian entrepreneurs financial assistance through guarantees, "round-off" capital and other debt instruments, where necessary.

6.4 ENCOURAGE HIGHLY SKILLED EXPATRIATES TO COME HOME

The BioPharma sector should capitalise on the "Indian Diaspora" by encouraging expatriate Indians with the relevant expertise to return to their home country. The IT sector has been very successful in attracting highly qualified professionals of Indian origin back to their homeland by holding "job fairs" in Silicon Valley and other regional clusters of IT expertise. The BioPharma industry should emulate these tactics.

The GOI can play a valuable role in helping organisations like ABLE and the Federation of Indian Chambers of Commerce (FICCI) create targeted seminars in high-density locations such as the UK, and US East and West Coasts, to advise professionals of the opportunities in India. The global consolidation of the industry – and the instability it is causing in some of these careers – should also weigh in India's favour.

6.5 LEVERAGE INDIA'S EXPERTISE IN HOLISTIC AND TRADITIONAL MEDICINE

Lastly, India should leverage its expertise in herbal and nutritional therapies, and other forms of traditional medicine. The Department of Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (AYUSH) and the Council for Scientific and Industrial Research (CSIR) have jointly created a database to codify this knowledge. The Traditional Knowledge Digital Library (TKDL) now contains nearly 805,000 Ayurvedic formulations, 98,700 Unani formulations and 9,970 Siddha formulations – information that could be invaluable in developing new biologics.

Conclusion

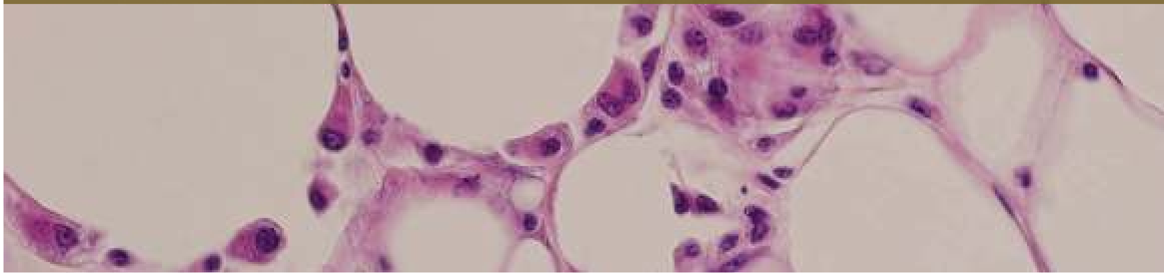
The future of medicine lies in biologics and diagnostics. Conventional drugs will continue to play an important role in the treatment of disease, but many of tomorrow's most innovative therapies will be biopharmaceutical packages comprising biologics and companion diagnostics.

India has a unique opportunity to make its name in this space. It can leapfrog over an entire generation of technologies and compensate for what it missed in the field of chemistry by making targeted investments to create the infrastructure required for researching, developing and manufacturing large molecules.

There are a few pockets of excellence in the public domain but, on the whole, the physical infrastructure for R&D remains minimal. There are critical gaps in the facilities required for everything from basic research to clinical development. So, if the domestic BioPharma industry is to succeed in securing 10% of the global biosimilars market and becoming one of the world's top five biosimilars producers – as we believe it should aim to do – it will need to make a substantial investment in new manufacturing capacity. However, the creation of physical assets cannot be seen in isolation from the ecosystem that is necessary to foster an expertise in biopharmaceuticals.

We have outlined the key steps we believe the GOI should take to help the country attain its vision of becoming the world's leading producer of biopharmaceutical products by 2020. Fulfilling this vision will take capital and commitment. But if India succeeds, it will not only generate significant additional revenues, it will also be serving a global population desperately in need of good, affordable medicines.

India has a unique opportunity to make its name in the biopharma space. It can leapfrog over an entire generation of technologies and compensate for what it missed in the field of chemistry by making targeted investments to create the infrastructure required for researching, developing and manufacturing large molecules.



Appendix 1: Exchange Rates

All currency conversions were calculated using the average exchange rate on 21 June 2010, as follows:

Australia	1 AUD	= 0.87335 USD
China	1 CNY	= 0.14666 USD
Euro Zone	1 EURO	= 1.23988 USD
India	1 INR	= 0.02178 USD
Israel	1 ILS	= 0.26221 USD
Korea	1 KRW	= 0.0008381 USD
Singapore	1 SGD	= 0.72314 USD
Taiwan	1 TWD	= 0.03113 USD

Appendix 2: The Global Pipeline for Biosimilars

The table below provides details of some of the biosimilars currently in development.

Biologic Drug Class	Biosimilar	Development Company	Stage of Development
Monoclonal Antibodies	Bevacizumab	Biocon	In development
	Trastuzumab	Biocon	In development
	Rituximab	GTC Biotherapeutics and LFB Biotechnologies	Preclinical
	Rituximab	Ranbaxy/Zenotech	In development, along with other monoclonal antibodies
	Various	Dr. Reddy's	Various biosimilars in development including monoclonal antibodies
	Four monoclonal antibodies against lung cancer, melanoma and pancreatic and breast cancer	Shantha Biotech	In development in the US
Insulin	Gensulin/SciLin	Bioton/SciGen	Marketed in Poland and Russia. Launches expected in Asian and Pacific markets, including China
	Human insulin and insulin analogues	Biocon	Marketed in India. Launches planned in Europe and the US

Biologic Drug Class	Biosimilar	Development Company	Stage of Development
Erythropoietin	JR-013	Kissei	Submitted for regulatory approval in Japan
	MK-2578 (darbepoetin alfa)	Merck & Co.	Phase 1 of clinical testing
Interferons	Biferonex (Interferon beta-1a)	BioPartners	Submitted to EMA as new drug but rejected
	CinnoVex (Interferon beta-1a)	CinnaGen	Marketed in Iran. Launches planned in European markets
	Interferon beta-1b	Merck BioVentures	In development
	Alpheon (Interferon alfa-2a)	BioPartners	Submitted to EMA but rejected
	Pegylated Interferon alfa-2b	Shantha Biotechnics	In development
CSF Drugs	Tevagrastim (filgrastim)	Teva	Approved by EMA in 2008. Launched in Germany, further launches expected including US in 2010
	Nufil (filgrastim)	Biocon	Launches planned in Europe and the US
	Nugraf (filgrastim)	Ranbaxy/Zenotech	Regulatory submissions planned in European markets
	Neukine (filgrastim)	Intas/Apotex	Phase I in Europe completed
	Filgrastim	Stada	Phase I
	Filgrastim	Wockhardt	In development
	INS-20 (pegfilgrastim)	Merck BioVentures	Phase I
	Neupeg (pegfilgrastim)	Intas/Apotex	In development
	Macrogen (molgramostim)	Ranbaxy/Zenotech	Regulatory submissions planned in European markets
Growth Hormones	Valtropin (somatropin)	LG Life Science/BioPartners	Currently marketed (South Korea) Approved (EU and US) Phase III (once-a-week, sustained release)

Source: visiongain

Appendix 3: Biologics with Patents due to Expire between 2010 and 2020

The following table lists all biologics with patents due to expire during the next decade. We have ranked them by date and, where the same product is made by more than one company, we have included details of the revenues generated by each company.

Product	Generic Name	Company	Therapeutic Subcategory	2009 Sales [$ millions]	Patent Expiry	Patent Expiry [ex US]
NovoSeven	eptacog alfa	Novo Nordisk	Anti-fibrinolytics	1,324	15-11-2010	28-02-2011
BeneFIX	nonacog alfa	Pfizer	Anti-fibrinolytics	98	11-02-2011	
Enbrel	Etanercept	Amgen	Other anti-rheumatics	3,493	23-10-2012	
Enbrel	Etanercept	Pfizer	Other anti-rheumatics	378	23-10-2012	28-02-2015
Neupogen	Filgrastim	Amgen	Immunostimulants	1,288	12-03-2013	22-08-2006
Humalog	insulin lispro	Eli Lilly	Anti-diabetics	1,959	07-05-2013	
Avonex	interferon beta-1a	Biogen Idec	MS Therapies	2,323	30-05-2013	31-12-2005
Epogen	epoetin alfa	Amgen	Anti-anaemics	2,569	20-08-2013	
Procrit/Eprex	epoetin alfa	Johnson & Johnson	Anti-anaemics	2,245	20-08-2013	
Cerezyme	Imiglucerase	Genzyme	Other therapeutic products	793	27-08-2013	
Rebif	interferon beta-1a	Merck KGaA	MS Therapies	2,142	31-12-2013	
NovoMix	insulin & insulin aspart	Novo Nordisk	Anti-diabetics	1,216	06-06-2014	31-12-2015
NovoRapid/ NovoLog	insulin aspart	Novo Nordisk	Anti-diabetics	1,825	07-12-2014	31-12-2017
Rituxan	Rituximab	Roche	Anti-neoplastic Mabs	5,620	31-12-2014	31-12-2013
Kogenate	octocog alfa	Bayer	Anti-fibrinolytics	1,238	31-12-2014	14-04-2009
Prevnar	Pneumococcal vaccine	Pfizer	Vaccines	287	01-01-2015	
Lantus	insulin glargine	Sanofi-Aventis	Anti-diabetics	4,293	12-02-2015	30-09-2013
Actemra	Tocilizumab	Roche	Other anti-rheumatics	44	07-06-2015	
Gonal-F/ Gonalef	follitropin alfa	Merck KGaA	Fertility agents	678	16-06-2015	
Neulasta	Pegfilgrastim	Amgen	Immunostimulants	3,355	20-10-2015	02-08-2015
Nimotuzumab	Nimotuzumab	YM BioSciences	Anti-neoplastic Mabs		17-11-2015	22-05-2016
Norditropin SimpleXx	Somatropin	Novo Nordisk	Growth hormones	824	15-12-2015	31-12-2017
Helixate	octocog alfa	CSL	Anti-fibrinolytics	563	31-12-2015	14-04-2009
Humira	Adalimumab	Abbott Laboratories	Other anti-rheumatics	5,488	31-12-2016	10-02-2017
Zostavax	herpes zoster vaccine	Merck & Co.	Vaccines	277	31-12-2016	
Tysabri	Natalizumab	Elan	MS Therapies	509	26-04-2017	
Tysabri	Natalizumab	Biogen Idec	MS Therapies	544	26-04-2017	
Provenge	sipuleucel-T	Dendreon	Immunostimulants		30-06-2017	23-12-2016
Victoza	Liraglutide	Novo Nordisk	Anti-diabetics	13	22-08-2017	
Cervarix	Human papillomavirus (HPV) vaccine	GlaxoSmith Kline	Vaccines	293	06-10-2017	31-12-2019

Product	Generic Name	Company	Therapeutic Subcategory	2009 Sales [$ millions]	Patent Expiry	Patent Expiry [ex US]
Avastin	Bevacizumab	Roche	Anti-neoplastic Mabs	5,744	26-02-2018	
Apidra	insulin glulisine	Sanofi-Aventis	Anti-diabetics	191	18-06-2018	
Forteo/Forsteo	teriparatide acetate	Eli Lilly	Bone calcium regulators	817	08-12-2018	
Remicade	Infliximab	Johnson & Johnson	Other anti-rheumatics	3,088	31-12-2018	
Remicade	Infliximab	Merck & Co.	Other anti-rheumatics	431	31-12-2018	
Xolair	Omalizumab	Roche	Other respiratory agents	572	31-12-2018	
Xolair	Omalizumab	Novartis	Other respiratory agents	338	31-12-2018	
Gardasil	Human papillomavirus (HPV) vaccine	Merck & Co.	Vaccines	1,119	24-01-2019	
RotaTeq	rotavirus vaccine	Merck & Co.	Vaccines	522	19-02-2019	
Erbitux	Cetuximab	Merck KGaA	Anti-neoplastic Mabs	971	13-05-2019	
Erbitux	Cetuximab	Bristol-Myers Squibb	Anti-neoplastic Mabs	683	13-05-2019	31-12-2016
Levemir	insulin detemir	Novo Nordisk	Anti-diabetics	978	17-06-2019	31-12-2018
Elaprase	Idursulfase	Shire	Other therapeutic products	353	03-09-2019	
Orencia	Abatacept	Bristol-Myers Squibb	Other anti-rheumatics	602	14-10-2019	31-12-2017
Herceptin	Trastuzumab	Roche	Anti-neoplastic Mabs	4,862	31-12-2019	
Pegasys	peginterferon alfa-2a	Roche	Interferons	1,528	31-12-2019	22-05-2017
PEGIntron	peginterferon alfa-2b	Merck & Co.	Interferons	149	31-12-2019	
Vectibix	Panitumumab	Amgen	Anti-neoplastic Mabs	233	08-04-2020	12-04-2022
Lucentis	Ranibizumab	Novartis	Eye preparations	1,232	30-06-2020	31-12-2022
Lucentis	Ranibizumab	Roche	Eye preparations	1,106	30-06-2020	
Botox	onabotulinumtoxinA	Allergan	Muscle relaxant, peripheral	1,310	21-07-2020	
Bapineuzumab (AAB-001)	Bapineuzumab	Pfizer	Nootropics		28-11-2020	
Synflorix	Pneumococcal vaccine	GlaxoSmith Kline	Vaccines	114	31-12-2020	31-12-2020
Replagal	agalsidase alfa	Shire	Other therapeutic products	194	31-12-2020	31-12-2020

Source: EvaluatePharma

A*STAR	Agency for Science, Research and Technology
ABLE	Association of Biotechnology Led Enterprises
ABT	American Board of Toxicology
ANDA	Abbreviated New Drug Application
AUD	Australian Dollars
AYUSH	Department of Ayurveda, Yoga and Naturopathy, Unani, Siddha & Homoeopathy
BLA	Biological License Application
CAGR	Compound Annual Growth Rate
CBER	Center for Biologics Evaluation and Research
CCH	Central Council of Homeopathy
CCIM	Central Council of Indian Medicine
CDER	Center for Drug Evaluation and Research
CDSCO	Central Drugs Standard Control Organisation
CEE	Central and Eastern Europe
cGMP	Current Good Manufacturing Practices
CHMD	Code for Human Medicines Directive
CNBG	China National Biotec Group
CNY	China Yuan Renminbi
CONICYT	Chilean National Commission for Scientific and Technological Research
CORFO	Chilean Economic Development Agency
CPCSEA	Committee for the Purpose of Control and Supervision of Experiments on Animal
CRO	Clinical Research Organization
CSIR	Council for Scientific and Industrial Research
CSIRO	Commonwealth Scientific and Industrial Research Organisation
CSL	Commonwealth Serum Laboratories
DABT	Diplomate of the American Board of Toxicology
DBT	Department of Biotechnology
DCGI	Drug Controller General of India
DIA	Drug Information Association
DNA	Deoxyribo Nucleic Acid
DoP	Department of Pharmaceuticals
DRA	Drug Regulatory Authorities
ELISA	Enzyme Lined Immuno Sorbent Assay
EMA	European Medicines Agency
EPO	European Patent Office
EU	European Union
FD&C	Food Drug and Cosmetic Act
FDA	Food and Drug Administration
FDI	Foreign Direct Investment
FICCI	Federation of Indian Chambers of Commerce
FMCG	Fast Moving Consumer Goods
GCP	Good Clinical Practices
G-CSF	Granulocyte Colony Stimulating Factor
GEAC	Genetic Engineering Approval Council
GLP	Good Laboratory Practices
GMOs	Genetically Modified Organisms
GOI	Government of India
HBSP	Hsinchu Biomedical Science Park
HPLC	High Performance Liquid Chromatography
HSA	Health Sciences Authority
HVAC	Heating, Ventilating and Air Conditioning
IBSC	Institutional Biosafety Committee
ICH	International Conference on Harmonisation
ICH-GCP	International Conference on Harmonization-Good Clinical Practices
IIC	Israel Investment Centre
ILS	Israeli New Shekel
INR	Indian Rupees
IP	Intellectual Property
ISM&H	Indian System of Medicine and Homeopathy
ISPE	International Society for Pharmacoepidemiology
IT	Information Technology
KFDA	Korean Food and Drug Administration
KRW	Korea (South) Won
LMOs	Living Modified Organisms
Mabs	Monoclonal Antibodies
MALDI	Matrix Assisted Laser Desorption Ionization
MCB	Master Cell Bank
MHLW	Ministry of Health, Labour and Welfare
NCBS	National Center for Biological Sciences
NDA	New Drug Application
NDRC	National Development and Reform Commission
NHI	National Health Insurance
NMR	Nuclear Magnetic Resonance
NRHM	National Rural Health Mission
OCS	Office of the Chief Scientist
OECD	Organization for Economic Cooperation and Development
PHS	Public Health Service
PwC	PricewaterhouseCoopers
Q-TOF	Quadruple Time Of Flight
R&D	Research and Development
RaaS	Research-as-a-Service
RCGM	Review Committee on Genetic Manipulation
RDAC	Recombinant DNA Advisory Committee
r-DNA	Recombinant DNA
SFDA	State Food and Drug Administration
SGD	Singapore Dollars
SME	Small and Medium Enterprise
STSP	Southern Taiwan Science Park
TGA	Therapeutic Goods Administration
TKDL	Traditional Knowledge Digital Library
TRIPS	Trade Related Aspects of Intellectual Property Rights
TWD	Taiwan New Dollars
US	United States
USD	United States Dollars
VAT	Value-Added Tax
WCB	Working Cell Bank
WFI	Water for Injection

We would like to express our gratitude for the input and help we have received from the following experts who so generously donated their time and effort to the project:

Amit Chatterjee	Managing Director, Sartorius India.
Anuj Goel	Chief Scientific Manager, Biocon.
Aparna Ganesan	
Arun Chandavarkar	Chief Operating Officer, Biocon.
Ashok Mundrigi	General Manager, R&D, Sartorius Stedim Biotech.
Dhananjay Patankar	Chief Operating Officer, Intas Biopharmaceuticals.
Helen Kay	Director, The European Communications Consultancy.
Jo Pisani	Partner, Pharmaceuticals & Life Sciences, Strategy PwC.
K. S. Rao	Associate Vice President - Corporate Development, Advinus Therapeutics.
K. V. Subramaniam	President & Chief Executive Officer, Reliance Life Sciences.
Kavitha Iyer Rodrigues	Director, Chief of Operations, Inbiopro.
Kiran Mazumdar Shaw	CMD, Biocon.
Lakshmi Annapurna	
M. S. Mahadevan	World Wide Manager, Biosimilars, Millipore Corporation.
Maria Kammerer	
Nandita Chandavarkar	Director - Operations, ABLE.
Narayanan Suresh	Group Editor, BioSpectrum.
Nisha Vishwakarma	Manager, India Pharmaceutical & Life Sciences, PwC.
P. M. Murali	Managing Director & Chief Executive Officer, Evolva Biotech.
Satya Dash	Chief Operating Officer, ABLE.
Shrikumar Suryanarayan	Hon. Director General, ABLE, & Chief Executive Officer, Translational Health Biosciences Cluster, Government of India.
Shriram Raghavan	Principal Coordinator, Operations, Evolva Biotech.
Simon Friend	Global Pharmaceuticals & Life Sciences Leader, PwC.
Sohang Chatterjee	Chairman & Chief Executive, Inbiopro.
Srinivasan Raman	General Manager - Manufacturing, Biocon.
Steve Arlington	Global Pharmaceutical & Life Sciences Industry Advisory Leader, PwC.
Sudhir Kant	President, Millipore Corporation.

The views expressed herein are personal and do not reflect the views of the organisations represented by the individuals concerned.

Selected

"India biotech Grows17% Registers Rs 14,199 cr." BioSpectrum (June 2010): 20-28.

"BioPharma to grow over 10% for next 5 years." BioSpectrum (June 2010): 36-38.

Burrill & Company. *Biotech 2009 – Life Sciences: Navigating the sea change*. 2009.

Datamonitor. *Biosimilars Series: Forecast Analysis. Beyond the Generics Horizon.* June 2009.

Europabio & Venture Valuation. *Biotech in the New EU Member States: An Emerging Sector. 2009.*

Evaluate Pharma. Top 100 Drugs (Biotechnology Worldwide Sales) - Ranked 2016. June 30, 2010.

"Overview of Singapore's Pharmaceutical and Biotechnology Industry." Singapore Biotech Guide 2010/2011.

Pisani, Jo. "Opportunities and challenges for biosimilars in the global market?" Paper presented at the World Generics Medicines Congress Europe, London, United Kingdom, February 23-26, 2010.

South East Universities Biopharma Skills Consortium Project: Final Report March 2010. http://www.reading.ac.uk/web/files/press/B01896_biopharma_skill_report.pdf

Visiongain. *Biosimilars and follow-on biologics: Global market out look 2012-2025.* 2009.

Wenzel, Richard G., ed. *American Journal of Health-System Pharmacy.* Vol. 65, Issue 14, Supplement 6 (July 15, 2008).

About ABLE

The Association of Biotechnology Led Enterprises (ABLE) www.ableindia.org is the apex body that represents Indian biotechnology. ABLE currently has a membership profile of over 200 organizations representing biopharmaceuticals, CRAMS, agribio, bioinformatics, clinical research and industrial biotechnology sectors as well as academic and research institutes, and service providers. The objectives of ABLE are to accelerate the pace of growth of the Biotechnology Industry in India through partnering with the Government, encouraging innovation, entrepreneurship and investment in the sector, providing a platform for domestic and overseas companies to explore collaborations and partnerships and forging stronger links between academia and industry.

About PricewaterhouseCoopers

PricewaterhouseCoopers' (PwC) Global Pharmaceutical and Life Sciences Industry Group is dedicated to delivering effective solutions to the complex business challenges facing pharmaceutical and life sciences companies. A global leader in serving the pharmaceutical and life sciences industry PwC has extensive experience working with companies on industry-specific strategic, operational, and financial issues. Our expertise includes assurance, tax and advisory services, as well as specialised capabilities in regulatory compliance, risk management, performance improvement and transaction support. In helping our clients, we draw on the full knowledge and skills of PwC's professionals. More than 163,000 people in 151 countries connect their thinking, experience and solutions to build public trust and enhance value for clients and their stakeholders.

Disclaimer

Design: Cariapa Design Associates, Bangalore. Print: Black White & Colour, Bangalore

chandru@strandls.com

Strand and Avadis

A Story of Technology Leadership from India in the New Millennium

September 2012

In late 2000, Strand Genomics Pvt Ltd had been launched from IISc as India's first example of sanctioned academic entrepreneurship. The promoters were 4 professors of computer science and the Society for Innovation and Development (SID) - the commercialization arm of IISc. It took about a year for the business plan of the company to be honed and taken to the investor community for both angel and Series A equity funding. During this first year Strand was operating as a consulting firm with biotech companies in the US as clients. Strand earned two and a half crores in revenues that year.

THE NICHE

The Human Genome Project had put out the first complete map of a human reference genome. Genomics had arrived. High throughput proteomics, cellomics, metabolomics, ... were just around the corner and the first signs of what is now called the "Fourth Paradigm or Data-Intensive Science" was already evident in research biology. Strand was going to build the platform that would enable life scientists to build a quantitative science, much in the same way as MatLab has empowered physicists and electrical engineers in their respective disciplines.

The team at Strand architected a modular software platform they called Soochika to denote that the strength of the platform would be to aid in data mining and classification. Data access, normalization and data cleaning were enabled for high-dimensional experimental data that that would be read off high throughput measurement instrumentation. State-of-the-art algorithms for clustering, machine learning, sorting and searching were programmed. Web crawlers and text mining were implemented for interpretation of results. Technology had to scale because a single experiment involving several samples, with several time stamps and high density scanners was already generating data sets with millions of readings and the trend was exponential. Stand knew how do all of this and with a core team of computer scientists of the highest caliber in the country and perhaps globally, Soochika was on its way. That Soochika had been built on the best-in-class algorithmics became evident a couple of years later when a team from Strand won the runner-up prize in the KDD Cup (the world cup of data mining).

The Scientific Advisory Board of Strand included Charles Cantor, Charles DeLisi and Eric Eastman – all three were thought leaders in genomics and suggested we build applications for the coming revolution in high density microarrays that could be used to study high throughput gene-expression, genotyping, sequencing etc. A brand new technology based on "flow through" microarray chips in an emerging microarray platform called "Metrigenix" was

our initial focus. We built primary (Sarani – for Oligo Design), secondary (Chitraka – for image analysis) and tertiary software (Soochika – for data analysis) for Metrigenix. Unfortunately Metrigenix never took off but our learning of this domain was immense. We quickly realigned our loyalty by building special features for the Affymetrix platform which was the clear market leader at that stage. However, Affy, like most instrument companies as we were to learn, held primary and secondary software as strategic and internally controlled software and encouraged third party software developers of tertiary analysis tools like Soochika access to their customers.

While Soochika had covered the "Access", "Analysis" and "Discovery" aspects of a bioinformatics workflow, it had not paid adequate attention to the data visualization component. Early feedback from scientists and advisors indicated that we needed to pay a lot more attention to visually guided analytics with dynamic linking of "views" of high-dimensional data in multiple renderings and effective GUI widgets for data filtering. We realized that to compete with the best out there we would need to plug this lacuna. However, this represented considerable engineering effort and needed a fresh look at our financial outlays.

DARSHEE and NMITLI

We heard about NMITLI (New Millennium Indian Technology Leadership Initiative) of CSIR after the first round of funding had been awarded. We took notice because a competitor had been awarded a large grant to work with academic bioinformaticians to create an Indian alternative to the successful platforms that global companies like Accelrys, Tripos, et al. had defined as the standard that structural biologists aspired to but could not afford at many institutions in India. This commendable project was called BioSuite.

What Strand required was much more modest. We needed resources to have a dedicated core team of implementors who could create a data envisioning platform that would do Edward Tufte proud (Tufte is a professor at Yale who is considered one of the great thought leaders in the field of data visualization). We wrote up a proposal and approached CSIR for a NMITLI award of around Rs 2 Crores. We were awarded a soft loan for this amount and after some painful negotiations around the Intellectual Property Rights of the resulting technology, Strand was on its way to adding serious "Visualization" capabilities to the Soochika platform.

With DARSHEE integrated into Soochika we had a platform that had a broader profile than just data mining. By mid-2003 it was time for rebranding. We considered many options but finally settled on AVADIS, an acronym for Access, Visualize, Analyze to DIScover. The analytics workflow for experimental biology from Strand was ready to be tested in the marketplace.

AVADIS in Japan

The Avadis Microarray product was initially targeted at research biology labs in the US market through distributors and application scientists and young business development managers from Strand. The going was tough and except for a few opportunistic sales, we ran into strong competition from open source as well as more established brands like GeneSpring, Spotfire and ArrayAssist.

Our first break, surprisingly, came in Japan. JETRO (the Japanese export and trade promotion office in India) offered us a beachhead in their Tokyo office and introduced us to a young agent who agreed to represent us on a small retainer plus commission basis. He had tried our technology and was familiar with the sector. The CTO/co-Founder of Strand, Prof. Ramesh Hariharan, traveled to Japan for conferences and gave several demonstrations of the technology. The Japanese are great connoisseurs of technology and seemed to really like Avadis. Hitachi and Kurabo both agreed to distribute our product in Japan with the latter also providing services using our product. Within about 6 months, Avadis had captured about 20% market share for microarray data analysis in Japan. The financial success was modest but the morale boost was immense.

The success in Japan led to an important business event for Strand. We attracted the attention of a Japanese strategic partner – MediBic. Yas Hasimoto, the MD-PhD founder of this fast growing and successful consulting firm in Japan, was interested in representing us in Japan. This was both as a distributor for the technology products based on the Avadis platform, as well as for some predictive QSAR (statistical models) for ADME (drug like properties of molecules) and Toxicity in drug discovery programs. The ADMET/QSAR modeling work had been driven by our Chief Scientist Dr Kas Subramanian and his team. MediBic also invested in Strand and Dr Hashimoto (a thought leader in pharmaco-genomics and molecular diagnostics) joined the Board of Strand. This event boosted the confidence, of our investors and shareholders, in the company and its future.

TDB funds AVADIS Enterprise

The Avadis platform had not stood still through this period. We made two major advances. The first was to recode major chunks of the platform to reach a modular architecture that would allow us to script workflows to capture a great deal of the drudgery and complexity in running analysis of data. This was also a way of convincing bench biologists to attempt to analyse their data without the use of a professional bioinformatician – a theme that serves us well even today.

The Avadis product had been created as a desktop client software that would work with Windows, Linux, MacOS and most flavours of Unix. However, we still found that there were some potential customers that we were losing because we lacked "enterprise application" support in the product. This was particularly true of large companies and core labs that preferred to hold their data at central servers and the client software tools like Avadis needed to be able to work in such environments.

We took the addition of enterprise architecture to Avadis as a commercialization proposal to TDB at DST. Since the desktop software was already functional and in the market, this seemed like an appropriate project for TDB to fund. We were given a soft loan again for about four and a half crores and set about building out this capability[1]. We also took steps to protect the Avadis brand and proceeded with registration of the trade mark as it went from Avadis™ to Avadis®. This was an expensive process for Strand and we spent well over US $100,000 to register the trade mark globally. This was prior to the Indian compliance with the Madrid Convention on Trade Marks and hence required distributed filings.

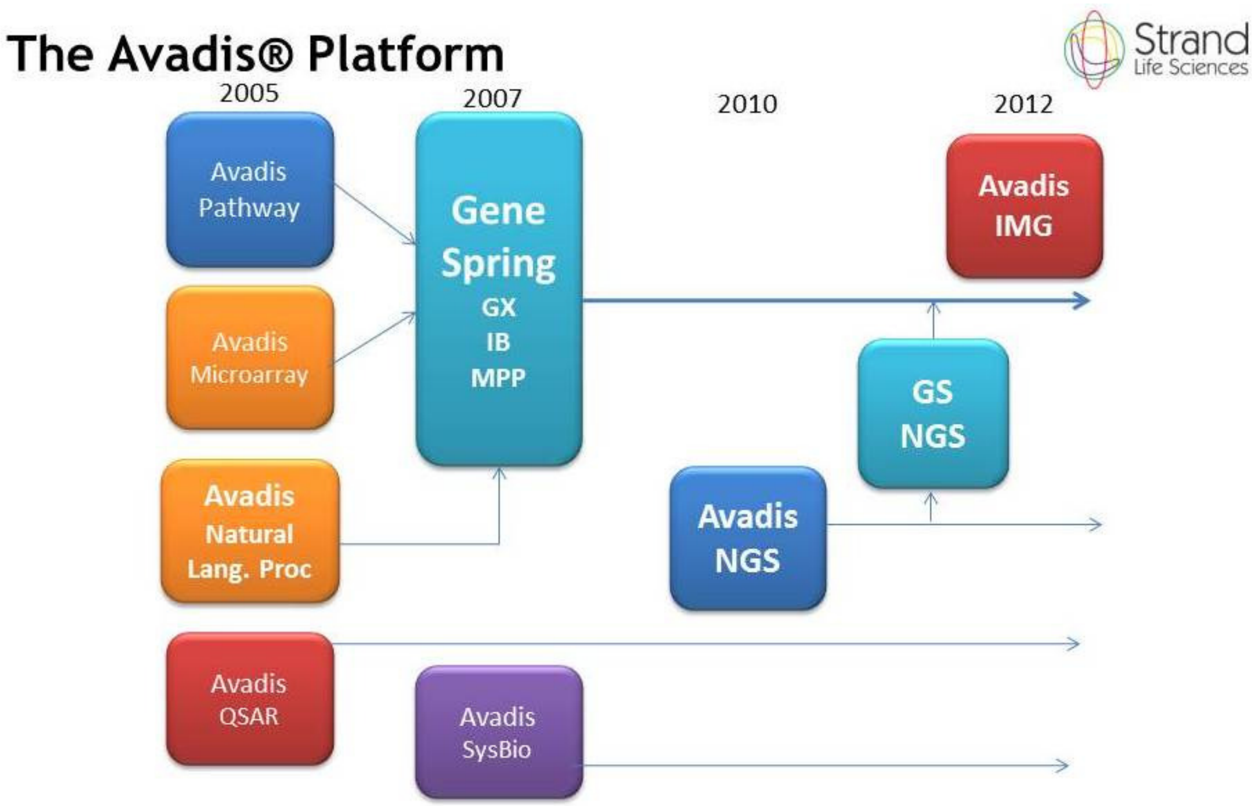
AVADIS® goes Global

In 2005, Strand was approached by Nick Roelofs of Stratagene with an offer to sell Avadis Microarray globally under the brand name of Array Assist®, a brand that they already owned. In addition, they wanted a new product called Pathway Architect® to be also built on the Avadis® platform. The latter required a completely new natural language processing engine that would read research documents (Pubmed/Medline abstracts and complete articles) and extract all gene, protein and ligand interactions and render pathway maps for visual interrogation of related research documents. This led to the creation of Grammatica – the NLP engine that formed the core of Avadis for Text Analysis and Mining.

History repeated itself in 2007, when Agilent (the original HP electronic and chemical / life sciences measurement company) acquired Stratagene. Agilent owned the GeneSpring® product which was the market leader at the time but needed an engineering overhaul (*software is the ultimate vain organism – if not constantly fussed over and tended to – it will wither away*). Agilent found it expedient to kill the Array Assist product and replace the innards of GeneSpring® with Avadis MicroArray and parts of Pathway Architect. Nick Roelofs, who had been the architect of the Avadis adoption at Stratagene was now head of life sciences at Agilent and had suggested a consideration of Strand and Avadis for GeneSpring.

[1] Strand recently repaid its last installment to TDB (in August 2012). The loan has been cleared from earnings.

How Soochika eventually became the world's market leader GeneSpring® is the story you just read. GeneSpring 12.0, released early this year, is perhaps the first true integrated biology (IB) platform that can read in all flavors of genomics data, proteomics data and metabolomics data and present integration through pathway constructs.



Avadis® has a life of its own outside GeneSpring as well.

I. Avadis NGS (www.avadis-ngs.com) is Strand's celebrated proprietary product for next generation (genomic) sequencing data analysis.
II. Avadis Chemistry (SArchitect) is our QSAR platform for use by structural, computational and medicinal chemists.
III. Avadis Systems Biology (Bio-Lego) is our quantitative systems biology platform that has been used for custom solutions delivered to global pharma and for our internal R&D on the Virtual Liver that Strand has developed.
IV. Avadis IMG (iManage) is our image management platform for microscopy images. This was developed in collaboration with researchers at UCSF and NCBS. The first commercial installation of this platform is in progress at Institut Curie in Paris and will be fully functional in Octber 2012.

Strand serves over 2000 laboratories around the world and, as the market leader in the commercial software business, has substantive market share. The quantification of market share of these technology products is tricky but here is an idea - we could just count the number of citations of GeneSpring vs the rest of the competition (including open source) and see how this works out. Thanks to Google Scholar this is easily done today and the table below captures the results. This may be a fairly crude analysis but it is likely that the trends are robust - as seen from the table GeneSpring dominates the commercial market and runs neck and neck with the leading open source platform Bioconductor.

Products	# of Publications		
	# as of Jul 2012	% of Commercial	% of Total
Commercial			
GeneSpring	14,800	59%	28%
Genomatix	4,730	19%	8%
Partek	3,570	14%	7%
GeneData	894		
GeneSight	435		
GeneSifter	635		
Free/Open Source			
Bioconductor	17,400		32%
SAM	7,860		15%
MeV	3,110		6%
Amada+	386		
ArrayTrack	312		
TOTAL	54,132		

We do not intend to comment on the reasons for the popularity of open source platforms but will note that there is often confusion between the requirements of “open access” and “open source”. While open source is often considered a low cost option, the price that biologists sometimes end up paying for adopting open source tools is often in productivity or for additional support. Perhaps most importantly, the dependence on professional bioinformaticians / computational biologists for even routine analysis is quite significant. One of the notable successes of the Avadis related suite of products has been in empowering biologists/biochemists to work with their own data - the endearing advertising tag line for Avadis is "analysis for the rest of us!"

Business Impact on Strand

The fiscal year 2007 turned out to be the turning point for Strand as we pulled to a break even position. Since then, Strand has grown on its own earnings and has now grown to a 150 employee strength. In FY 2012, Strand had a turnover of approximately 30 Crores with profit (EBITDA) at a 20% margin.

Strand has just over 30 Phds and MDs mostly trained in the life sciences and a few in computer science. It has around 40+ alumni of IITs from EECS backgrounds. Strand also has an enviable gender balance with around 48% women employees. Strand headquarters and development centre are in a contiguous 2,800 square meters of rented commercial office space in the Kirloskar Business Park which adjoins the Astra Zeneca Research Centre and the Columbia Asia Hospital, in Hebbal, Bangalore.

Strand invested 5 crores in a public-private partnership to help create a state-of-the-art genomics laboratory called Ganit Labs http://www.ganitlabs.in/ which has the facility to sequence whole genomes and has had some early success with sequencing the Neem genome and tongue cancer genomes. Strand also has a lab for liver enzyme assays (for studying hepato-toxicity of drug molecules) and for clinical diagnostics under contruction in Hebbal (currently incubated at C-CAMP, a DBT centre in the NCBS campus).

Private Equity Capital Investment in Strand is INR 26 Crores (USD 5.4Mn). Consider the capital efficiency!

What Next

The future challenges for Strand and Avadis will be in taking the analytics from the bench to the bedside (to the physician at the bedside). Presenting the molecular profile of the patient's biology as information the physician can actually use for clinical decision support is the challenge for the future. "Translational Omics" is certainly the playing field for Strand to compete in and Avadis will get manifested in fascinating new avatars (with new branding) as Electronic Health Records accommodate genome browsers with digital pathology and radiology images and annotations that indicate pharmacogenomic and therapeutic strategies that may suit the patient best.

Will it be technology leadership from India that delivers on these promises of the holy grail of personalized medicine? Why not indeed!

Strand 3.0: Harvesting a Dream for a Billion People

I am sure many of you saw the article about Ramesh Hariharan that accompanied his listing among the 30 most innovative CEOs to watch for. In case you missed it – here is the link of the article with the title of this note. http://www.insightssuccess.com/dr-ramesh-hariharan-the-scientist-harvesting-a-dream-for-a-billion-people/

We are at the cusp of the new year and Ramesh has been steering the ship for the last 18 months.

- The traditional bioinformatics products & services global business of Strand1.0 remains stable and an important profit centre of the company. GeneSpring and Strand NGS continue to be important platforms that the global research community uses and appreciates. With the addition of the RNA Seq capabilities to Strand NGS and cloud deployment, new levels of engagement are beginning to emerge.

- Our global bioinformatics business has now been complemented by our home grown clinical capabilities to address the needs of instrument companies and diagnostic centres across the globe with Strand Smartlab Solutions which includes our novel panels and assays, the Ramanujan informatics stack and our expert curation and interpretation. Our ability to achieve global accreditation for our tests also positions us as one the handful of solutons providers who can help a diagnostics lab anywhere in the world with end-to-end proprietary solutions as a SmartLab.

- Our clinical diagnostics service business in India has grown at a steady clip while maintaining a great brand of as the most scientifically trustworthy and reliable company in this line of high complexity genomic tests. In oncology we are clearly the leaders at the pole position and perhaps the only player in the region that is capable of innovating to improve affordability and access to tests. In rare diseases, the opinion of leading geneticists around the country is that our quality in testing, clinical interpretation and counselling of patients is unmatched and we have a great opportunity to grow our market share.

- Our R&D has delivered two important results for the company during this year – the breakthrough in liquid biopsy as well as a number of innovations to bring down the costs of our tests to squeeze out a healthy profit margin for the testing business. The latter will help us in sustainability and the former can be a game changer and a growth driver for the company as we get adoption first in monitoring and companion diagnostics and eventually in screening applications.

The imminent infusion of fresh capital in the New Year will see Strand's footprint of laboratories in India growing to cover a far greater geographical spread and provide a wide range of testing options for physicians and hospital partners. We are also committed to rolling out more products and services in reproductive health in the New Year. This is all going to mean that we have to be extra focused on execution and acceleration of growth.

Strand 3.0 has to be the story of scale and impact – harvesting a dream for a billion people. Let us be bold and determined because "Boldness has genius, power, and magic in it."

Wishing each of you and your loved ones the very best for 2018.

Warm regards
Chandru

strand New Generation Healthcare

July 5, 2015

Strand 2.0: The Healthcare Pivot

In September 2012, on a request from the Council for Scientific and Industrial Research in India, I authored an essay entitled *"Strand and Avadis® – A story of technology leadership from India in the new millennium".* The essay ended with a speculation of what was to come next from the Strand innovation engine and I wrote:

"The future challenges for Strand and Avadis will be in taking analytics from the proverbial bench to bedside (to the physician at the bedside). Presenting the molecular profile of the patient's biology as information the physician can actually use for clinical decision support is the challenge for the future. Translational Omics is certainly the playing field for Strand to compete in and Avadis will get manifested in fascinating new avatars (with new branding) as Electronic Health Records accommodate genome browsers with digital pathology and radiology images and annotations that indicate pharmacogenomic and therapeutic strategies that may suit the patient best.

Will it be technology leadership from India that delivers on these promises of the holy grail of personalized medicine? Why not indeed! "

In the second half of 2011, the founders of Strand held a series of brainstorming meetings that were moderated by L.R.Narayanan (a friend who is an investment banker in the life sciences and healthcare domain). The choices ahead for us were clear.

The first easy choice was to call it a day as a computational and systems biology company, sell the asset and move on. It would have been an end to a 12 year journey that had set out to build world class analytical tools for discovery research in life sciences. There were good reasons to feel satisfied with reaching that goal - we had become a world leading toolmaker, with bioinformatics products that set the gold standard for the slice of the world of science that was our addressable market. We had served up technologies and solutions for over 1400 research laboratories around the globe as clients (or about 30% global market share). Granted that our earnings were modest, but with a PPP (purchase power parity) factor taken into account, we were like a $25M revenue company in the west - a profitable, mid-sized company with close to 100 employees.

The second choice was to find an application that excited us enough to pivot the company to address the application. We first toyed with the well tested idea in India of transforming ourselves into a Contract Research Organization (CRO) with a potential collaborative drug discovery route with Pharma companies as our clients and sponsors. We went on this thread for a month or two and then abandoned the idea. While we had considerable knowledge about disease targets and a good handle on computational chemistry (predictive ADMET), we really did not have the medicinal chemistry expertise needed for drug discovery in house to justify this step. With the turbulent disruptions in discovery research in Pharma companies worldwide at the time – it also seemed a long shot to be able to enter co-development deals with Pharma companies.

In 2011, we had also entered the space of NGS (next generation sequencing) in a wholesome way. The Avadis® platform was extended to work with NGS raw sequence data. This needed alignment algorithms, specialized genome browsers and variant calling algorithms for identifying variants in the aligned genome when compared against the human reference genome. The Avadis NGS product (now called Strand NGS) was in the market and gaining decent sales traction. Strand had also invested close to a million dollars in a public-private partnership with the state and federal government funding agencies in India to help create a state-of-the-art genomics laboratory (including NGS instruments) called Ganit Labs http://www.ganitlabs.in at a local academic centre for biotechnology IBAB. By late 2011, Ganit labs had sequenced and analysed some paired normal samples of tongue cancer cells. In principle, we had a couple of building blocks in place to launch into personalized medicine.

Dr Meeta Patnaik, an MD based in Southern California with considerable experience in molecular diagnostics companies like Specialty labs, Lipomics, LabCorp, etc. was a good friend and we engaged her as a consultant to help us formulate a plan to build out an NGS based molecular diagnostics practice in India. Meeta arranged a number of calls and meetings and provided us with some rudimentary understanding of the specifics of the molecular tests that would be considered a menu to draw from, the regulatory landscape for these tests and the reimbursements. Both Binay Panda who heads Ganit Labs and Preveen Ramamoorthy who now heads diagnostics at Strand, were introduced to us by Meeta. Based on this emerging understanding and going out on a limb, we formulated a business plan to pivot into a healthcare company and went out to investors early in 2012.

In April 2012, we had a term sheet from an investor in Mumbai who did not seem to really believe that Strand could successfully pivot to healthcare and thought there was an opportunity for the legacy bioinformatics business to scale if we applied ourselves to building the right sales engine. It was evident that personalized medicine was not in the radar of the investment community in India at that point of time. So we looked westwards and found a more receptive investment community.

Since early 2013, we have raised close to 12M$ of funding from Biomark Capital and other investors to build out clinical diagnostic labs in India and the US and launch various diagnostic tests for cancer genome profiling and familial risk analysis for various inherited disorders. Strand is in the middle of a pivot from its foundation as a pioneering scientific intelligence company to a new generation healthcare company. Our legacy innovation business in bioinformatics software products and solutions continued to be a profitable business unit of the company.

Over the course of 2013, we built two labs in Bangalore – the production testing lab in Hebbal in the space vacated by the Centre for Human Genetics at the UAS alumni building and the second R&D lab within the translational centre run by the Mazumdar-Shaw Medical Foundation in the Narayana Health City south of Bangalore. In the fall of 2013, as the Hebbal lab was rolling out the first validation runs, we brought in a boutique marketing firm called Scion to help us with rebranding the company and announcing the pivot to the Indian healthcare establishment. The Scion team gave us a title "Strand Centers for Genomics and Personalized Medicine", a new logo seen at the top of this page, and the tag line "Deeper Insights. Better Decisions.".

It was not just a matter of branding and market messages, Strand needed to add a whole new layer of skills on the clinical and laboratory front. Starting with knowledge curation and engineering for better clinical interpretation, laboratory regulatory compliance skills, medical genetics, genetic counselling, pathology and a clinical sales force. Strand grew from a workforce of 100 to 230 in the course of 18 months – the HR team was kept busy. This was largely built organically through references from existing employees and with a couple of senior part-timers. The Strand senior management team went through a huge learning experience spending 16-18 hour workdays, countless hours in discussions, reading, visiting hospitals and physicians around the country. We brought in a senior medical director Dr Sudhir Borgonha MD, to lead the India Clinical business in early 2015.

Under Sudhir's guidance we now have a fresh product strategy and a crack at defining new guidelines for standard of cancer care using multi-genic testing in the country.

By mid-year 2014, we had built corporate relationships with a number of hospitals and clinicians in India and had built a complete workflow automation platform *StrandOmics* for going from raw sequence data to a clinical report. Our reports were well received by key opinion leaders and it was clear that we would eventually dominate this nascent service market in India, since we had some the best scientific minds and engineers in the country doing the R&D and delivery. Eventually this would amount to staggering volume of patients as the market for personalized medicine in India developed.

However, the market in which we needed to build future growth in the near term is the US which in spite of the complexity of the regulatory agencies and the reimbursement from private and public insurance, still remains the value driver in personalized medicine. We were indeed fortunate to attract Scott Storrer to Strand. As an executive with over two decades of experience in the payer, provider and diagnostics healthcare sectors, Scott was brought on board in July 2014 as President of Strand.

As President of Strand Life Sciences and CEO of our US subsidiary Strand Genomics Inc., Scott has delivered on several milestones and has positioned Strand as a fast follower to Foundation Medicine in the vertical of precision medicine with cancer related pharmacogenomics. Our CLIA registered lab in Aurora, Colorado is up and running since March 2015 and the company is well poised to ramp on sales in the US market. We recently launched new products in oncology – a 152-gene StrandAdvantage panel for solid tumors in the US and the StrandAdvantage tissue specific tests based on the 48-gene panel for standard of care and investigational use in treating cancer patients in India.

After having worked together for a year, it was time for passing on the baton of executive leadership of Strand, and as you all know Scott Storrer succeeded me as global CEO of Strand as of July 1st, 2015. The next piece of history of Strand will be created under Scott's leadership and we will have better scribes who can document "Strand 3.0 – a Global Personalized Medicine Company!"

PostScript:
The launch of the StrandAdvantage 152 test for somatic profiling was followed by sales momentum that got us at a volume of close to 70 tests a month by March 2016. However, the financial viability of the US business came into question as challenges in reimbursement by Medicare and the private payers (insurance) for these advanced genomic panels was under threat. Strand considered a reverse merger to acquire a listed company on Nasdaq with a fund reverse but had to unwind that as well due to public market risks. Scott left the company in early December and the founders had to step back into the pilot cabin and steer the mothership into calmer waters. The personalized medicine business in the US has been converted into a channel business where we license our IP and provide back end support to labs that deliver tests in that market. This is reminiscent of how we succeeded in discovery informatics (The Avadis Story – Epoch No. 2). Concurrently, we have also seen that the market in India for clinical genomics is moving forward more rapidly than anticipated. We already have the pole position in the oncology vertical and an exciting pipeline of breakthrough products in our R&D labs. So we will focus on this as a major growth driver in Strand 3.0. As of July 1st 2016 Professor Ramesh Hariharan, a co-founder of Strand, has taken on the CEO responsibility for Strand and I will go back to being Chairman and Managing Director.
Vijay Chandru, (Bangalore, August 7th 2016)

The Simputer: Technology with Heart[1]

Vijay Chandru
Professor, Indian Institute of Science
Trustee, The Simputer Trust
Founder Director, PicoPeta Simputers Pvt Ltd

This is an inside and personal account of the Simputer project written by one of the founders of the project, an academic and a technologist turned entrepreneur.

Technology with a social conscience

I have always admired Professor Amulya K N Reddy and in many ways have tried to emulate him.

In my childhood, I had heard stories of Professor Reddy and his wife Vimala as close friends of my parents, when he was a doctoral student in England. My next memories are from the late 1970's when I read some of his papers in EPW on energy and development. I did then and even now have maintained a deep interest in model based planning for development. I was a student at MIT working on my doctorate in applied mathematics with a minor in economics. To keep my social conscience and relevance to India alive I attended reading seminars offered by progressive faculty. Prof. AKN Reddy's papers were part of the reading. I was impressed that a powerful electrochemist at the Indian Institute of Science had made such a substantive transition into social sciences. Prof Reddy was certainly considered a leading intellectual and the pioneer in appropriate technology in India. In a personal encounter with Ivan Illich in Boston around 1980, I recall hearing him talk of Amulya Reddy with great admiration and affection.

In the 80's I had several opportunities to interact with Prof Reddy and his team at ASTRA in IISc. It was striking that while the development issues being addressed were quite elementary and sometimes carried out under very stressful conditions, the quality of the science and technology innovation was world class and certainly a source of great pride to the Indian Institute of Science. Somewhere around 1997 I recall having a conversation with Prof Amulya Reddy about the relationship of information and development and why there ought to be lessons that we could draw from his work on energy and development. He sent me home with a simple piece of advice: "You will have to work it out". This meant an AAA journey beginning with Analysis, leading to Advocacy and eventually to Action. The Simputer Story is really about this journey.

ANALYSIS ⟶ ADVOCACY ⟶ ACTION

- **(ANALYSIS) information and development (June-October 1998)**

  Prof Roddam Narasimha, the director of NIAS (the National Institute of Advanced Studies) invited me to join a group that was looking at Information Technology and Society in the context of the IT revolution that had gripped the city of Bangalore and the seeming disconnect

[1] This paper was prepared in the context of discussion meetings that were held in 2005 on the occasion of the 75th birthday celebrations of Professor AKN Reddy (1930-2006).

with the rural countryside even a few kilometers outside the city. The late Prof M N Srinivas, the eminent social anthropologist, Sanjay Dasgupta, the then secretary for information technology of the government of Karnataka (who is tragically also no more), a group of academics from computer and information sciences (I had also pulled in my colleague and long time collaborator Prof Swami Manohar), as well as a few industrialists (including Vinay Deshpande, a founder of Processor Systems India and NCore Technologies) were pulled together.

The discussions of a "Global Village" vision enabled by information technologies were lively and gradually a plan was hatched to hold a conference in November that would bring together a few outstanding individuals who we identified as the early thinkers and "doers" in tune with our concerns. The plan was also to run this conference in parallel with the trade show "BangaloreIT.com" and the launch of the ITPL tech park in Whitefield. Sanjay seemed to be playing a balancing act of the buzz around the new technology czars (Infosys, Wipro, etc.) and the social scientists looking for more well rounded development.

After a few sessions of the brainstorming, Manohar and I felt that we should write up the stream of consciousness emanating from the discussions of the global village group. This was an absorbing exercise for us and took about a month of long discussions, many walks around the East Campus of IISc and some spells of reading and debating on issues. What emerged was the "White Paper" entitled "The Global Village: Aspirations And Opportunities For Developing Economies"[2]. We shared the white paper with several colleagues and friends and amused and perhaps mildly irritated Prof Srinivas with the mixed metaphors that popped up all over the document (the naivete of scientists writing style!).

- **(ADVOCACY) The Bangalore Declaration[3] (November 4th, 1998)**

The white paper led to the idea that the conference would debate and adopt a declaration which eventually came. to be known as "The Bangalore Declaration on Information Technology for Developing Countries in the Global Village." Manohar and I drafted the declaration with lots of inputs from Prof M N Srinivas and Prof Roddam Narasimha. The draft was discussed through the global village conference and we incorporated the suggestions of the close to one hundred participants who engaged in several lively interactive sessions. The preamble of the declaration begins with

"Information Technology presents developing countries with a historic window of opportunity that enables them to create national wealth and break the cycle of poverty and dependence they have been caught in and leverage their wealth of human resources finally to secure a rightful place for themselves in the Global Village."

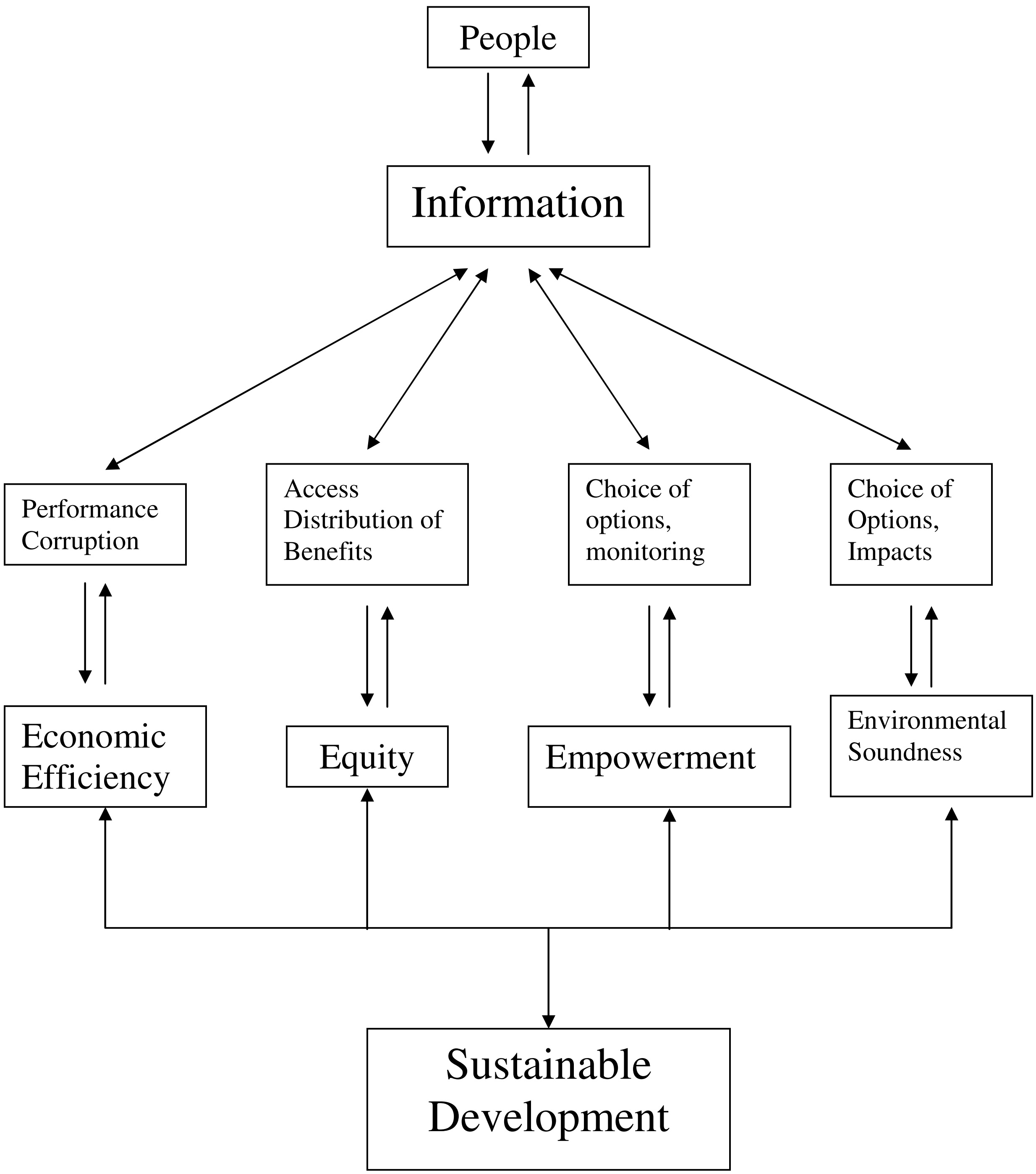
And ends with

"From all perspectives - humanitarian, moral, ecological, economic and social - Information

[2] The white paper can be accessed at http://www.csa.iisc.ernet.in/bangit/global/papers/chandru-manohar.html
[3] Published as a NIAS report and on line at http://www.csa.iisc.ernet.in/bangit/bangdec/bangdec.html

Technology has the potential to elevate the quality of human life everywhere. It is imperative that we, the citizens of this Global Village, seize this historic opportunity."

AKN was away on one of his frequent visits with his collaborators in IEI and I sent a copy of the white paper to him for comments. He returned in November called Manohar and me over, congratulated us for the effort and suggested the following diagram for developing our thoughts further. He was now willing to help us translate his experiences with energy to the field of information.

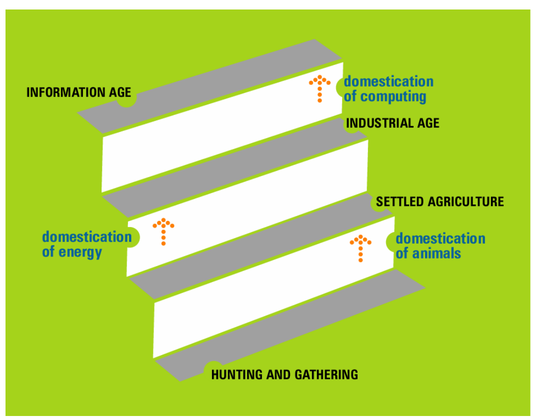


This could have led to a collaboration with AKN on some conceptual and theoretical investigation of information and sustainable development. But Manohar and I had already embarked on the next stage of "Action" and our patience with analysis and advocacy had run out. The Bangalore Declaration had many calls to action and we had already responded to one such call.

"Facilitate the rapid and equitable development of infrastructure for I.T., and in the process, ... facilitate the availability of a range of computers, including cable-TV, modified telephone instruments, and **rugged, low-cost, hand-held devices***, along with traditional computers, for purposes such as information gathering, communication and transaction processing."*

(ACTION) The Simputer

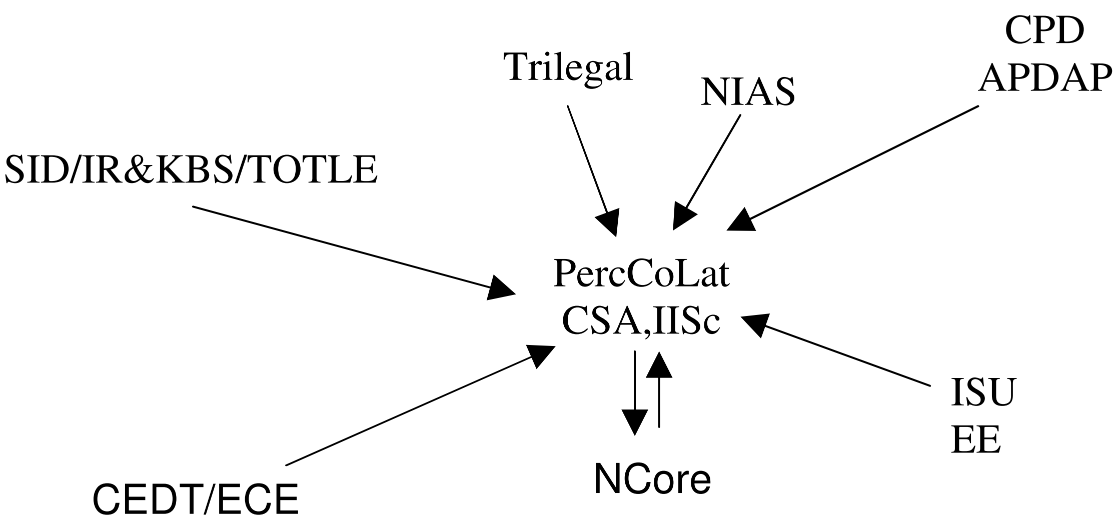
As technologists are wont to, Manohar, Vinay Deshpande and I soon tired of all the talking and declaring that was going on and resolved to actually work on some solutions. We met a couple of times in October 1998 and decided that we should work on a small handheld computer that could penetrate the Indian landscape the way transistor radios had. Since we could leverage the telecom revolution that had already been seeded by C-DoT and Sam Pitroda's vision, rural connectivity was no longer an issue. Vinay Deshpande had a paradigm application in mind – that of rural banking (micro-finance). The idea of a smart card integrated handheld was an early commitment. From a techno-historical perspective, we also believed that building a computer from scratch was a necessary rite of passage that Indian technologists needed to make any reasonable claims of having entered the information age.



Manohar, the prolific blogger (well before blogging became such a widespread phenomenon), quickly put together a high level spec document and christened the device "The Simputer". As he pointed out, the name Simputer obviously suggests that it is an abbreviation for "simple computer". It is also an acronym that expands to **S**imple, **I**nexpensive, and **M**ulti-lingual **P**eople's comp**UTER**. The framing of the Bangalore Declaration on Information Technology for Developing Economies clarified and fortified the concept of the Simputer and its role in information technology driven development by design. By November 1998, the Simputer team had expanded to include Professors V Vinay and Ramesh Hariharan from IISc and Mark Mathias and Shashank Garg (an IISc alumnus) from NCore. We pulled in Rahul Matthan as a legal counsel and formed the non-profit Simputer

Trust with Vinay Deshpande, “the elder” of the team as the Managing Trustee. The group began regular meetings.

The rest of the story is one of determination and camaraderie within the Trust. The hardware responsibility was given to the NCore team. The Alpha board had some minor problems and the debugging of the board had some interesting twists and turns and finally got resolved at IISc in PerCoLat (Perceptual Computing Laboratory) with Ramesh spending a night in the lab staring at a memory dump in Hexcode to pattern recognize that two wires had been switched. Porting the Linux kernel onto the board was the next challenge and the duo V Vinay – Swami Manohar pulled it off. The VV-SM pair went to make a number of huge software strides with IML etc. Ramesh was focused on getting the text to speech system “Dhvani” squaking in Kannada, Hindi, English and Tamil, while Vijay worked on the box with the product design students and the rapid prototyping centre APDAP at IISc. The following diagram gives an interesting view of the many departments and individuals at IISc/NIAS/Trilegal/NCore who were pulled together to make the Simputer a reality. The fact that this was all done with no funding at all from IISc or any other agency (we were deliberate in this frugality as we wanted the Simputer to be completely free of encumberances) is perhaps a testimony to the academic goodwill enjoyed by the trustees.

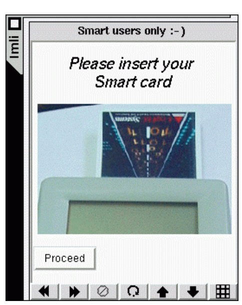


The Simputer was designed to be a low cost and portable alternative to PCs, by which the benefits of information technologies can possibly reach across the digital divide. It was to have a special role in developing economies because it offered the possibility that illiteracy is no longer a barrier to handling a computer, by using icons and text to speech technologies for multiple languages. One key to bridging the digital divide is to have shared devices that permit truly simple and natural user interfaces based on sight, touch and audio. The Simputer met these demands through a browser for a new XML language defined at IISc known as the Information Markup Language (IML). IML had been created to provide a uniform experience to users and to allow rapid development of solutions on any platform. The Simputer was to offer conversion of unicode text to speech. Demonstrating, its universality, the Simputer was to read and speak in several Indian languages in its initial release.

The Simputer, a hand-held computing device, was to be about the size of a palmtop though much more powerful. In its power and features it fit somewhere between a palmtop and a laptop computer. It was very clear to us, right from the outset, that the Simputer had to run on the Linux operating system. The Open Source movement had become very close to home. IISc probably had the largest install base of Linux amongst all universities in India. The CSA (Computer Science Automation) department had hosted Richard Stallman in 1993 when the Dunkel Draft of GATT-TRIPS (ultimately WTO) was being debated. Faculty of the CSA department and the National Law School pulled together an advisory conference with a view to warn the Indian Government of the consequences of accepting the Dunkel Draft. After 1993, the GNU products and the eventually Linux had almost universal acceptance. So, when it came to a choice of operating system for the Simputer, we stuck to Linux and ensured through the structuring of the license that all Simputer derivates would always stay with an open source operating system. The open source meme was persistent.

The projected cost of the Simputer was about Rs 9000 (US$ 200) at large volumes – this price point was arrived at as a compromise between engineering specs and the market's ability to pay (the price of a colour television set in 1998 was used as the benchmark). But even this, we knew, was beyond the means of most citizens. The Smart Card integrated feature of the Simputer enabled the Simputer to be shared by a community.

A local community such as the village panchayat, the village school, a kiosk, a village postman, or even a shopkeeper should be able to loan the device to individuals for some length of time and then pass it on to others in the community. The Simputer, through its Smart Card feature allowed for personal information management at the individual level for an unlimited number of users. The impact of this feature coupled with the rich connectivity of the Simputer can be dramatic. Applications in diverse sectors such as micro banking, large data collection, agricultural information and as a school laboratory is now made possible at an affordable price. We used the slogan "Radical Simplicity for Universal Access!" to describe the vision of the Simputer Trust.



The Simputer General Public License

The Simputer Trust is the coming together of academics and technologists from industry with a broad imperative of harnessing the potential of the Simputer for the benefit of all sections of society. The vision of this non-profit trust is to promote the Simputer, not as an end product but as an evolving platform for social change.

An innovative licensing mechanism has evolved through intense discussion within the Simputer Trust. We acknowledge the influence of the Free Software movement in this regard. However, the Simputer General Public License (SGPL) is more complex in many ways, partly due to the nature of hardware and partly to ensure that there are sufficient incentives for continuous innovation on top of the Simputer platform. Some clarifications on the similarities and differences between GPL and SGPL are presented in < http://www.simputer.org/simputer/license/sgplvsgpl.php >.

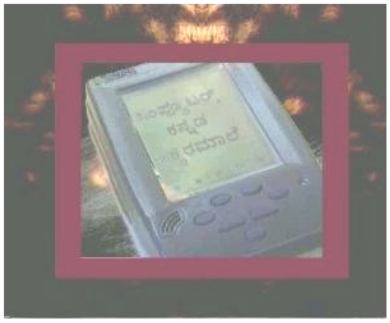
The Proliferation of Simputers: Making it happen

Rapid growth of knowledge can only happen in an environment which admits free exchange of thought and information. Indeed, nothing else can explain the astounding progress of science in the last three hundred years. Technology has unfortunately not seen this freedom too often. Several rounds of intense discussions among the trustees convinced them that the only way to break out of the current absurdities is to foster a spirit of cooperation in inventing new technologies. The common mistake of treating cooperation as a synonym of charity poses its own challenges. The Simputer Licensing framework is the Trust's response to these challenges.

The system software of the Simputer, since it is Linux based is under GPL. We have been working on a license similar to the GPL, but applicable to hardware. We realised, after considerable discussions, that hardware has significant differences that precludes the possibility of using a simple extension of the software GPL. We now have the Simputer General Public License (SGPL) that we believe to be a practicable license which at the same time facilitates the rapid spread of Simputers. The full text of SGPL can be accessed at the Simputer website www.simputer.org. We invite your comments on the SGPL. The hardware specifications of the Simputer can be downloaded only under SGPL. The SGPL permits anyone to build devices out of the downloaded specification. However, once a product is ready for commercialisation, one of two possible licenses need to be obtained from the Simputer Trust. These are

The Simputerised Device Manufacturing License
The Simputer Device Manufacturing License

The Simputer manufacturing License referes to a Core Simputer Specification, a functional description of the Simputer to be specified by the Simputer Trust and which evolves with the development of the Simputer. The highlights of SGPL are compiled in the appendices.



The Proptotypes are Launched (April 25th, 2001)

The Trustees shown below and Rahul Matthan (Trilegal), the legal counsel for the Simputer Trust who played a key role in defining the Simputer General Public License, received the nations hghest honour for innovation in information technology, the Dewang Mehta Award, the first time it was awarded in 2002.

Standing L to R: Swami Manohar, V Vinay, Vijay Chandru, Vinay Deshpande
Sitting L to R: Mark Mathias, Shashank Garg, Ramesh Hariharan

Courtsey: Frontline Magazine

The Simputer prototypes were launched by the Trust on April 25th, 2001 and the complete design details of the Simputer have been made available on the web site (www.simputer.org).

The Simputer was profiled by BBC's Click OnLine show, made the cover of Business Week, was written up in Wired, Technology Review and New York Times (below). The Simputer became a meme – a persistent idea that would represent technology innovation that uses "radical simplicity for universal access". The image below is of children in a school in Chhattisgarh using Simputers as part of their school curriculum. This lab was part of an educational initiative funded by the South Asia Foundation and implemented by PicoPeta Simputers in 2002-2003. The Simputer in the hands of a child is an idea that Nicholas Negroponte has re-invented in a Western idiom as the OLTPC project (one lap top per child) – while the Simputer was designed to be shared, the OLTPC is designed to be "personal". While the former is computing "as Gandhi would have invented", the latter is computing as donated by the invisible hand of Adam Smith! Recall: "The reason that the invisible hand often seems invisible is that it is often not there" - Joseph E. Stiglitz.

The most significant innovation in computer technology in 2001 was not Apple's gleaming titanium Powerbook G4 or Microsoft's Windows XP. It was the Simputer, a Net-linked, radically simple portable computer, intended to bring the computer revolution to the third world.
…
This is computing as it would have looked if Gandhi had invented it …. India has already succeeded in localizing cinema, satellite communications,cable television and radio. The Simputer is meant to do the same for the Internet.
…

Bruce Sterling, NY Times Magazine, December 9, 2001 "The Year in Ideas"

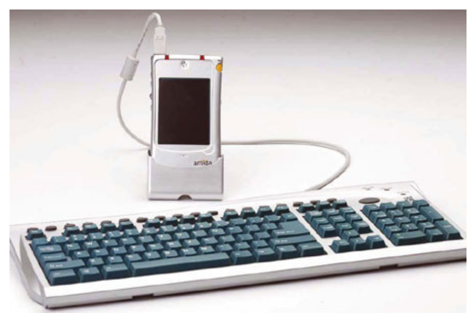
Epilogue

After the launch of the prototypes in April 2001, the trustees from IISc were empowered by IISc to launch a commercial venture to realize the proliferation of Simputers. This bold step by the Governing Council of IISc will be remembered as a turning point in the history of the institute. The Institute also granted full rights for the Simputer to the Simputer Trust which had the licensing scheme worked out under SGPL as described above.

As of this writing there are two licensees of the Simputer technology are taking the Simputer forward on a commercial basis. These are PicoPeta Simputers Pvt Ltd (http://www.picopeta.com/) which spun out from IISc, launched the Amida Simputer (cf. Appendix A4) in 2004.

The Amida Simputer

Introducing the Amida Simputer. It's time to Rewrite.



And Encore Software (http://www.ncoretech.com/) with the Encore Simputers licensed by the NCore trustees (NCore Technologies had been reinvented as a public company with a new name Encore Software).

Encore Simputers – Launched October 30, 2002

The story of this commercialization and its aftermath is still unfolding and will be told by others. This author moved on to guide another venture in scientific computing that was also blessed by the IISc Council – Strand Genomics Pvt Ltd.

APPENDICES

Appendix A1: Technology Choices and Challenges Tech Speak: The Simputer Specs

Hardware
- CPU 32-bit Strong Arm SA-1100 RISC CPU running at 200MHz
- 32 MB of DRAM
- 32 MB Flash for Permanent Storage (DOC)
- Display I/F 320x240 Monochrome LCD Display Panel

Interfaces
- Touch-panel Overlay on LCD Display used with a plastic stylus (Pen)
- Speaker and MIC Jacks Smartcard Connector
- RJ-11 Telephone Jack
- USB Connector

Dimensions
- Approximately 8cm x 13 cm x 2 cm
- Power Supply 3 AAA-sized NiMh batteries

System Software
- Operating System: GNU/Linux
- Soft-Modem Algorithms V.34/V.17 Data/Fax Modem Technology
- Perl/Tk scripting environment

Application Software
- imli: an IML browser
- Tapatap: Input method
- Internet access (Browser, Email, etc.)
- Dhvani: Text-to-Speech Software
- MP3 Player

Information Markup Language (IML) is an XML (Extensible Markup Language) application for describing the content and applications handled by a Simputer. The goal of IML is to enable handheld computers, that have limitations in the display size and input capabilities, to access and render content from the Internet.

Appendix A2: Highlights of SGPL

- Any individual or company can download the hardware specification, PCB layout details, the bill of materials, etc., henceforth called "Specifications" free of charge. The act of doing so binds the individual or company to the SGPL.

- Any derivative work has to come back to the Trust to allow for further dissemination. To allow the commercial exploitation of the derived work, a one year delay in putting back the derived work is permitted. This does not however preclude others from independently engineering a similar derivative work during this period.

- Any derivative work is subsumed as Specifications and hence, they are also governed by this same license.

- The word "Simputer" is trademarked and cannot be used without the permission of the Trust. If an individual or company is interested in using the word "Simputer" in conjunction with their products, they can do so only if their product conforms to certain rules that will be put up on the trust website (and which may undergo periodic revision). The product has to provide a visual clue to attest it being a Simputer by way of displaying a logo issued by the Trust.

- While recognizing the possibility of using the Specifications in application other than as a Simputer, the License deems that such derived work be called "Simputerized" products. The product description should state that the product is "Simputerized" and provide a visual clue on the product by way of displaying a logo issued by the Trust.

- Any commercial exploitation of the Specifications (whether Simputer or Simputerized) involves a nominal one time payment to the Trust. The payment will be $25,000 for companies in developing countries and $250,000 for developed countries.

Appendix A3:

Prof AKN Reddy's "TALK TO SIMPUTER TRUST" (15 MAY 2001)

- Your launch was one of the most inspiring events in my recent life.
- What inspired me was
  - Your teamwork
  - The technical excellence
  - The balanced emphasis on hardware and software
  - The attention to detail
  - The interaction with users
  - The social concerns
  - The articulation of a perspective
- Like many others, I felt that we were at the cutting edge of history and watching the salvation of India
- But the real reason for my excitement was that the whole atmosphere was reminiscent of the launch of ASTRA in 1974 over 25 years ago
- ASTRA was
  - An informal group of like-minded scientists
  - Inspired by a shared vision that our deep concern for deprived rural masses could be implemented through the instrument of S & T
  - Based on the synergy between social concerns and technical excellence
  - A group with intellectual excitement, sense of innovation, frontier spirit and harbingers of hope rather than prophets of despair
- ASTRA had a meteoric rise
  - It very soon became a world leader in almost every area it touched It acquired an international reputation
  - It was flooded with visitors including the PM
  - It was publicised, e.g., the BBC film "West of Bangalore"
  - It defined a new paradigm for R & D -- identify topics through a study the immediate environment rather than following the West
- After about 15 years, the meteor burnt out
  - The movement became a department
  - The fervour became routine
  - The voluntary commitment became jobs and career
  - The openness of learning system became the closed mindset of a group avoiding peer review
  - The social concerns disappeared
  - The partnership with villagers became a hierarchical bureaucratic top-down approach
  - The power of the market was extolled without seeing the limits of the market
  - The onslaught of LPG could not be, withstood

- At my Keynote Address to the 25 year review of ASTRA on July 20, 2000, I revealed the changes in ASTRA through the SWOT analyses of 1974, 1989 and 2000
  - I argued that a sustainable institution was characterised by relevance, excellence, governance and financial stability.
  - I urged ASTRA to re-affirm its socioeconomic objectives, to induct new people, to strengthen old areas and initiate new areas
  - I identified IT and Biotechnology as the new areas
- Everyone listened spellbound... but nothing has been done
- Then came the Simputer launch
- But it must he noted that the Simputer came from outside ASTRA which missed the significance of the Simputer -- "none so blind as those who will not see"
- The Simputer is a break-through because
  - It bridges the digital divide
  - It is a new road to rural empowerment in which illiteracy is not a road block (cf. my 5 year-old granddaughter Aqeela)
  - It links IT and Sustainable Development
  - It forges the badly required bond between advanced S&T and the people - in which there is flow of knowledge from the people and to the people
  - It is a world leader
- It also throws new light on the pro- and anti-LPG positions and shows a new path
- Someone should do the theorizing. This is the only lacuna 1 see in the Simputer effort -- there is no sustainable development analyst. Had I been 20 years younger, 1 would volunteer. Now my advice is: don't turn to any gurus with entrenched positions. Get a person from Bangalore as young as you -- from the National Law School or from IIMB or an economist
- A word of caution
  - as you become more successful and as you grow, you will have to institutionalise
  - institution's are built by charismatic pioneers
  - make sure that you identify all the elements of charisma and creativity and inspiration and institutionalize those elements
- My golden rule is listen carefully to criticism and switch off praise
- I am breaking that rule deliberately
- The stone-breaker story

Appendix A4

PRESIDENT APJ KALAM'S COMMENTS AT THE LAUNCH OF AMIDA SIMPUTER

26-03-2004 : Bangalore
Tele-Conference from New Delhi

I am glad you are launching the indigenously developed simputer with the academy, industry and academy developed entrepreneurs. This joint venture between Indian Institute of science, Bharat Electronics, Pico-peta Ltd., is the first example of such partnership in the country. I greet the entrepreneurs led by Dr. Swami Manohar, and from industry Dr. GopalRao, the Chairman and Managing Director, BEL, and from Indian Institute of Science Prof. Balakrishnan and other participants of this programme. I recall a presentation by Dr. Swami Manohar and his team on the development of Simputer, where Prof. Balakrishnan took me. What I like most about AMIDA Simputer venture is, the academic institution has generated entrepreneurs, which is the need of the hour for the country. It was a visionary job done and now I am happy to note that the product is getting launched and has a large market potential. The partnership will ensure a high tech cost effective product for multiple applications such as e-governance and citizen centric services in language independent environment using open source operating system. I am sure you will keep pace with the technological improvements taking place in the field and also work for providing wireless connectivity to the simputer, which will enhance its application potential.

I have one message for you, you have to constantly improve the product and make it competitive. The competitiveness has three dimensions, one is cost effectiveness, second is quality that what you specify that is what you deliver and third is deliver just-in time when the market needs.

I congratulate all of you on this mission and I wish you all success in bringing out many more cost effective products for taking the technology to the common man especially to the rural India.

REFERENCES

The Simputer Trust website www.simputer.org contains a lot of material related to the technology of the Simputer, the SGPL (license) details as well as documentation of the evolution of the Simputer project. The Wikipedia discussion of the Simputer is also a well compiled source of information http://en.wikipedia.org/wiki/Simputer

The Simputer was also profiled in the first Development by Design conference held by the MIT Media Lab in 2001 http://www.thinkcycle.org/tc-filesystem/?folder_id=16173

ABOUT THE AUTHOR

Vijay Chandru.

Vijay Chandru (PhD, MIT 1982) is an academic turned entrepreneur. Vijay's academic career as a teacher and researcher in computational mathematics has spanned over two decades – from 1982-1992 at Purdue University and 1992-2005 at the Indian Institute of Science. Vijay was elected to the Indian Academy of Sciences as a fellow in 1996 and served as an associate editor of *Resonance*, the journal of science education published by the academy. As an inventor of the Simputer, a novel hand-held computer, Vijay received the Dewang Mehta Award the first time it was awarded (it is India's highest award for innovation in information technology).

In October 2000, Vijay and three colleagues (the Simputer gang) from IISc also founded Strand Life Sciences, India's first *in silico* life sciences company. He currently serves as Chairman & Chief Executive of Strand. The company is well established as a leading provider of bioinformatics and cheminformatics products and solutions to biotech and pharmaceutical companies.

Report of the Expert Committee on Innovation and Entrepreneurship

August 2015, NITI Aayog, New Delhi

No. M-13040/11/2015-CIT&I
Government of India
NITI Aayog
(CIT&I Division)

Sansad Marg, New Delhi
Date: April 2, 2015

Office Memorandum

Subject: Constitution of an Expert Committee on Innovation & Entrepreneurship.

1. The Hon' ble Finance Minister in his Budget Speech 2015-16 announced the Government' s intention to establish the Atal Innovation Mission (AIM) in NITI and stated that initially a sum of Rs. 150 crores would be earmarked for this purpose. The overarching purpose of this mission is to promote a culture of entrepreneurship and innovation in India. In the years to come, entrepreneurship and innovation is expected to be an ever more important source of growth and job creation. New technology also has the potential to find solutions to pressing economic and social problems. The government sees the AIM as critical to expediting the entrepreneurial process in India. Key objectives of the AIM will be:

a. To create an umbrella structure to oversee innovation eco-system of the country;
b. To provide platform and collaboration opportunities for different stakeholders;
c. To study and suggest best and novel practices to be adopted by different stakeholders in the innovation chain;
d. To provide policy inputs to NITI Aayog and various Government Departments and Organizations.
e. To create awareness and provide knowledge inputs in creating innovation challenges and funding mechanism to government; and,
f. To develop new programmes and policies for fostering innovation in different sectors of economy.

2. The Government' s commitment to catalysing the entrepreneurial process is evident from its establishment of the SETU (Self Employed and Talent Utilization). SETU (as stated in paragraph 50 of the Finance Minister' s Budget Speech) will be a Techno-Financial, Incubation and Facilitation Programme to support all aspects of start-up businesses and other self-employment activities, particularly in technology driven areas. As stated in the Budget Speech (2015-16), initially a sum of Rs. 1000 crores would be set aside for the purpose.

3. Accordingly, NITI Aayog has constituted an Expert Committee for working out the detailed contours of the Atal Innovation Mission (AIM) and SETU to achieve the stated objective, with the following composition:

(i)	Prof. Tarun Khanna	Chairman
(ii)	Dr. Rukmini Banerji	Member
(iii)	Shri Binod Kumar Bawri	Member
(iv)	Shri Vallabh Bhansali	Member
(v)	Shri Pramod Bhasin	Member
(vi)	Prof Vijay Chandru	Member
(vii)	Shri Mukesh Chatter	Member
(viii)	Shri Ashish Dhawan	Member
(ix)	Dr. Swati Piramal	Member
(x)	Shri Manish Sabharwal	Member
(xi)	Prof. Goverdhan Mehta	Member
(xii)	Prof. Devang Khakhar	Member
(xiii)	Prof Ashish Nanda	Member
(xiv)	CEO, NITI Aayog	Member
(xv)	Dr. C. Muralikrishna Kumar	Convener

4. The Terms of Reference of the Expert Committee are as follows:

i. To review the existing initiatives aimed at promoting innovation and entrepreneurship in India, especially those efforts that result in widespread job growth and the creation of globally competitive enterprises;

ii. To make short and medium term recommendations for actionable policy initiatives aimed at creating an innovation and entrepreneur-friendly eco-system such elements as creation of world-class innovation hubs and digital SMEs, and innovation driven entrepreneurship in such sectors as education and health;

iii. To address any other related issues.

5. The Expert Committee will be serviced by the Communications, Information Technology & Information Division, NITI Aayog. The Committee will submit its report within 3 months.

6. The Chairman may co-opt additional members to the Committee as may be required for achieving the overall objectives.

7. The meetings of the Expert Committee would be held in NITI Aayog, New Delhi.

8. For attending meetings of the Expert Committee, the Chairman and Members would be entitled to Travelling Allowance & Dearness Allowance as under:

i. Officials would draw TA/DA as per their entitlement from their respective Ministry/ Department/ Institution.

ii. Non-officials from India will be entitled to TA/DA as admissible to Grade-I Officers of Government of India (travel by economy/ executive class as per entitlement by Air India) and this expenditure will be borne by the NITI Aayog. Members unable to travel from abroad may participate via videoconferencing.

9. This issues with approval of Vice Chairman, NITI Aayog.

R.K Gupta
Research Associate (CIT & I)

Foreword

Late in the spring of 2015, I was privileged to be asked to chair a committee on Entrepreneurship and Innovation, constituted under the auspices of NITI Aayog. The committee was pulled together in rather short order—a testimony to the good spiritedness of my colleagues who were willing to carve out time from their incredibly busy schedules—and we met formally to initiate the work on May 1, 2015, in the NITI Aayog offices in New Delhi.

Subsequently, there has been a whirlwind of activity, a near-continuous virtual-meeting to brainstorm and write the report these past months, punctuated by occasional physical meetings in New Delhi, Bangalore, and Boston (my adopted home).

We took care to ensure the committee was drawn from very diverse walks of life, to include entrepreneurs (with track records of building for-profit and social enterprises), financiers, scientists and academics, heads of academic institutions and, finally, several who have experience liaising with government. Each member has, in turn, helped us access an incredibly broad network of individuals not on the committee, both in India and in parts of the Indian diaspora, who have all unfailingly and generously given of their time.

Through all this, my colleagues in NITI Aayog, led by Arvind Panagariya and his able team, and, informally, colleagues in the Prime Minister's Office, have been nothing short of spectacular.

Our approach has been to take a big-tent view of entrepreneurship, to think of the innovation and creativity that underlie successful entrepreneurship in a variety of walks of life, far-transcending the exciting but ultimately rather limited remit of current entrepreneurial hotspots in information technology and e-commerce. These hotspots have given us much to be proud of, the "re-Bangalorization of the world" as the media sometimes puts it, but we cannot rest on these (limited) laurels. This has meant that our report is intentionally broader than past reports that have tended to emphasize venture capital and private-equity as handmaidens of entrepreneurship. There is an important role for such financing, and we suggest how to nurture it. However, at the end of the day, such financing is but one of multiple factors that need attention.

That said, we surely stand on the proverbial shoulders of giants. The committee laboured hard to acquaint itself with the myriad reports related to entrepreneurship produced by prior government committees and bodies, as well as numerous superbly qualified groups from the private sector and civil society. We are mindful of learning from these, as well from some other groups of people working in parallel in India on, for example, skill-development and securities markets issues.

Finally, consistent with India's desire to reimagine its future as a world leader, we seek to learn from the experiences of others and to contribute to these. So the reader will find an openness to experiments from far and wide. Chile, China, Israel and the United States, are sites of some of these that receive mention in these pages.

All this has led us to a conceptual model of an Entrepreneurial Pyramid that offers a prioritization of the many efforts that will propel entrepreneurship forward. First, we recognize that success breeds success, so, in the interests of building a constituency for change, we must identify actions that yield short-term payoffs. The so-called 'top layer' of our pyramid model does this. In particular, it identifies a dramatic upgrading and broadening of the incubators that pepper the Indian landscape already, a commitment to using competition and prizes to encourage grass-roots innovation, and the initiation of a symbolic but also substantive national entrepreneurship movement. The use of novel methods to stimulate innovation—including a special focus on science and technology—will affect the social inclusion that is sorely needed to empower the disenfranchised.

From there, we sequentially identify longer-range actions that will yield longer-term and more systemic payoffs, and will amplify the efforts of subsequent short-term initiatives. There is no use pretending that these investments–often in the form of public goods that have received short shrift from Indian society in recent decades—will yield results immediately, but there's also no option but to embrace the need for these changes.

Through all this, we repeatedly emphasize the importance of some behavioural traits that we'd all do well to embrace—a bias to action, a willingness to be accountable with output rather than input metrics, an openness to ideas from wherever they emerge, and a partnership ethos, the last especially important to overcome the huge trust deficit that exists between the public and private sectors. These 'can-do' attitudes will, we are confident, ensure that our report does not gather dust in some cabinet.

As the Honourable Prime Minister said in his Independence Day address on August 15, 2015, to the people of India, "Startup and Stand-up". I couldn't agree more. Let's do it *sans* partisanship and with team spirit.

Tarun Khanna
New Delhi and Cambridge

Acknowledgements

On the behalf of the committee, the chairman would like to thank the following organizations that contributed to this report:

- Aavishkaar Venture Management Services
- Aspiring Minds
- Association of Biotech Led Enterprises (ABLE)
- Axilor Ventures
- Biotechnology Industry Research Assistance Council (BIRAC)
- Ernst & Young India
- Harvard Business School India Research Centre (HBS - IRC)
- Harvard Business School South Asia Institute
- Honey Bee Network
- InnAccel
- McKinsey India
- Pratham India
- Seedfund India
- The Indian Institute of Management, Bangalore (Prof. Chirantan Chatterjee)
- The Global Education & Leadership Foundation
- The KAMMA Incorporation
- The National Science & Technology Entrepreneurship Development Board (NSTEDB)
- Think India

The Chairman would also like to thank the many experts from industry, academia, and government who attended and contributed to the brainstorming sessions organised by the NITI Aayog in New Delhi; IIM Bangalore professor Chirantan Chatterjee, and HBS professors Prithwiraj Choudhury, Ramana Nanda, and William Kerr, for their valuable inputs; and Radhika Kak (HBS - IRC) and Aarthi Gunnupuri for helping with the research and drafting of this report.

Table of Contents

Executive Summary

The purpose of this paper is to put forth expert committee recommendations to boost entrepreneurship and innovation in India. Recommendations are outlined within the AIM pyramid framework - a tool to holistically address the entrepreneurship challenge. As one moves from the base to the pinnacle of the pyramid, initiatives are designed to deliver more immediate payoffs. Conversely, the base symbolises actions with long gestation periods, but with profound effects. The fundamental premise here is that all layers of challenges to entrepreneurship need to be systematically addressed to bring about systemic change. In that sense, this report more comprehensively examines the entrepreneurship challenge than most recent analyses, which tend to focus primarily on the role of venture capital and private equity in promoting entrepreneurship.

Over the last few years, many expert committees and key government agencies such as the Planning Commission, the Department of Biotechnology, the Department of Scientific & Industrial Research and the National Innovation Council have released reports relevant to entrepreneurship and innovation in India. The committee recognises the contribution of these reports and their many excellent recommendations (see Appendix A for a literature review). However, despite exceptional ideas and the best of intentions, many of these recommendations were either not taken on board, or not implemented with full vigour. Therefore it is important to have good recommendations backed by good governance and an outcome-driven and measurement-based culture, the impact of which could be revolutionary.

The formulation of the Atal Innovation Mission (AIM) could be a defining moment in India's economic history, and the idea must be unfettered and allowed to flourish. To ensure that AIM kick-starts immediate innovation while setting into motion long-term changes, the committee has devised a pyramid framework that has allowed the report to include recommendations ranging from the short- to the long-term, which could have a transformative impact on the economy on this country. The committee also emphasizes the need to establish clear systems to monitor implementation, execution, and impact. Accountability and good governance form the backbone of successful policy. The government intends to set up a monitoring committee under the NITI Aayog. This is a welcome development, and can play a key role in ensuring execution of planned projects if implemented effectively.

The committee also recognizes the urgent need for disruptive change; that bold reforms are the order of the day. India needs to generate 115 million non-farm jobs over the next decade, to gainfully employ its workforce and reap its "demographic dividend". Given this context, encouraging and promoting self-employment as a career option for young people will be of highest importance. The culture of entrepreneurship should be inclusive and focus on a variety of enterprises, such as young & innovative technology firms, upcoming manufacturing businesses and rural innovator companies. Entrepreneurs should also be encouraged to help solve pressing socio-economic problems. There is a dire need to find financially viable solutions for the challenges of the disenfranchised. This will be of paramount importance, so as to ensure long-term social and economic stability, India needs a model that pulls along the 350-400 million people that currently reside outside mainstream society.[i] Social inclusion not only fulfils higher altruistic purposes, but can also be financially viable. Furthermore, measures taken to enhance social inclusion would inevitably result in the opening up of a significant new market.

The first section of this report dwells on the role and importance of entrepreneurs in society. Entrepreneurs play an important role in the economic development of a country. Successful entrepreneurs innovate, bring new products and concepts to the market, improve market efficiency, build wealth, create jobs, and enhance economic growth. The ability of entrepreneurs to create jobs is certainly relevant to India, given the need for incremental new jobs. While we celebrate the success of several new ventures in e-commerce, information technology and mobile telephony, these will not likely suffice to deliver our aggregate growth expectations. The agricultural sector remains moribund, the rural economy neglected, and vast sectors starved of capital and talent, constrained as these are by our collective underinvestment in the requisite supporting institutions. Similar institutional inadequacy challenges bedevil investments in so-called social enterprises as well. India will have to encourage creation of new SMEs focused on manufacturing and innovation, while spurring rural innovation and growth. Hence, putting entrepreneurship at the forefront of the economic agenda is the need of the hour.

The second section of the report explains the AIM pyramid framework and its components in detail. The committee used this framework to make short, medium, and long-term recommendations to boost innovation. The pinnacle of the pyramid represents areas where action can be taken relatively quickly to deliver almost immediate payoffs. The middle of the pyramid highlights medium-term issues—those that are not simple enough to be solved immediately, but that can be addressed within a 5-7 year time frame if remedial steps are taken now. The base of the AIM pyramid symbolizes changes which are likely to have long gestation periods, but will lead to a profound transformation in the entrepreneurial fabric of the country.

In the third section of the report, we examine the current state of entrepreneurship in India, within the context of the AIM pyramid framework. The top layer identifies low-hanging fruit. If these are addressed, immediate results follow. These include: a lack of incentives for innovation at the national level, a lack of efficient business incubators, and insufficient access to capital. The intermediate layer sheds light on more structural challenges to innovation like an archaic education system, lack of work-related skills amongst the youth, and a disabling business environment. The base layer deals with fundamental challenges to entrepreneurship related to culture and mindset, which might well take a generation to address, assuming we start now.

In the fourth section of the report, we put forth short, medium, and long-term recommendations proposed by our expert committee to accelerate the entrepreneurship agenda. In the short-term, the committee proposes: introducing competitions and prizes as tools to encourage innovation, encouraging corporates to fund research and development at the university level, improving the efficiency of business incubators, and fostering a national entrepreneurship and innovation movement. In the medium-term, the committee proposes using digital platforms to encourage innovation, reforming the educational system to encourage creativity and upskilling workers to make them more employable, improving the ease of doing business, and strengthening intellectual property rights. Finally, the committee also proposes a number of measures to change cultural biases and attitudes towards entrepreneurship in the long-term, including attaching entrepreneurship to large scale economic and social programs, promoting new high-potential sectors via the government' s " Make in India" campaign, fostering a culture of coordination and collaboration, attempting to redefine cultural notions of success, and tying entrepreneurship with the social inclusion agenda.

In the fifth section, we outline a structure for a proposed " AIM organisation" -a body to catalyse and energize the entrepreneurship agenda. This organisation must be empowered to deliver against its mandate and not be hamstrung by an array of vested interests that have bedevilled the laudable efforts at implementation of the reports of past committees.

Finally, in the last section, the committee uses the example of the bio-economy as one which has tremendous potential for innovation and growth. The potential of this sector to transform the economic outlook is akin to that of the software industry in the early 1990s. Experts say that biotechnology may have an even greater impact in a shorter period of time. The expert committee makes recommendations to unleash the potential of this sector. These include introducing competitions and prizes to bring forth innovations, adapting and re-orienting course curricula at the school and college level to enhance skill in biotechnology, and supporting early-stage ventures in this area. While these recommendations are substantive in their own right, given the likely importance of the sector in the next generation, their more parochial purpose here is to serve to illustrate the pyramid model in the report.

For this report, the committee has gathered data on a range of issues pertaining to entrepreneurship and innovation from several excellent government and non-governmental agencies, academic institutions, and consulting firms.[1] The committee is particularly grateful for their research and findings. It has also taken insight from experiments and developments taking place in many other countries. To substantiate and supplement publicly available data, the committee commissioned a few small-scale surveys to seek information on specific areas of entrepreneurship. Comprehensive findings from these surveys are presented in the appendices at the end of the report.

Numerous companies, non-profits, government departments, and individuals gave generously of their time to prepare this report speedily. For this the committee expresses its heartfelt gratitude.

[1]Not all data provided and used in the report has been validated by the expert committee.

Why is Entrepreneurship Critical to India's Development?

Entrepreneurs play an important role in the economic development of a country. Successful entrepreneurs innovate, bring new products and concepts to the market, improve market efficiency, build wealth, create jobs, and enhance economic growth. De novo firms that unleash creative destruction shift surpluses from rent-seeking large producers to consumers and broader society. Joseph Schumpeter, one of the greatest economists of all time, put innovation at the heart of economic theory and capitalism. He proposed that innovation was the process by which economies were able to break out of their static mode and enter a path of dynamism. It was his theory of "creative destruction" that first highlighted the importance of innovators in revolutionizing the economic structure, leading to the creation of new products, services, and markets, and the decay of the old. Just as boosting entrepreneurship can lead to growth and job creation, failing to promote entrepreneurship can lead to stagnation, and social and economic inertia.

Bringing about innovation has never been as important as today, as the global economy shifts away from the industrial economy towards the innovation economy. Traditional manufacturing is becoming increasingly commoditized while intellectual property is the need of the hour. What is heartening is that recent economic theory suggests that government investment in R&D, knowledge-creation, and technological progress does have a role to play in fuelling innovation, productivity, capital creation, and therefore growth.[ii] This thinking highlights the scope for appropriate government policy and investment to enable entrepreneurship and innovation.

Many studies and data show the importance of entrepreneurs in creating jobs and up-skilling. In the United States of America (US), new businesses were responsible for creating on average one million jobs annually as compared to 300,000 by ten year old firms.[iii] In fact, companies less than a year old have created an average of 1.5 million jobs per year over the past three decades in the US.[iv] Israel saw its unemployment rate fall from 9% in 2000 to 5.5% in 2011 as new businesses grew 23% over the same period.[v] In contrast, Japan has lost two decades partly due to stagnation in entrepreneurial activity. Up and re-skilling is another important service that new firms can provide. As the global economy moves towards automation, firms will require a very different skill-set from what workers currently have. The education system can only do so much. Agile de novo firms can play a role in providing workers with the skills required in the new economy. More efforts need to be made to encourage startups. India ranks in the fourth quartile among the G20 countries in a ranking of ecosystems that boost entrepreneurial activity.[vi] A recent survey of Indian entrepreneurs revealed that only 18% of respondents felt that the government had taken satisfactory measures to nurture startups as compared to 37% in China.[vii] Similarly, a survey of entrepreneurial ecosystems found that only 37% of Indian respondents believed that there was an availability of factors conducive for innovation.[2]

While supporting young firms in technology and other new-age innovative sectors is important, India also needs to develop an ecosystem that encourages innovation at more mature enterprises across the industrial spectrum—across the existing manufacturing, export, and rural and social enterprise sector. This segment has the capacity to generate a large number of jobs. For instance, in the EU, 60% of jobs on average across the region are accounted for by SMEs.[viii] The success of the Mittelstand in Germany highlights that small and mid-sized family-run businesses can lead to job creation in an economy. Indeed, roughly 80% of jobs come from SMEs in Germany. In Korea, 90% of jobs are generated by SMEs. In India, the SME sector employs only 40% of the countries workforce, and is plagued by low productivity. This segment needs a boost. Diego Comin has conducted interesting research on scaling up SMEs in Malaysia, which could provide a blueprint.[ix]

The ability of entrepreneurs to create jobs is particularly relevant to India given its employment crisis. India's demographic dividend has been much touted-while a substantial portion of post-independence India's population consisted of young children, by 2020, 63% of India's population will be of working age.[x] McKinsey estimates that India's working-age population will grow by 69 million between 2012 and 2022.[xi] Cashing in on this dividend will require India to create 69 million additional appropriate jobs, as well as jobs for those that are currently unemployed. Estimates indicate that to pursue an inclusive reform agenda, India needs 115 million additional non- farm jobs in the next decade.[xii] Creation of new businesses will therefore be an important avenue for absorption of these workers. Therefore, developing and sustaining a vibrant entrepreneurial fabric is one policy option that

---

[2]Pillars measured included accessible markets, human capital, funding & finance, regulatory frameworks & infrastructure, education and training, major universities as catalysts and cultural support.

should be part and parcel of any economic development plan. India has seen a wave of successful entrepreneurship previously, which started during the time of the Swadeshi movement. Amongst these entrepreneurs were Jamshedji Tata who founded the first iron and steel company, P.C. Roy who founded Bengal Chemical Works, V.O. Chidambaran Pillai who founded the Steam Navigation Company, and Khwaja Hamied who founded Cipla, a pharmaceutical company. These firms have played an important role in alleviating the employment crisis over the years.

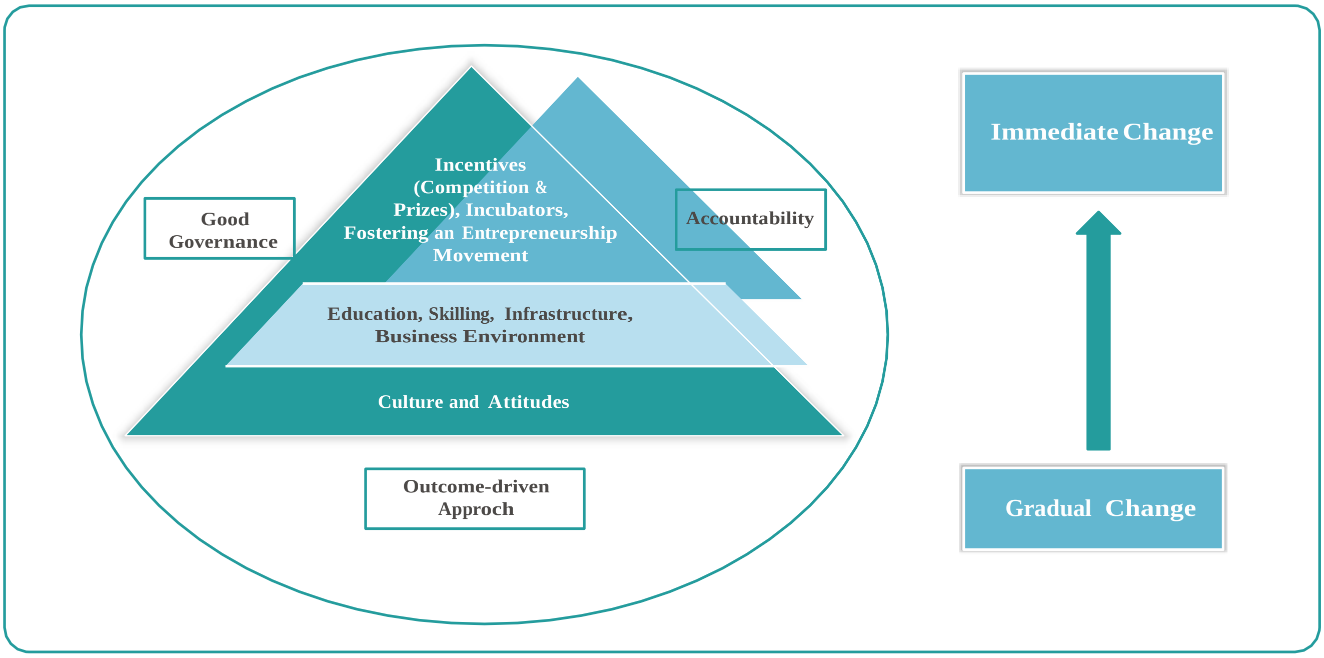
The entrepreneurial culture in India is picking up. Bangalore has been listed within the world’ s 20 leading startup cities in the 2015 Startup Genome Project ranking. It is also ranked as one of the world’ s five fastest growing startup cities.[xiii] Nevertheless, much of this entrepreneurship is limited to the IT, e-commerce, and m-commerce sector. Furthermore, the number of entrepreneurial ventures remains small relative to India’ s population. Only 0.09 companies were registered for every 1,000 working age person-among the lowest rates of G20 countries in 2011.[xiv] The Global Entrepreneurship monitor that tracks entrepreneurial activity, found that new business ownership rate for India in 2013 was the same as in 2008.[xv] To create new jobs, India must move beyond its reliance on IT achievements and the industrial conglomerates that drove earlier post-liberalization growth. For example, India needs to develop technological capabilities to serve the requirements of its core industries—capital goods used in manufacturing industries are mostly imported, as are electronic goods.

There is tremendous scope to boost entrepreneurship in India. Some sectors immediately provide opportunities for growth. For example, the auto components sector is expected to see substantial growth as India moves from being the world’ s seventh largest automobile manufacturer in 2014 to the fourth largest in 2015.[xvi] Sectors like IT infrastructure, biotechnology, healthcare and education, too are poised to grow several times in size over the next couple of years.[3] There is huge scope in the field of social inclusion. Bringing the economically disenfranchised (including the dalits, scheduled castes, scheduled tribes, and other backward castes) and women into the economic mainstream[xvii] not only serves a higher purpose; there is also a strong economic and social justification for the same. It would lead to greater stability in society in the years to come (a benefit across all socio-economic strata), and would also open up a significant new market for firms to tap. As such, it would substantially increase the proportion of the economy able to engage in productive activity. Of course, these projections are predicated on “ business as usual” assumptions regarding entrepreneurial growth of new enterprises; were the latter to take off, the growth might be much greater, especially in sectors like life sciences and automobiles with a global “ wind in the sails” . Government policy that favours innovation can have significant impact on growth and job creation in the economy, as indicated by economists that show innovation and productivity to be endogenously generated. Furthermore, India has a latent science and engineering talent pool, which may be particularly advantageous in a context where fewer graduates in Western countries are opting for STEM (Science, Technology, and Engineering Majors) coursework. Indeed, representatives from Google, General Electric, and IBM have noted that conducting world-class R&D in India is seen as a major opportunity to serve both domestic and international markets.[xviii] This
strength should be capitalised to generate indigenous intellectual capital.

[3] India’ s growing IT infrastructure is another huge opportunity for tech-focused businesses, especially those that used app-based platforms since smart phone penetration in India is expected to grow 45% in 2014 from 117 million users in 2013. India’ s biotechnology sector is expected to grow 2.5 times over 2011 to 2015; health care is projected to grow 7 times from 2010 to 2020: Higher education is poised to grow five times by 2020.

The AIM Pyramid Framework for Entrepreneurship

Figure 1: The AIM Pyramid Framework to examine entrepreneurship



The AIM pyramid framework examines entrepreneurship within the context of short, medium, and long-term obstacles and opportunities. The fundamental premise here is that all layers of challenges to entrepreneurship need to be systematically addressed to effect systemic change. The pinnacle of the pyramid represents issues for which there are relatively quick fixes, wherein measures taken today should deliver almost immediate payoffs. The middle of the pyramid highlights medium-term issues—those that are not simplistic or superficial enough to be tackled immediately, but that can be addressed within a 5-7 year time frame if remedial steps are taken now. The base of the pyramid presents measures to tackle fundamental impediments to innovation. These are likely to have long gestation periods, but will likely lead to profound changes in the entrepreneurial fabric of the country. The AIM pyramid presents a multi-layered framework to tackle the entrepreneurship agenda in a holistic sense. Most recommendations rely on collaboration between the public and private sector to unleash full entrepreneurial potential of the country. This is further highlighted through the fact that the AIM Pyramid is encased in a Good Governance circle, which the committee strongly feels is the backdrop of a successful entrepreneurial and innovative economy. A Development Monitoring and Evaluation Office is soon to be set up under the NITI Aayog to monitor progress on the government' s flagship programs. This is certainly a welcome step if implemented and executed effectively.

A. The Top Layer: This layer seeks to highlight measures with quick payoffs that can expedite the development of an entrepreneurial and innovative culture in India. It consists primarily of measures designed to provide support to early stage startups and to encourage innovation in the manufacturing and rural sector. First-time entrepreneurs have difficulty initially due to the lack of mentoring facilities, and technical and financial support. This problem is also present in the manufacturing-based SME sector and amongst rural enterprises, many of which are focused on traditional arts and crafts and have great export potential. A lack of incentives also plays a role in inhibiting would-be entrepreneurs. Various steps can be taken to encourage early-stage ventures. This includes encouraging new entrepreneurs through competitions and prizes for innovation, encouraging corporates to fund research and development at the university level, enhancing the scope and efficiency of business incubators, and celebrating and recognizing the importance of entrepreneurs at the national level. Measures taken at the policy level could provide the catalyst to unlock private sector potential. Israel, for instance, presents a great example of how public-private partnerships (PPP) can effectively encourage entrepreneurship. Amongst other measures, the Israeli government introduced partial loss guarantees on venture capital funds, with crucial accounting mechanisms in place, that have enabled new ventures to come to market.

B. The Intermediate Layer: The intermediate layer consists of fundamental factors that come in the way of entrepreneurship. These institutional inadequacies prevent the consummation of productive enterprise—for example, absence of entities that can help assess and locate talent handicap growing enterprises searching for managers and engineers, and have long been the subject of policy interest. Amongst others, these voids consist of gaps in physical infrastructure, and in the soft infrastructure required to enforce contracts expeditiously. More generally, they can be thought of as deficiencies in the intermediation structure necessary to enable buying and selling in product, labour, and capital markets.[4] Measures taken to effectively improve the institutional framework of the economy have the potential to significantly alter the entrepreneurial landscape over the medium term.

Various measures can be introduced to address medium-term challenges. These include embracing digital platforms to encourage the public and private sectors, reforming the archaic education system and putting emphasis on work-related skills, improving the ease of doing business by simplifying regulations, and putting more emphasis on a stringent intellectual property rights regime. Setting the foundation of a strong business environment would unleash private sector investment and entrepreneurship.

C. The Base Layer: The base layer consists of deep-rooted cultural impediments to entrepreneurship. Tacking these obstacles will require fundamental changes in societal thinking. This may only be possible with generational change.

The biggest cultural obstacle to unleashing entrepreneurial potential is middle-class India’ s scepticism about entrepreneurship. Notwithstanding the exuberance around the IT, e-commerce, and m-commerce sector at the moment, in the “ mainstream” of the economy, there continues to be suspicion, even distaste, evinced by the idea of entrepreneurship. Of course there are certain sections of Indian society (Gujaratis, Marwaris, Parsis) which have long track records of success in entrepreneurship. Hence it would be unfair to generalize across communities. Nevertheless, there is room for much greater percolation of the entrepreneurship spirit through the interstices of Indian society. Young creative individuals are often held back from starting new ventures due to inherent scepticism amongst family members. Given this cultural context, any blueprint that intends to effectively encourage entrepreneurship must promote and embrace a risk-taking and creative mindset. A related cultural obstacle to innovation is the limited tolerance for failure in Indian society. Experimentation and failure are integral parts of creating and building new ventures. Therefore, a societal tolerance for failure as part-and-parcel of the learning process needs to be inculcated. Failure should not be *per se* chastised.

Apart from the overarching theme of a generalized aversion toward risk-taking and failure in India, other more subtle cultural impediments also present a challenge to entrepreneurship and innovation. These include: a general sense of apathy and untimeliness in law enforcement, aversion to change within the bureaucracy, the culture of rent-seeking at several layers of governance, a cultural skepticism toward intellectual property rights, and a general, pervasive and deeply corrosive sense of mistrust between the government and the private sector. These factors slow down the private sector in India. Other cultural impediments include a general de-prioritization of soft skills in favor of hard technical knowledge, and a bias by employers against graduates hailing from non-elite institutions or non-engineering backgrounds, amongst others. Given this cultural context, even measures enlisted within the intermediate layer may take longer to implement and see impact. Conversely, addressing these mindset issues consistently and with gusto over the years to come will enhance the efforts of the middle and top layers of our pyramid.

[4] The idea of ‘ institutional voids’ is explored at length in Tarun Khanna & Krishna G Palepu’ s book (with R. Bullock), “ Winning in Emerging Markets: a Road Map for Strategy and Execution,” Harvard Business Press, 2010.

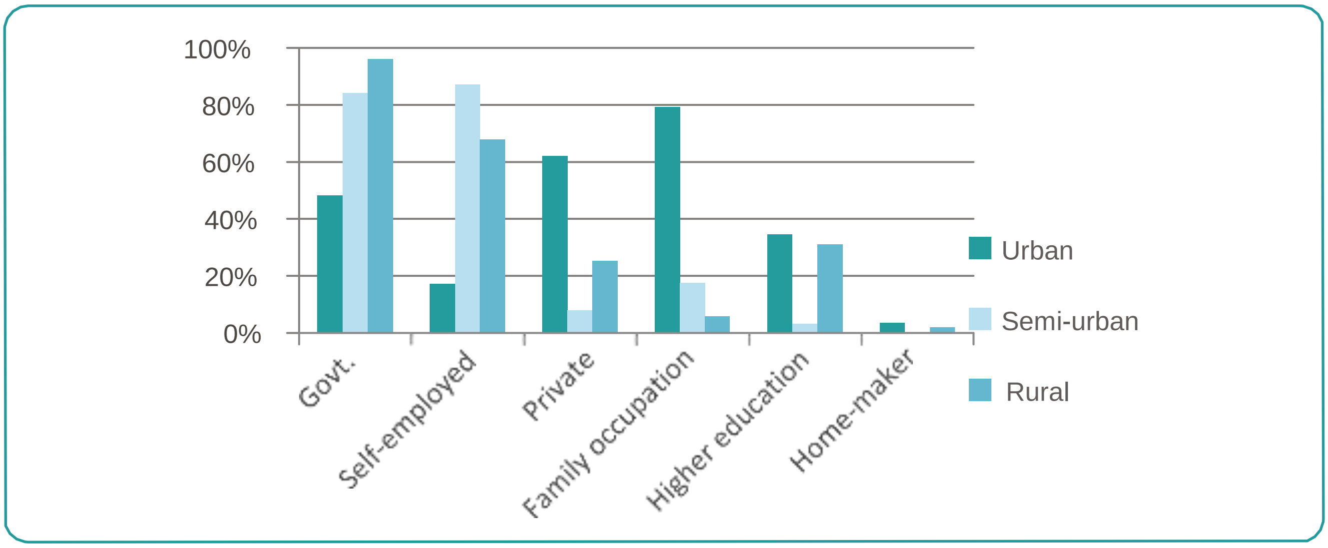
The State of Play

The Top Layer: Inadequate Support to Early-Stage Startups

1. Lack of incentive structures for entrepreneurship: Incentives have long been used as a tool to encourage innovation in the developed world. Grand Challenges for instance, have a long history, dating all the way back to the 1700s, when the British Crown announced a "Grand Prize" for finding a way to measure longitude. In recent times, Grand Challenges have been on the rise after the Ansari X-Prize, which called for innovation in spaceflight, was introduced in the US in 1996. In 2014, the British Government announced Grand Challenges—to promote innovation in science and technology.[xix] In the US, through the challenge.gov platform, the Federal Government has run over 400 competitions through 75 agencies.[xx] More recently, in China, state-backed groups and institutes use the challenge-approach to widen the debate on innovation and find solutions to problems. China Association of Construction Enterprise Management, for instance, gives over 100 awards annually for innovation in construction.[xxi] China's Ministry of Education oversees several national competitions in areas ranging from advertising to electric design to cloud computing.[xxii] The US also has several reality TV shows that focus on business plan competitions, including "The Profit", "Project Runway", "Shark Tank", "Crowd Rules", and "The Apprentice". Conversely, India traditionally has not had structures and incentives in place to encourage innovation. As our patent system is cited as relatively weak and ineffective, there is room to use competitions and prizes as tools to encourage entrepreneurship.

In India's context, competitions could be used in particular to expedite rural development and social inclusion. Though this mechanism is by no means an infallible method to encourage entrepreneurship, it can help in incentive-creation for entrepreneurship in the absence of a fully functional patent system.[xxiii] We have a strong scientific knowledge base that should be harnessed to solve the country's most dire socio-economic problems. If India is able to successfully achieve this, the model can be replicated across the developing world. There is a strong case to use local talent to develop solutions to rural problems.[5] Organisations like the Honey Bee Network are focused on finding and encouraging innovations stemming from rural regions. It has documented several hundreds of these over the years. Adequate training, mentorship, access to capital and a supportive ecosystem could unleash a rural "startup" boom. Further, rural youth are also keen on self-employment. A survey conducted on 195 economically backward students revealed that rural youth are also strongly inclined towards entrepreneurship—87% of respondents from semi-urban areas and 68% of respondents from rural areas expressed interest in self-employment, versus only 17% in urban areas (see Figure below and Appendix D for more details).

Figure 2: Responses from a survey conducted on economically-backward youth: The question asked was "Which of the following areas of work do you see yourself engaged in after 3 years?"*



*Survey respondents were allowed to choose multiple options

[5] With the assistance of an extensive volunteer base, Ahmedabad-based Honey Bee Network has mobilized more than 170.000 ideas, innovations and traditional knowledge practices from around 545 districts. The Network is driven by the philosophy that economically poor people should not be treated as sink of public aid or merely as bottom of the pyramid consumers, rather as a source of ideas, innovations and institutional arrangements with which formal public and private institutions can engage.

2. Lack of adequate business incubators to support early-stage ventures: Incubators play an important role in supporting early-stage startups. Typically, they provide infrastructural support, a platform for networking, management assistance, and other support services. In comparison with traditional measures used to promote industrial development, incubation is found to positively impact the economy and create jobs with much less investment.[xxiv] An NYU Stern School of Business economic impact study found that in a span of five years, three incubators linked to the Polytechnic Institute of New York University, generated $251 million in economic activity and created 900 jobs (in 2013).[xxv] In South America, a dynamic business incubator has put a relatively small country on the global map of innovation and enterprise building—the activity and curiosity generated by the state-promoted Startup Chile has been unprecedented.[xxvi] Israel and Singapore have also made much progress in this arena (see Figure below).

In India, the concept of business incubation is still at a nascent and experimental stage and focussed on urban areas. Much can be done to strengthen the scale, scope, and efficiency of business incubators. One area where there is a clear need for support is in mentoring and networking—a survey (commissioned by the committee) conducted on several traditional clusters in India reveals that approx. 50% of respondents feel the need to improve formal networking within their sector through associations, and to enhance their ability to work cooperatively with the government. Approximately 60% of respondents stated that access to global markets needed to be strengthened—another area where incubators can help. Surveys also highlight that widespread investment, rather than focussed investment in urban areas, is needed to trigger entrepreneurship (see Appendix B).

Our incubators have tended to operate as silos. Though they are housed within Universities in some instances, none of them build on meaningful links with the research climate in the universities, neither really benefiting from the research, nor contributing to it. Nor are the incubators adequately connected to the established corporates, so their identification of needs suffers from inadequate exposure to ‘ real world’ problems, and the ability of the incubators to generate productive insight relevant to existing firms is rather limited. Finally, the incubators do not learn from one another’ s experience. The end result is that, as an ensemble, the incubators that we have punch way below their weight. By investing dramatically in them—not just in the per incubator allocations, but also in the institutional linkages that prevent them from each operating in isolation—we can dramatically upgrade their contribution.

Conversely, we must introduce a degree of accountability, those incubators that prove unequal to their mandate must cease to draw on society’ s resources. The committee was bedevilled in its attempt to collect meaningful output measures, and concluded that there are insufficient incentives currently for incubators to hold their own proverbial feet to the fire. Underperformance therefore persists.

Figure 3: Examples from the World

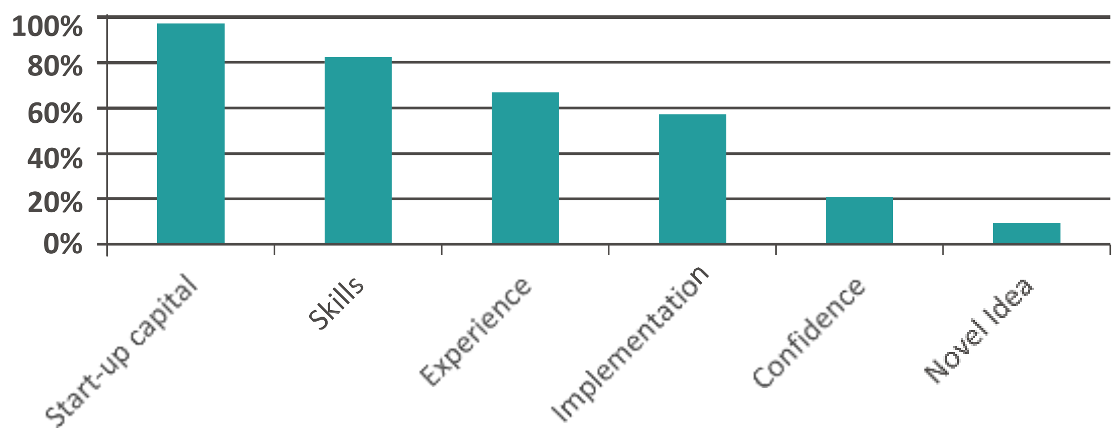
Chile	Startup Chile is a program created by the Chilean Government, that seeks to attract early stage, high-potential entrepreneurs to bootstrap their startups in Chile, using it as a platform to go global. The end goal of the accelerator program is to convert Chile into the definitive innovation and entrepreneurial hub of Latin America. The program, which was launched in 2010, provides Rs. 24 lakhs ($40,000) to entrepreneurs who move to Chile for six months and start a new business. At its inception, it was one of the few initiatives in the world that offered bootstrapping entrepreneurs financial support without taking equity. The program has since inspired similar initiatives, including Startup Peru and Argentina’ s Incubar. Although, retention numbers are poor and nearly 80% of participating entrepreneurs left Chile after the first six months, Startup Chile is helping foster an entrepreneurial spirit in the country: only 10 % of the initial applicants to Startup Chile were Chileans. Now, 29% of the entrepreneurs are citizens of the country.

Figure: Examples from the World

Israel	The government's YOZMA program helped boost the VC industry in Israel, by using Rs. 180 crores to set up 10 private sector venture funds. It also spent Rs. 40 crore as direct investments in high-tech enterprises. Further the government required each fund to partner with a foreign VC and an Israeli investment company or bank. The funds had the option to buy out the government's share after five years. In addition to this catalytic funding, the government also exempt non-Israeli investors in a VC fund from capital gains tax and allowed pension funds, endowment funds and High Net worth Individuals, among others, to invest in startups. The Israeli government also absorbs a percentage of losses that institutional investors may suffer from VC investments. To boost R&D, the government gives grants to companies that employ immigrants with advanced degrees as well as returning residents.
Singapore	The government has introduced a tax deduction of up to 50% on the investment for approved angel investors that commit a minimum of Rs. 37 lakhs. This is subject to a maximum limit of Rs. 93 lakhs with a holding period of two years.

3. Inadequate access to capital and insufficient impact assessment: Access to patient capital is a critical ingredient for startup success. In India, entrepreneurs struggle to obtain funding. Entrepreneurs spend an average of four-to-five times more effort raising funds as their American counterparts.[xxvii] Access to funds is even more constrained for the economically deprived and for women. In a survey conducted on economically backward youth, 97% of respondents stated that access to capital was a barrier to entrepreneurship (see Figure below, and Appendix D for more details). In 2012, female entrepreneurs had access to only 27% of needed debt capital from formal lenders, versus a 70% corresponding figure for male entrepreneurs.[xxviii]

*Figure 4: Factors limiting entrepreneurship for the economically-backward**



* Survey respondents were allowed to choose multiple options

One issue is that much of invested private equity and venture capital (VC) funds are directed towards the e-commerce sector. As a result, large swathes of entrepreneurs struggle to secure capital.

Another issue is the lack of funding available at the seed funding and angel stage. Angel investments in India comprise only 7% of early-stage investing as compared to 75% in the US. VCs in India tend to prefer later-stage type funding i.e. companies that are already generating revenues, which would ideally attract PE investments. They also prefer to make large investments in a handful of companies, in the range of Rs.3 crores to Rs.5 crores. Consequently, smaller startups, that need funding in the range of Rs.25 lakh to Rs.50 lakh, are often overlooked. One key reason for inadequate PE/VC investment is tough regulatory restrictions on domestic investors. Consequently, the industry as a whole is heavily funded by foreign capital. While foreign VC investors have no specific tax exemption, many investors avail the benefits of the Double Taxation Avoidance Agreement (“ DTAA”), which India has with several countries. While foreign capital is welcomed for the VC/PE industry, an excessive reliance on these investments may simply mimic solutions that exist in the developed world and not necessarily encourage businesses and innovations that offer solutions to India’ s needs. Further, foreign investments tend to be procyclical, varying with macroeconomic conditions. This is a risky phenomenon, as was shown through the experiences of Asian economies during the East-Asian crisis. The absence of corporate venture capital (CVC) firms accentuates the problem. CVCs are an important source of financing. Apart from capital, they also offer expertise, reputation and technical knowledge when they invest in startups that have synergies with their companies. Research has shown that where such synergies exist, corporate venture capital can be more efficient than independent venture capital.[xxix] While Indian corporates have entered venture funding, such as Future Ventures, the investment arm of Future group, such instances are few.

Debt funding is difficult to come by. Banks tend to lend very conservatively, and demand cash flow or collateral for financing. These are two assets most young entrepreneurs do not possess. The International Finance Corporation (IFC) estimated that in 2012 there existed close to a $58.6 million (Rs.352 crores) of debt gap for MSMEs.[xxx] Venture debt recognizes entrepreneur’ s need for flexibility, and provides cash flow between rounds of investment, without requiring the dilution of equity. However this avenue of funding is underdeveloped in India.[6]

Inadequate government funding of research and entrepreneurship is also an issue. Government-funded research can lead to technological breakthrough. A substantial set of commercialized innovations have stemmed from the work of publicly funded research institutions and universities in the US.[xxxi] For instance, Google was based on an algorithm developed as part of a research project supported by the National Science Foundation. The Internet and the GPS were breakthroughs by teams initially pulled together by the Defence Advanced Research Projects Agency.[xxxii]

Over and above the need for more capital, there is also a need to ensure that funds invested are generating impact. As with other areas of governance, there is perhaps inordinate emphasis on reaching numerical targets and too little emphasis on ensuring that funds are utilized well. Purely illustratively, consider the 2% CSR target imposed on large corporates by recent modifications to the Companies Act. If complied with, this law has the capacity to generate $2.5 billion per annum—a sizeable corpus that can be used (at least partially) to fund new ventures. But there is vast amount of wastage already inherent in the dispensation of CSR funds. One is that firms have little to no knowledge on related activities of other organizations. This leads to a redundant duplication of efforts. Information sharing via an e-platform could result in augmentation of existing efforts, and therefore greater impact. Furthermore, rather than disparate small investments made at the firm level, a more effective strategy could be to pool firm CSR resources and direct them to established Impact Investors. These agencies have developed skill and expertize in lending to new ventures through years of experience, and would arguably be best placed to deploy resources efficiently, and monitor impact.

The Intermediate Layer: A Disabling Environment for Innovation

1. Gaps in Education and Work-Readiness: The Indian education system is perhaps among the most unprepared for the new innovation economy. A World Bank report suggests India is seen as a country facing skills shortage caused by a lack of skills development and an unresponsive education system.[xxxiii] A survey of entrepreneurial eco-systems found that only 22% of Indian respondents believed that education and training were adequate in the country, the lowest score amongst all the regions surveyed.[7] The Global Innovation Index 2014 ranks India as the lowest among all the BRIC nations,[xxxiv] and pinpoints India’ s state of education as its most acute challenge.

---

[6] While institutions like the Small Industries Development Bank of India, KKR Financial services, the non-banking financial company arm of PE fund Kohlberg Kravis Roberts, Silicon Valley Bank India Finance have begun providing venture debt in India, much more needs to be done.

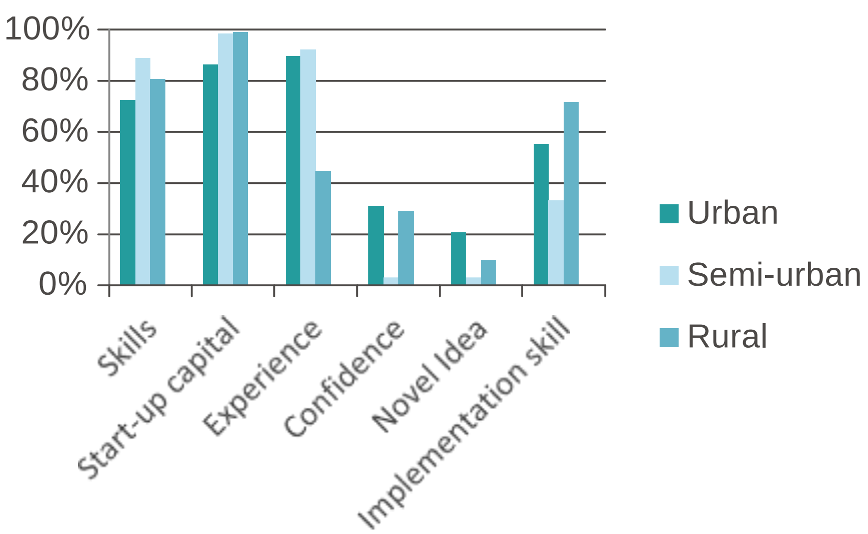
[7] Comparative countries included the US, UK, Australia, Mexico, Australia, Mexico, Pakistan, Singapore, Spain, Ireland, Switzerland.

The biggest problem is that the Indian school system is geared towards a rigid overly prescribed curriculum and an emphasis on rote memorization. Soft skills—collaboration, interpersonal engagement, creativity and critical thinking—are ignored in favour of single-minded test preparation. In a survey of Indian teachers, 92% of respondents were of the opinion that due to exam preparation, students have reduced focus on academic, cognitive as well as vocational skills. Supporting the results, a survey of Indian graduates revealed that no more than 25% could apply abstract concepts to solve a real world problem in the fields of finance and accounting. In comparison, while US students do not fare that well in mathematics and science,[8] they have access to a more diverse and multi-disciplinary curriculum that cultivates creativity, risk-taking and a broader perspective on job possibilities. Experts say that by the time students in India reach higher education levels they lack the creativity, critical thinking skills and open-mindedness required to become successful innovators and entrepreneurs. Think about the now-iconic iPhone. Its design required not just " hard" engineering knowledge, but also a " softer" aesthetic sense to redefine the way individuals worldwide engage with the ubiquitous device in their pockets. India' s rigid system makes it overly difficult for such intellectual symbioses to occur.

A 2008 National Knowledge Commission survey revealed that one in three entrepreneurs considered it " somewhat difficult" or " very difficult" to find employees with the right skills. According to the government's own analysis, of the 12 million universities and technical college graduates produced every year, only 10% have the skills needed to join the workforce. According to a survey by Aspiring Minds, 47% of graduates are not employable in any sector of the knowledge economy—less than 20% engineers are employable for software jobs and 7.5% are employable for core engineering jobs.[xxxv] This deficit in employability is accompanied by a significant demand for manpower of on average 18.4 million people per sector.[xxxvi] To address this gap, the government under its National Skill Development Mission has set a target of training 500 million people by 2022. Though this target in theory marks a step in the right direction, effective and timely monitoring will be required to ensure compliance. Progress towards this numerical target should be monitored on at least an annual basis, and industry experts should be roped in to assess the quality of skill development on a yearly basis.

The problem is compounded by the fact that only a small number of elite institutes, like the Indian Institute of Management, Bangalore; Indian School of Business and SP Jain Institute of Management & Research, offer coursework in entrepreneurship. The limited number of faculty with either practical or academic experience with entrepreneurship makes it difficult for young entrepreneurs to find mentors. This explains why 70% of respondents in a survey across manufacturing sectors agreed that access to relevant knowledge needs to be strengthened. Simultaneously, over 80% of economically backward youth highlighted the need for technical and entrepreneurship training to allow them to become self-starters (see Figure below, and Appendix D for more details).

Figure 5: Factors that would enable economically-backward youth to become entrepreneurs, by location*



*Survey respondents were allowed to choose multiple options

[8] As per the 2012 PISA results for 65 countries, for 15 year-old US students, the average mathematics literacy scores were lower than 29 other countries and the average science scores were lower than 22 other countries (see http://nces.ed.gov/).

2. A Disabling Business Environment: India's business environment is decidedly unfriendly and not conducive to corporate investment. In the World Bank Ease of Doing Business Survey for 2015, India ranks last amongst BRIC countries, and 142 out of 189 countries.[xxxvii] Lack of institutional and infrastructural support could be the reason that only 28% of respondents in a survey of over 25 manufacturing and export sectors stated that they felt they were globally competitive (see Appendix B).

One of the issues is a complicated tax regime arising from India's federal structure. Twenty-eight states levy 28 sales taxes and have different procedures to obtain a sales tax license. Any business that covers more than one state must navigate multiple tax systems. While the proposed Goods and Services Tax (GST) would go a long way in simplifying business administration, its future is far from certain.[xxxviii]

Another issue is the time and cost required to shut down a failing business. Even companies that have minimal assets take on average about six months to close down under the so-called "fast track" procedure. Because of the number and expense of procedures, many owners choose to keep their defunct businesses alive on paper. This is costly in the long-term. The government has set up a committee to consider bankruptcy laws that allow entrepreneurs to easily close down unviable businesses. Progress here is urgent.[xxxix]

India's intellectual property regime is weak, and a deterrent to innovation. India ranks at the bottom of the US Chamber of Commerce's Global Intellectual Property Center's ranking of 25 countries, in terms of its intellectual property environment.[xl] This perception is largely driven by weak enforcement of intellectual property rights, rather than the laws themselves. Experts estimate that it takes on average five to seven years to process a patent application, and in some cases even more than a decade, as compared to two years in the United States.[xli] India had 201 patent examiners in 2012-2013. This is not enough to deal with the quantity of patents. In 2005, there were 56,171 pending patent applications; this has increased to 194,000 in 2013.[xlii] A weak IPR regime on patents and trademarks reduces incentives for entrepreneurs to invest in innovation and research. The number of patents filed in India is rising, but the numbers from other countries such as China and Korea are rising exponentially. According to a World Intellectual Property Organization (WIPO) study, India's patent application numbered 1400 at the same time that China filed 24,000, Japan 45,000 and the US 63,000.[xliii] Even in traditional areas of strength like chemistry, China's patents have far exceeded that of India's.

India also needs a regulatory framework that favours innovation. In every field that requires licenses the laws are old and obsolete, many of which were written over 50 years ago. There is no regulatory pathway to facilitate Indian scientists and regulators working on new technologies. For example, in the case of new medicines, there is often an insistence on replicating studies done abroad which sometimes increase the cost unnecessarily.

In short, India needs an innovation strategy. Countries like Korea, Canada, Israel are focussing on cutting-edge technologies and encouraging both private and public sector research. These include semiconductors and molecular imaging (Korea), rare metals (China), biomaterials (Canada), and Information Technology (Israel) to name a few. India is behind the curve on this front.

Labour laws are complex and cumbersome. Currently there are 44 central labour laws and 160 state laws pertaining to labour. There are many overlaps in jurisdiction. The Industrial Disputes Act is particularly rigid as it requires that companies employing more than 100 workers obtain government permission to fire a worker or close down a plant. Inflexible labour laws are a deterrent to doing business in India, and serve as a disadvantage versus South-East Asian countries.

Finally, India's well-documented infrastructure deficit creates barriers for entrepreneurship. While 21st century startups are not often associated with brick-and-mortar, they require infrastructure—road, rails, and internet—that join them to their markets. Amongst the most significant issues are connectivity, low bandwidth availability, and the lack of electricity in cities and towns outside of major metro areas or satellite towns Typically, businesses have to incur additional costs to overcome these infrastructure deficits. For instance, frequent power outages need to be countered by investment in expensive diesel-powered electric generators to run manufacturing units, factories, shops and even offices. Land acquisition is a notoriously time-consuming and costly process, and makes it difficult for new companies to set up in India, and existing companies to physically expand their operations. Rural microenterprises find it costly and sometimes impossible to transport their goods to more urban areas, due to poor physical infrastructure.

Apart from these issues, business leaders often complain of a slow and inefficient bureaucracy, which leads to delays in decision-making and implementation. This serves as a deterrent to public-private partnerships.

While this litany of problems affects all firms, larger established corporates have the wherewithal to roll with the proverbial punches; for de novo entrepreneurs, these collectively sound the death knell far too often.

The Base Layer: A Cultural Context that Favours Stability over Risk-Taking

1. A cultural affinity for stability: The prevailing culture in middle-class India emphasizes stability and job security over risk-taking. Students have been conditioned to pursue high status and well-paid positions with reputable national and multi-national corporations as these are the main indicators of success. The salary and stability of such jobs are perceived as critical for young (especially male) professionals preparing for marriage and older professionals who are beginning to consider the costs of caring for elderly or retired parents. Therefore, young workers or students prefer not to turn to the risky and financially uncertain world of entrepreneurship. In an employment survey, only 6% of engineers had startup companies as their first job preference.[xliv] In a survey conducted on economically-backward youth, 60% of respondents stated they would prefer the security of steady income from a job, even though most felt that they could earn more money if they succeeded as an entrepreneur. Nearly all who said they would not choose entrepreneurship as a career, stated the fear of monetary risks as a hindrance. Almost 70% cited family pressure to seek stable employment as a deterrent (see Appendix D).

This cultural bias plays out in a disproportionate preference for careers like engineering and management. In 2011-2012, of the total students enrolled in higher educational institutions, 17.5% were enrolled in commerce/management and 16% in engineering as compared to only 1.8% in law and 3.6% in education.[xlv] India also needs urban planners, demographers, architects, artists, movie producers and many others. The Indian education system does not promote these careers on the same footing as more traditional careers. Neither is there a system of talent assessment and associated counselling that could steer young people towards different educations.

This notwithstanding, there are sections of Indian society, like the Gujaratis, the Marwaris, and the Parsis, that are culturally disposed to entrepreneurship. Many "business families" within these communities have built businesses that span the globe. However, even these successful families are sometimes averse to adopting modern techniques of financing or modern management techniques; this inadvertently impedes their growth.

2. Stigma over failure: The prevailing culture in the country tends to stigmatize failure. As a result, 60% of entrepreneurs that have failed with their first venture return to large companies instead of trying out new ideas or improving the original.[xlvi] The World Economic Forum survey of entrepreneurial eco-systems found that only 17% of Indian respondents stated that the so-called "cultural support pillar" was available in the country. This was the *lowest* score amongst all the regions surveyed.[xlvii]

3. Bias against hiring non-engineers and those from non-elite universities: Many graduates face discrimination based on the ranking of universities they graduate from and the assumptions made by the labour market about such graduates. Job candidates for engineering positions from a tier-3 college are 24% less likely to get the job than a tier-1 student of equal merit. Should the tier-3 candidates get the job, they will receive substantially lower salaries. There also tends to be a bias against hiring non-engineers. This creates an inequality in the job market and reduces firms' access to a large set of meritorious students.[xlviii] Indeed, in 2014, 41% of employable graduates in accounting hailed from colleges outside the top 30% of ranked schools.[xlix] Similarly, 70% of employable engineers graduate from lesser-known colleges.[l]

In diametric opposition to this bias, research on software firms in India has demonstrated that employees hired from remote geographic areas actually outperform employees hired from non-remote locations. Further, survey results from the Koramangala start-up cluster in Bangalore (a survey commissioned by the expert committee), showed that 85% of entrepreneurs came from lesser-known colleges, whilst only 16% came from elite universities. 44% of entrepreneurs hailed from non-engineering backgrounds. Some companies are realizing this, and proactively devising systems to leverage this high-ability under-utilized talent.[li] Encouraging entrepreneurship among this subset of meritorious students could be one solution to the employability problem.

4. Loss of top talent to other countries: Top talent from India has traditionally sought overseas opportunities both for education and jobs. India needs to devise mechanisms to leverage its diaspora to enhance knowledge inflows and attract talented Indians working abroad. This is especially important as return migration is often concentrated among the most successful and highly educated migrants.[lii] Recent optimism surrounding India' s growth prospects has brought back some highly skilled talent to the country, but not in big enough numbers to create impact. Bringing about reverse migration will be a long-term theme as it involves offering would-be returnees not only attractive opportunities and pay, but also a western-style lifestyle. Significant effort will be required to improve urban infrastructure (roads, transportation, education), and the perceived " liveability" of cities.

5. Trust deficit between the government and private sector: Business and government have to work hand-in-hand to deliver economic growth and employment. Unfortunately in India, a culture of mistrust between the public and private sector has developed due to years of non-cooperation. Corporates complain of a general sense of apathy and untimeliness in law enforcement, aversion to change within the bureaucracy, the culture of rent-seeking at several layers of governance, and an erratic and mercurial political context. For instance, sectors like real-estate, defence, and manufacturing require a multitude of licenses, which adds unnecessary cost and time to complete. The licensing department has no incentive to simplify procedures as the opaque requirements allows for graft. On their part, policymakers and bureaucrats complain that even meeting with senior executives from big corporate houses bring forth allegations of corruption and crony capitalism. Hence, despite their desire to help, officials are wary of being seen as partial to business, lest they be accused of wrongdoing. This fear has assumed such proportions that senior business leaders in India now complain of a total lack of engagement with policymakers. A lack of collaboration between industry and the government is damaging for investment and growth. This relationship must change to one of cooperation and collaboration to achieve mutually beneficial goals. In this regard, recent successful efforts to bridge the sense of mistrust must be celebrated and encouraged.

Committee Recommendations

Committee recommendations range from the shortest-term measures that can be quickly taken to encourage entrepreneurship, to those that are most foundational, that will manifest over the long haul, but will increase the efficacy of short-term efforts over time. We emphasize that measures at each layer of the AIM pyramid will need to be undertaken to fundamentally and sustainably boost entrepreneurship in India.

Top Layer: Providing Adequate Support to Early-Stage Ventures

1. Introducing Competitions to Solve Pressing Economic and Social Problems: Incentivising technology breakthroughs can lead to disruptive innovation and viable (low-cost) solutions to tough developing-world problems. Introducing prizes and competitions could encourage more young creative minds to go down the path of entrepreneurship, and provide a foothold even for entrepreneurs that do not win. For instance, through the course of UTV Bloomberg's "The Pitch", Zipdial, a company that made it to the final round, secured funding from Mumbai angel investors, despite not winning the competition. There is plenty of evidence that entrepreneurial talent and capability does exist. The indigenous development of the Simputer, a handheld low-cost alternative to a personal computer, is one such example of this. Simputers have already been used effectively to automate land record procurement processes in Karnataka, for e-education in Chattisgarh, and more recently for tracking of traffic offenders by the police. India's Mars mission—ten times less expensive than the corresponding US mission—was another example of India's technological capability. Costs were maintained by prioritizing home- grown components and technologies over expensive foreign imports.

In India, innovation is particularly needed to expedite the sluggish rural development process. About 74% of India's population lives on the fringes of economic development, mired in deep-rooted multigenerational poverty.[liii] They are most vulnerable to the lack of good infrastructure, electricity, clean water, quality healthcare, and education; even as problems related to rapid development, like waste disposal and pollution continue to rise. While scientific innovation and technology offer solutions to many of the problems plaguing India and much of the developing world, these solutions are often unaffordable to the citizen or the government. For instance, a small percentage of India's population uses Reverse Osmosis (RO) and other filtration systems for clean water but these are out of reach for the majority of the country. Similarly, the technology to build roads and seamless electricity grids exist but is prohibitively expensive for the government to invest in all at once; meanwhile millions remain excluded and vulnerable.

Among India's Most Pressing Problems: Water, Hygiene, Housing, Electricity.

- 594 million people or over 50% of India's population defecates in the open.[liv]
- 21% of India's communicable diseases are water related.[lv]
- 77 million households use kerosene for lighting, while 44% of households in rural India lack access to electricity.[lvi]
- Only 19% of the rural population lives in pucca houses.[lvii]

The committee recommends a "Grand Prizes" approach to finding ultra-low-cost solutions to India's most intractable problems. Incentivized innovation has worked around the world in stimulating innovation. In the US, XPrize is giving tens of millions of dollars to those who can provide solutions to major technological challenges. For instance, the Automotive XPrize, run from 2007 to 2010, was successful in getting teams to build vehicles that were minimally 100MPGe efficient, producing less that 200g/mile of CO2, and built for the mass market. In India, the Infosys Prize of Rs. 65 lakh, given for outstanding achievement in research in six categories[9] to contemporary researchers and scientists, is a good example.

[9] The six categories are: Engineering and Computer Sciences, Humanities, Life Sciences, Mathematical Sciences, Physical Sciences, and Social Sciences.

Through the Incentivize Innovation in India (i3) programme, the AIM plans to announce a challenge and a Grand Prize (substantial monetary rewards) for ultra-low-cost solutions. The spirit of the scheme is to encourage the use of technology to empower the disenfranchised. Grand Challenges and the associated Grand Prizes could also have a ripple effect on innovation and problem-solving in the country. The substantial cash prize will be greater motivation for researchers, students, and even amateur innovators in the country to find solutions for the nation' s most pressing challenges. The corporate sector could be roped in to substantially supplement the prizes associated with whichever competitions it finds relevant.

AIM Funds to Boost Entrepreneurship Through "Grand Challenges":

In his budget speech, the Hon'ble finance minister allocated Rs. 150 crores for the AIM. The committee's recommendations for utilization of these funds are based on an assumption that the AIM budget can be turned into an annual allocation. The committee also recommends that the aggregate funds for AIM be revised upwards over time.

The committee recommends that the AIM budget be used entirely to award up to 12 grand prizes annually. The topics for such prizes, such as those suggested on page 27, and in Appendix E of the report, are to be set by the governance structure described in Section V of the report. AIM will seek input from the appropriate ministries, including the Ministry of Science and Technology to vet the decision to run specific Grand Challenges. Each challenge should carry a prize of between Rs. 10 cr and Rs. 30 cr for achieving a significant precisely-defined target over a precisely defined target over a precisely defined time horizon. The AIM should also consider setting aside part of the prize money to place orders for the products and services that are generated by winners. This would contribute towards creation of a market for winners' products, thereby bringing their innovations to life.

These amounts are deemed adequate to attract the best talent worldwide to dedicate its energies to the requisite innovation. The committee believes that talent from all over the world should be welcomed to work on problems relevant to India' s social and economic development. Any team with at least one Indian citizen or NRI will qualify; this lowers the odds of excluding significant pools of talent that have hitherto no connection with India, but might have the knowhow that can be put to use.

The plan is to provide the prize money in two progress-contingent phases. In the first phase of each challenge, five or fewer promising proposals will be shortlisted and they will share 25 percent of the prize money assigned to that challenge. The remaining 75 percent will only be given at successful completion of the challenge. It will take much of a year to finish phase 1. An implication is that, in the first AIM allocation, only a quarter of the allocated funds will be dispersed in year 1, with the remainder dispersed in subsequent years contingent on success. It should be noted that many challenges are unlikely to be solved in the stipulated time, by the very nature of their ambition and intractability. Therefore, some method will be devised to bank the parts of the prize money that is unclaimed for appropriate reinvestment into the country' s innovation fabric.

Another method to scale funds available for such competitions is to implement schemes similar to the Small Business Innovation Research (SBIR) program in the US. Through this program, 11 federal agencies in the US set aside 2.5% of their extramural R&D budget exclusively for SBIT awards. Each year these agencies identify various R&D topics representing scientific and technical problems that require innovative solutions. These topics are then released to the public for proposals from interested small businesses. Through this program, the SBIR aims to competitively award grants to small high-technology firms which are technically sound and commercially promising but unproven ideas. Funding generated is significant, and is sometimes scaled up by being used in conjunction with venture capital and angel funding. Though similar such schemes exist in India, they could be better implemented and monitored to ensure impact.

Illustrative Examples of Potential "Grand Challenges":

Problem: Economically-viable electricity usage

Challenge: DESIGN A BATTERY FOR INDIA

A novel and environmentally safe energy storage that delivers about 150 Wh for minimum six hours, at half the cost of a typical corresponding lead-acid battery, and has at least 5 times the life cycle of the prevalent lead-acid batteries. Other than routine maintenance, the battery should essentially be maintenance free. This will enable economically viable off-grid electricity usage without having to wait for the government' s distribution grid to be built. A poor rural household will be able to operate two bulbs, a fan, a TV, a small refrigerator and charge a mobile phone. This will directly impact the quality of education and study hours of rural students. The battery should also be linearly scalable for larger loads. With incremental power one will be able to build cold storage units requiring 24x7 electricity, thus reducing the large scale produce wastage. Small scale industries such as dairy, food processing, machinery repair, handicraft, ancillary units will be enabled. All of these will substantially enhance the average income of a rural household. Usage of lead acid battery in inverters commonly used in urban households could also be replaced thereby reducing the highly toxic lead contamination.

This challenge is expected to be solved within 18 to 24 months.

Problem: Cheaper Solar Panels

Challenge: CAN YOU MAKE SOLAR CHEAPER?

A panel that costs 33% of the existing widely used solution while maintaining the same specifications in terms of power delivered, durability, and more. Solar panels combined with the battery described above would go a long way to make solar power economically attractive and primed for large-scale adoption.

This challenge could take 2 to 3 years.

Problem: Pure Drinking Water

Challenge: CLEAN DRINKING WATER FOR EVERYONE

A fast, cheap way to purify drinking water is a must. A solution would have to be able to deliver 10 litres of potable water for under 2 rupees. A solution would not only remove 99%+ pathogens; it must also remove pollutants, which often end up in the water supply due to fertilizers and pesticide run-offs, and various other reasons. Centralized solutions so far have high procurement, maintenance, and delivery costs, thus individualized solutions such as a tablet that can be dissolved in 10 liters of water is a preferred choice.

This challenge is expected to be solved within 18 to 24 months.

For a more elaborate list of i3 Grand Prize and Challenges, refer to Appendix E.

2. Harnessing Corporate Funds to Finance R&D: In the knowledge economy, universities as a source of knowledge have become far more important than in the past.[lviii] In recent years, they have acquired a crucial " third mission," of contributing to economic development after teaching and research.[lix] Among the developed countries, the United States is feted for offering entrepreneurs many structural advantages, among them close linkages with universities. Many universities have incubators, technology parks, and venture funds within their sprawling campuses.[lx] Similarly, in Cambridge, UK, engagement of the faculty with industry has spawned many " millionaire dons" .[lxi] The private sector can be tapped to fund research and development at universities. The expert committee proposes various recommendations to build links between large corporates and research at universities. These include:

A. 1% of corporate profit could be directed towards research labs in universities and/ or industry-university collaborative research. The government could provide some tax benefits against this. Monitoring of this rule should focus not only on the absolute amount channelled into universities, but also on the efficiency of spending, that is, it needs to be output-rather than input-oriented. The idea here is that universities become the breeding ground for new technology/ ideas that can be used by the corporate sector. Firms would implicitly be outsourcing R&D financing development of products/ services that can be bought by them. In that sense, this financing would be perceived as absolutely core and fundamental to a firm's operations, rather than as a CSR-related activity. Though the actual development of R&D may take some time, beginning the involvement of the corporate sector in the financing of universities could be achieved relatively quickly.

B. Encouraging top Indian firms to set-up research and education wings at universities

C. Introducing a "Make in Universities" program which would involve setting up 500 tinkering labs with one 3D printer per institute and trained people to operate this, across the country, including in smaller cities, to boost the spirit of production and collaboration. This concept could be introduced in competition format. Corporates would stand to gain as winning in such competitions would create brand value within the university that it contributes to.

D. A percentage of corporate profit could be directed towards corporate venture capital funds, for the purposes of investment in start-ups and/or incubators. The government could offer tax credits against this.

E. All contracts with foreign defence companies above $5B should include a clause for 5% of contract value to be directed to establish research centric universities with strong emphasis on its core product areas in particular and broadly focused on the related areas in general. Aggregated contracts with companies below $5B should include a clause for 5% of the contract value to be directed to establish a research center associated with a university, focused on the corresponding product space. In the committees view, in order for such a scheme to be implemented effectively in India, the responsibility of setting-up, staffing, and managing these institutes should also be that of the company for at least 10 years. This would allow India to develop a number of top-class research institutes with high-quality faculty within a few years, and would kickstart the process of industry-university collaboration in a meaningful way. Companies should also clearly define ex-ante how these funds should be used, and guidelines should be as specific as possible to prevent misuse of funds. The practice of encouraging foreign investors to partake in local R&D development is not entirely new. Over the past two decades, for example, western multinationals operating in China have come under increased pressure to undertake technology transfer to local personnel as a quid pro quo for accessing their domestic market. In India, Bosch's Rs. 140 crore investment in a research center at the Indian Institute of Science is an example (though it is not a defence company). The purpose of this investment is to promote applied research in a range of new-age areas, including but not limited to cyber security, mobility solutions, renewable energy, and the like.

3. Improving the Efficiency of Incubators: Some progress has been made in developing the business incubation industry in India. Since the late 1980s, the National Science & Technology Entrepreneurship Development Board (NSTEDB), under the Department of Science & Technology, has invested about Rs 100 crores in incubators across the country. Privately-owned incubators have recently emerged, located mostly in tier-1 cities like Bangalore and Mumbai. The Startup Village in Kerala, which won a national award for technology and business incubation in 2014, is the country's first PPP (Public-Private Partnership) technology incubator.[10] However this is not enough.

Much more can be done to improve the efficiency and scope of incubators. Firstly, more funds need to be channelled into incubators. The total value of investments in incubators remains miniscule in relation to demand. While the efforts of existing incubators, especially those set up with public money, are much to be celebrated, creating 40,000 jobs (see table below) over 30 years is an embarrassingly low target with which it is hard to be satisfied. Secondly, more emphasis needs to be put on monitoring the efficiency and impact of incubators. It is not enough to simply create these clusters. Active supervision and revision is required to ensure value creation. The AIM organization should have the mandate and governance to deliver.

*Figure 6: Major outcomes of the Incubators backed by the NSTEDB since 1982[11] (all figures are approximate) *Evaluated at 12 to 36 months.*

Start-ups promoted each year	500
Technology jobs created since launch	40,000
Companies/graduates since launch	800
Turnover of graduates & incubatees	Rs.4000 crore
Total Value of Companies*	Rs.1500 crore

[10]Stakeholders: The National Science and Technology Entrepreneurship Development Board, The Department of Science and Technology of the Government of India, and Kochi Technopark and MobME Wireless.

[11]As per National Science & Technology Entrepreneurship Development Board (NSTEDB), July 2015

The committee makes the following recommendations on business incubators:

A. Increasing the Amount of Funding Going into Business Incubators: The total amount of funding going into business incubators is miniscule in relation to demand for financing. A target of up to INR 200 crore per year should be set for public investment in incubators in the initial years. Efforts should also be made to rope in private sector funding. CSR funds, for instance, could be at least partially directed to incubators. Apart from the need for more funding, there is also a need to monitor efficiency of spending and impact. We discuss this in more detail in the following points.

B. Creating Virtual Incubators: Curated sites should be set up to provide entrepreneurs with access to advisors, mentors, and experts. These sites should also include information on how to access funding, how to navigate the regulatory landscape, and e-education. The purpose is to raise the odds that even those in remote/ inaccessible areas to launch their own businesses.

C. Keep Incubators Up to Date: Incubators must be able to provide services and facilities most in demand with current market and business conditions. Key decision-makers and managers must routinely study and survey the needs of the entrepreneurs, as these are constantly evolving, and offer the best they can to help the incubated ventures succeed. For example, while a few years ago, high-level guidance was most sought after by entrepreneurs, at present, young startups are increasingly seeking specific, actionable assistance and skill development, such as creating sales proposals, hiring strategy, introduction to influencers and investors. One way to kick-start these linkages would be to connect incubators with pre-existing networks of entrepreneurs – for example, through TiE or NASSCOM – so as to exchange ideas and build partnerships.

D. Link Funding with an Institutionalized Annual Ranking: Incubators must be ranked every year according to a set of stringent guidelines, whilst offering space for failure and risk-taking. Increased funds and resources help incubators provide better facilities. However, greater resource commitment must be linked to performance, which could be identified through annual rankings recommended here.

E. Exit Non-Performing Incubators: Incubators that perform poorly beyond a certain timeframe must be shut down to channel resources that enable relatively successful incubators to do even better. A formal ranking process and annual reviews will help identify these non-performers.

F. Introduce Specialized Sector-based Incubation Services: Since different sectors throw up unique challenges, the one-size-fits-all approach does not work when it comes to incubation. For example, currently, e-commerce ventures receive greater interest and funding offers. Therefore incubators focused on e-commerce must strive to improve the quality and mortality of these start-ups. On the other hand, sectors like social inclusion, healthcare, education and clean technology do not attract sufficient attention and incubators should work on increasing awareness, and getting more entrepreneurs & investors involved, which will also boost innovation across these crucial sectors. Separate incubators are required to help the manufacturing SME sector innovate.

G. Mould a Supportive Incubation System to Encourage Disruptive Innovation: Incubators must strike a balance between offering stable environments for incremental innovations but also permitting creative and disruptive innovation. To encourage this, rules and procedures must be minimal, while offering a supportive and empathetic system.

H. Strengthening Links between the Corporate Sector and Incubators: Currently, incubators operate in silos rather than within an overall ecosystem. This dilutes impact. To maximise impact, links between incubators and the corporate sector need to be strengthened. The corporate sector should be encouraged to provide more finance and support for start-ups, via incubators. This money could partially come from corporate venture capital funds. This could serve a dual purpose. Start-ups would get access to much needed funding and mentoring, while corporates would implicitly be supporting ventures that could later add value to their own business. For example, Indian pharmaceutical companies could have an interest in supporting drug discovery firms. This system is well established in the US, where corporate buy-outs present an important source of funding for technology/products generated by incubators.

SETU Funds Should Be Used to Upgrade Incubators and Set-Up Tinkering Labs:

In his budget speech, the Hon'ble finance minister allocated Rs. 1000 crores for Self Employed and Talent Utilization (SETU). This is a one-time allocation, intended to jump-start skilling for the better utilization of talent.

The SETU funds should be used to jumpstart innovation through two concrete initiatives at scale. We recommend, for example, that half of the Rs. 1000 crores be spent in upgrading the system of incubators already in place in the country. Within six months, a system can be created to survey and rate the incubators on objective criteria, of the sort the committee has not be able to obtain to its satisfaction currently. At AIM' s discretion, though always transparently, criteria should be spelled out to codify this rating function, and then to award the majority of the Rs. 500 crores to the best ten incubators, publicly or privately funded, with the proviso that they will be held accountable for the efficacy with which the funds are spent. Subsequent years' allocation of AIM funds should similarly be used to continue to rate the incubators and to reward those producing exemplary results

The remaining Rs. 500 crores SETU money should be used to set up Tinkering Labs. These are centers that will permit aspiring entrepreneurs to experiment to create products that address local problems. A number of so-called Tinkering Labs could be based in engineering colleges, with the state and the academic institute sharing responsibility for maintenance. They should be equipped with basic engineering design equipment and with a 3D printer, as well as staffed with appropriate technical personnel. They will need young leaders who have strong hands-on experience in technology problem-solving. While designing training programmes for the convenors of these Tinkering Labs, inspiration could be sought from organisations like JED-I (The Joy of Engineering, Design and Innovation), focused on quality in engineering.

Some experimentation is needed to decide the format and associated economics of the Tinkering Labs. For example, it' s quite clear that such a lab will not exist in a vacuum. It will have to be part of the local societal fabric. How to fashion that link in different local circumstances is an exercise in creativity. Further, we envisage that the labs will develop differing sectoral expertise depending on their local contextual circumstances.

4. Fostering a National Entrepreneurship & Innovation Movement: Celebrating and recognizing entrepreneurship at the national level could go a long way in raising the profile of entrepreneurs, shedding light on their role and importance in society, and encouraging more young people to consider entrepreneurship as a career.

A. Institute a National Entrepreneurs' Day: The committee recommends instituting a National Entrepreneurs' Day during which entrepreneurial success from different entrepreneurship programs is celebrated. On this day, winners of competitions such as those between entrepreneurs from different state initiatives, or business plan
competitions run by incubators, could be honored.

National Entrepreneurship Day – USA

- The National Entrepreneurs' Day is held in November since 2010 in the USA. It celebrates the contribution of American new businesses and entrepreneurs in job and wealth creation.
- Celebration of the Entrepreneurs' Day started as recently as 2010. The proposal for the idea came from two young entrepreneurs and the idea was turned into a Presidential Proclamation, with now efforts being made to turn it into an official Holiday.
- In 2012, the President declared November of that year the National Entrepreneurship Month, which included the Entrepreneurs' Day.

B. National Action Brigade: The committee recommends setting up a platform to get youth and others interested and involved in multiple ways. The Action Brigade could be a network of volunteers helping with the " back-end" of setting up online entrepreneurial platforms, volunteering for events such as the National Entrepreneurs' Day, and helping build other assets for the Atal Innovation Mission. One could think along the lines of the " Teach for India" or even the National Cadet Corps (NCC).

C. National Knowledge Infrastructure: Developing powerful knowledge infrastructure is just as crucial as building world-class institutions and organizations. Without highly-skilled intellectuals, academics, managers and leaders, even the most progressive physical infrastructure and facilities will remain under-utilized. The country needs to invest and nurture its talent from academia and industry, as well as tap into the network of accomplished Indians globally. Two simple yet effective recommendations to kick-start the building of Knowledge Infrastructure follow:

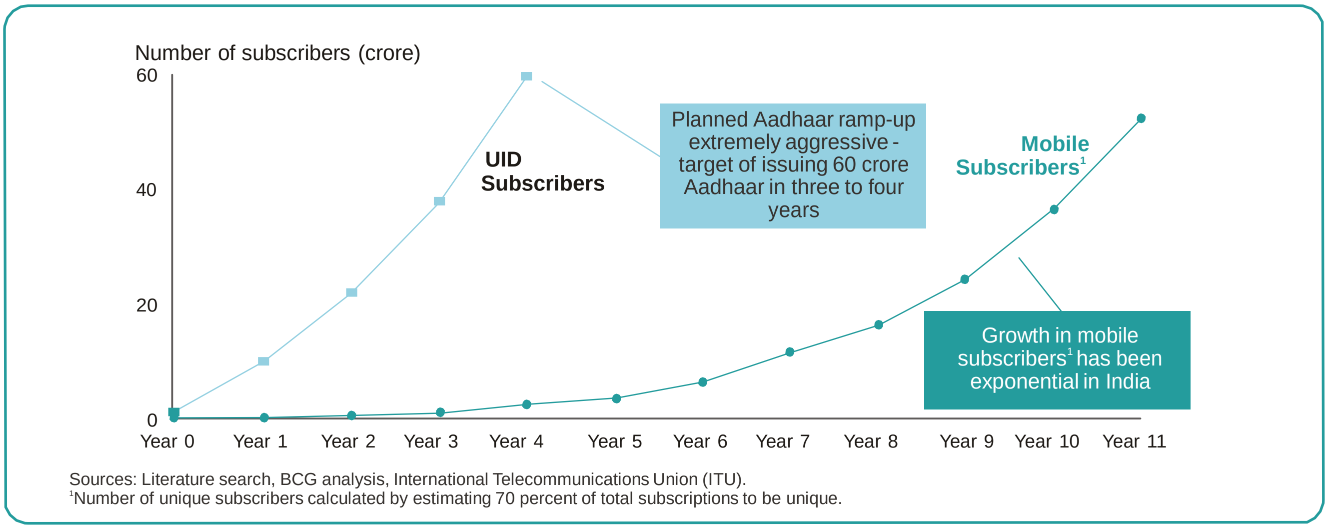
Harnessing the NRI talent pool: We are fortunate that members of the Indian diaspora have attained much professional and personal acclaim in several countries around the world. For example, the Indian-American community is among the most educated and affluent in the US. Roughly a third of new ventures in Silicon Valley are promoted by entrepreneurs of Indian origin. This massive knowledge pool of NRIs, from all over the world covers all the major advanced technology sectors and could be harnessed quickly. China, with a laser-like focus on tapping into its own diaspora, has shown the way here over the past three decades. Members of the diaspora could be offered time-bound assignments, opportunities to attend/conduct seminars, lead short or long-term courses at universities and more.

Faculty Fellowships for Indian Academics: The committee recommends encouraging professors to be stakeholders/partners of the entrepreneurial and incubation eco-systems in universities by offering " faculty entrepreneurial fellowships" . In addition to monthly stipends, the perks could include time-off from teaching responsibilities, talent resource from the student pool and more.

Intermediate Layer: Creating an Enabling Environment for Innovation

1. Embracing the Platform Mindset: The committee recommends creating digital platforms, similar to the Unique Identifications Scheme (Aadhaar), to inspire innovation & entrepreneurship. The Aadhaar initiative was introduced to provide identification for each resident across the country, thereby providing legitimacy to vast numbers of people living in rural areas, who currently have no proof of identity. One of the key objectives of the program was to encourage innovation and provide a platform for public and private agencies to develop Aadhaar-linked applications. The Aadhaar platform has already been used by the government to initiate Aadhaar-based direct benefit transfers for cooking gas, and Aadhaar-based biometric attendance systems. More recently, in June 2015, the Unique Identification (UIDAI), in collaboration with Khosla Labs, Nasscom, and AngelPrime, launched an " Aadhaar hackathon" , an initiative to create useful applications based on the platform. More than 5000 participants (from India and abroad) were part of the 48-hour coding marathon with the purpose of building applications in areas like financial services, payments, healthcare, FMCG, and social inclusion. Winners received prize money of up to Rs. 2 lakh. The winner developed an app to use Aadhaar for verification of student identity. It provides a central database for all exam results, enables online registrations for exams, and prevents impersonation during exams. A previous such hackthon winner developed an app to link Aadhaar with medical history. The full socio-economic impact of the UIDAI project is yet to be seen, as the platform continues to enroll Indians at a pace faster than mobile phone subscription in India.[lxii] Though launching and implementing new digital platforms cannot be done overnight, it will not require generations either. Projects should be conceived to develop new platforms that can be implemented in the medium-term.

Figure 7: Aadhaar Scheme has seen a rapid take-up



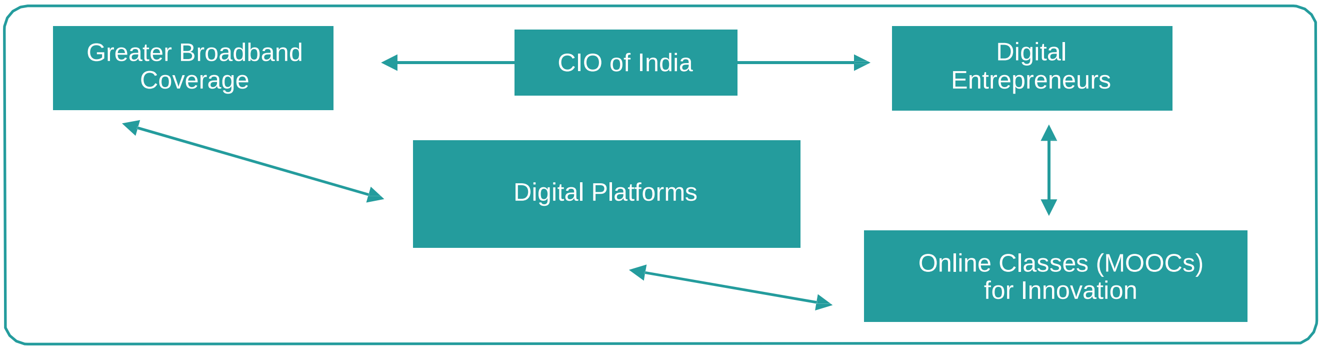
Source: Boston Consulting Group

The committee makes the following recommendations:

A. The AIM should focus on building sector-specific searchable digital platforms that could lead to the creation of new enterprises, digital entrepreneurs and extensive digital innovation. For instance, one of the first initiatives should be building a platform to store Health Care records for all individuals, starting with new born babies, and then building it out for major chunks of the population like government employees etc. This would create opportunities for entrepreneurship across the health care, insurance, pharmaceutical, and training space. Similar platforms could be built to cover pensions, insurance, and education.

B. The AIM must be empowered to appoint India' s Chief Information Officer in order to manage the Digital
Platforms programme.

Figure 8: Entrepreneurship & Innovation centred around Digital Platforms



C. Related initiatives centered on the digital platform, such as online learning (Massive Open Online Courses, MOOCs), and more generally digitization of services, together with greater broadband coverage, could have a multiplier effect on enterprise creation and innovation. For instance, basic digital training and online information on specific topics relevant to agriculture (e.g. appropriate fertilizer usage) or and health care (e.g. vaccination use, or policies to address maternal mortality), could help improve the quality of life and health of the disenfranchised. Access to e-information could also allow villagers across India to access new markets, find customers, and therefore improve economic freedom.

D. Digitization of government processes could help drive faster procurement, processing, approvals, project management, and payments. These could dramatically improve efficiency for the public distribution system, pensions, taxes, agricultural subsidies and more. For instance, a public registry linked to Aadhaar that offers a searchable database of beneficiaries of various public distribution/welfare schemes could help curtail exploitation of the system. The digitization space also offers tremendous opportunities for entrepreneurs who can get involved to help deliver these services.

2. Reforming the Education System and Upskilling Workers: The creation of the Ministry of Skill Development and Entrepreneurship is a welcome initiative and the National Policy for Skill Development and Entrepreneurship 2015 report has many excellent recommendations, which this committee acknowledges. The committee supports the Ministry's focus on the steady re-engineering of the education system in the country to prepare our youth for the new innovation economy and provide our young enterprises with a large pool of highly-employable workforce. A multi-layered approach will be required, including overhaul of existing school and college curricula, change in existing teaching techniques, better monitoring of school and faculty standards, better access to e-education facilities, and better targeted skilling and training to ensure employability of youth. In the section below, the committee makes a further number of recommendations that it views as pivotal.

A. Reforming school curricula and examination methodology: The curricula and examination format of the Indian central and state boards need to be overhauled and reoriented from rote-based to application-based learning and testing. The focus should be on testing higher-order skills like reasoning, analysis, lateral thinking, creativity, and judgement, rather than memorization alone. Until testing standards change, teaching and learning methods are unlikely to change. The International Baccalaureate (IB) curriculum, for instance, could be used as a benchmark. The IB is known to cover a broader spectrum of subjects that leads to all-round development. IB examinations test students' understanding rather than their memory/ speed. It is also known to equip students with tools needed to succeed in higher education and in the workplace, like self-confidence, preparedness, research skills, and presentation skills. Teaching methods that use real world problems to explain concepts and theories should be encouraged. Students should be exposed to broad needs of society—health, education, hygiene and cultural development—to promote business and entrepreneurial solutions to public policy issues.

B. Annual assessment of schools and faculty: Schools across the country should "pass" an annual assessment based on a standardised exam that their students must take in basic science, maths and literacy. Care must be taken to ensure that the exam cannot be cracked by rote learning or formulaic study methods. The Pratham Annual Status of Education Report (ASER) is a good example of a non-governmental annual survey of children's basic learning skills in arithmetic and reading in rural India. In 2014, the ASER survey covered 577 rural districts, surveying children from 341,070 households, and covering over 500 institutions. Even in many of the best urban and international schools, the quality of teaching is poor and learning is based on rote. To ensure a high quality of teaching, teaching standards need to be monitored. One way to do this is to set up an expert committee consisting of qualified NRIs to carry out random quality checks. The emphasis in these checks should be on how well teachers are able to communicate to explain complex and abstract issues, rather than on textbook teaching. This could serve as an effective deterrent to lax teaching quality standards. Simultaneously, a revised and updated minimum nation-wide standard/ exam for faculty should be introduced, to ensure that teachers are suitably qualified to effectively impart learning.

C. Providing access to entrepreneurship courses: Students in secondary and tertiary levels must have easy access to entrepreneurship education courses and programs.

D. Build a virtual platform that documents experiments and innovations in education around the world, and provides an ecosystem for education innovation in the country: The platform would help draw together a community interested in education and innovation, which looks at the challenges bottom-up rather than top-down and encourages, rewards and supports these initiatives. The platform must be accessible to all, allow screening, categorization, review, data analysis and evidence-based evaluation of these innovations by the broad community and global experts. The Government of Singapore has encouraged many such experiments, for example, from which there is much we can learn.

E. Encourage a focus on technology-based solutions to education and open up the market to global education providers: International or home-grown education innovators (both private and public) should be allowed access to schools and universities in India. These places of learning could become "virtual laboratories" where education experimentation is encouraged and monitored.[12] All universities in India should be allowed to offer online education, available across states. The use of MOOCs and online education is a fast-moving space and can ensure access to those in remote and rural locations, and aid the process of social inclusion.

---

[12] Students and teachers should collectively and individually have the option to opt in or out of the innovation programs.

F. Create a National Education Service Program, after drawing best practices from the world over: Inspiration could be derived from the National Service systems of Israel (IDF), Korea and Singapore, but the program must be designed specifically to meet Indian needs. The Indian program could have several tracks such as practical education, education and health, education and security, education technology, education and skilling and entrepreneurship education.

G. Train workers for specific job-related skills: The committee recommends creating special education curriculums, vocational, and skilling programs to fill large-scale vacancies in certain careers that have a big and direct impact on entrepreneurship and innovation. At the moment, the limited incubation-based skilling programs present in India are focussed on e-commerce-related skills. More broad-based skilling programs need to be developed to cater to the needs of unskilled, semi-skilled, and highly-skilled workers. Such programs could be implemented on a PPP-basis, with some sort of funding/ fiscal incentives from the government, and expertize/ management from the private sector.

The unskilled category, consisting of those with only a high-school degree, could be trained to fill vacancies in low-skill jobs. This includes jobs within gardening, dry cleaning, women’ s tailoring, women salons, taxi driving, carpentry, and construction, amongst others. In smaller towns in India, where many low-skilled women need work, there is no training available in fields that may be suitable for them—such as beauty, tailoring, embroidery, design and more (this is not an exhaustive list and certainly women have wide-ranging interests). The rapid growth of the private cab industry in cities has led to many unskilled workers finding jobs in the sector, thanks to a semi- organised driving school industry. Similarly, India is experiencing a construction boom unlike ever before in its modern history. Construction Schools across the country could elevate lakhs of manual labourers to specialists in plastering, tiling, or foundation laying. After training, these labourers would be eligible to earn close to double of their current daily wages. SMEs could benefit from trained workers in this category and there should be dedicated SME training camps that could potentially have a bigger impact on business than the current trend of incubators dedicated to ecommerce and mobile phone applications. The expert committee recommends setting a target to introduce 5000 SME training camps/ schools within 3 years. If each takes 3 months to train a batch of students and
500 students are admitted per batch, 10 million students could graduate in one year.

Similarly, there are lakhs of young Indians who fall into the semi-skilled category, those who have completed basic undergraduate college degrees, but are not trained to take up highly-skilled jobs. Efforts can be made to train and match these graduates with suitable jobs. To this purpose, skilling programs should be devised after a careful assessment of the job market and the supply gaps that these workers could fill.

Finally, even highly-skilled workers i.e. those with a technical undergraduate college degree, an advanced degree, or vocational training, need help finding suitable work. For this category, there could be special, nation-wide entrance exams and training programs that enlarge the pool of trained professionals in a short timeframe. For instance, two areas where highly-skilled workers are immediately required are teaching and patent examination. Both these areas have a direct impact on innovation. For instance, in 2010, India was short of 12 lakh teachers. While the Right to Education requires 1 teacher for every 30 students, some government schools function at the ratio of 1 teacher for over 100 students.[lxiv] It is not just the number of teachers which is important; quality is equally important. The B.Ed curriculum should be overhauled and reoriented towards practical rather than textbook teaching. Examinations should place emphasis on how well teachers can explain complex and abstract concepts, rather than textbook teaching. Similarly, the speed of patent approvals has a direct impact on innovation and the morale of the innovators. As of 2013, there were 194,000 pending patent applications and only 201 patent examiners in the country.[lxv] Quality skilling programs need to be introduced to train workers to fill vacancies in patent examination. The number of patent examiners needs to be increased at least tenfold by 2018.

Over and above this, the efficiency of existing government initiatives need to be improved. The number of apprentices need to be increased to 1 crore from the current 4 lakh, and employment exchanges need to be converted to career centers (last year, 1200 exchanges led to only 3 lakh jobs for the 4 crore people registered). Loan programs for skilling need to be scaled up. One option is to fund students directly rather than institutions. In this context, the government’ s recent launch of the Skill Loan scheme is a step in the right direction. Under this scheme, loans ranging from Rs 5000-1.5 lakh will be made available to 34 lakh youth seeking to attend skill development programmes over the next 5 years. More generally, the Skill India initiative, under which the government aims to train 40 crore workers by 2022, is an excellent initiative, and efforts should be made to monitor impact annually, to ensure effective execution of the program.

H. Promoting internship and coop programs to improve "work-readiness": The corporate sector should be required to develop clearly-defined internship and coop programs. The requirement on the number of internships/ coop programs could be made proportional to firm revenues. This would be useful for students, who would get a sense of different career options and the skills/ inclinations required for each. Graduates would thereby be able to make informed career choices. It would also help firms in getting a head start in identifying and training talent that they can later employ. Through " testing" graduates within a trial period, firms would also be able to limit the number of hiring mistakes. The internship and coop systems are well-developed in the West, and have proved to be effective in helping match students with firms.

I. Establishing two separate sets of regulations for universities: There is a case to have two separate regulatory regimes—one for small research universities (focused on knowledge creation, innovation, and global rankings), and large vocational universities (focussed on employer connectivity and delivery via distance education and apprenticeships).

3. Strengthening the Intellectual Property (IP) Rights Regime: As a fast growing economy, India will have to establish that it is serious about rewarding innovation. A strong and robust IP ecosystem will not just boost India' s image globally but also help spur domestic innovation and investment in R&D.

A. Enhanced Enforcement: IP Laws in India are compliant with Trade-Related Aspects of Intellectual Property Rights (TRIPS) administered by the World Trade Organization (WTO). However, the laws are poorly enforced, as innovation and protection of intellectual property are not prioritised adequately by enforcement officials. An important first step is spreading awareness and sensitising all the relevant authorities involved in enforcing laws, including the police and the judiciary. Further, specialised training must be provided to members of the judiciary on Intellectual Property and innovation, given the complexity of the cases involved.

B. Dedicated IP Courts: Due to the limited jurisdictional reach of the Intellectual Property Appellate Board and the rise in the number of patent litigations, India needs dedicated IP courts to manage specialised IP cases, as well as to improve the efficiency and speed in IP judgement. The IP courts must be stipulated to give their judgements within 2 years, with no more than 3 adjournments between hearings, and no more than 10 adjournments in total throughout the trial.

C. National Virtual IP Platform: AIM could oversee the establishment of a National Virtual IP Platform that offers a forum for the stakeholders and all those interested in Intellectual Property Law to discuss, innovate and collaborate. Further, the Virtual IP Platform could contain a database of all the resolved IP cases in India, as well as details of those under litigation. Over time, an electronic case management system can be integrated into the Virtual IP Platform for quick resolution of IP-related disputes and issues.

D. Increase Number of Patent Examiners: New innovations are often time-sensitive and a large number of pending patent applications severely dampens the spirit of innovation in the country. The shortage and attrition of patent examiners at the Patent Office must be addressed to resolve the issue of pending applications. There should be a concerted effort to introduce a large number of high-quality patent examiners, with a ten-fold increase from the current number of examiners by 2018 being a reasonable goal.

4. Improving the Ease of Doing Business:

A. Digitization of government permits: The central government must require all government departments to have all registrations, permissions, and licenses to be online within two years. States must be given incentives and ranked based on electronic, single window compliance within a deadline. The process for name, approval, and company incorporation needs to move completely online. There could be a deadline of three days free, and one day with payment of expedited services. All new laws should be mandatorily born digitally native. Automation of as many government processes should take place, and discretion of government officials should be reduced as much as possible, to reduce red-tape. Innovation in governance is critical. Without this, even well-intentioned policies are doomed to fail.

B. Create a Central ID for enterprises: Every enterprise in India, whether company or not, is assigned multiple numbers. A single number for enterprises is a key ease of doing business initiative and the basis for the " online" portal for enterprises proposed above. The AIM should set up a committee to chart a 1-year roadmap for all enterprise numbers to converge on the Central ID.

Among the Multiple Identities Issued to Indian companies are:

- Corporate Identity Number (21 digit alphanumeric) from ROC
- Tax Payer Identification Number for Commercial Taxes (11 digit numeric)
- Service Tax Number (15 digit alphanumeric)
- PAN Number (10 digit alphanumeric)
- Central Excise (PAN Number +2 characters)
- Provident Fund Number (11 digit alphanumeric)
- Profession Tax Registration Certificate (9 digit numeric)
- Profession Tax Enrollment Certificate (9 digit numeric)
- Tax Deduction and Calculation Number (10 digit alphanumeric)
- ESIC number (17 digit numeric)
- Labour Department Registration Number (13 digit alphanumeric but varies from state to state)
- Importer Exporter Code (10 digit numeric)
- Shops and Establishments Act Registration (20+ digits alphanumeric)
- CLRA Registration (15+ digits alphanumeric)
- Labour Welfare Board (5 digit numeric)

C. Revisit the Companies Act: The Act needs to distinguish between closely held private companies, public companies, and publicly listed companies. Recommendations specific to the Act follow:

- Regulation of listed companies should be left to stock exchanges and SEBI norms.
- Streamline the more than 15 filings required
- Revisit liabilities of key management or raise the threshold to Rs. 50 crores.
- The Act bans ESOPS to Independent Directors of unlisted companies. Further, it confers on these Independent Directors more powers that come with additional responsibilities. Such rules which deter talented and experienced individuals from taking up these positions that come with all of the challenges but none of the rewards. As a result of this ban, Indian companies are finding it increasingly difficult to recruit Independent Directors, who often play a crucial role in offering strategic guidance to new and small enterprises.

D. Tax compliance: All tax law, government notices related to taxation, and responses to taxpayer queries should be available online. Any tax law that cannot be codified should be reviewed to avoid misuse. Digitization will require a radical review of the organization structure, culture and incentives of the Central Board Of Direct Taxation (CBDT). The government should prescribe a clear timeline for complete digitization of taxation within the next 24 months. There is also an urgent need to revisit Section 56 of the Income Tax Act that has greatly impacted fair market valuation norms on angel investments.

E. Move service tax back to actuals rather than accruals: The move to service tax becoming due when payments are received rather than when they are invoiced has created a huge cash flow problem for all enterprises but particularly the small ones.

F. Improving access to capital: To structurally lower interest rates, the following measures are needed: more bank licences, development of the corporate bond market, a level playing field for offshore and onshore private equity and venture capital, and a stable and transparent regulatory and taxation regime. The RBI has recently approved 11 applications for payments bank licenses, and intends to approve applications for small finance bank licenses shortly. The latter would further the government' s financial inclusion agenda by making large swathes of the population " bankable," and provide a meaningful source of funds for the micro-enterprise sector both in urban and rural areas. Approval of more bank licenses will increase competition within banking and lower interest rates over a period of time. Quick implementation of the proposed bankruptcy law will also be key to improve access to capital. The law aims to allow faster closure of troubled businesses and give creditors easier and faster exit options. Therefore, it should expedite the cleansing of bank balance sheets, and allow banks to undertake fresh lending. Delays in dealing with bad loans can be hugely detrimental to economic growth, as we have seen in Japan in the 1990s, and in the European periphery more recently. Development of a healthy corporate bond market too will require a quick resolution process in favor of secured bond holders (within 90 days) in the event of corporate default. India could also take note from foreign countries that provide fiscal incentives for entrepreneurship. Some examples are: In Israel, non-Israeli investors are exempt from capital gains tax on investments in VC funds and tax rates on capital investments has been lowered for certain priority areas. The government absorbs a percentage of losses that institutional investors might suffer from VC investments, and also gives grants to R&D centres that employ high-salary staff. Singapore provides tax deductions of up to 50% of angel investment into new ventures. In New Zealand, the government co-invests with angel groups. Similar measures should be considered to boost entrepreneurship in India. We note that the Securities and Exchange Board of India (SEBI) has recently set up the Alternative Investment Policy Advisory Committee to advise the Board on growing the alternative investment industry in the country. Some of their recommendations, due later in 2015, are expected to bear on financing methods relevant to entrepreneurs, for example, the provision of risk capital (angel capital).

G. Labour market reform: The committee recommends the following:

- The central government should implement the 3 employee choices for pensions and health insurance announced in the Union Budget 2015-16.
- Complete consolidation of 44 central labour laws into 4 labour codes.
- Create an employer choice of complying under Factories Act or Shops and Establishments Act for all non-hazardous enterprises.
- Amend the Trade Unions Act to make them more representative, and amend the act to reduce if not remove the role of outsiders because the politicisation of trade unions is toxic for the Make in India campaign.
- Set a deadline for all central government labour approvals to be live on a single online portal with a 90 day deadline.
- Encourage states to use Article 254 (2) of the constitution to amend labour laws.

H. Creation of an online nationwide real estate registry: An online portal should be set up for the registration of all land purchase/ sale deals by all entities and their beneficiaries. Registration for all real-estate transactions should be made mandatory within 48 hours, with strict penalties laid out for non-compliance. This would serve as a deterrent to corruption in land deals. Simultaneously, the land registration fee should be lowered to facilitate a more dynamic and transparent real estate market. The Registry should have the " search" feature, and search should be enabled by name, address, company, across companies, chain of relatives etc. This database should allow cross referencing with Income tax and other tax departments. Any piece of urban or rural land that is sold or provided at nominal value (including land given to nonprofits) by a government entity without an open bidding process must be publically listed on this portal. Any land acquired by non-profits should also be listed on this portal, as this is a common route for corruption.

I. Creation of an online nationwide lien registry: An online nationwide Lien Registry should be set up to record details on all borrowings and pledgings of all persons and entities (without exception) across India, and their beneficiaries. This portal should include a search function, to enable searching lending data across banks, thereby providing a comprehensive view into a borrower's existing borrowing, collateral pledges, and credit history.

J. Creating an online portal to aggregate information on funding to entrepreneurs: Information on all state and central incentive programs for entrepreneurship, including loans, grants, subsidies, venture funds, assistance to minorities, and other such measures should be available on one online portal. This would ensure that entrepreneurs have access to information on funding options and assistance available to them. Lack of visibility on government venture funds, for instance, is a key problem.

K. Creation of an AIM Entrepreneurship Index: AIM should work towards creating an Entrepreneurship Index that measures entrepreneurial activity in India, and thereby help stakeholders track improvement. Each year, the Index could update the number of start-ups in India, the states and cities attracting the most number of entrepreneurs, the share of female, minority, and dalit entrepreneurs, and other important factors. In the US, the Kauffman Index has become the authoritative indicator and predictor of entrepreneurial activity.

L. Creation of a separate regulatory category for new business: According to the World Bank 2015 Ease of Doing Business Survey, India ranks in the bottom quartile in the "Starting a New Business" category, at 158th place out of 189 countries. According to this survey, it takes close to a month and more than 10 procedures to start a new business in India.[lxvi] India's New Business ranking is even lower than the country's overall "Ease of Doing Business" Ranking, which is at a dismal 142nd place. The committee recommends introducing a "New Entity" category that exempts new businesses from heavy regulatory compliance. New businesses should only have to follow a set of bare minimum regulations and procedures. This new category could exclude businesses operating in sensitive areas, large-scale businesses and where foreign capital is involved.

M. Legal reforms: The legal system is notoriously inefficient and slow. There is an urgent need to improve timeliness in adjudication. For instance, the government could set a rule stipulating that there can be no more than 3 adjournments between hearings and no more than 10 adjournments in total through the course of each case.

N. Creating an enabling environment for social enterprise: Creating the relevant infrastructure to promote social enterprise is important. Though we have several successes in this space, efforts need to be scaled up to have an impact at the macro level. In the following box, we discuss what can be done to promote social entrepreneurship.

Encouraging Social Enterprise: Social entrepreneurship is the recognition of a social problem and the uses of entrepreneurial principles to organize create and manage a social venture to achieve a desired social change. Social entrepreneurs are commonly associated with the voluntary and not-for-profit sectors, but this need not preclude making a profit. Social entrepreneurship has progressed significantly in India over the last decade. Companies like Narayan Hospitals, Arvind Eye Hospitals, and Vaatsalya Healthcare are well-known and recognized for improving access to affordable healthcare. Pratham and Aakanksha are well-recognized for improving access to education amongst the underprivileged. Several enterprises have been set up to deliver high-quality affordable solar solutions for families living off-grid. Of late, many SMEs have been set up to encourage rural craftsmanship. Impact investing is becoming increasingly better established in India, with over $1.6 billion invested in over 220 enterprises since 2000. However, there is still a long way to go. Though several ventures are creating impact on a local/ regional basis, the impact at the macro level appears small.[lxvii] Furthermore, 60% of all impact investment was channelled to just 15 out of the 220 enterprises, and 70% of investment was directed towards microfinance and financial inclusion. Areas like water and sanitation, affordable housing, energy, education, and healthcare need funding and attention. Much can be done to encourage these ventures. Some measures that would encourage social enterprise are:

A. Involving the corporate sector to fund social enterprise: The corporate sector could have much to gain from partnering with social entrepreneurs. This includes reaching untapped markets through entrepreneurs who have spent decades refining innovation means of bringing excluded groups into the marketplace. It also includes attracting more " millennials" , for whom social impact is a key component of job satisfaction. The 2% CSR " tax" (estimated to amount to $2.5 billion per year) on corporates presents a significant pool of domestic funds that could be used to fund social enterprise. To be effective, funds need to be utilized efficiently. Improved transparency on CSR spend is much required. This could be achieved through collation of company-level CSR information on an e-platform. This could help firms augment each others' CSR efforts, rather than duplicate them. Coordination of CSR can be done by industry bodies. A common fund to pool CSR resources to make sure the corpus is spent intelligently could also be useful. One option is for firms to aggregate CSR funds and channel them to Impact Investors (like Aavishkaar, Acumen, and Elevar Equity), as these organizations have the skill and expertise to generate and monitor impact.

B. Encouraging the setting up of business incubators for social impact: UnLtd, Dasra, and Khosla Labs, are amongst the most prominent incubators for social enterprise. Many more such incubators are needed to provide expertise, technical help, and financial support for social enterprise. A quick way to achieve impact is to set up virtual incubators i.e. curated sites to provide social entrepreneurs with access to advisors, mentors, and experts, remotely.

C. Promoting a culture of "giving back": The government should encourage philanthropy amongst high net-worth individuals to increase funding. One way is to provide tax credits to HNIs that invest in early stage VCs or to incubators and angel investors. Involve students in finding solutions to practical problems could promote a culture of innovation and may increase the number of professionals working in the social impact space.[xlviii]

Base Layer: Addressing the Cultural Context

Changing cultural attitudes towards entrepreneurship is likely to be a long-term exercise, and may require generational change. Recognizing that change needs to occur is an important first step for cultural attitudes to be modified. As the first generation in many families will soon be educated and will have moved up the income ladder, they may be more open to the idea of their children being self-employed. A revamp of the school system and curriculum could also play a rule in encouraging entrepreneurship in the generations to come. There are some sections of Indian society (the Gujarati' s, Parsis, and Marwaris, for instance) which are culturally more disposed to entrepreneurship than others. One way to expedite the entrepreneurship agenda could be to conduct exchange programs wherein the youth from some parts of the country spend time learning in another, and vice-versa. Europe has a well-formed such exchange program to facilitate cross-country learning, termed the " Erasmus" program. Changing the culture of apathy, corruption, and scepticism towards intellectual property rights within the bureaucracy, is also likely to take time. Consistent trust-building over time can help address the general sense of mistrust between the corporate sector and the government. The government' s commitment to avoid retrospective changes in law going forward is a positive first step. To expedite cultural changes, the expert committee proposes the following recommendations:

1. Attach Entrepreneurship to Large Scale Economic and Social Programs: To spread the spirit of entrepreneurship and innovation widely, AIM should use various important government economic and social programs as vehicles of change. For instance, " Swacch Bharat" could be used to encourage and promote social entrepreneurs focused on the areas of cleanliness, hygiene and civic responsibility.

2. Promote New, High-Potential Sectors via the "Make in India" Campaign: New-age sectors within manufacturing with the potential to create future jobs, enterprises and economic growth should be encouraged. AIM should set-up incentive structures to develop a few chosen high-potential industries as part of the " Make in India" initiative. Knowledge flows into these sectors must be supported to foster India' s competitiveness in the world. A focused and systematic approach towards some of the niche sectors could improve export income, create industry growth and enable talent development.

3. Foster a Culture of Coordination and Collaboration: Working in silos has an insidious effect on the innovation ecosystem; while there are many noteworthy entrepreneurship and innovation-boosting initiatives emerging from different sections of the government, private and non-profit sectors, the lack of coordination does not help these programs scale and create long-lasting, long-term impact. A culture of coordination between ministries, departments, incubator cells across the country, enterprises, and more should be strongly encouraged and institutionalised. To monitor progress and measure impact, an annual survey of start-ups and early-stage ventures should be commissioned. The purpose of this survey should be to receive regular feedback from entrepreneurs on problem areas and areas of strength in coordination and collaboration with various stakeholders. This relates to an earlier point on monitoring impact of incubator cells and using feedback as a tool to reform policy and improve efficiency.

4. Redefine Success: While encouraging a healthy tolerance for failure at the societal level, AIM must push for a new culture that redefines success in crucial governmental bodies. R&D, entrepreneurship and innovation-centred organisations within the government system must be allowed to pursue projects and experiments that are high risk and may fail. Different measures of accountability and success must be outlined.

5. Make Entrepreneurship Part of the Social-inclusion Agenda: With economic growth and national progress, India will remain focused on greater social inclusion and mobility for decades to come. AIM must capitalise on this predictable trend and make entrepreneurship part of the larger social agenda, bringing in more women, dalits, rural population and the urban underprivileged into the fold of new Indian entrepreneurs and innovators.

Proposed Structure for the AIM

The time is ripe for India to embrace a new path of enterprise building and disruptive innovation; a path that may involve risks but also high rewards, because status quo will only yield incremental results—and for a country where more than 350 million people are below 24 years in age, that simply isn't good enough anymore. The committee welcomes the vision to create the Atal Innovation Mission (AIM)—a catalyst for change and the vehicle that will steer the construction of a new and strong innovation ecosystem in India.

Below, the committee has outlined the recommended objectives of the Atal Innovation Mission and provides a brief overview of the recommended structure.

Objectives of the Atal Innovation Mission

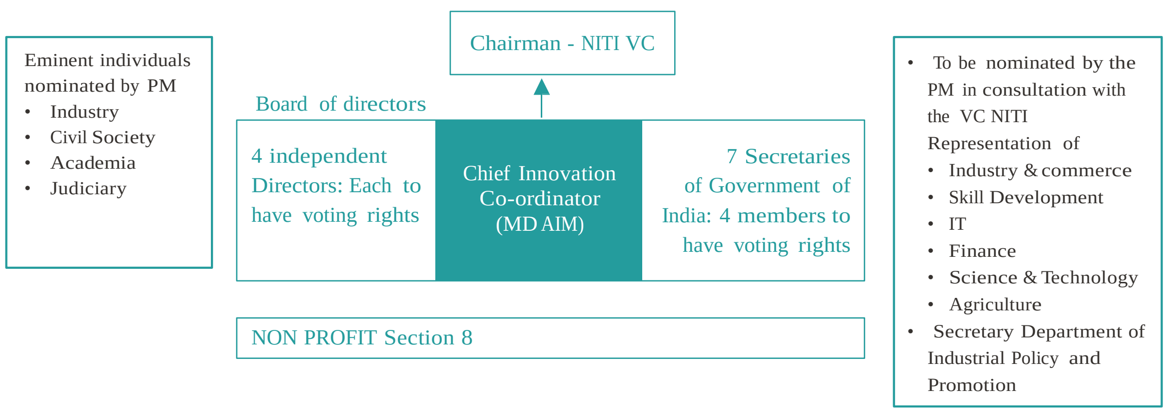
AIM is conceived as a self-sustaining organisation that will play the following key roles:

- Facilitate growth and development of an innovation eco-system through policy advocacy and, in select cases, offer active implementation support
- Identify, recognize and reward innovations with demonstrated or clearly visible potential and impact
- Innovations can be of technology, business model, commercialization, working innovation, scaling-up or any other type
- Provide support for scaling up of solutions based on proven and promising innovations
- Institutional collaboration/support, tie-ups with global institutions, funding, (direct funding, or fund of funds model), other resources/support, support of policy for innovation success
- Identify problems of national importance, and facilitate innovation focus by orchestrating institutional priorities, funding and other levers
- Establish itself as an apex governance body regarding co-ordination of innovation agenda being executed through different elements of the eco-system
- Authority to measuring outputs and value of innovation (without changing reporting)
- Facilitator of internal collaboration and tie-ups with external agencies (including public-private partnerships)
- Must have the authority to request and receive information in a timely manner

Figure 9: Recommended Structure of the Atal Innovation Mission

Governance of Atal Innovation Mission

The high-level governance architecture of AIM is shown in the diagram below. The details of different components are elaborated further.



1. **Chairman** – VC NITI Aayog shall serve as the chairman of AIM board.

2. **7 Government Functionaries** would serve as Directors on the board of AIM. These functionaries shall be no-less than Secretary level from the following Ministries:

   a. Secretary Skill development
   b. Secretary Industry and commerce
   c. Secretary IT
   d. Secretary finance
   e. Secretary science and technology
   f. Secretary Department of Industrial Policy and Promotion
   g. Secretary Agriculture

Only 4 of these Directors would have voting rights

3. **4 Independent Directors** chosen from outside of Government would serve on the board of AIM. Each of these Directors would have voting rights. With this structure in place, private and public sector Directors would have an equal number of votes.

For more details on the structure, competencies required for AIM personnel, organizational metrics, please refer to Appendix E.

Case Study: Potential of the Bioeconomy Sector

This section draws special focus to the biotechnology industry that currently generates close to $5 billion a year in India, and is projected to continue to grow at just under 20% CAGR for the next decade.[lxix] The economic potential of biotechnology is comparable to the software industry in the early 90s; indeed experts say biotechnology can have a more wholesome impact because of its relevance to domestic and global markets.[13]

Biotechnology presents technology driven levers for addressing health, food, fuel and environment—the four largest challenges faced by the planet. As a nation we need to embrace biotechnology and thereby create a thriving bio-economy just as we have an information economy. As former Prime Minister Atal Bihari Vajpayee used to say " IT stands for India Today and BT stands for Bharat Tomorrow" .

After a brief review and exploration of the sector, including its history and current state, industry-specific recommendations are offered below. The recommendations are structured along the lines of the AIM pyramid model, and classified into short, medium, and long-term measures.

Industry Overview

1. Dropping costs and rising speed of innovation: PCs and Smart Phones have transformed our lives at a rapid pace and continue to do so, reflecting Moore' s Law which posits that as computing technology increases in power, its relative costs decrease. But biotechnology is poised to take us even further, as technology unlocks the potential of biology at inconceivable speed and cost. For instance, the costs and speed of genome sequencing is outpacing Moore' s law. In 2003, less than a decade ago, genome sequencing cost US $2.5 billion. Just a year back, a high- end genome sequencer machine that cost US $10 million brought the cost of genome sequencing down to $1000. More recently, during the Ebola outbreak, a hand-held genome sequencer was deployed to check for mutations; the cost of the device was just $1000.[lxx] These dropping costs of measurement of molecular signatures opens a strategic opportunity for India to play in the " software" needs of genomics globally – bioinformatics analytics and genome variant interpretation.

2. A Brief History of Biotechnology in India: India' s contribution to global biotech began in the 1950s. Shortly after the double-helix structure of the DNA was discovered, Indian scientists Dr. G.N. Ramachandran proposed a triple-helix structure of collagen using X-Ray diffraction, which led him to develop the Ramachandran Plot—an essential tool in protein confirmation even today.[lxxi]

In 1982, the Department of Biotechnology (DBT) was set up, which established and funded research labs. Meanwhile, India' s pioneering private biotech company, Biocon, had become the first Indian company to receive funding from the US for proprietary technologies.[lxxii]

In recent years, Biotechnology Industry Research Assistance Council (BIRAC), a section 8 non-profit under the DBT, has been charged with fostering innovation and entrepreneurship in the sector. So far, BIRAC has supported over 150 young Entrepreneurs, 300 startups and SMEs. BIRAC also provides incubation space to over 100 young ventures. The challenge for this sector is whether this early stage support is followed up with right levels of capitalization. Biotechnology is a risk capital intensive sector with longer gestation times for entrepreneurial success. It therefore requires a complex financial ecosystem that provides either risk capital that is patient or speculative public markets like Nasdaq that value the innovation potential of intellectual property far ahead of full commercialization of the innovation.

Figure 10 : Ranking of Biotechnology Firms in India

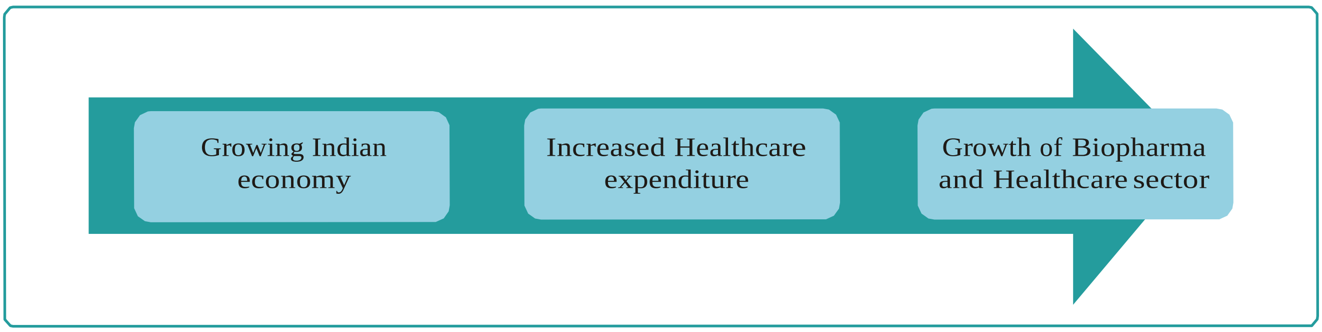
Rank	Company	2013-14
1.	Serum Institute of India	3,340.00
2.	Biocon	2,218.00
3.	NovoNordisk	793.00
4.	Nuziveedu Seeds	775.00
5.	Syngene Internatinoal	715.00
6.	Reliance Life Sciences	700.00
7.	Kaveri Seeds	647.00
8.	GVK Bio	599.00
9.	Indian Immunologicals	442.00
10.	Eli Lilly	430.82

Source: ABLE Spectrum Survey 2013-14

[13]Communication with members of Biotechnology Industry Research Assistance Council and Association of Biotechnology Led Enterprises, July 2015

3. State of Play-Biotechnology: In 2014, the Indian biotech industry was worth $4.66 billion and was growing at a modest rate of 6.98% perhaps due to regulatory challenges that the industry faced in the areas of clinical trials and GMO based agriculture.[lxxiii] In 2013, the industry had registered growth of 15.08% and for the preceding five years, Indian Biotech had grown at an average rate of 18%.[lxxiv] With a vast technology-trained talent pool and adequate infrastructure, biotechnology in India is on a high-growth trajectory. It holds immense potential to contribute in a big way to " Make in India," raise the country' s profile as a manufacturing hub with a difference, and contribute to enterprise creation, innovation and economic growth. Its growth potential is comparable to that of the IT sector in India in the early 1990s.[14] Like the software industry, biotech can play a crucial role in driving India' s economic growth in the future. The Indian economy[lxxv] is set to grow to US $4 trillion by 2025, and Biotech could become a significant contributor to the Indian economy. Importantly, as the economy grows, India will contend not just with diseases borne out of poverty, but also with chronic and acute diseases of affluence. The sector must rise to the challenge, provide innovative yet affordable solutions, and seize this opportunity for growth.

Figure 11: Potential for Growth



Source: Indian Biotechnology Roadmap to the Next Decade and Beyond, May 2012, ABLE & DBT Report.

Over the years, several Indian biotech firms have devised innovative solutions and scaled up the production of cutting-edge products that are low in cost and high in quality. These include Serum Institute, Bharat Biotech and Shantha Biotech that have developed and supply more than 50% of vaccines to international organizations such as WHO and UNICEF.[lxxvi]

There are also several bio-clusters spread across the country, in Maharashtra, Karnataka, Uttar Pradesh, Andhra Pradesh, NCR and Gujarat. However, Bangalore could be considered the country' s biggest hub with 137 biotechnology companies, which makes it 40% of the total 340 such units in India.[lxxvii] The biotech industry includes biopharma manufacturing, bioservices, bioagriculture, bioindustrial and several sub-sectors, with biopharma being the largest revenue generator.

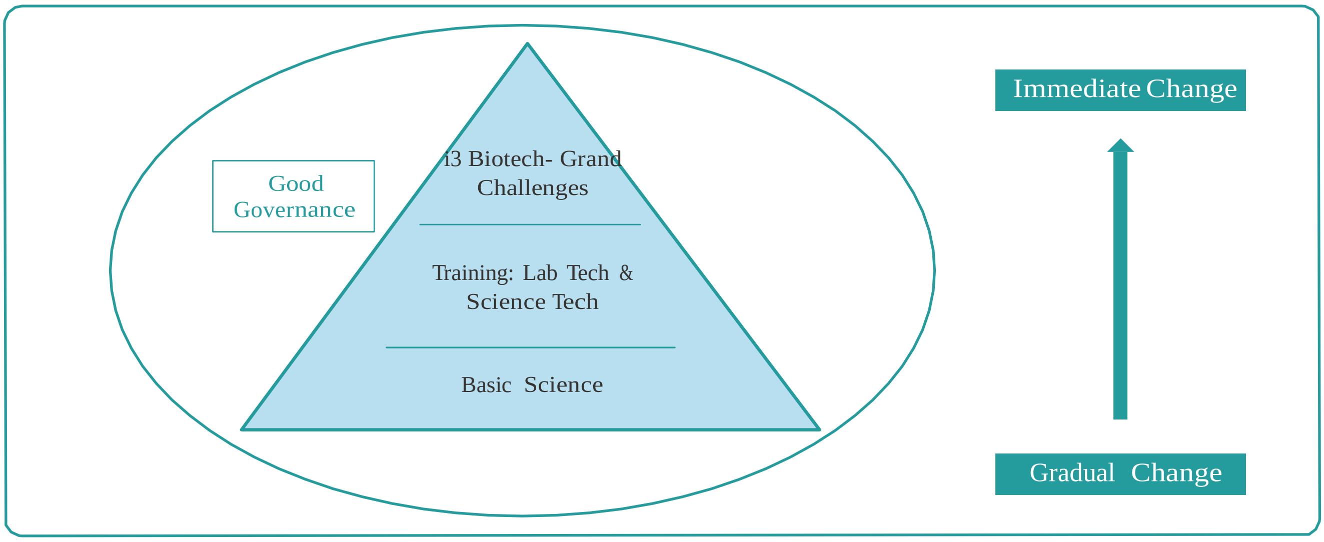
Recommendations to Unleash a Biotech Boom

Biotechnology stands at the threshold of enormous growth and could have a direct impact on the economy and offer wide-ranging societal benefits beyond just public health. The industry has attracted talent from different parts of the country, as well as qualified non-resident Indian scientists and other biotech professionals. While biotech innovation exists in India, there is scope for greater improvement.

The expert committee has highlighted recommendations to boost the Biotech sector that range from the short-term to the long-term. As in the main recommendations section of the report, long-lasting but profound changes are outlined in the base layer of the pyramid, measures that will have an impact in the medium-term are outlined in the intermediate layer, and measures in the top layer offer payoffs that will be quick, but perhaps not as substantial. The Biotech Pyramid is also encased in the very crucial good governance circle.

[14] Communication with members of the Biotechnology Industry Research Assistance Council and Association of Biotechnology Led Enterprises, July 2015.

Figure 12: Biotech Pyramid



Top Layer Recommendations

In general, biotechnology innovations have long gestation periods. In the biopharma sub-sector, for instance, some drug discoveries may take close-to a decade to be launched as products, and the entire process cycle may involve millions of dollars of investment.[lxxviii] However, it is still possible to create an inspiring and vibrant environment, and draw attention to the sector in the short to medium timeframe.

1. Incentivize Innovation in India (i3) - Biotech-focused Grand Challenges

The committee recommends that the Mission set aside part of its resources to structure and run biotech focused i3 challenges.[15] Medical device innovation and Med Tech lend themselves well to the i3 format, where the potential of the innovation can be tested relatively easily in a shorter timeframe. Similarly, interplay of informatics in the life sciences and healthcare sectors is also a domain, where Grand Challenge innovations can come from.

i3 Ideas to Overcome Skill Shortage

A medical device plugs the skill gap in Indian Healthcare today, where often, poorly trained or semi-skilled professionals deliver healthcare solutions.

Grand Challenge 1: Non-invasive tool for automated diagnosis and screening of TB, particularly the drug resistant strains. This challenge could also be applicable to other key public health concerns such as diabetes, oral and cervical cancers (saliva and urine tests) and high-risk pregnancies.

Grand Challenge 2: A robotic surgery platform that can perform basic laparoscopic surgical procedures. The surgery platform should be affordable (cost less than Rs. 2 Crore) with low procedural costs, easily operable by a trained MBBS physician and perform a set of relatively easy, high-volume, minimally invasive procedures.

i3 Ideas to Drive Innovation in Bioinformatics and Medical Informatics

We believe there are many interesting grand challenges that can be posed in this interface domain. It is interesting to note that genomics alone is likely to be the source of the world' s largest data type in the next 5 to 10 years.[lxxix]

Grand Challenge 3: Akin to the KDD Cup challenges[16], benchmark sets of raw sequence big data sets of say cancer genomes, rare disease genomes, plant genomes and microbial genomes of South Asian origin are created and contestants are given a restricted time to analyze and deliver interpretation reports. A substantive KPO service sector for processing global genomics data can result from the innovators who rise to the challenge.

[15] i3 is discussed in greater detail in the main recommendations section

Grand Challenge 4: A mobile friendly *Electronic Health Records* platform in the cloud that can work with data standards of hospital based instrumentation (medical scans, for instance) and with readings from wearable/portable devices used by ASHA workers in remote primary healthcare settings. Interfacing medical records with payment systems (insurance, public health schemes) is usually a difficult challenge that might be facilitated in the Indian context with the Aadhaar platform.

2. Support Early Stage Investments

The gestation period for biotech start-ups is between 5 to 10 years, while investors prefer shorter timeframes. This, among other factors, has led to an imbalance in early stage investments, where most investors look for opportunities in e-commerce and IT-enabled services rather than high technology startups. Funding vehicles like the BIRAC AcE Fund will need to be adequately supported to help boost the sector in the short- to medium-term.

Intermediate Layer Recommendations

Biotechnology is a multifaceted field and encompasses different areas like biopharma, biofuel, biosimilars, medical equipment and more. It requires a vibrant pool of talent employable by the different sectors of biotechnology. However, the pool of skilled workers available to the sector isn’ t nearly diverse enough.[lxxx] In the main section of this report, the committee recommended embracing the platform mindset. Members are of the view that this is applicable to biotechnology as well.

1. Responsive Course Curricula: Public academic institutes offering biotechnology training must adapt course curricula to reflect the diverse nature of the industry and look beyond large sub-sectors such as biopharma. The institutes should also offer skilling programs that bridge the common gaps in biotech entrepreneurs or employees of biotech startups. For instance, scientist-entrepreneurs should have access to tailor-made management training programs. Karnataka’ s Biotech Finishing Schools have now completed close to 5 years of operations and should be evaluated in arriving at a model curriculum.

2. All India Biotech Digital Platform: The relevant stakeholders in the Biotech ecosystem must come together to set up a Digital Platform that boosts linkages, promotes collaborative innovation, mentoring and more. The Digital Platform could include the following:

A. Database of Current Biotech Ideas: As the biotech community grows in India it is important that a database be created that is dynamic and reflects the growth of the sector India. The database should contain the technologies emerging from various start-ups, academia and research institutes. This should help investors, potential collaborators and other stakeholders easily identify strategically important technologies and extend support or partnership.

B. Virtual Clusters: The platform could integrate forums, virtual meets and e-events into the system to enable greater interaction among the diverse group of scientists, academicians, entrepreneurs, institutes, private companies and investors from the biotech sector. These virtual clusters, integrated within the Digital Platform, would reduce the investment in building physical infrastructure, and the funds could be put into funding ventures, skilling programs and more.

C. Accessible Bioinformatics Databases: India has the potential to create a large pool of excellent bioinformaticians. Hands-on experience with a number of open source bioinformatics tools and big data representative of real experimental data will make a lot of difference. There are many global datasets in the public domain made available by institutions such as NCBI/EBI, which can be used to begin with. Eventually more context specific (Indian) datasets should be brought into these environments to seed locally relevant innovation.

---

[16] http://www.sigkdd.org/kddcup/index.php

3. Reform Regulatory Impediments to Biotechnology Innovation:

Innovation in biotechnology is an R&D intensive enterprise and several incentives for R&D institutions in India have been created for tax advantage, special funding and special privileges for import of equipment etc. In order to avail these incentives, the enterprise is required to qualify as a " DSIR Certified R&D Enterprise" . Some of the requirements for such qualification is that the enterprise has to be at least 3 years old, it needs to have equipment of a certain quantum in value, library facilities and so on, many of which are challenges for early stage biotechnology ventures. This regulatory issue needs to be looked at and suitable mechanisms evolved for a more flexible approach in this sector, so young and dynamic biotech firms are able to capitalise on these incentives and privileges.

Another serious regulatory impediment is the regulatory burden imposed by the " Biological Diversity Act of 2002" which has introduced certain immutable restrictions on biotechnology ventures that have any international funding or foreign national director/s that need to access biological material in India. The process for obtaining approvals is cumbersome and causes serious delays in research and is a deterrent to innovation. The general wisdom on this impediment is that the Act itself may need to be amended to arrive at a solution.

Base Layer Recommendations

As with the AIM Pyramid in the main section of the report, the base layer of the Biotech Pyramid seeks to highlight long-term changes that will have a profound impact on innovation and enterprise-creation in the Biotech sector.

1. Basic Science in Schools and Colleges: Innovative and hands-on science teaching in schools and colleges will help contribute to biotechnology in the future. Given the projected growth of the industry and its potential for job creation, a well-rounded education with science, maths and creative thinking will help prepare the students of today for the jobs, enterprise-building and innovation opportunities of tomorrow.

2. Indians Among the World's Top 50 to 100 Computational Genomics Experts: Over the next five to ten years, computational biology and genomics must receive a big push in engineering colleges to meet the demand of future professionals, help raise the profile of the sector on the global map, and inspire innovation within the country.

3. Addressing Negative Connotations of Biotechnology: Over the last five years, biotechnology has suffered from negative publicity associated with ethical issues in clinical trials, genetically modified organisms and so on. The positive impact of biotechnology research and investment on public health and the economy should be brought into the narrative of public discourse and help address the issue.

Appendix A
Literature Review of Past Reports on Entrepreneurship

1. Twelfth Five Year Plan - Chapter 9 – Innovations[17]

India has unique challenges and large unmet needs across diverse areas such as health, education, skills, agriculture, urban and rural development, energy and so on. We also have significant challenges of exclusion and inequitable access due to multiple deprivations of class, caste and gender—all of which require innovative approaches and solutions, and looking beyond the conventional way of doing things. Innovation is going to be central to providing answers to the most pressing challenges and for creating opportunity structures for sharing the benefits of the emerging knowledge economy. Affordable solutions, innovative business models or processes which ease delivery of services to citizens can enable more people to join the development process.

India is also uniquely poised to reap the advantages provided by a nation of a billion connected people, with over 800 million mobile phones, and global leadership in Information and Communication Technology (ICT) and software. This connectivity as well as ICT talent is changing the nature of processes, business, industry, governance, education and delivery systems: and our innovation thinking also has to leverage the unprecedented advantages provided by this changing landscape of connectivity and collaboration.

Government has a critical role to play in strengthening the innovation ecosystem. It must provide the enabling policy interventions, strengthen knowledge infrastructure, improve inter-institutional collaborations, provide a mechanism for funding business innovations at all levels especially small and medium scale enterprises (SMEs) and provide vision through a national-level road map for innovations.

Innovation can play a very important role in the development discourse, because it can offer a new approach to a system that is currently over-burdened by the multiple demands and has limited resources at its disposal. Enhanced focus on innovation can have an impact much beyond the realm of S&T in diverse areas such as health and education delivery, governance, enterprise development and much more. Collectively, this can herald a generational change in the country and can lay out a chart for a more sustainable and inclusive growth paradigm.

2. Initiatives of S&T Departments:

2.1 Department of Science and Technology (DST)

2.1.1 Science, Technology and Innovation Policy-2013[18]

Shaping the Future of an Aspiring India

Science, Technology and Innovation (STI) have emerged as the major drivers of national development globally. As India aspires for faster, sustainable and inclusive growth, the Indian STI system, with the advantages of a large demographic dividend and the huge talent pool, will need to play a defining role in achieving these national goals. The national STI enterprise must become central to national development.

Changing Phases of National Policies in S&T

India's Scientific Policy Resolution (SPR) of 1958 resolved to "foster, promote and sustain" the "cultivation of science and scientific research in all its aspects". Technology was then expected to flow from the country's established science infrastructure. The Technology Policy Statement (TPS) of 1983 emphasized the need to attain technological competence and self-reliance. The Science and Technology Policy (STP) of 2003 brought science and technology (S&T) together and emphasized the need for investment in R&D. It called for integrating programmes of socio-economic sectors with the national R&D system to address national problems as well as creating a national innovation system.

The Need for a Science, Technology and Innovation Policy

Scientific research utilizes money to generate knowledge and, by providing solutions, innovation converts knowledge into wealth and/or value. Innovation thus implies S&T-based solutions that are successfully deployed in the economy or the society. It has assumed centre stage in the developmental goals of nations. Paradigms of innovation have become country and context specific. India has, hitherto not accorded due importance to innovation as an instrument of policy. The national S&T enterprise must now embrace S&T led innovation as a driver for development.

---

[17] 12th Plan Document Chapter 9 – "Innovations"
[18] Science, Technology and Innovation Policy, Department of Science and Technology, Government of India 2013

India has declared 2010-20 as the " Decade of Innovation" . The Government has stressed the need to enunciate a policy to synergize science, technology and innovation and has also established the National Innovation Council (NInC). The STI Policy 2013 is in furtherance of these pronouncements. It aims to bring fresh perspectives to bear on innovation in the Indian context.

STI Policy: A New Paradigm

Science, technology and innovation can exist separately on their own in disconnected spaces. But, it is their integration that leads to new value creation. India' s global competitiveness will be determined by the extent to which the STI enterprise contributes social good and/or economic wealth. There is, therefore, the need to create the necessary framework for enabling this integration in identified priority areas by exploiting endogenous resources, strengths and capacities. New structural mechanisms and models are needed to address the pressing challenges of energy and environment, food and nutrition, water and sanitation, habitat, affordable health care and skill building and unemployment. " Science technology and innovation for the people" is the new paradigm of the Indian STI enterprise. The national STI system must, therefore, recognize the Indian society as its major stake holder. Global innovation systems tend to bypass large sections of the community. Innovation for inclusive growth implies ensuring access, availability and affordability of solutions to as large a population as possible. Innovation, therefore, must be inclusive. The instruments of the STI policy will enable this to be realized. The policy will drive both investment in science and investment of science-led technology and innovation in select areas of socio- economic importance. Emphasis will be to bridge the gaps between the STI system and the socio-economic sectors by developing a symbiotic relationship with economic and other policies.

Capturing Aspirations

The key elements of the STI policy are:

- Promoting the spread of scientific temper amongst all sections of society.
- Enhancing skill for applications of science among the young from all social strata.
- Making careers in science, research and innovation attractive enough for talented and bright minds.
- Establishing world class infrastructure for R&D for gaining global leadership in some select frontier areas of science.
- Positioning India among the top five global scientific powers by 2020.
- Linking contributions of science, research and innovation system with the inclusive economic growth agenda and combining priorities of excellence and relevance.
- Creating an environment for enhanced Private Sector Participation in R&D.
- Enabling conversion of R&D outputs into societal and commercial applications by replicating hitherto successful models as well as establishing of new PPP structures.
- Seeding S&T-based high-risk innovations through new mechanisms.
- Fostering resource-optimized, cost-effective innovations across size and technology domains.
- Triggering changes in the mindset and value systems to recognize, respect and reward performances which create wealth from S&T derived knowledge.
- Creating a robust national innovation system.

Investment in Research and Development

Global investments in science, technology and innovation are estimated at $1.2 trillion as of 2009. India' s R&D investment is less than 2.5% of this and is currently under 1% of the GDP. Increasing Gross Expenditure in Research and Development (GERD) to 2% of the GDP has been a national goal for some time. Achieving this in the next five years is realizable if the private sector raises its R&D investment to at least match the public sector R&D investment from the current ratio of around 1:3. This seems attainable as the industrial R&D investment grew by 250% and the sales by 200% between 2005 and 2010. Increased private investment is necessary for translating R&D outputs into commercial outcomes. While maintaining current rates of growth in public R&D investments, a conducive environment will be created for enhancing private sector investment in R&D.

The gross budgetary support for the science and technology sector has significantly increased during the last decade. The impact of such increase is becoming evident. India ranks ninth globally in the number of scientific publications and 12th in the number of patents filed. The Composite Annual Growth Rate (CAGR) of Indian publications is around 12±1% and India' s global share has increased from 1.8% in 2001 to 3.5% in 2011. But the percentage of Indian publications in the top 1% impact making journals is only 2.5%. By 2020, the global share of publications must double and the number of papers in the top 1% journals must quadruple from the current levels. The citation impact of Indian publications must improve and match at least the world average. Initiatives under the new policy should enable these macro indicators of research to be achieved by 2020.

According to the Global Science Report of the UNESCO, India' s current global ranking is commensurate with its number of Full-Time Equivalent (FTE) of R&D personnel. It is imperative that the total number of FTE of R&D personnel increases by at least 66% of the present strength within the next five years.

Promoting Excellence and Relevance in R&D

Nourishing the Roots

Ensuring sustainable pipeline of talented youth for science is a challenge. India has mounted some significant initiatives for attracting talent to science and careers with research. Empowering stakeholders for local actions is a key element of these initiatives. The policy framework will further enable school science education reforms by improving teaching methods, science curricula, motivating science teachers and schemes for early attraction of talent to science. Also special incentive mechanisms will be devised to stimulate research in the universities and develop young leaders in science and engineering.

Excellence and Relevance

Investment in basic research will be enhanced for fostering excellence against global benchmarks and focusing on relevance for addressing national challenges.

Gender Parity

Participation of women in STI activities is important. New and flexible schemes to address the mobility challenges of employed women scientists and technologists will be put in place. A broad scope for re-entry of women into R&D and facilitation mechanisms for special career paths in diverse areas will be sought.

Inter-University Centres

The few inter-university centres that have been set up have proved the concept to be a successful and viable one. Such centres need to be multiplied in different fields to enable a wider cross section of university researchers access advanced research facilities and equipment which are otherwise not available in university environments. These will be discipline-specific as well as multi-disciplinary, including humanities, to address the grand challenges in S&T and its applications.

Participation in Global R&D Infrastructure and Big Science

Modern science is increasingly becoming resource intensive. It has become necessary to create high-cost global infrastructures in some fields through international consortia models. Indian participation in such international projects will be encouraged and facilitated to gain access to facilities for advanced research in cutting edge areas of science. This will also enable the Indian industry to gain global experience and competitiveness in some high-technology areas with spin-off benefits.

Performance-Linked Rewards and Investments

Transparent centrally implementable Performance Related Incentive Scheme (PRIS), based on past and proven track record in research, will be put in place to enable grant-based investments in such performers. For R&D leading to technology development and knowledge services, the criteria would, however, be institution specific. Centrally instituted incentives to public-funded R&D centres for outcomes leading to public

National Agenda and the STI System

Macro indicators of R&D do not really reflect the innovation capability of a nation. Appropriate indicators, which integrate measures of excellence and inventiveness with relevance and affordable innovation, are necessary for evidence based policy actions. Supply side interventions have hitherto been the main strategy for public investment in R&D. This needs to change. There should be equal emphasis on both supply side interventions and demand based investments.

Around 10 sectors of high impact potential will be identified for directed STI intervention and deployment of requisite resources. Enabling policy instruments that facilitate both institutional research and R&D enterprises to focus their efforts in these areas will be put in place.

The complex value chain of innovation – from idea to market – often calls for STI intervention at all levels: research, technology inputs, manufacturing and services. In the priority areas of socio-economic importance, the policy will enable a holistic approach to intervention, support and investment. Measures taken in this direction will be in consonance with the programmes initiated by the NInC.

R&D policy for agriculture is articulated by the Indian Council of Agriculture Research (ICAR). Integration of the agriculture R&D policy with the national R&D system and the STI policy will be brought about.

STI inputs to the manufacturing sector can lead to enhanced employment generation. The innovation ecosystem for the sector, however, depends on the nature and size of the enterprise and the context. India’ s share of global trade in high technology products is at present only around 8% and the present technology intensity of the sector is a low of 6-7%. The aim is to double these through greater technology inputs from R&D. A strategic selection of some industry sectors, where India can aspire for leadership, would be made for stepping up R&D intensity and increase India’ s share in high-technology trade. Small and Medium Enterprises (SME) generally have low R&D intensity. Special schemes, to support R&D as well as related services at the firm or collective level, will be devised and put in place.

The R&D intensity of the service sector is generally low. This needs to be enhanced considerably and the skill base also expanded significantly. For rapidly accomplishing the tasks of modernization of technology-based services, missions in some select service sector areas, will be identified. Deployment of technology-led services for transparent Government machinery will also be supported.

Climate variability and change is of global concern and India has articulated a National Action Plan for Climate Change (NAPCC) and identified several national missions. The STI system will have an active role in these missions. It will also serve as a source of strategic knowledge to cope with the challenges of climate variability and change as well as to meet equity-based differentiated and shared responsibilities of India.

Attracting Private Sector Investments in R&D

Public funds for partnerships with the private sector for social and public good objectives will be earmarked as a new policy initiative. A National Science, Technology and Innovation Foundation will be established as a Public-Private Partnership (PPP) initiative for investing critical levels of resources in innovative and ambitious projects.

The focus of the policy will be:

- Facilitating private sector investment in R&D centres in India and overseas.
- Promoting establishment of large R&D facilities in PPP mode with provisions for benefits sharing.
- Permitting multi stakeholders participation in the Indian R&D system.
- Treating R&D in the private sector at par with public institutions for availing public funds.
- Bench marking of R&D funding mechanisms and patterns globally.
- Modifying IPR policy to provide for marching rights for social good when supported by public funds and for co-sharing IPRs generated under PPP.
- Launching newer mechanisms for nurturing Technology Business Incubators (TBIs) and science-led entrepreneurship.
- Providing incentives for commercialization of innovations with focus on green manufacturing.

Delivery Systems for STI Outputs to Stake Holders and Society

Diffusion of scientific outputs and technology interventions into social systems is a multi-layered process. Except for the mission-oriented strategic sectors, the delivery mechanism involves a large number of intermediaries both from the public and private sectors. This requires strengthening of linkages between the scientific and socio-economic sectors. The STI policy will leverage the R&D allocations of socio-economic ministries through a shared vision, mission-oriented approach and adoption of new delivery models with provisions for accountability. The state governments constitute important stakeholders. Measures will be taken to ensure that state-specific S&T vision and plans are informed and guided by the new STI Policy towards which State S&T Councils/Boards will be strengthened. NGOs will be accorded a pivotal role in the delivery of STI outputs, especially rural technologies, to the grassroots level.

Ecosystem Changes for Science, Technology and Innovation

Special and innovative mechanisms for fostering academia-research-industry partnerships will be devised. Mobility of experts from academia to industry and vice-versa will be facilitated. Success stories in S&T-based innovations from Indian experience would be replicated and scaled up. Regulatory and legal framework for sharing of IPRs between inventors and investors will be put in place. Measures to close gaps in the translation of new R&D findings and grass root innovations into the commercial space will be taken.

Rigidity of centrally developed plans for investments often does not suit frontline science, technology development and innovation. A flexible approach that allows for fine tuning the Five Year Plan schemes in response to rapid changes in S&T would be put in place with speed, scale and sustainability as key governance parameters. “Risks” are an integral part of a vibrant innovation system. Risk sharing by the government will significantly increase private sector investment in R&D and technology development. New financing mechanisms for investing in enterprises without fear of failure and options for foreclosing unsuccessful ventures are essential part of an enabling innovation ecosystem. A public procurement policy that favours first of its kind products developed through indigenous innovation and measures to promote such products globally are necessary.

General rules of expenditure control of publicly funded institutions do not suit non-linear growth sectors like science and technology, and more so the innovation sector. Auditing principles should be more aligned to “performance” than “compliance to procedure”. The system should be able to differentiate between genuine failures and process deficits.

Specifically the policy will focus on:

- Prioritizing critical R&D areas like agriculture, telecommunications, energy, water management, health and drug discovery, materials, environment and climate variability and change.
- Promoting inter-disciplinary research, including traditional knowledge.
- Fostering the delivery and use in the society of innovations in the strategic sectors with civilian application potential.
- Promoting mechanisms such as “small idea-small money” and “Risky Idea Fund” to support innovation incubators.
- Establishing of a Fund for Innovations for Social Inclusion.
- Leveraging traditional knowledge through modern science for finding solutions to national challenges.
- Supporting STI driven entrepreneurship with viable and highly scalable business models.
- Investing in young innovators and entrepreneurs through education, training and mentoring.

Gaining Global Competitiveness through Collaboration

Open source approaches for public and social goods form interesting innovation systems. Knowledge commons is an emerging theme for managing IPRs created through multi-stake holder participation. The STI Policy will seek to establish a new regulatory framework for data access and sharing as also for creation and sharing of IPRs. The new policy framework will enable strategic partnerships and alliances with other nations through both bilateral and multilateral cooperation in science, technology and innovation. Science diplomacy, technology synergy and technology acquisition models will be judiciously deployed based on strategic relationships.

Public Awareness and Public Accountability of Indian STI Sector

Public understanding of science is an important dimension for introducing and reaching the benefits of modern science and technology to the people. The civilizational aspect of science, or scientific temper, needs to be promoted across all sections of the society systematically. Effective science communication methods, by using tools such as the National Knowledge Network, will be initiated.

Public and political understanding of science should be based on evidence and debates with open mind. People and decision makers must be made aware of the implications of emerging technologies, including their ethical, social and economic dimensions. White papers on mission-oriented programmes, with specific deliverables and timelines, will be published. Mechanisms for assessing the performance of the national STI enterprise through an autonomous and robust evaluation system, which includes social scientists, will be established. The national science academies will be accorded a major role in this endeavour of public accountability.

Policy Vision

The guiding vision of aspiring Indian STI enterprise is to accelerate the pace of discovery and delivery of science-led solutions for faster, sustainable and inclusive growth. A strong and viable Science, Research and Innovation System for High Technology-led path for India (SRISHTI) is the goal of the new STI policy.

2.1.2 In addition to the above policy initiative, the following activities have also been taken up by DST:

i. DST has established new institution, namely: National Innovation Foundation (NIF), Ahmedabad which provides support to grassroots innovators and outstanding traditional knowledge holders from the unorganized sector of the society. NIF helps India becoming an innovative and creative society and a global leader in sustainable technologies by scouting, spawning and sustaining grassroots innovations. Its mandate include: evolution and diffusion of green grassroots innovation to meet the socio-economic and environmental needs of our society, provide institutional support for transition of grassroots green innovations to self-supporting activities and create a knowledge network to link various stakeholders through application of information technology and other means.

ii. DST is implementing Institutional Program on Innovation and Entrepreneurship with an aim of orienting job seekers to be the job generators and translating knowledge to wealth creation of globally competitive and innovative ventures.

iii. Technology Business Incubator (TBIs) have been established for nurturing technology based new ventures/start-ups by incubating TBIs hosted generally by academic and R&D institutions. Around 2,000 new ventures have been incubated so far. These enterprises have cumulatively generated approximately 32,000 jobs. The incubated start-ups have generated around 450 parents/ copyrights.

iv. Innovation and Entrepreneurship Development Cell (IEDC) have been promoted in education institutions to create entrepreneurial culture amongst the faculty and students.

v. Scheme on Innovation-Science and Technology Entrepreneurship Development for promoting micro-enterprises based on technologies relevant at local area level is being implemented.

vi. DST has also taken initiatives for Awareness, Training and Capacity Building on entrepreneurship to help self-employment as well as creation of enterprises.

vii. Under the PPP Initiative for supporting Innovation and Entrepreneurship, various programmes such as India Innovation Growth Program (IIGP) with world-class commercialization strategies and the business development assistance which helps accelerate innovative Indian technologies into the global markets, “ The Power of Ideas” Program for scouting, selecting and recognizing innovation across various sectors and across the country and India Innovation Fund- A SEBI registered, PPP venture capital fund that invests in innovation led, early stage Indian firms. These programmes will helps in making enterprises globally competitive and innovative. In addition, value added support to Entrepreneurs and the seed supports to TBIs are also being provided.

viii. DST has also established the Technology Development Board (TDB) as per the provisions of the Technology Development Board Act, 1995. The Act enables creation of a Fund for Technology Development and Application to be administered by TDB. The Fund receives grants from the Government of India out of the Cess collected by the Government from the industrial concerns under the provisions of the Research and Development Cess Act, 1986, as amended in 1995. Any income from investment of the amount of the Fund and the recoveries made of the amounts granted from the Fund is also credited to the Fund. The donations are also received by the fund. The Finance Act, 1999, enabled full deductions to donations made to the Fund for income tax purposes.

2.2 Department of Biotechnology (DBT)

i. BIRAC (Biotechnology Industry Research Assistance Council), a non-profit, section 8, Public Sector Undertaking set up under the aegis of Department of Biotechnology, has been a pioneer in promoting the innovation research and ecosystem in the country. BIRAC has scaled up its operations to promote the innovation research for product development in biotechnology. DBT, through BIRAC has laid emphasis on strengthening the funding, mentoring and infrastructural support to the bio-entrepreneurs for creating a sustainable bio-economy in the country.

ii. DBT initiated the establishment of bio-incubation hubs across the country and since its inception, BIRAC has been in the forefront in providing the infrastructural access to the innovators and entrepreneurs through its BISS (Bio-incubator Support Scheme) programme. The BISS programme aims at strengthening and up-gradation of the existing Bio-incubators and also to establish New World Class Bio-incubators. These Bio-incubators will provide the incubation space and other required services to start-up companies for their initial growth.

iii. In consideration of the Government’ s vision to promote the innovation hubs in the country, BIRAC has undertaken following initiatives:

- Supported and strengthened the Innovative R&D infrastructure in 15 bio-incubators across the country
- 1,24,000 sq feet of incubation space created, support extended to 199 start-ups/entrepreneurs
- Rs. 100.00 crore sanctioned under the programme
- 5 University Innovation Clusters have been initiated to promote the innovation entrepreneurship at university level
- 3 Bio-industrial facilities supported

iv. BIRAC, also provides funding support through programmes such as Biotechnology Ignition Grant (BIG), Small Business Innovation Research Initiative (SBIRI), Biotechnology Industry Partnership programme (BIPP), Contract Research Services (CRS) and Social Innovation Programme (SPARSH). The impact of BIRAC’ s initiatives is as follows:

- BIRAC has been managing challenge oriented calls from the idea to pre-commercialization stages, under which the support has been extended to 270 companies through 360 projects
- Fund commitment of USD 225 Mn
- Outcome of the BIRAC funding programmes – 17 affordable products, 11 New Technologies and Intellectual Property

2.3 Department of Scientific & Industrial Research (DSIR)

The thrust of the Department of Scientific & Industrial Research (DSIR) is to promote industrial research, technology development and transfer to enable India to emerge as a global industrial research and innovation hub. Emphasis is on attracting industrial research in the country through industry and institution-centric motivational measures and incentives, creating an enabling environment for development of new innovations to channelize benefits to the people. DSIR has launched four new schemes during 12th Plan such as

i. Promoting Innovations in Individuals, Start-ups and MSMEs (PRISM)—wherein innovative proposals of MSMEs are being supported; existing network of TePP Outreach Centres are being expanded; proposals from individual innovators/incubates are being supported;

ii. Scheme on "Patent Acquisition and Collaborative Research and Technology Development" (PACE) - wherein support is being provided to Indian industries to acquire Intellectual Property at early stage from overseas or within the country and add value to the acquired IP; and focus on Public-Private- Partnerships (PPP) to create enabling environment for collaborative research between Industry and Universities/Public Funded Research Institutions;

iii. "Building Industrial R&D and Common Research Facilities" (BIRD)—wherein R&D in Industry are being encouraged and supported; and support is provided for creation of Common Research Facilities for Small and Micro Industries;

iv. Access to Knowledge for Technology Development and Dissemination (A2K+)—wherein science, technology and innovation related international journals from major publishers are made accessible to 1500 in-house R&D units of industry and 600 Scientific & Industrial Research Organisations (SIROs) and techno-entrepreneurs, besides conducting studies/conferences on Industrial Status in the country

2.4 Council of Scientific & Industrial Research (CSIR)

During Twelfth Five Year Plan, focus of CSIR is on: achieving science and engineering leadership; developing innovative technological solutions; practicing open innovation initiatives; developing and nurturing human resource in trans-disciplinary areas; facilitating science based entrepreneurship.

i. CSIR has launched Open Source Drug Discovery (OSDD) programme which has emerged as a new platform for innovation in the domain of healthcare. This CSIR-led "Team India" consortium with global partnership has more than 4500 researchers from over 100 countries as registered participants. Distributed Organic Chemical Synthesis (DOCS) programme envisages building a national repository of
400,000 small molecules by the end of the Plan through open source.

ii. Science 3.0, an initiative for open innovation and knowledge-ware development through crowd sourcing would endeavour to engage a large number of engineering institutions to identify the most vexing problems and attempt to provide solutions on issues like attaining energy efficiency, reduction in materials use, minimizing waste generation and developing business and financial models to increase productivity and profitability of the units.

iii. CSIR Outreach Centres are envisaged to be operated and managed through CSIR-people partnership mode (CPP), and implemented either through mobile kiosks or pre-fabricated self-inclusive containers placed at identified locations. The Centres would also have close coordination and networking with the Cluster Innovation Centres of the NInC-CSIR initiative. The Innovation Complexes are envisaged to consolidate and sustain the value chain of R&D within the CSIR; consolidate the CSIR brand and make CSIR R&D accessible to society at large; catalyse regionally balanced economic development; and promote entrepreneurial culture among the scientific community. The CSIR Initiative for Inverted Innovation is a unique paradigm where children/young engineers invent, CSIR laboratories mentor and industries commercialize.

iv. CSIR has also undertaken a scheme entitled New Millennium Indian Technology Leadership Initiative (NMITLI). NMITLI seeks to catalyze innovation centered scientific and technological developments as a vehicle to attain for Indian industry a global leadership position, in selected niche areas in a true " Team India" spirit, by synergising the best competencies of publicly funded R&D institutions, academia and private industry. NMITLI is the largest public-private-partnership effort within the R&D domain in the country. It looks beyond today' s technology and thus seeks to build, capture and retain for India a leadership position by synergising the best competencies of publicly funded R&D institutions, academia and private industry. The Government finances and plays a catalytic role. It is based on the premise of consciously and deliberately identifying, selecting and supporting potential winners. NMITLI has carved out a unique niche in the innovation space and enjoys an excellent reputation. NMITLI has so far evolved 60 largely networked projects in diverse areas viz. Agriculture & Plant Biotechnology, General Biotechnology, Bioinformatics, Drugs & Pharmaceuticals, Chemicals, Materials, Information and Communication Technology and Energy. These projects involve 85 industry partners & 280 R&D groups from different institutions. Approximately 1750 researchers are engaged in these projects. These 60 projects cumulatively have had an outlay of approximately Rs. 550 crore. It is also plan to establish NMITLI innovation centres in the country.

2.5 Department of Electronics & IT (Deity)

i. Technology Incubation and Development of Entrepreneurs (TIDE) scheme was initiated by Department of Electronics and Information Technology (DeitY), in the year 2008 with an outlay of Rs 23.40 crores for supporting 15 TIDE centres over a duration of 4 years with the following objectives:

- Setting up and strengthening Technology Incubation Centre in institutions of higher learning,
- Nurture Technology Entrepreneurship Development for commercial exploitation of technologies developed by them,
- Promoting product oriented research and development,
- Encourage development of indigenous products and packages and bridging the gap between R&D and commercialization.

The scheme was expanded to include additional 12 TIDE centres in the year 2009 with an additional outlay of Rs 25.934 crores. The outlay included cost for scheme publicity, project monitoring unit and creation of virtual incubation centres. The current tenure of the scheme is till March, 2017.

ii. DeitY has initiated a scheme to support International Patent Filing in January 2015 and duration of the scheme is 5 Years with a budgetary outlay of Rs 1846.62 lakhs. So far DeitY has received 25 applications for reimbursement for international patent filing. The expert committee scrutinised the applications and recommended 5 applications for support. For conducting IPR Awareness workshops, 6 applications have been received and 1 workshop has been organized successfully.

iii. DeitY has also taken up Multiplier Grants Scheme. The aim of the scheme is to encourage industry to collaborate with premier Academic and Government R&D institutions for development of products/packages. The idea for collaborative research should originate from industry/industry consortium, and academic institution(s)/R&D bodies undertaking industry specific research will submit the project proposal jointly with the industry/industry consortium to DeitY under the scheme. The proposal should be for innovation in modules/ products/ packages/ services in the area of Electronics and IT. The proposals envisaging prototyping to package for commercialization may also be considered under the scheme. The objectives of the scheme are:

a. Establish, nurture and strengthen the linkages between the Industry and Institutes;

b. To promote industry oriented R&D at institutes;

c. Encourage and accelerate development of indigenous products and packages; and

d. Bridge the gap between R&D / Proof-of-concept and commercialization / globalization.

3. Initiatives undertaken by the erstwhile Office of Adviser to PM on Public Information Infrastructure and Innovation (PIII).

Hon' ble President of India declared 2010-20 as the " Decade of Innovation" . To take this agenda forward, the former Prime Minister set up the National Innovation Council (NInC) under the Chairmanship of Mr. Sam Pitroda, former Adviser to Prime Minister on PIII with the following Terms of Reference:

(i) Formulating a Roadmap for Innovation for 2010-2020

(ii) Creating a framework for

- Evolving an Indian model of innovation, with focus on inclusive growth
- Delineating policy initiatives within the Government, required to spur innovation
- Developing and championing innovation attitudes and approaches
- Creating appropriate eco-systems and environment to foster inclusive innovation
- Exploring new strategies and alternatives for innovations and collaborations
- Identifying ways and means to scale and sustain innovations
- Encouraging Central and State Governments to innovate
- Encouraging universities and R&D institutions to innovate
- Facilitating innovations by SMEs
- Encouraging all important sectors of the economy to innovate
- Encouraging innovation in public service delivery
- Encouraging multi-disciplinary and globally competitive approaches for innovations

3.1 Sectoral Innovation Councils

In order to drive innovative strategies in key sectors and create multiple roadmaps, the NInC encouraged the creation of multiple Sectoral Innovation Councils aligned to Union Government Ministries for enhancing innovation capabilities in respective sectors. The Sectoral Innovation Councils are expected to drive the innovation agenda in the country across various sectors and harness the core competencies, local talent, resources and capabilities to create new opportunities taking into consideration cross-cutting themes that impact the sector and work collaboratively with other Councils. The focus is on undertaking activities that improve the innovation quotient of the sector going forward, with a special emphasis on inclusive and sustainable innovation.

In all, 26 Ministries have constituted the Sectoral Innovation Councils. Out of these, Innovation Roadmaps have been submitted by 8 Ministries/Departments.

3.2 State Innovation Councils (SInC)

Further, NInC has also encouraged the creation of State Innovation Councils (SInCs) to replicate the efforts of the National Innovation Council to nurture an innovation ecosystem at the State level. The innovation ecosystem consists of five critical ingredients namely – (a) providing a conducive policy framework; (b) offering institutional platforms for inter-agency collaborations; (c) strengthening and expanding ICT connectivity; (d) fostering innovations in the education system; (e) and setting up a regime of incentives and rewards to encourage innovations. The SInCs are expected to carry forward the innovation movement on all these five fronts and present a Roadmap or Innovation Action Plan (IAP) for the State.

So far, 28 States and UTs have constituted the SInCs. Most of the State Governments which have formulated the Councils have focused on the multi sectoral nature of these platforms and anchored them in the Planning Departments of the State Governments, with the Departments of Science & Technology and Education as the nodal pivots to support the campaign. The focus of most of the Councils has been on areas such as Planning, S&T, Education (both School, and Higher education including Technical education), Industry, Agriculture, Rural Development, Urban Development and Information Technology.

The National Innovation Council' s initiative to set up multiple innovation Councils, both at the State and Sectoral level, was a first of its kind Government effort is to galvanize a large constituency around the idea of innovation. The Councils have been tasked to recommend strategies and policies for driving innovation and create an innovation movement by expanding the discourse on innovation and empowering domain experts across areas. The State Innovation Councils are also expected to sensitize State Governments to the idea of innovation and to ensure that innovation becomes a focus area in the long term development agenda of the States. Some State and Union Territory Innovation Councils have taken impressive initiatives.

3.3 India Inclusive Innovation Fund (IIIF)

To promote inclusive innovation and entrepreneurship, focusing on the needs of the citizens who lie at the bottom of the economic pyramid, the India Inclusive Innovation Fund was launched jointly by NInC and Ministry of Small and Medium Enterprise. The Fund aims to fill a presently un-serviced area of venture capital funding for enterprises that create social impact and also generate a modest financial return. The IIIF is now being operated by the Ministry of Small and Medium Enterprises (MSME).

3.4 Innovation in Education

Recognizing the fundamental role of education in nurturing and fostering an ecosystem of Innovation, NInC also took a series of initiatives to encourage innovations in existing educational Institutions, as well as promoting new educational models and innovative platforms for knowledge creation, dissemination and application. Some of the key proposals of the NInc in this domain include:

i. Creation of a separate scholarship stream of **National Innovation Scholarships** analogous to the National Talent Search Scheme.

ii. Setting up an **Innovation Centre in each DIET** (District Institute of Education and Training) to enhance teacher training and enable them to become facilitators of creativity and innovative thinking.

iii. **Mapping of Local History, Ecology and Cultural Heritage** by High School students to create critical thinking on their local environment.

iv. Creation of a **National Innovation Promotion Service** to replace/add to National Service Scheme in Colleges to use college students to identify local innovations.

v. Setting up a **Meta University**, as a new model for a 21st Century University to offer students a collaborative and multidisciplinary learning experience.

vi. Setting up twenty **Design Innovation Centres** co-located in institutes of National importance.

vii. Setting up of 20 **University Innovation Clusters** across the country where innovation would be seeded through Cluster Innovation Centres.

viii. Igniting Youth Innovation with **Tod Fod Jod** Centers at Schools and Colleges to foster Innovation at an early stage and to create an innovative mindset in the youth.

4. Report of the Committee on Angel Investment & Early Stage Venture Capital[19]

A Committee on " Angel Investment and early Stage Venture Capital" was constituted by the erstwhile Planning Commission on 17th October, 2011 under the Chairmanship of Shri Sunil Mitra, former Revenue Secretary, Government of India. The committee gave its report in June, 2012. The executive summary of the report is given below:

---

[19] Report of The Committee on Angel Investment & Early Stage Venture Capital, June 2012, Planning Commission, Government of India

4.1 Executive Summary

India needs to create 1- 1.5 crore (10-15 million) jobs per year for the next decade to provide gainful employment to its young population. Accelerating entrepreneurship and business creation is crucial for such large-scale employment generation. Moreover, entrepreneurship tends to be innovation-driven and will also help generate solutions to India' s myriad social problems including high-quality education, affordable health care, clean energy and waste management, and financial inclusion. Entrepreneurship-led economic growth is also more inclusive and typically does not involve exploitation of natural resources.

Large Indian businesses—both in the public and private sector - have not generated significant employment in the past few decades and are unlikely to do so in the coming decade or two. Public sector and government employment has declined in the last few years, and is expected to grow very slowly in the coming years. Large private sector firms have also n slow in generating employment, which is unlikely to change due to increasing automation, digitization, and productivity gains. For example, the banking sector in India has recorded almost no employment growth in the last two decades despite multifold growth in its revenue and assets. Agriculture employs nearly a half of India' s work force but employment is likely to decline in this sector, due to improvements in productivity.

India is an entrepreneurial country, but its entrepreneurs have had to struggle to create and grow their business ventures. There is, however, a growing group of first-generation Indian entrepreneurs - the founders of companies such as HCL, Cognizant, Infosys, Bharti and others—that have generated large scale employment and significant wealth. They and others such as IndiaBulls, Makemytrip and Naukri have also demonstrated value creation through a public listing. These successes have encouraged a new breed of entrepreneurs especially in the internet and e-commerce space.

The Committee believes India' s entrepreneurial growth can be accelerated by creating more conducive conditions - a catalytic government and regulatory environment, adequate capital flows (both debt and equity), support from businesses and society, and availability of appropriate talent and mentoring. Such an environment could replicate the success demonstrated by the Indian Information Technology (IT) and IT enabled services (ITES) industry over the last two decades. Starting in the early 1990s, the industry now directly employs 28 lakh (2.8 million) and indirectly an additional 89 lakh (8.9 million) people. Its revenues have grown from Rs. 350 crore in 1990 to Rs 4.5 lakh crore ($88 billion) in FY 2012[20]. Several leading companies in this industry started as first-generation entrepreneurial ventures. By 2020, the IT and ITES industry is projected to contribute 9% of GOP with a revenue of over Rs 12 lakh crore ($225 billion) and direct and indirect employment to 30 million people[21].

India has the potential to build about 2,500 highly scalable businesses in the next 10 years - and given the probability of entrepreneurial success that means 10,000 start-ups will need to be spawned to get to 2,500 large-scale businesses. These businesses could generate revenues of Rs10 lakh crore ($200 billion) - a contribution to GOP and creation of employment at the same scale as projected for IT and TTES industry. Experience across countries suggests that vibrant entrepreneurial activity significantly improves social harmony, living standards, and quality of life.

However, there are significant roadblocks that hold back and dampen entrepreneurial activity in India. The country ranks low on comparative ratings across entrepreneurship, innovation and ease of doing business. The ecosystem for starting and running new ventures has many gaps. Regulations and procedures are restrictive and time-consuming and add significant cost for an emerging venture. Banks and financial institutions are wary of lending to first-generation entrepreneurs and to MSMEs in general, due to various norms like tangible asset coverage, DER etc., even though such enterprises make a major contribution to the economy, employment, and exports. This imposes constraints on their credit absorption capacity and consequently, growth. Established businesses have generally been passive in engaging with emerging ventures. Educational institutions are yet to actively promote entrepreneurship over careerism. Lack of collaboration between all stakeholders leads to further roadblocks.

---

[20]NASSCOM Strategic Review 2012
[21]NASSCOM Perspective 2020:Transform Business, Transform India

The entrepreneur' s journey proceeds from idea generation to venture formation to scaling up a business. Typically, the steps are:

- An entrepreneur begins with an idea and works towards a prototype with either his own funds or those taken from friends and family. He/ she then seeks to create a more formalized venture, often with the aid of an incubator that provides him real estate and other services, which may also include introduction to potential investors. Angel investors play an important role at this stage by both funding and mentoring the entrepreneurs.
- Further scale-up then requires higher amount of capital, which is typically provided by venture capital funds.
- Apart from equity capital, the venture also needs debt for working capital. In the absence of formal sources of debt, the entrepreneur is often compelled to use equity capital for working capital, which limits the ability of the entrepreneur to scale the business.
- In addition to funds, the entrepreneur needs support from a range of other players in the ecosystem - educational institutions, large businesses and government - as also access to skilled labour, research, and a market. In other words, an environment that eases this journey.

Creating a vibrant entrepreneurial ecosystem will require strong capital flows to entrepreneurial ventures- the Committee expects that Rs.3 lakh crore (or around $55 billion) of capital would be needed over the next decade with around half of this in form of debt.

Currently, early stage investing in India is inadequate:

- Angel investors drive significant early-stage investments in countries with high entrepreneurial activity. Apart from capital, Angel investors also provide mentoring and network access to entrepreneurs. They play a critical role in scaling up businesses to make them attractive for institutional investors such as venture capital funds. Angel investing is just beginning in India. In 2011, Indian angels, constrained by regulations that make both investing and exits cumbersome, invested only about Rs 100 crore (approximate.ly $20 million) in around 50 deals as compared to Canada[22]- where angels invested Rs 2,000 crore ($390 million). Even as a proportion of early-stage investing (investing in the initial or growth phase of the venture), angel investments in India comprise around 7 per cent as against around 75 per cent in the US. We estimate that a well-developed ecosystem would expand these investments to around Rs. 3,500 crore ($700 million) am1ually in next 10 years.
- India also lags in early-stage venture capital investing[23]. Annual investments are around Rs.1,200 crore ($240 million) as against Rs. 29,000 crore ($6.3 billion) in the US and Rs. 3,000 crore ($700 million) in China. Around 90% of the early stage venture funds in India come from offshore sources rather than from domestic investors. We estimate that these investments could increase to around Rs. 14,000 crore ($3 billion) annually in the next 10 years as both the number of funds and propensity to invest increases with a larger entrepreneurial base.
- Financing of MSMEs, particularly service sector enterprises is also below par because these industries often do not create tangible assets. Capital is required for product development, R&D, marketing, hiring professional teams, etc. Thus, access to adequate and timely finance continues to be an issue for MSMEs. Early stage companies in innovation-driven industries such as IT services, business process outsourcing, healthcare, etc., especially, if promoted by first generation entrepreneurs, also find it difficult to access funds from banks.
- Impact investing, or investing for both social and financial return, is an imperative in India. This kind of investing seeks out businesses and social ventures that can deliver measurable social and/ or environmental impact as well as appropriate financial returns in sectors such as education, healthcare, sanitation, environment, and infrastructure. In 2011, around Rs 400 crore ($80 million) of such investments were made by impact investors. Impact investing needs to grow dramatically to support businesses that target India' s 900 million strong base of-pyramid population.

---

[22] Investment Activity by Canadian Angel Groups. 2010 Report, NACO

[23] Early stage investments defined as investments of up to $10 Million in early stage ventures

Sectors with higher potential for rapid entrepreneurship-driven growth include manufacturing (IT hardware and electronics, automotive components, food processing), software, technology and telecom, affordable healthcare, clean technology (alternative energy, clean water, and sanitation) and personal care amongst many others. These sectors are seeing growing consumer demand and have already received some investment from both angel investors and venture capital funds albeit to a somewhat limited extent.

4.2 The Committee has identified five major drivers for creating a vibrant entrepreneurial ecosystem. Given below are the key recommendations of the Committee:

4.2.1 **Catalytic government policy and regulatory framework:** The government and its agencies could play a proactive role in facilitating entrepreneurship. Following are key steps that should be taken:

a. **Facilitate investments:** Recognition and promotion of early-stage investments and early stage investors such as angel investors, venture and seed funds, and impact investors through development of appropriate policy measures and fiscal incentives. The Committee recommends the following definitions:

i. **Angel investing:** An angel investor is an individual who invests his own money directly in a seed stage venture in which there is no family connection. The angel investor could be acting alone or in a formal or informal angel group. Such investment in the venture for the individual is less than Rs 5 crore and in aggregate for a group is less than Rs. 10 crore. Seed stage venture is defined as a business which (a) has a turnover below Rs 25 crore; (b) is unlisted and (c) is not promoted, sponsored or related to an Industrial Group whose group turnover is in excess of Rs. 300 crore. The limits on investments and turnover threshold for the seed stage should be indexed to inflation.

ii. **Early-stage venture capital investing:** Investments in an early stage venture by an entity which is registered with the appropriate financial regulatory authority (such as appropriate AIF or FVCI). Early-stage venture is defined as a business which (a) has a turnover of less than Rs 50 crore; (b) is unlisted; and (c) is not promoted, sponsored or related to an Industrial Group whose group turnover is in excess of Rs. 300 crore. The limits on investment and turnover threshold for the early stage should be indexed to inflation.

iii. **Impact investing:** Investments in businesses and social ventures with the intention to generate measurable social and environmental impact alongside a financial return and which target a range of returns from below market to market rate. Impact investing is based on the conviction that such investments play a crucial role in addressing social and environmental challenges.

b. **Enhance and scale-up venture incubation programs:** Incubators are a very critical part of the entrepreneurial ecosystem:

i. Currently there are 120 incubators in India -almost all are government sponsored and largely affiliated to educational institutions. The number of incubators needs to be exponentially increased. In the next decade we should aim to have 1000 incubators covering most of Tier I and II cities in India.

ii. In addition to physical infrastructure, Incubators must enhance their services to include mentoring, providing access to business networks, and investors.

iii. For rapid scale up of the Incubators and enhancement of services, participation of the private sector will be critical—whether through PPP or other appropriate models including for-profit models.

iv. We also need to establish incubators that focus on socially relevant businesses such as the renewable-energy dedicated arm of CIIE (IIM Ahmedabad’ s incubator).

c. **Ease entrepreneurial activity:** Regulations and processes for setting up, operating, and exiting a business in India are time consuming and complex. Governments and their agencies can at all levels - central, state, and local - reduce transaction time and costs through measures such as single-window clearance and access to well-developed industrial clusters. Also, a mechanism where businesses of a certain scale can sell-certify regulatory compliance would be useful. A model along the lines of Software Technology Parks of India (STPI) could be created for early stage ventures which could get affiliated to "entrepreneurial hubs" that enjoy similar facilities as STPI units.

d. **Ease exits for investors:** Policy framework for easier exits will encourage early stage investments by Angels and others:

i. Provide appropriate fiscal incentive on capital gains to Angels and other early stage investor).

ii. Simplify IPO requirements including permitting overseas listing without requirement of domestic listing and exclusion of such investors from "lock in" provisions.

iii. Enable preferential treatment of such investment in liquidation.

e. **Establish expeditious procedures for closing of businesses:** Closing down or winding up a business in India is a complex and time consuming process. Entrepreneurial ecosystem recognizes that business ventures fail and such failure is neither a stigma nor a constraint on future entrepreneurial activity. The complexity of closing down a business is a disincentive to setting up a business in the first place. Cost and complications of retrenchment of workers is a prime disincentive for establishment of labour intensive businesses.

4.2.2 Easy access to equity capital and debt

a. **Remove regulatory hurdles that inhibit domestic fund raising:** Permitting pension funds, insurance funds and provident funds to invest a small part of their corpus in early-stage venture funds could significantly improve capital flows. Special incentives such as tax credits could be provided to HNIs, corporates and institutions that invest in early stage venture funds or to incubators and to angel investors. Banks must also be encouraged to invest in early-stage venture capital funds by treating such investments as " priority sector" funding without capital market exposure and provisioning norms being applied.

b. **Government could establish a "fund-of-funds (FOF)" to seed other early stage venture funds:** With a corpus of Rs. 5000 crore this FOF will invest as an anchor investor, in a number of Alternative Investment Funds. These AIFs will raise capital from other sources - domestic and foreign - thereby creating a multiplier effect. This could result in capital flow of up to Rs. 25,000 crore over the next 10 years. The India Opportunities Venture Fund, set up under SLDBI by the Government of India should also play this role.

c. **Develop and scale-up debt offerings:** Debt is critical to meeting working capital requirements. Traditional debt providers, however, do not lend without collateral. As such, early stage ventures cannot meet lender requirements. Measures to mitigate this can include:

i. Expand the lender base by incentivizing banks to offer SIDBI like schemes to early stage ventures. Banks to create capacity and capability for lending to such ventures.

ii. SIDBI can expand its role in the area of venture debt by providing funds to specialized NBFCs focused on venture debt (similar to Silicon Valley Bank).

iii. Establish and promote UNIDO like mutual credit guarantee schemes which have played an important role, particularly, in Europe and Brazil. These measures would require the support of Ministry of Micro, Small & Medium Enterprises and the banking system to work successfully.

iv. Improve- credit rating models and their coverage; SMERA, CRISIL and other rating agencies empanelled under " Performance and Credit Rating Scheme" of the Ministry of MSME need to actively work towards improving quality and reach of their credit models for small businesses. This would be a key enabler in allowing banks and other financial institutions to provide debt capital to such ventures.

4.2.3 **Businesses as entrepreneurial hubs:** Greater engagement of established businesses with emerging ventures is needed.

a. **Private sector could participate in setting up and operating incubators in** PPP or other appropriate models including for profit. Such incubators would have access to high quality managerial and technical resources of the sponsoring private enterprise.

b. **Industry bodies and chambers of commerce could drive greater collaboration** between established and emerging businesses, leading to:

i. Greater collaboration both as a buyer and a supplier: Established businesses can work with small businesses strategically to source innovation and technology. They will also better understand buying needs of emerging firms and hence develop solutions and products specifically tailored for them.

ii. Greater M&A activity: Established businesses need to also proactively look at entrepreneurial ventures as sources of inorganic growth - both intellectual property (IP) and non-IP related. Greater engagement with new ventures will improve established businesses' understanding of such ventures and encourage acquisitions as a preferred mode over organic development, with the benefits of the access to passion- driven talent and reduced time-to-market.

4.2.4 **Culture and institutions which encourage entrepreneurship over careerism:** There is a mutually reinforcing relationship between thriving entrepreneurial activity and a culture that supports entrepreneurship - risk-taking and greater tolerance of failure. Measures taken by educational institutions greatly assist in creation of such a culture over time.

a. **Upgrade courses and programs**: India needs more entrepreneurship courses (including social entrepreneurship) and programs across institutions of higher learning. Such programs will be a valuable complement to programs that have traditionally focused on technical and theoretical learning.

b. **Enhance linkages between educational institutions and entrepreneurial ecosystem:** This can be done through a sustained engagement of entrepreneurs, angel investors, and venture capitalists as faculty and mentors.

c. **Promote innovation and commercialization:** India has several institutions dedicated to advanced research. However, the commercialization of research is lacking. Engagement with entrepreneurial ecosystem can provide such institutions with a mutually beneficial and reinforcing relationship leading to opportunities of commercializing the IP through appropriate licensing arrangements.

d. **Celebrate success stories:** The media - TV, print and online - should disseminate entrepreneurial success stories to inspire and encourage entrepreneurship.

4.2.5 **Adequate and effective collaboration forums:** A vibrant entrepreneurial ecosystem needs forums, virtual or physical, where all stakeholders can come together to share experience, expertise and develop symbiotic relationships. There are several models of developing such forums including:

a. **Develop online portals:** Creating an online portal that provides comprehensive information to a new entrepreneur is highly valuable. ln addition to providing information on government policies and regulations, such portals can also provide means for virtual mentorship support. The government can play a role by aligning all arms - various ministries, regulators, etc - to help develop such a comprehensive information portal which will host all relevant information on statutory compliances for setting up and operating businesses. Collaborative forums could also participate in creation of such a portal by providing inputs and encouraging its use.

b. **Encourage creation of collaborative forums** along the lines of TiE and MentorSquare to make mentorship networks available to a wider audience of entrepreneurs.

c. **Set up Innovation Labs (iLabs):** India should develop a network of 15-20 iLabs that can serve as the focal points for collaboration across the region. Such iLabs could also serve as a hub for incubators, accelerators and other enablers in the region.

4.3 Key recommendation: National Entrepreneurship Mission

The Committee has made extensive recommendations that are relevant to a number of stakeholders both within the Governmental and Regulatory fold and those outside their immediate purview. The Committee believes however, that to build a vibrant entrepreneurial ecosystem leading to significant employment and wealth creation in the country, there has to be a sustained and continuous focus on the simultaneous and coordinated implementation of these measures.

Towards this objective, we recommend that the Central Government set up a National Entrepreneurship Mission (the "Mission"), whose sole focus will be to establish a vibrant entrepreneurial ecosystem in India. The Mission' s mandate, as one single entity within the Governments both at the National and State levels, will require it to pursue exclusively, the task of facilitating entrepreneurs and entrepreneurship. The Mission' s key roles will be:

i. The Mission will collaborate and work with all other entities, within Government and outside it, with the following objectives:

   a. Ensure that the promotion of entrepreneurship is continuously high on the agenda of all stakeholders

   b. Educate & inform all best practices globally & put forward well researched recommendations and action plans that would facilitate entrepreneurship

   c. Create appropriate measurements, methodologies and systems to track performance across various industries, in this area. A few of these for example, could be India' s global ranking in entrepreneurship, ease of doing business

ii. The Mission would work closely with Government ministries/ departments of Finance, MSME, HRD, Industry, IT, etc. at both National and State levels, many of whom have developed strategic plans of their own and seek to help them strengthen the element of entrepreneurship in those plans.

iii. It would similarly work with Regulators, Banks, financial Institutions, Angel investors, Venture Capitalists, industry bodies & Chambers of Commerce and educational institutions, both public and private, with the objective of regulatory outcomes which promote and facilitate entrepreneurship.

iv. While the general approach would be to work in an enabling and coordinating capacity, it would have the lead role in the area of driving the financing part of the eco system which is the most critical component. In this area it would need to have appropriate empowerment whilst engaging with other stakeholders. In the area of financing, the Mission would be the sole recommending authority to the Government of India and counterpart bodies set up at the State levels.

v. This Mission would derive its unique strength and importance from the fact that it would be the most knowledgeable entity in India on the subject of creation and development of an entrepreneurial eco system that will foster levels of innovation, enterprise and employment that the country needs, on a sustainable basis. It would therefore, be able to achieve a vast majority of its objectives without having an overarching mandate over other entities of Governments.

vi. It would also become the nodal point for an entrepreneurship movement and in that capacity, articulate and disseminate the view point of the entrepreneurs amongst all the stakeholders within Government and outside- a capacity that is lacking today.

vii. The mission will develop a clearly defined plan of action, ownership of initiatives, key dependencies, resource-requirements for research as well as designing, devising, driving, tracking, and monitoring progress of the initiatives and plans.

viii. The mission should ideally be set up under the Prime Minister' s office which will give it the ability to exercise adequate influence without necessarily a statutory authority.

ix. The mission would set up appropriate mechanisms and metrics that will allow it to track its impact on the entrepreneurial eco system in the country.

x. It would similarly help all other stake holders in drawing up mechanisms to measure their impact on increasing entrepreneurial activity.

5. Assessment of India's Innovation Policies (RIS Discussion Paper)[24]

This paper, authored by Shri Biswajit Dhar and Sabyasachi Saha, presents a detailed overview of the innovation policy framework in India in order to assess its role in innovations and enterprise development in the Indian industry. Over the decades, India' s innovation strategies have been guided by the S&T policy statements, while industrial policy resolutions/statements have given direction to the development of manufacturing enterprises. These twin processes have tried to ensure that India is able to develop a sufficiently robust manufacturing base and at the same time build a sound S&T infrastructure and create a high-skilled manpower base. The authors have distinguished between eras of closed and liberalised economy in India and accounted for recent policy overtures. The authors have also closely examined the Indian scenario with respect to technological capability of its industry and have drawn suggestive international comparisons. Substantial attention has been devoted to the emerging issue of innovations in the SME sector in India and technological interventions in two traditional industry clusters in India have been discussed in detail. Finally, the existing bottlenecks in India' s national innovation system have been highlighted. In this paper the authors have noted that the existing policy paradigm does not draw upon immediate innovation challenges that may be specific to India, particularly when developmental priorities are overwhelming. The paper suggest that while, economic policies should ensure sustained demand for innovations, innovation policies in India at this juncture should cater to two definite goals. First, streamline availability of broad-based skills to seize opportunities of specialization, industrial development and knowledge economy. And, second, achieve frontier R&D focused on pro-poor innovations, niche knowledge and green technologies.

This paper has relevant references which could be of interest to the expert committee.

[24] Research and Information System for Developing Countries Discussion Paper # 189, by Shri Biswajit Dhar and Sabyasachi Saha

Appendix B
Results from Survey on Manufacturing Clusters

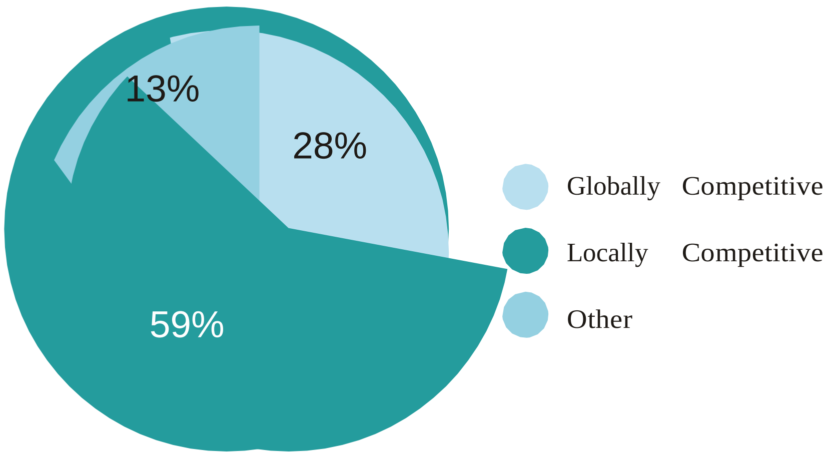
The expert committee commissioned several surveys to understand entrepreneurship as it currently exists, the challenges facing existing entrepreneurs and the barriers to entry for new business owners. Taken together, these surveys offer key insights into various factors hampering or slowing down the growth of entrepreneurship for different sections of the population, and provide a way to look at solutions for the future.

In this section, we outline the main results from a survey on manufacturing clusters. The survey was conducted on 462 participants from 28 independent industry clusters across different geographic areas.

1. Sector competitiveness:

While 59% of the respondents from over 25 different manufacturing and export sectors felt they were locally competitive, only 28% felt they were globally competitive.

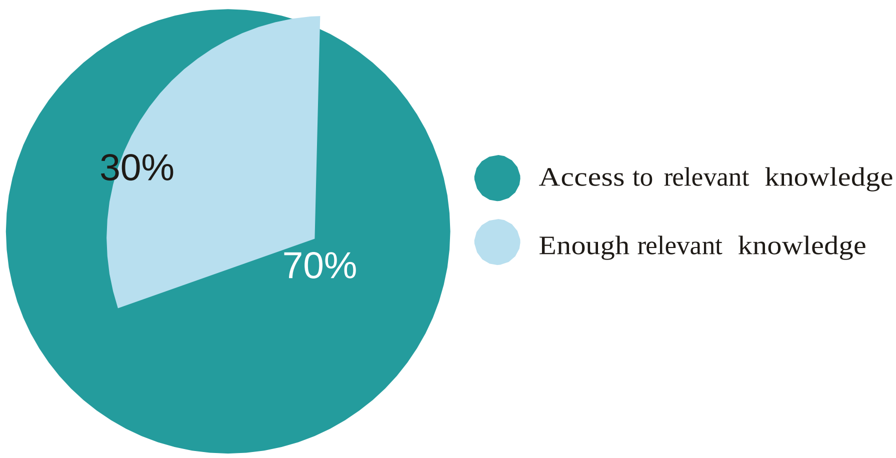
Figure 13: How Healthy is Your Business?



2. Access to relevant knowledge

An astounding 70%, across various manufacturing sectors, agreed that access to relevant knowledge needs to be strengthened. In some sectors, such as among machine knitters, garment manufacturers, rice millers and pulses manufacturers, 100% of the respondents felt that access to knowledge needed to be firmed up.

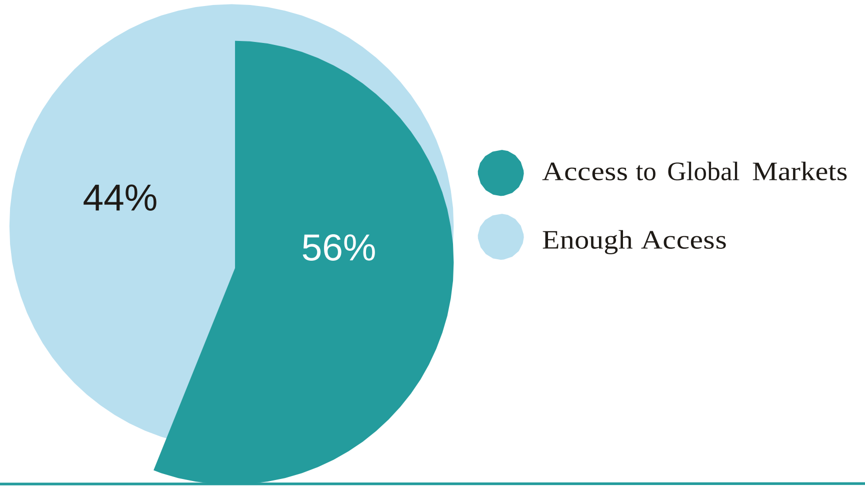
Figure 14: Benefit to Strengthen - Knowledge Access



3. Access to global markets

An average of 56% of the respondents across all the sectors surveyed felt that access to global markets needed to be strengthened. In some sectors such as among brass manufacturers and yarn merchants, over 80% of the respondents felt this was a benefit that needed to be strengthened.

Figure 15: Benefit to Strengthen - Global Market Access

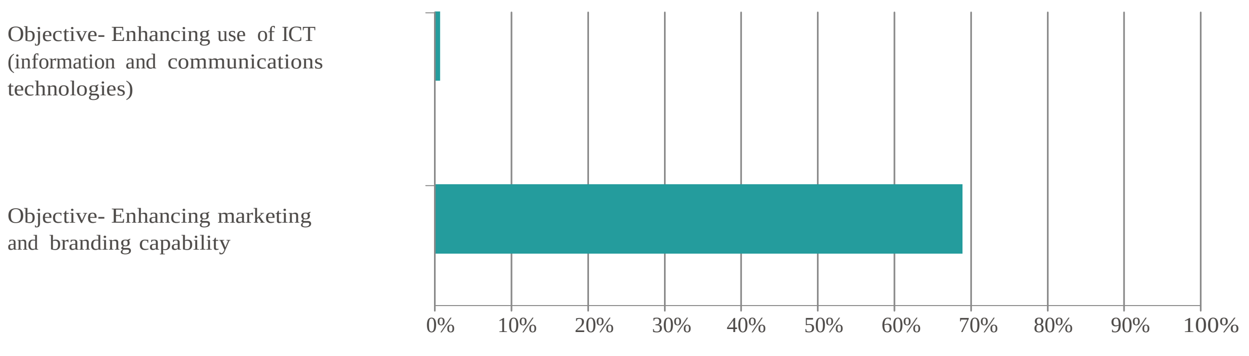


4. Marketing and branding capability

69% of respondent felt that enhancing marketing and branding capability was needed. In fact, in many clusters such as rice millers, machine knitters, garment manufacturers, yarn manufacturers, 80 percent or more correspondents felt that this was an area to focus on. In contrast, most clusters felt that knowledge of technology or ICT was not holding them back.

Product design, packaging and brand creation help manufacturers capture and retain more value from their products. Marketing allows manufacturers to see their product and its usage in a new light, thereby opening up new markets locally, nationally or even globally. Understand consumer behaviour and needs leads to future innovation. As the economy opens up and manufacturers understand competition or aspire to plug into the global economy, it is natural that marketing and branding capabilities would be viewed as a necessary skill to expand and move up the value chain. While design thinking and product innovation needs to be absorbed over time, in the short term interventions are possible to enable these clusters to learn about marketing and quality control.

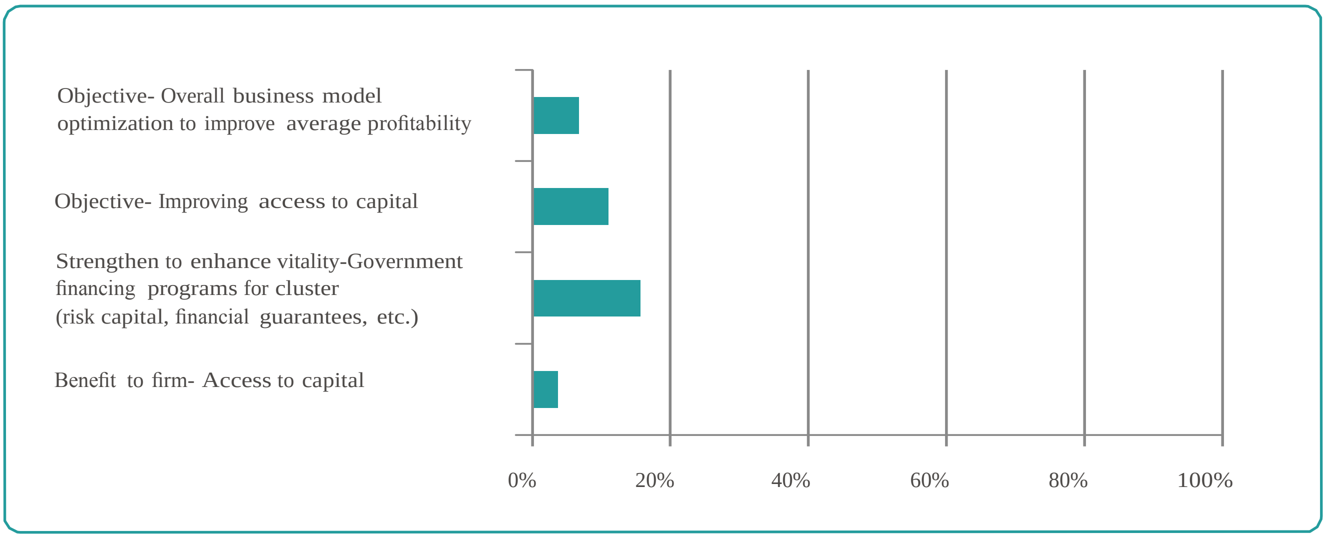
Figure 16:Technical and marketing capability in manufacturing clusters



5. Access to capital or business knowledge is not perceived as a barrier

It is interesting to note that most clusters did not find elements related to the business model or access to capital as a deterrent. This may be because the government has various schemes to support MSME clusters and established manufacturing outfits could be adequately aware of sources of funding needed to support their businesses.

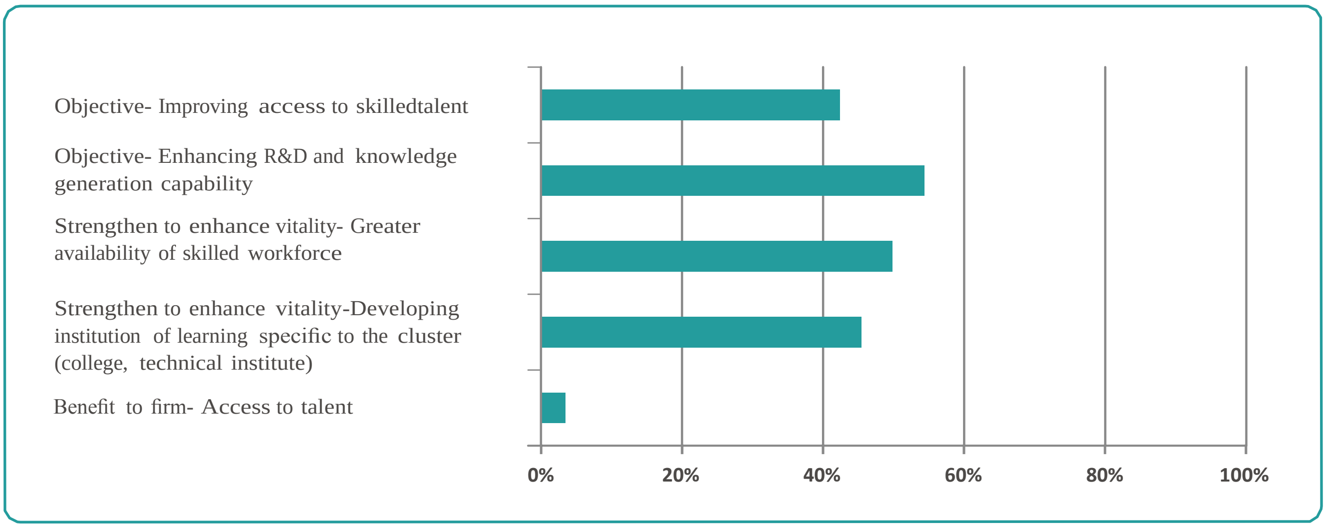
Figure 17: Access to Capital in manufacturing clusters



6. Knowledge and Technical Knowhow

Nearly 50% of clusters are looking for availability of skilled workforce. The critical elements to revitalise and skill the workforce could be through cluster-specific technical institutes, as suggested by 45% of respondents, and an ability to research and innovate in areas related to the cluster, as suggested by 54% of respondents. This indicates a need, not for broad-based education, rather for targeted skilling of large number of people in distributed clusters to support the specific requirements of each sector.

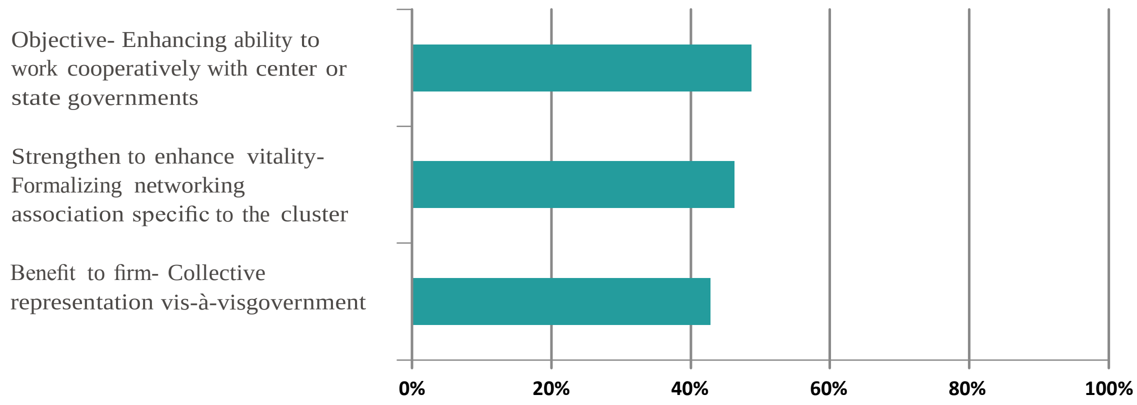
Figure 18: Need for skills & knowledge in manufacturing cluster workforce



7. Platforms for Cluster Networking and Government Interconnect

49% of respondents surveyed felt the need to improve formal networking within the sector through associations. 46% of respondent felt the need to enhance their ability to work cooperatively with gover nment at the centre or state levels. While industry bodies and associations already exist and are active at the national and zonal levels, there may be a need to cascade the activities of such networks down to the district and cluster levels. Clusters also perceive they could benefit from collective representation with the government.

Figure 19: Need for organization in clusters



Appendix C

Results from Survey on Start-up Clusters

In this survey, the online profiles of 900 start-up innovator/ entrepreneurs were analyzed in clusters located in Bangalore. Our study reveals that these entrepreneurs come from a variety of backgrounds and also different parts of the country. Quest for innovation and the demand for talent are increasingly drawing a pool of skilled individuals from diverse backgrounds to cities like Bangalore and Mumbai. While Internet and e-commerce-based ideas dominate the start-up " scene," there is a lot that these hot-spots reveal to us that we can apply to larger programs related to entrepreneurship and innovation. But before that, a quick look at some of the key figures related to start- up clusters in Mumbai and Bangalore:

Both clusters demonstrate a year-on-year compounded annual growth rate of around 30%

40 start-ups raised a median amount of $2.5 million

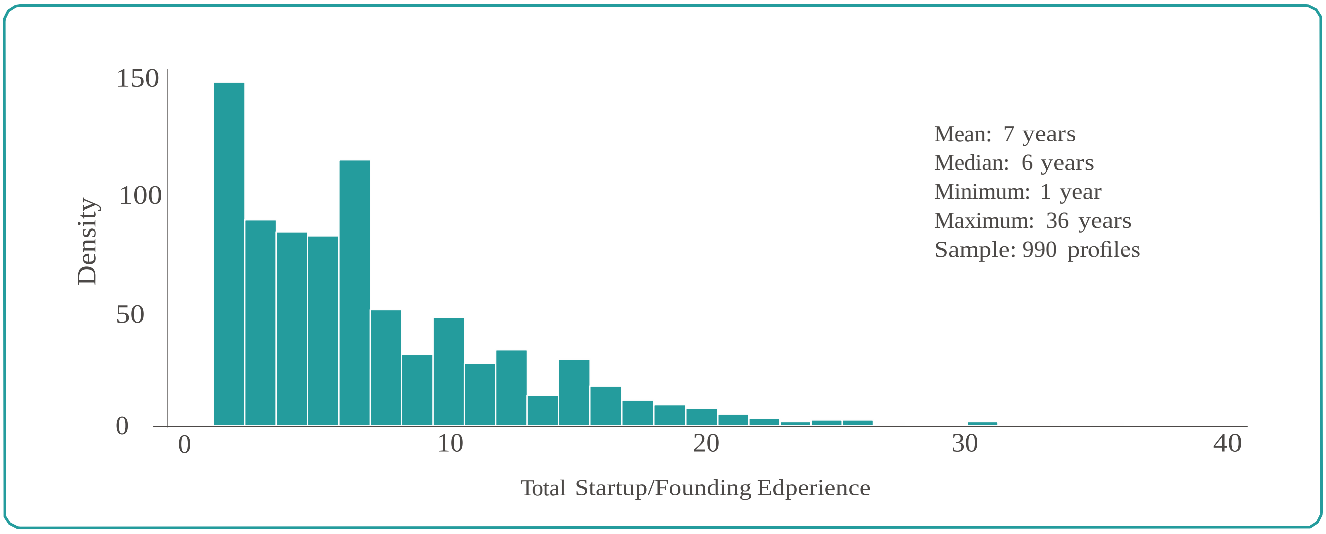
There are approximately 180 investors, with 15 to 20 serial investors

In terms of immediate relevance, the Koramangala Cluster study offers some interesting insights about the experience level of start-up entrepreneurs:

1. Work Experience

A look at the experience level of start-up entrepreneurs reveals that on average, most have worked for a few years thereby acquiring skills necessary to found and grow their own businesses. These highly educated individuals find easy employment in India' s fast growing new economy firms, before branching out to start their own ventures, after having seen process orientation, implementation procedures, and entrepreneurship in action.

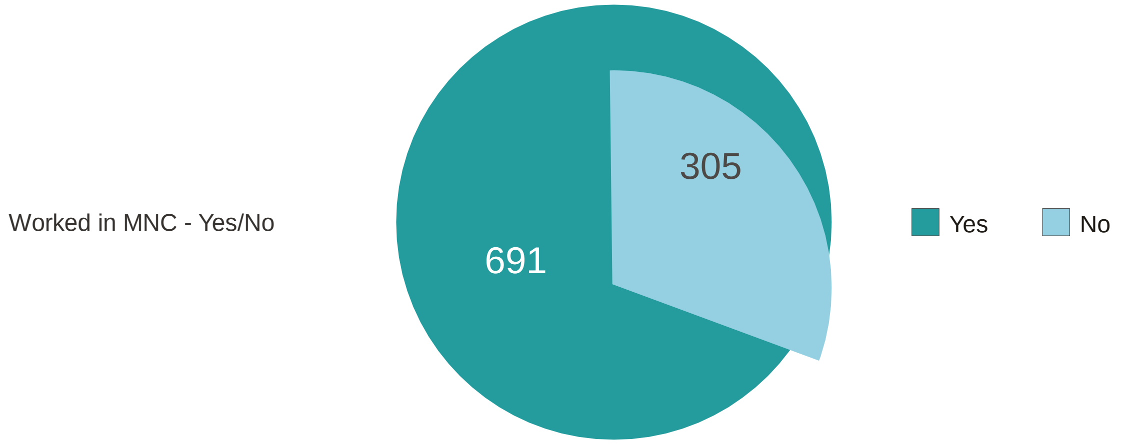
Figure 20: Start-up founder experience



2. Work Segments

It is interesting to note that digital economy entrepreneurs gain their work experience in MNCs or Indian firms, indicating that working in more structured environments with early on-the-job training and development interventions offered by these organizations could play a role in the development of entrepreneurial skills, knowhow and networks.

Figure 21: Start-up founder experience

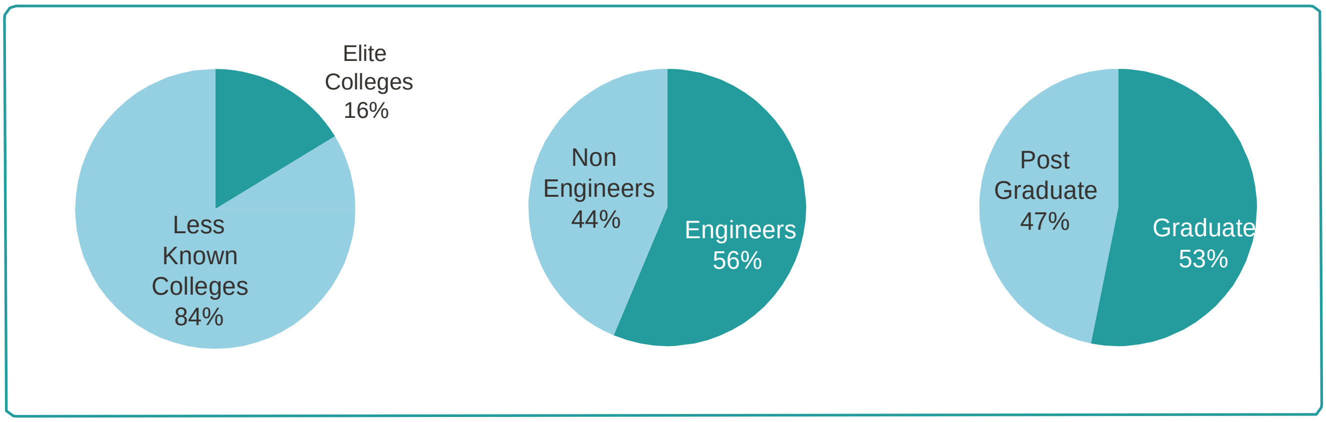


3. Education

One of the most interesting findings from the Koramangala Cluster study was that start-up clusters foster a culture of Inclusive Innovation. What seems to count here is talent, skill and experience, rather than a degree from a national institute of importance. While all entrepreneurs were highly educated with graduate or post-graduate degrees, they came from a mix of engineering and non-engineering backgrounds and only 16% came from institutions of national repute.

Figure 22: Start-up founder education

Figure 23: Education of start-up founders



The above finding is relevant given the fact that the degrees offered by most institutes in India are not valued as highly as say the top twenty or so national institutes of importance. This leaves out a large talent pool from the mainstream, partly because the logistics and expenses involved in recruiting talent from these vastly spread out institutes outweigh the benefits for employers. Now, clusters seem to make opportunities accessible to talent, and draw them out of wherever they are – be it Bareilly or Asansol! This suggests a need for focus on quality education and awareness in a larger number of institutions spread across the nation, rather than focusing inordinate resources on fewer elite colleges, to lead towards wider spread of the entrepreneurial culture.

Survey results also highlight a trend of entrepreneurs moving back to their home provinces—in the North East or Goa or Orissa—to start ventures, after having worked in urban clusters of Bangalore or Mumbai. Reasons cited were: desire to cater to their local areas, desire to create a national footprint while being closer to the source of their raw materials, and desire to employ a local workforce. This type of distributed growth is essential to enable every part of the country to move ahead and prosper. It might even be required for long term stability that the aspirations of youth in every segment of society and in every part of the country are met.

Appendix D

Results from Survey on Economically Backward Youth

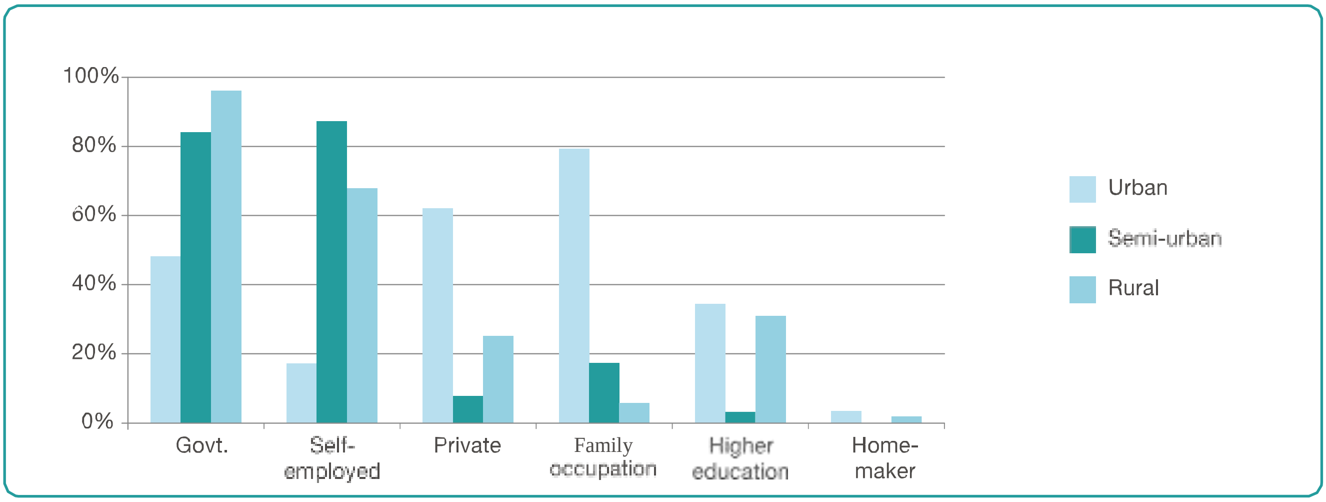
In this survey, 195 participants were surveyed across 10 Pratham Institutes in the states of Maharashtra, Delhi, Andhra Pradesh, Chhattisgarh. They included economically challenged youth in urban, semi-urban and rural areas. The purpose of the survey was to investigate the propensity and challenges for self-employment and entrepreneurship amongst the economically-backward youth. Respondents hail from the bottom 30% income strata, and are undergoing vocational training in sectors of hospitality, healthcare, construction, and beauty and wellness. Most respondents have completed education only till Class 10 or 12. Various multiple choice questions were asked, to which respondents in many cases chose more than one option. In this appendix, we outline the most interesting results from this survey.

1. Desire for entrepreneurship is high

The desire for entrepreneurship varies by location and gender. Only 17% of lower income youth in urban areas expressed interest in entrepreneurship or self-employment, as compared with 87% of respondents in semi-urban areas and 68% in rural areas. Opportunities for formal or informal sector employment in private or family occupations are lower in semi-urban and rural areas, potentially leading to this desire.

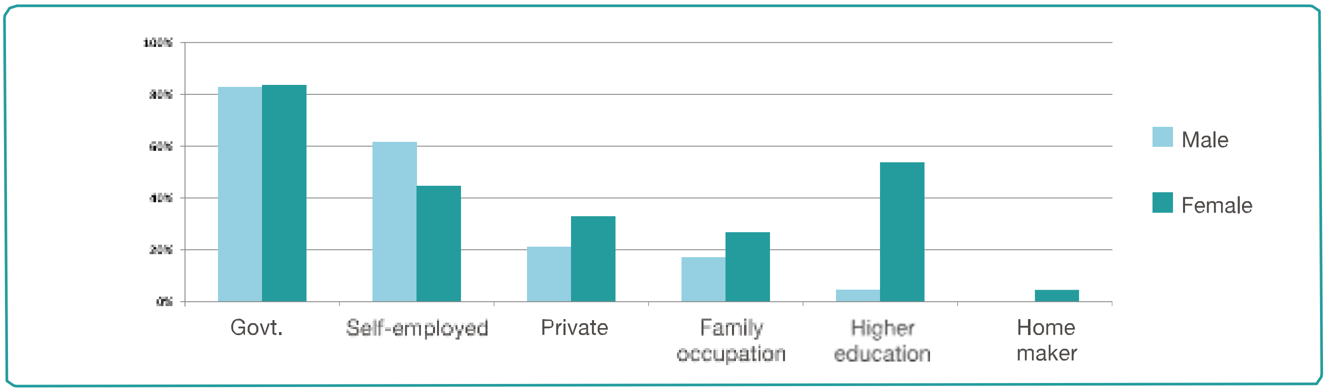
Over 80% of respondents in semi-urban and rural areas also aspire for government jobs, whereas urban youth envision themselves working in private or family businesses.

Figure 24: 3 year employment aspiration of youth, by location



Female respondents were more inclined towards self-employment (62%) as opposed to the male respondents (45%) and also responded saying they were more likely to opt for higher education (54%) as opposed to their male counterparts (5%). This indicates that much could be done in the short term to encourage education, vocational education and self-employment in women.

Figure 25: 3 year employment aspiration of youth, by location



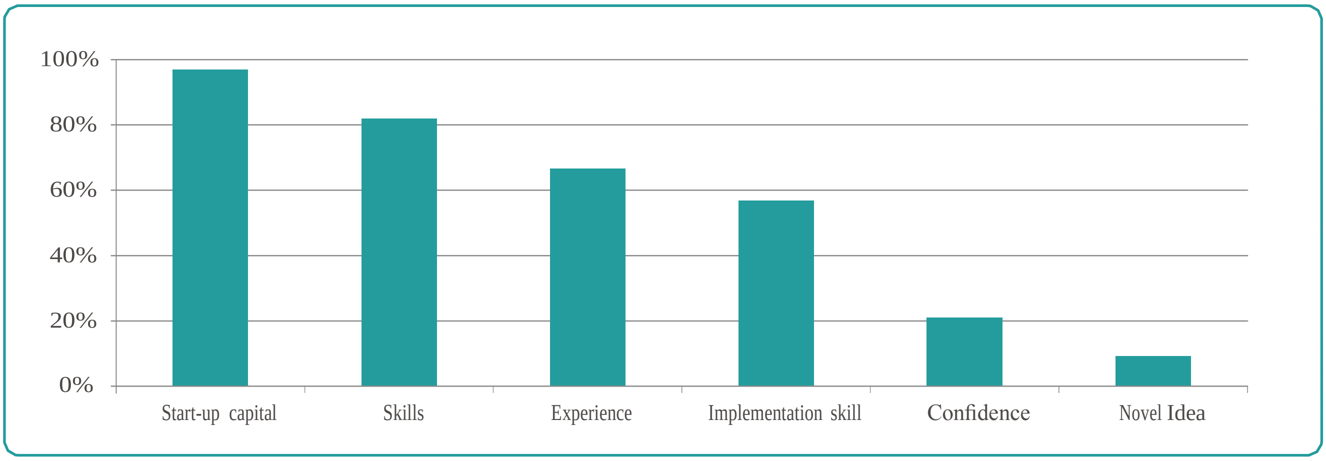
Nearly 55% of respondents in rural areas and 60% of respondents in urban areas would prefer to start an enterprise in areas closer to their homes. The openness to migration is highest in semi-urban areas.

2. Access to start-up capital is perceived as biggest barrier

Across all geographic areas, 97% of the youth perceived that access to start-up was a barrier to entrepreneurship. Looking at detailed responses, nearly 75% of respondents in semi-urban areas and over 90% of respondents in rural areas indicated that they would opt for entrepreneurship if a government scheme provided support. *It is interesting to contrast this with the response from the manufacturing clusters, where less than 10% of respondents felt that lack of capital, government financial support schemes or business knowhow was a deterrent to their business.* Similarly, 60% to 90% of respondents in urban or semi-urban areas would opt for entrepreneurship if independent sources of capital were accessible.

All this points towards an interesting information gap that can be bridged in the short term to make lower income youth become aware of existing government programs, policies and schemes to support entrepreneurship, as well as non-governmental funding and patient capital sources available to support business or self-employment, particularly in rural or semi-rural areas.

Figure 26: Factors limiting entrepreneurship



3. Skills, Experience and Implementation Skills

Lower income group youth uniformly express the need for skills by location, whether in urban, semi-urban and rural areas, and by gender. The need for experience was higher in urban (84%) and semi-urban (85%) youth compared with rural youth (44%), and was balanced across male and female respondents. In contrast the need for implementation skills was felt the most by rural youth (75%) and female respondents (88%).

Over 80% of youth in all locations and both genders feel that technical and entrepreneurship training would allow them to become self-starters. 46% think that they need to get specialized sector based knowledge and 46% would like to see entrepreneurship being taught in school and college curriculum.

Figure 27: Factors that would enable youth to become entrepreneurs, by location

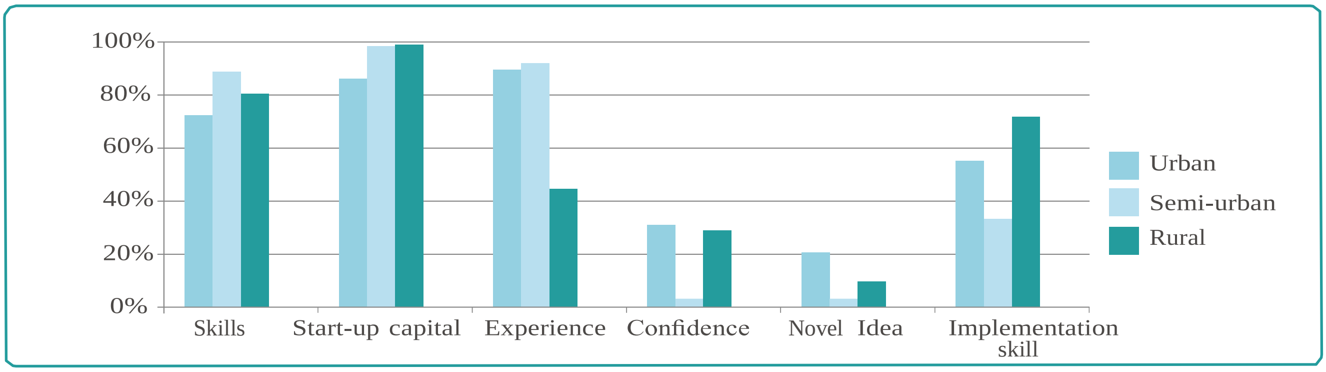
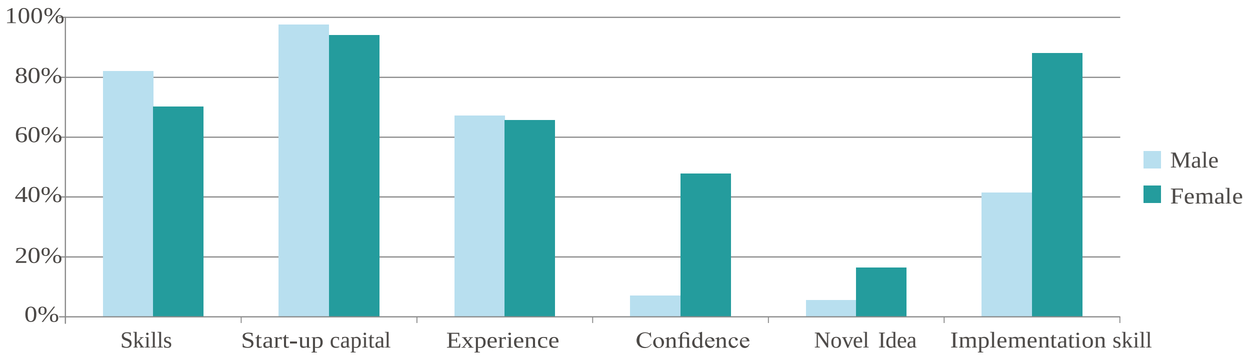


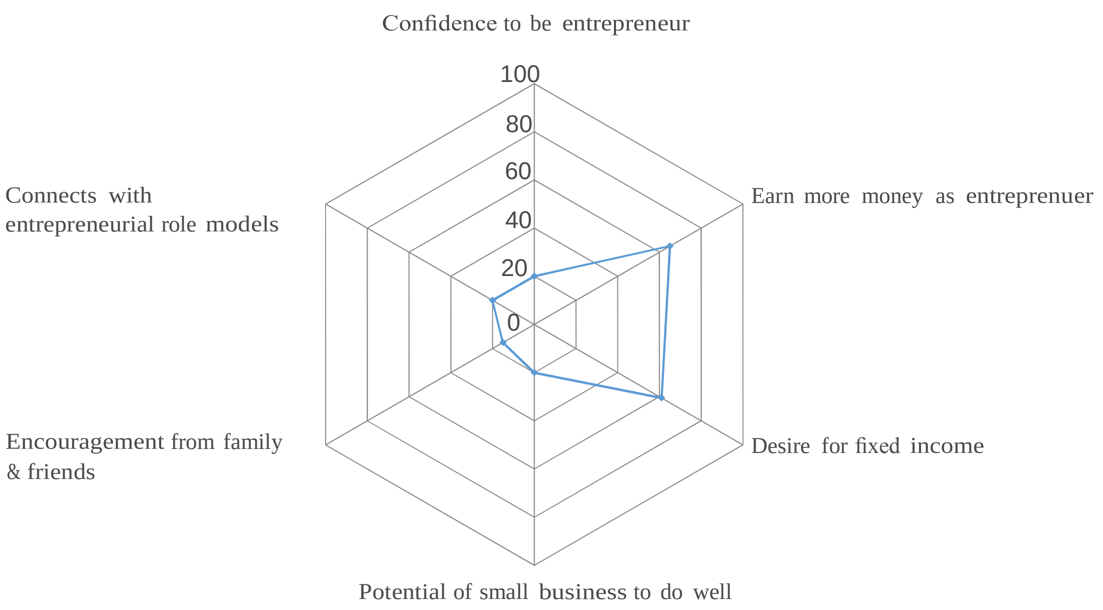
Figure 28: Factors that would enable youth to become entrepreneurs, by gender



4. Cultural impediments to entrepreneurship:

The Pratham survey highlights societal preferences for job security. While respondents felt that they could earn more money if they succeeded as an entrepreneur, they have a high desire to get a fixed income. They can neither envision a small business doing well in their hometown, nor do they have sufficient confidence, encouragement from friends and family or access to role models to venture into self-employment or entrepreneurship.

Figure 29: Attitudes that deter entrepreneurship amid low income youth



i. Desire for security: Over 60% of the respondents across location, whether in urban, semi-urban or rural areas, would prefer the security of steady income from a job. The ratio of male respondents preferring steady income (40%) is higher than female respondents (20%). Youth in semi-urban and rural areas see greater potential for a small business doing well in their hometown, compared with 45% of urban youth who are sceptical of entrepreneurial success. This is perhaps reflective of the opportunity space in semi-urban and rural areas; given fewer jobs in the formal sector, they see more potential in entrepreneurship.

ii. Confidence: Entrepreneurship requires the ability to take an initial risk and stay the course, both of which require a large measure of confidence. Urban (31%) and rural (29%) youth had higher response to “ lack of confidence” in being successful as an entrepreneur, whereas only 3% youth in semi-urban areas indicated a lack of confidence. Further, more women expressed “ lack of confidence” (48%) as a barrier to starting their own business compared with male respondents (7%). This justifies the effort being given to promote girls education and their participation levels in social and government institutions.

iii. Family background: Of respondents whose parents are entrepreneurs, only 65% aspired for government jobs, compared with 87% for the overall group. While nearly all youth selected “ fear of the monetary-risks” as a reason for not wanting to become an entrepreneur, a high 67% also chose “ Family pressure to seek stable employment.” Barring 8% of the respondents, who wanted to follow the family tradition of becoming an enterpreneur, the others prefer the stability of a government job to private sector job. The highest proportion of “ aspiring entrepreneurs” come from agrarian families – the “ self-employed” nature of their family profession seems to incline these youth towards entrepreneurship as well. These results suggest that families play a huge role in supporting career choices, and that the change would happen not just by skilling the youth, but by challenging the belief systems and changing the mindsets of families and communities at large.

iv. Access to role models: 48% of respondents in urban areas and 30% of males indicated that meeting some entrepreneur role models with similar backgrounds would help them in starting their own businesses. Very few respondents knew an entrepreneur in their immediate friends or family circle. However, 98% of male and 69% of female respondents knew an entrepreneur in their village or home town. Looking by location, 72% of urban, 94% of semi-urban and 89% of rural respondents knew an entrepreneur from their home town or village. Clearly many respondents knew of successful entrepreneurs, but outside their immediate circle. To help them understand how others from a similar background succeeded as business owners, these youth need platforms and organized opportunities for interaction. Such interactions demystify the entrepreneurial process and give confidence to youth to take risks. Local incubators or association chapters could play a role in making these interconnections and help aspiring entrepreneurs meet with and learn from role models.

Appendix E

Detailed Explanation on Grand Challenges/ Incentivized Innovations in India (i3)

1. Why Incentivised Innovations in India?

While the country has made significant progress in the last two decades, the population remains the tale of two economies - 74% of the population remains mired in deep poverty according to Social, Economic, and Caste Census of 2011. On the other hand about 15% to 20% of the population seems to be doing well. About 69% of the population lives in rural India and their contribution to the national GDP is lagging far behind. The average income of the 75% of the rural households is Rs.33 per person, per day. A high percentage of the rural population suffers from unemployment, low wages, lack of industrialization, and unsuitable housing. Basic services such as functioning roads, 24/7 electricity, and clean water are typically missing. Same applies to rudimentary healthcare and quality education. Many of these issues also plague day-to-day lives of the poor urban population. Additional challenges such as waste treatment, food adulteration, severely weakened old infrastructure, and high pollution remain unsolved.

While state-of-the-art technology can address a number of these challenges, the existing answers have been out of reach due to the excessively high price tag. This document highlights eleven key burning problems and recommends grand prizes to anyone who delivers in a timely manner disruptive technology solutions that are ultra-low cost, low maintenance, and durable. Wide spread adaption and deployment of these disruptive solutions will result in economic transformation of the bottom 70% and beyond of the population and elimination of poverty. An additional objective is to further energize the local scientific and engineering community/academic institutions and engage them in innovative research and development towards finding novel solutions. Yet another objective is to make India a source of innovation and novel products to address similar problems faced by about 5 billion people worldwide and in the process also accelerate our own economy.

2. Brief Summary of i3

An organization structure that is part of the suggested AIM organization and a process are proposed for managing the grand prizes. The head of this organization will work with corresponding subject matter experts for each challenge. Anyone who is an Indian citizen is eligible to participate. Same applies to those of Indian origin such as NRIs, as long as there is a credible technical counterpart in India working in close collaboration. Those receiving the prize money are mandated to make the product in India. Each challenge will end in three years from the date of the grand prize challenges' announcement. It is anticipated that several of them should have solutions within 12 to 24 months. 25% of the prize money is provided in phase one to promising technical approaches. Remaining 75% of the prize money is paid only at successful delivery, validated by rigorous testing against detailed specifications. The prize amount for a challenge will range between Rs. 10 crore to Rs. 30 crore. Assuming a budget of Rs. 150 Crore, the first year outlay will be Rs. 37.5 Crore. It will be followed by Rs. 56 Crore per year for the subsequent two years assuming every single challenge has been successfully addressed. There is a possibility that no novel solution is found within the stipulated time frame for some of the challenges. However, if solutions are delivered even for half of the challenges, that will be a substantial success. The exact amount for each challenge is To Be Determined and is a function of importance of the project and the level of technical difficulties involved. The award amount will be decided after consultation with the Ministry of Science & Technology and others suggested by Niti Aayog. In the future, other government and non-government entities can also announce their own grand challenges or add to the prizes announced, under this umbrella. It is recommended that AIM collects 3% to 4% royalties on the gross sales from the recipients of the grand prizes.

Finding desirable answers to these challenges dubbed as " i3 grand prize challenges" will assist in quickly meeting the goals of several major government campaigns such as Make in India, Housing for all, Health assurance, Swacch Bharat, and Rural industrialization.

(More details on implementation, eligibility and process can be found below.)

3. Overview:

The factors holding back rural and semi-urban India are lack of 24/7 electricity, roads that are usable round the year, clean water, suitable housing, access to basic healthcare, quality education, and employable skills. From the outset providing these services is quite expensive and is further hampered by sub-par execution, resulting in sluggish deployment across the board. All of these lead to rampant poverty, unemployment, and quality of life that leave a lot to be desired. These issues are also forcing a significant migration from rural to urban areas that to begin with are falling apart at the seams in terms of their ability to deliver services. Meaningful rural industrialization and inclusion in supply chain cannot be developed without addressing these key challenges. Without a thriving rural industry, it is not possible to uplift the large majority of the country' s population. This is more important than ever considering that the average farm land holding is progressively getting smaller to the point where most small farmers cannot produce enough to support themselves and their families. Additionally there are significant issues that impact everyone regardless of rural or urban settings such as waste management, the rampant pollution, and food adulteration that by some estimates are the worst in the world.

4. The Key Problems:

- About 400 million people in India are without electricity, mostly in rural areas. Additionally, in the areas that do have power, electricity is unreliable and intermittent at best. Off-grid solar power remains economically unviable. Diesel and other environmentally unfriendly options are used to generate expensive back-up electricity and the impact of such practices on the environment is massive. It is also a noteworthy drain on foreign exchange. Lead acid batteries are also widely used either as a primary or back-up source. At the end of such battery' s life, lead is routinely disposed of in the ground by scrap merchants causing a serious ground/water lead contamination making its way into food we eat, and thus a source of crippling diseases. Lack of electricity hurts every aspect of economic activity and human life; from the lack of rural industrialization resulting in high unemployment, to rural students who cannot study after dark and are permanently relegated to being second tier citizens, to health clinics without any ability to test, to large amount of produce that rots, farmers who cannot operate pumps etc. Even those who are employed, their wages are meager: 670 million people on average earning Rs.33 per person, per day.

- Most people do not have consistent access to clean, usable drinking water. It is filled with pollutants, pathogens, and toxic substances. In rural areas the problem is even worse caused by run-off from extensive and unconstrained usage of chemical fertilizers and pesticides. The wealthy either buy bottled water, or set up personal water filtration systems, while the average individuals that are over 80% of the population are forced to ingest all manner of harmful substances dissolved in the water. The net result is a large number of water-borne diseases, serious kidney, liver, and gastrointestinal issues, and a rapidly growing cancer rate, human suffering at a mega scale with no escape so far.

- One of the biggest bottlenecks in rural industrialization is the lack of round the year quality road access in many parts of the country. This directly impacts the livelihood of the rural poor. While low cost manpower and cheaper land makes rural areas ideal for small industries and ancillary units, the lack of roads remains a serious impediment. How can we, as a nation, march forward when 69% of our population and the goods produced are unable to move in a timely manner? A data point to note, 51% of the rural households make a living by manual labor.

- Indian farmers in spite of their hard work lose the highest percentage of milk and produce in the world due to the lack of affordable cold storage. This is pathetic considering that 90% of the rural household make less than Rs.10,000 per month, so this is money straight out of their pockets. Furthermore, small fruit and vegetable street vendors are forced to sell their goods at pennies on the dollar at the end of the day due to lack of cold storage, thus denying them a fair return on their hard work – less money in their hands.

- There is a serious lack of affordable housing. About half of rural population (that is about 435 million people) still lives in Kuccha house. They are poorly protected against weather be it rain, cold or heat. Many of them routinely suffer serious damage during monsoon season. Furthermore, most generally lack running water or sanitation facilities forcing men, women, and children to defecate outdoors.

- The poor population routinely suffers from easily treatable illnesses due to a lack of access to rudimentary healthcare including cost effective diagnostics such as basic blood tests. Most rural areas lack the equipment and effectively force a common person to suffer with diseases much longer or go to the cities. Places where the equipment does exist, the tests themselves are cost prohibitive, and are often hindered by erratic power supply.

- Food adulteration is a serious problem across the board. A recent India Today article called it a " ticking food bomb." Milk adulteration is a serious health hazard. Urea, detergent, ammonium sulphate, paint etc., when ingested are toxic and carcinogenic, are routinely mixed in milk. The existing lactometers are not meant to test for the presence of such chemicals. Everyday usage of such toxic milk is leading to major diseases in the population-at-large from children to senior citizens.

- The cities have a huge problem with waste removal and decomposition, which is another source of debilitating diseases. Most urban facilities are woefully inadequate while in rural areas these are almost non-existent. Even if a village or a town could collect all the waste, what to do with it remains unanswered in most cases. Most incinerators typically burn the waste, in some cases the heat is used to generate electricity, and the gases are released in the air further compromising the already poor quality of the air. Approximately one ton of carbon dioxide is emitted for every ton of municipal waste incinerated. The solid residue is hazardous and is not properly stored creating additional health risks.

- A significant percentage of the population still uses solid or liquid fuel for cooking, resulting in deforestation and a considerably increased carbon footprint. Liquid fuel such as kerosene is also drain on foreign reserve. There are serious health consequences as well especially for women who primarily do most of the cooking.

- The strength of the existing buildings, bridges, and dams is in a state of disrepair and the structures built today typically last only half of their anticipated design life. Global climate change is further accelerating the deterioration. There are about 125,000 major bridges used by the Indian railways, most are over 50 years old. The current allowable loads far exceed that existed during construction. Furthermore, weakened infrastructure is also at a much greater risk of collapsing during earthquakes and other natural disasters including fires, cyclones, tsunamis, flooding, etc. Such situation can cause large scale disasters, threaten national security, and can derail progress for years to come. Unfortunately current strengthening solutions cost over 50% of the replacement cost.

- Air-conditioning usage is rapidly rising in the country. It has been estimated that if air conditioning was as prevalent in Mumbai as in New York, it will consume the same amount of electricity as 25% of the total electricity used for air conditioning in the United States. This has tremendously increased the demand on new power generation, most of that is coal-fired - a recipe for environmental disaster.

It should be noted that the problems for which research and development is already underway elsewhere and substantial sums are being spent are not included in the above list. An example is the extensive work being done by Gates Foundation on developing low-cost toilets. Besides, the award amounts would pale in comparison to the budget of such on-going efforts. Similarly, problems for which cost-effective solutions already exist but are constrained by the implementation efforts have also not been included.

In many cases, the technology addressing the above mentioned pressing problems does exist; however, in every instance they are too expensive and thus economically unviable in the Indian context. There is an urgent need to develop novel disruptive technologies that are ultra-low cost, economically affordable and customized to the local conditions. It should be noted that while on the surface these may appear to be easy challenges, one only needs to dig deeper to find out that ultra-low cost answers have been out of reach so far either due to lack of awareness or effort, or, lack of capital, or the inherent challenge in finding the right answer, or all of the above. We need to place these on top of the national agenda and draw attention of the community of scientists/engineers/entrepreneurs to quickly find out-of-the- box answers to these difficult challenges.

Disruptive innovation in addressing these problems and incentivizing technology breakthroughs is the way to move forward. By providing substantial cash rewards to individuals or teams that can come up with ultra-low cost sustainable novel solutions to the problems that face the common person in India, there will be greater motivation to find the answers to what the nation faces today. This enables the government to focus on change that will benefit everyone's progress. Additionally, this will act as a catalyst in encouraging local research and development activity at all levels from entrepreneurs to academic institutions. Rewards will only be given out at success to solutions that have been proven to work, and part of the innovation process must include extensive and thorough testing against well-defined detailed specifications.

The incentivized innovation challenges should be clearly defined along with detailed specifications and method of verification, and allow people to be as creative as they wish in devising the solution. The solution should be feasible by a single individual or a small team. It should not require a large group. The problem solving should drive investment in the corresponding field and related industries. And lastly, the innovation should provide vision and hope.

Incentivized innovation has worked around the world in stimulating innovation. In the US, XPrize is giving tens of millions of dollars to those who can provide solutions to major technological challenges. The Automotive XPrize ran from 2007 to 2010 and required teams to build vehicles that were minimally 100MPGe efficient, produce less that 200g/mile of CO_2, and able to be made for the mass market. There were two subdivisions within the competition: one for mainstream cars, and the other for alternative cars. The mainstream prize of $5M was awarded to a team for its four-passenger Very Light Car, obtaining 102.5 MPGe. Another team won the $2.5 million Alternative Side-by-Side competition with their aerodynamic Wave-II electric vehicle achieving 187 MPGe. Yet, another team won the $2.5 million Alternative Tandem competition with their 205.3 MPGe faired electric motorcycle.

Another XPrize competition that had a great impact was the Ansari XPrize which was awarded to the team that could create a manned spaceship that could go 100KM above the Earth's surface. Additionally it would have to be able to make the trip twice in 2 weeks, and hold at least 3 people. Mojave Airspace Ventures won this prize in 2004. This contest led to a great deal of innovation and started the concept of sustainable space travel. However, it should be noted that while XPrizes have focused on finding solutions where none exist, the i3 grand prize challenges are focused on finding ultra-low cost, durable and low maintenance variants.

5. Recommended projects for i3 grand prize challenge:

Below are essential ultra-low cost, low maintenance, and durable innovations customized to local conditions that will considerably increase rural employment and industrialization putting substantially more money in the pockets of rural/semi-urban households, improved health care, cleaner environment, strengthen infrastructure, and vastly improve the quality of life across the board.

Battery: A novel and environmentally-safe energy storage that delivers about 150 Wh for a minimum of six hours, at half the cost of a typical corresponding lead-acid battery, and has at least 5 times the life cycle of the prevalent lead-acid batteries. Other than routine maintenance, the battery should essentially be maintenance-free. This will enable economically viable off-grid electricity usage without having to wait for the government's distribution grid to be built. A poor rural household will be able to operate two bulbs, a fan, a TV, a small refrigerator and charge a mobile phone. This will directly impact the quality of education and study hours of rural students. The battery should also be linearly scalable for larger loads. With incremental power one will be able to build cold storage units requiring 24/7 electricity, thus reducing the large scale produce wastage. Small scale industries such as dairy, food processing, machinery repair, handicraft, ancillary units will be enabled. All of these will substantially enhance the average income of a rural household. Usage of lead acid battery in inverters commonly used in urban households could also be replaced thereby reducing the highly toxic lead contamination.

o *This challenge is expected to be solved within 18 to 24 months.*

Solar Panels: A panel that costs 33% of the existing widely used solution while maintaining the same specifications in terms of power delivered, durability, and more. Solar panels combined with the battery described above would go a long way to make solar power economically attractive and primed for large-scale adoption.

o *This challenge could take 2 to 3 years.*

Water Purifier: A fast, cheap way to purify drinking water is a must. A solution would have to be able to deliver 10 liters of potable water for under Rs. 2. A solution would not only remove 99%+ pathogens; it must also remove pollutants, which often end up in the water supply due to fertilizers and pesticide run-offs, and various other reasons. Centralized solutions so far have high procurement, maintenance, and delivery costs, thus individualized solutions such as a tablet that can be dissolved in 10 liters of water is a preferred choice.

o *This challenge is expected to be solved within 18 to 24 months.*

Cheap and Durable Rural Roads: Roads that will cost 33% of today' s and last for a minimum of 20 years. This will enable much rapid road build-out and in turn expedite rural industrialization and generate employment at higher wages. Quality roads accessible round the year will bring in ancillary units into the villages. This development along with 24/7 electricity will be a key factor in economic transformation of the rural economy and thereby of the country. A supply chain stretching from farms to factories will accelerate rural industrialization.

o *This challenge is expected to be solved within 12 to 18 months.*

Refrigeration: An affordable small cold storage option capable of preserving perishable fruits, vegetables, and dairy products, and costs no more than 33% of the prevailing cost, and consuming only half as much electricity when compared to the existing solutions, would enable scalable and viable rural industries in dairy and produce. It will allow farmers to put more money in their pockets by greatly reducing the amount of spoiled goods. It would also allow them to amass more production before selling it. Furthermore, similar units in urban settings will allow small fruit and vegetable street vendors to share such facility at a very low cost and considerably increase the money they make.

o *This challenge could take up to three years or potentially even longer.*

Affordable Housing: Small 200 to 300 square feet self-standing homes that could be built at the cost of Rs. 300/square feet using locally available materials and are optimized for local weather conditions, while meeting the building codes. Thus 200 square feet home will cost about Rs. 60,000. The homes will be wired for daily electric usage, running water and equipped with sanitation facilities. This should be feasible using emerging low-cost construction and roofing materials, and doors and windows built using man-made materials. Standardization of dimensions for windows, doors, sinks, cooktops etc. will enable high volume production at low cost further enhancing the affordability. Multi-story designs could reduce the cost further. Development of 3 or 4 standardized layouts to choose from will also assist in reducing the cost.

o *This challenge could be solved within 12 months.*

Blood Tests: Diagnostics are a fast, easy way to determine a plethora of health problems. Diagnostic equipment that is not only affordable in price but also low in operating cost can help save millions of lives. Diagnostic tests and screening protocols for both chronic and infectious disease exist. However, the dependence of these tests on central labs and trained laboratory technicians make it untenable for any cost-effective screening program to be implemented at scale. There is a need to create a new class of diagnostic devices without compromising the quality, which cover a basic blood test panel for Diabetes, Cardiovascular Disease, Malaria, TB and Anemia screening/management. For example (Hemoglobin, CBC, Glucose, Creatinine, HbA1c, Salts (Sodium, Potassium), Lipids, Liver Function). It should be able to provide instantaneous results at the point of care and cost less than Rs. 15 each or 25% of the existing cost. Untrained or semi-trained nurses or community health workers should be able to use it. The device will only require a single capillary blood draw using simple lancets. The device should be able to work with smart phones for data collection and telemedicine, and cost less than Rs. 3,000. Smart phone connectivity will make telemedicine a reality and will allow 100s of millions of people access to quality health care.

o *This challenge is expected to be solved within 18 months to 3 years.*

Milk Chemical Analyzer: Equipment that could analyze milk' s chemical components and detect presence of commonly used adulterations such as urea, detergent, ammonium sulphate, paint, oil etc. Within three minutes at Rs. 4 or less. This will allow the testing of the milk right at the last mile – the collection point at which milk is procured for dairies. Same equipment can also be used at the dairy itself. The equipment itself should cost under Rs. 10,000 and preferably plugs into a smartphone. This will help eliminate the curse that has been plaguing a billion plus population.

o *This challenge is expected to be solved within 18 to 24 months.*

Waste Treatment: Develop solutions at half the prevailing cost where the three incineration by-products, namely heat, ash residues, and carbon dioxide are entirely utilized to produce not only inexpensive thermal power but also environmentally friendly and durable products useful for infrastructure and other sectors (The desired cost point is To Be Determined.)

o *This challenge could take up to three years or potentially even longer.*

Solar Cooktops: An efficient cooktop that can deliver 1000 WH for at least three hours which will enable cooking without solid or liquid fuel. The product should be able to store heat and deliver on demand without going through the conversion of solar heat to electricity and then back to heat. The entire system including solar elements must be economically viable and priced to be less than Rs. 9,000 (equivalent cost of cooking using natural gas or kerosene for two years, assumed to be about Rs. 4,500 per year).This will go a long way to minimize deforestation and usage of kerosene in the country. Please note that solar cooktops are different than solar cookers. Another option is to integrate this solution with gas based cooktops allowing usage even during continuous monsoon days. An additional benefit is the reduced dependency on imported natural gas.

o *This challenge could take up to three years or potentially even longer.*

Strengthen the Existing Structures: An efficient strengthening technology needs be developed that could considerably strengthen the buildings and civil infrastructure at 10% of the replacement cost. This will not only extend the life of the existing structures, but will also bring them within the safety margin. It will also provide protection against natural disasters such as earthquakes, tsunamis, floods, and cyclones, and reduce the risk to national security.

o *This challenge is expected to be solved within 18 to 24 months.*

Efficient air conditioning: If power consumption of a room and/or industrial air conditioners can be reduced by 60% with a novel innovation while maintaining the price parity with existing solutions, it will reduce the overall (and rapidly growing) power generation requirements by several gigawatts hour per year. To avoid any confusion, the expectation is to have identical functionality and same or better remaining specifications as the existing conventional AC units. Furthermore, it will have a dramatic impact on the carbon footprint.

o *This challenge could take three years or potentially even more.*

Each of the above challenges when solved will also have a huge export potential roughly 4 times the size of the Indian market.

6. Grand Prizes Management Mechanism:

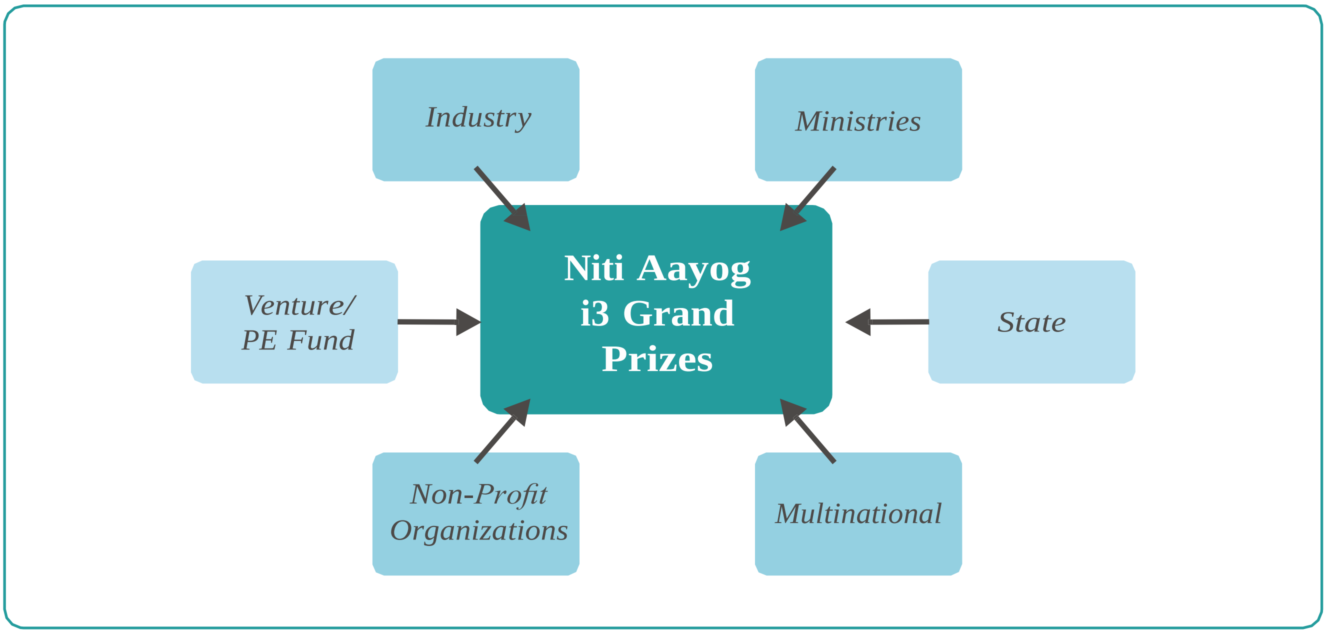
Ideally, " i3" , should become a focal point for grand-prizes be it individual ministries such as Science and Technology, Bio-technology, MoD, or private companies both domestic and multi-national, or Non-profits. States should also be able to add their own grand prize challenges as well, as shown in the figure below. Furthermore, non-government entities should be encouraged to increase the government announced award amounts. The initial list that is supported by the government money is recommended above. Having one focal point will help create broad awareness about the efforts being made to solve the key problems to the two key constituencies;

- The public at large who will benefit from it; and
- The scientific/entrepreneurial community interested in working on these serious challenges.

A common point also enables a consistent approach to administer the process. The award amount will be determined for each project subject to its potential impact and the likely difficulty in addressing it. There should be a minimum threshold in terms of award amount so that the i3 Grand prize administrative entity is not over-burdened.

The process from clearly defining the problem to rewarding the best solution has to be laid out. The detailed specifications, milestones, and time lines must be defined beforehand. It should take no more than 3 years from start to finish. Smaller amounts of prize money are awarded earlier in the program named phase 1 to provide funding to the most promising teams. One of the key goals is that only those teams with ideas that have a meaningful shot of success within 26 months are awarded in phase 1. Eventually a group of 1 or 2 finalists would be chosen for final testing, and the team that meets the detailed specifications after undergoing rigorous testing would win. The total award amount per grand challenge could range from Rs. 10 crore to Rs. 30 crore.

Figure 30:



7. Process

Proposals will be invited for each project and shortlisted by the subject matter committee as described below. In phase one, up to 5 proposals per project will be short listed and 25% of that project' s corresponding prize amount will be distributed equally to those short listed. In phase two, a winner will be picked (maybe 2) who meet the requirements laid out and the remaining 75% of the prize money is provided to the winner (s). The expected completion is within 3 years of the i3 grand prize announcements including the testing. It is expected and understood that some projects may finish earlier than three years, and whereas, some may not succeed within the stipulated time frame. An overall 50% hit rate should be considered a decent success for this i3 program. Those who are not short listed but wish to continue their work at their own time and expense remain eligible to submit their final product for consideration. In fact, there could be some who choose not to participate in proposal stage at all but may independently work on it or already be working on it should also be eligible to submit their final product within the guidelines.

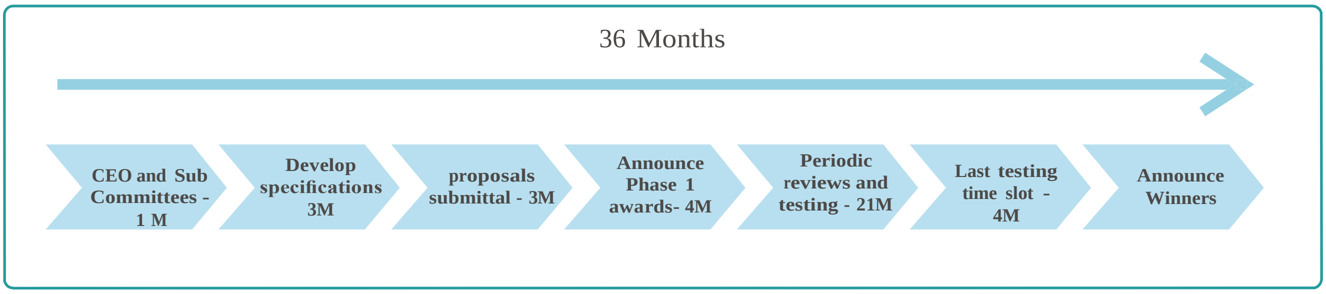
The timelines are shown below in the figure below and the steps are as follows:

- The CEO and the sub-committees are formed within the first 30 days of the i3 grand prize award announcements.
- The specifications by the respective sub-committees should be completed within subsequent 3 months.

The proposals should be submitted to the i3 organization within next 3 months.

- Review and short list selection based on the technical feasibility/strength of the proposals will happen in next 4 months thus completing phase 1.
- Within the next twenty one months the final solutions are submitted by the teams selected and/or by others who may not have been selected, or chose initially not to participate. For the clarity purposes, it should be noted that a team can submit their final solution anytime within the 21 month window. Testing is performed after submittal against the specifications in the subsequent 4 months. The first one to meet the defined goal is the winner.
- Maximum time allowed for completion from beginning to end is 3 years.

Figure 31:



8. Eligibility:

Eligibility requirements described below are applicable for the government funded grand-prizes. Eligibility for the private sector funded prizes is determined by the entity funding the award but at minimum will be as follows. Anyone with " Indian Origin" is defined as an Indian citizen or of Indian descent (NRI) regardless of the passport he/she carries. The criterion is as follows:

- Any individual of " Indian Origin" ; or
- Any start-up/enterprise based in India; or
- Any group of individuals in which the project lead person is of Indian Origin as defined above; or
- Any university in India or abroad but the project lead person is required to be of Indian Origin; or
- Any enterprise that is based overseas in which majority is owned by person(s) of Indian Origin.

If the project is being worked overseas then there should be at least one credible technical person based in India, actively participating in the research/development of the project.

Anyone associated with the i3 grand prize organization as a board member, CEO, employee, subject matter expert, advisor, or consultant or their immediate family can neither participate nor can have any financial interest in any team.

9. Commercialization of the Winning Solutions:

By definition there is an urgent need to solve the problem in a manner that is economically viable. Hence, there should not be shortage of funding to further commercialize these successful solutions from the private sources including venture funds, private equity, corporate funds, or other philanthropic funds. The solutions will have wider applicability across the globe potentially helping 5 billion people and thus sources of funding can be from anywhere. The projects that are funded by the government money could be further assisted by initial volume procurement by the government but should not be necessarily guaranteed. Those projects for which the award is funded by the government are mandated to manufacture the products in India further augmenting " Make in India" campaign. It is recommended that AIM collects 3% to 4% royalties on gross sales from the prize recipients.

An active effort should be made to encourage private sector to add their grand-prizes for their favorite projects.

Intellectual property so generated should belong to the inventors; whether an individual, or a group or someone at an enterprise that came up with the IP.

10. Proposed Administrative Structure:

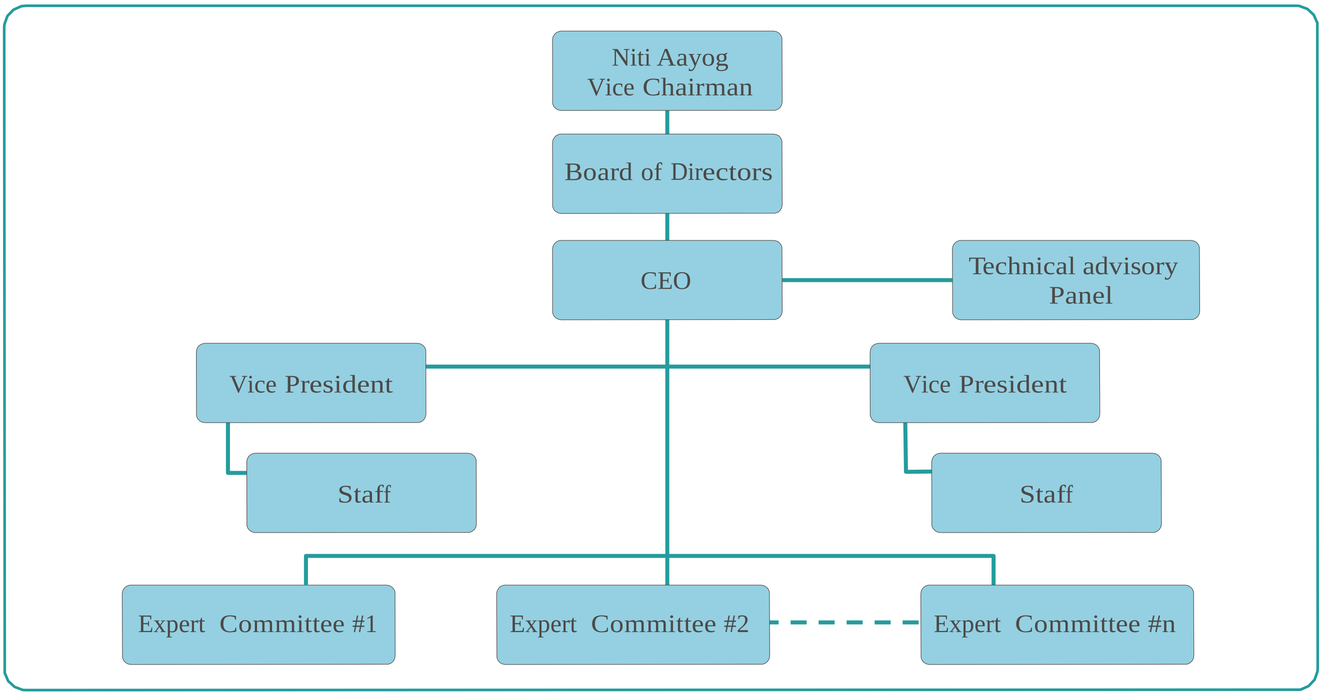
Please see the proposed organization chart in the figure below. The program should be headed by a full-time CEO who is responsible to manage the day-to-day operations. The CEO is helped and advised by a board of directors with a macro view of the pressing needs and aspirations of the people. The CEO should have a strong science/engineering background and ability to work closely with the ministry of science and technology, other government departments, and large number of top-notch scientists, engineers and entrepreneurs. Ideally, this person should not be a government employee.

The board will consist of 7 members, 3 from the outside, CEO, one person each from Niti Aayog, the Department of Industrial Policy and Promotion, and Ministry of Science & Technology. The board will work with the CEO to determine the prize amount for each challenge for which the Niti Aayog is providing the money. In general, the entity that offers the prize determines the amount.

The CEO will be assisted by a small technical advisory panel that will help in bringing in subject matter experts for each project. This advisory panel could also include representatives from prominent consulting companies working on pro-bono basis. Each grand challenge will have a sub-committee of three subject matter experts who will draft the detailed specifications, test methodology, and verification mechanism. They also will evaluate the proposal received and short list the most attractive ones. The subject matter experts can be in India or abroad and of any origin - Indian citizens, NRIs of non-Indian citizenship or neither. One of the key functions of the subject matter experts is to draft a thorough, accurate, and unambiguous set of specifications for their corresponding challenge. Any decisions in the sub-committee will be made by a majority vote. The selection of the members of the sub-committee is one of the most critical functions that will determine the success of the grand prize program. The members of this sub-committee are not only top technical experts in the chosen project area but also are expected to be held at the highest standards, avoid any controversies and/or any conflict of interest.

Initially, i3 can be housed at a university such as the Indian Institute of Science, Bangalore, and later on can have its own space or be co-located at a campus.

Figure 32:



11. Campaign to Create Awareness:

It' s important that a significant awareness campaign is carried out to the key constituents:

- Public-at-large to whom will be immensely helped by addressing these problems
- The scientific and entrepreneurial community including researchers, students, or start-ups.

We not only need to address the serious challenges but also create a sense of self-confidence that we can solve our problems and foster a long lasting environment in which innovation becomes an integral part of the thinking.

The following is proposed :

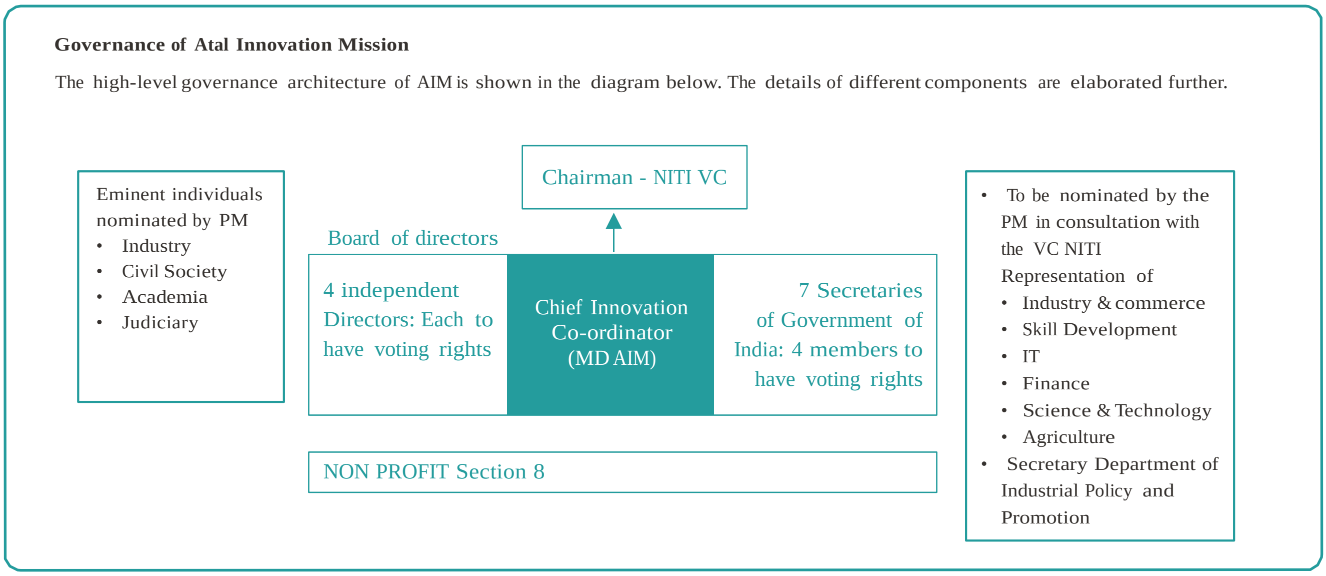
- An easy to use website with relevant information.
- A reputable PR firm should be hired to promote the campaign mostly in India and some outside targeting both broader audiences and the scientific community. This outreach should be made to English, Hindi and regional language media outlets. The target should cover print, TV, and social media.
- An extensive outreach by the committee and the CEO to all major technology institutes to popularize and answer questions at the grass root level to both faculty and students.
- At least once a quarter hold well-advertised conferences at a regional level inviting the scientific community, private sector, and investors.
- Engage state governments to communicate the message down at the high school level.
- Couple of brand ambassadors with wide name recognition and reputation will provide additional reputation.
- It may be beneficial to get pro-bono services from glamor world celebrities to bring about a broader change along the lines of " Innovate, invent, invest in India" . Their job is to make it fashionable to " Innovate" !
- Prime Minister calling on the scientific community at appropriate occasions will help considerably.
- Lastly, extensively promote the successes and the teams behind them as heroes.

Appendix F

Detailed Explanation of the Atal Innovation Mission Organization

The high-level governance architecture of AIM is shown in the diagram below. The details of different components are elaborated further.

Figure 33: Recommended Structure of the Atal Innovation Mission



1. **Chairman** – VC NITI Aayog shall serve as the chairman of AIM board.

2. **07 Government Functionaries** would serve as Directors on the board of AIM. These functionaries shall be no-less than Secretary level from the following Ministries:

   a. Secretary Skill development

   b. Secretary Industry and commerce

   c. Secretary IT

   d. Secretary finance

   e. Secretary science and technology

   f. Secretary Department of Industrial Policy and Promotion

   g. Secretary Agriculture

3. **04 Independent Directors** – chosen from outside of Government would serve on the board of AIM.

4. **Chief of Innovation/MD (competence, experience, abilities):**

   a. Proven capability to run an organization to deliver results with very strong people skills

   b. Worked with the Government system and or has experience in Government working

   c. Global experience– 5-7 years in a global company, fund or research institution

   d. Commercial/entrepreneurial orientation – experience of working in, starting up or leading a successful business or service line

   e. Institution-building experience – Experience in managing high quality professional teams and/or building an institution

   f. Academic qualifications – at least a post-graduate degree in a professional field such as science, engineering, management/business/commerce, law or medicine

5. 04 Advisors to the Chairman or Chief of Innovation/MD:

Individuals with large experience of innovation, wealth creation, intense knowledge of technological changes, very highly respected as individuals, large contribution by innovation and entrepreneurship to the country and or the world, figures to whom people look up for advise guidance and help.

6. Besides this let us have a number of advisors to the board on subject like:

a. Finance and Financing
b. Legal
c. System and processes
d. Culture and team building
e. Intensive knowledge of environment

7. Working Groups: AIM would have four working groups to execute the different functions to perform various activities related to innovation:

a. Idea and Networks: This group shall undertake outreach activities identifying innovations across the spectrum of sectors, screening and shortlisting for scalability, handholding and building entrepreneurial ecosystem for selected ideas. It shall institutionalize collaboration among national and international academia and civil society. This group shall be responsible for organizing and creating " grand challenge" for Indian needs of innovation.

b. Angel Funds: A fund of funds, ensure effective venture capital and angel funding for innovations/disruptive ideas. This fund is used to build scale of enterprises with innovations which have transforming impact.

c. Governance Innovation: Group responsible for bringing leading international practices of good governance and encourage intrinsic innovation for solving daily governance challenges.

d. Missions (or Special Initiatives) group: Group will take ideas from concept to implementation, with targeted outcome based approach with support from all levels of governance. Each mission may draw members from other AIM groups to ensure continuous learning at AIM. To help build credible AIM brand as benchmark to Government secretariats, they will also be responsible for identification, assessment and giving necessary support for creation of Indian Googles and or GEs through innovation, research and creative destruction.

8. Empowerment:

a. Culture of inter-disciplinary, cross-functional teams designing initiatives end-to-end

b. " Lab" culture – bring cross-functional expertise together to brainstorm and agree on a solution, not solve problems in a sequential and hierarchical way

c. Externally oriented organization – combining innovators from universities, consultants and private sector enterprises with innovators in government

d. Output orientation – capacity created for defining and measuring indicators of success for AIM' s initiatives and for other bodies/agencies

e. Focus on 2-3 big initiatives at a point in time

f. For special initiatives: " Performance contract" jointly for Chief of Innovation along with relevant Secretary; Budgetary powers delegated to AIM; AIM works with Ministry on a Detailed Action Plan; AIM ensures cross-functional Implementation Teams created; Steering Committee architecture set up to make decisions cutting across different silos; AIM ensures measurement, monitoring and review on a monthly basis, and reported to the Steering Committee.

9. Principles of working:

a. by and large do not execute themselves

b. create ability to identify root causes

c. enable organization capability of the Department where this is being done

d. work with measurements of outputs

e. teach to fish do not give free fish

f. giving the right systems and processes for working improvement

g. Enable innovation – " handholding over hands-on"

h. Identify, nurture ideas to be institutions/enterprises of future

I. Engage industry and academia internationally

j. Facilitate resolutions to enabling initiatives with government

k. Bring industry, academia, civil society, and Government together

l. Harmonizes existing innovation initiatives

m. Learn from global – build local on what exists – do not reinvent

10. AIM work:

There are multiple aspects of the work area where AIM would be able to create value towards the objectives of economic growth and increased prosperity. It is envisaged that AIM shall undertake two broad types of work – Advisory and Implementation.

In Advisory nature of AIM work, the organization shall identify ideas and provide recommendatory inputs to the Ministerial secretariats or state governments. Here the role of AIM shall be that of a policy shaper, idea facilitator, and the actual responsibility for outcomes shall vest with the relevant ministry or government body.

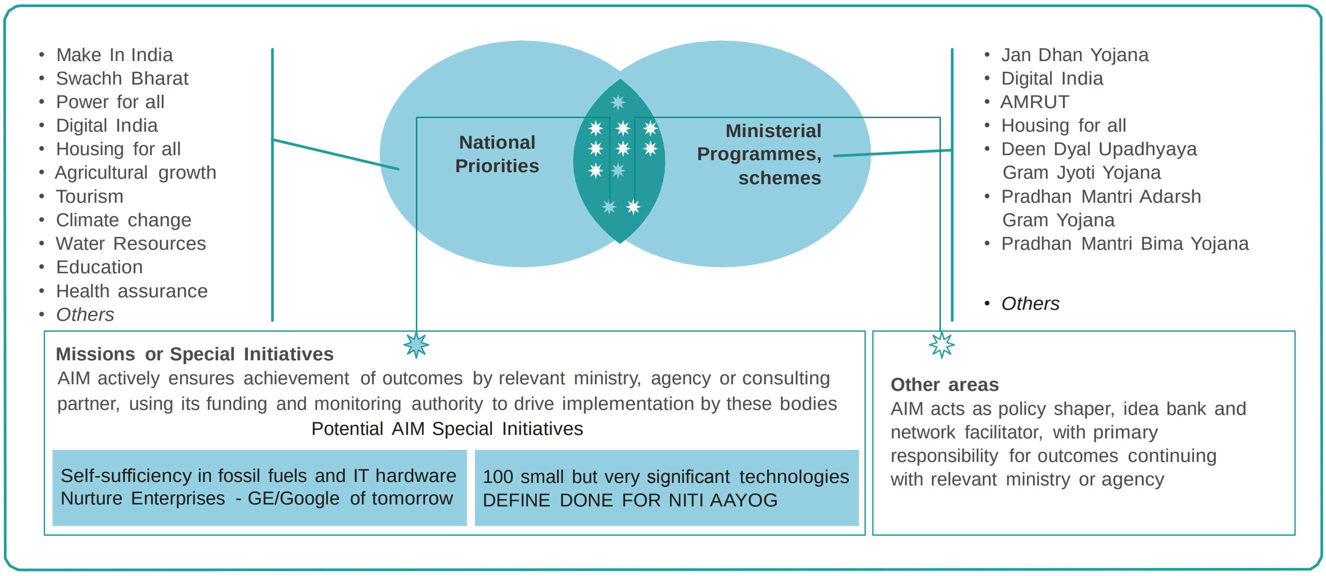
In Implementation aspect of AIM work area, the organization shall identify critical path breaking innovations where it would actively pursue implementation in coordination with the NITI Aayog and the Prime Minister' s office. Some examples of these special initiatives are:

a. Identify and encourage indigenization of technologies for self-sufficiency in fossil fuels and IT hardware manufacturing

b. Nurture an idea to be an enterprise tomorrow equivalent to an Indian GE or a Google through innovation, research and creative destruction

c. Identify 100 small and significant technologies for rural upliftment in Indian model

Figure 34:



11. Measures:

Organizational Metrics

a. Innovative ideas seeded, funded, and mentored until private interest – their impact in terms of output, jobs and value to citizens

b. Missions undertaken – achievement of outcomes/outputs

c. Number of global experts in network

d. Global events convened

National Innovation Metrics

a. Global recognition e.g., Ranking on global Innovation Index

b. Top research citations for Indian work

c. Number of Patents per capitas

d. Number of successful Indian GEs and Googles promoted

e. New enterprises funded

f. Number of ideas translated from laboratories to markets

g. Private investment in R&D

h. Adoption rate of technology (for select focus areas)

I. Productivity outcomes (for select focus areas)

j. Impact on Job growth

k. Public awareness and acceptance within India

l. Number of successful grand challenges with large impacts

Endnotes

[i] Tarun Khanna, "The Business of Inclusion," *Forbes on the Web, May* 29, 2010, http://forbesindia.com/article/ideas-to-change-the-world/tarun-khanna-the-business-of-inclusion/13682/1, accessed August 2015.

[ii] Paul M. Romer, "The Origins of Endogenous Growth," Winter 1994. Journal of Economic Perspectives—Volume 8, Number 1—Winter 1994, Pages 3-22, http://www.eco.unibs.it/~palermo/PDF/romer.pdf, accessed August 2015.

[iii] Tim Kaene, "The Importance of Startups in Job Creation and Job Destruction," *Kauffman Foundation Research Series: Firm Formation and Economic Growth*, July 2010, http://www.kauffman.org/what-we- do/research/firm-formation-and-growth-series/the-importance-of-startups-in-job-creation-and-job- destruction, accessed August 2015.

[iv] Tim Kaene, "The Importance of Startups in Job Creation and Job Destruction," *Kauffman Foundation Research Series: Firm Formation and Economic Growth*, July 2010, http://www.kauffman.org/what-we- do/research/firm-formation-and-growth-series/the-importance-of-startups-in-job-creation-and-job- destruction, accessed August 2015.

[v] Committee on Angel Investment & Early Stage Venture Capital, Planning Commission, *Creating a Vibrant Entrepreneurial Ecosystem in India*, June 2012, http://www.socialimpactatbain.com/Images/Indian_Planning_Commission_Report_2012.pdf, accessed August 2015.

[vi] Ernst & Young, "Home—Services—Strategic Growth Markets—The EY G20 Entrepreneurship Barometer 2013," Ernst & Young Web site. http://www.ey.com/GL/en/Services/Strategic-Growth-Markets/The-EY-G20-Entrepreneurship-Barometer-2013, accessed August 2015.

[vii] FICCI and Nathan Associates, "Nurturing Entrepreneurship in India," (August 2014), http://www.ficci.com/publication-page.asp?spid=20432, accessed August 2015.

[viii] European Commission—Enterprise and Industry, "SBA Fact Sheet Germany," European Commission, http://ec.europa.eu/enterprise/policies/sme/facts-figures-analysis/performance-review/files/countries- sheets/2008/germany_en.pdf, accessed August 2015.

[ix] V. Raja, "We Need a Sharp Policy Intervention," *Hindu Business Line*, October 8, 2014, http://m.thehindubusinessline.com/opinion/we-need-a-sharp-policy-intervention/article6478803.ece/?secid=11682, accessed August 2015.

[x] Gupta et al., "From Poverty to Empowerment: India's Growth Imperative for Jobs, Growth, and Effective Basic Services," McKinsey & Company, http://www.mckinsey.com/insights/asia-pacific/indias_path_from_poverty_to_empowerment, accessed August 2015.

[xi] Gupta et al., "From Poverty to Empowerment: India's Growth Imperative for Jobs, Growth, and Effective Basic Services," McKinsey & Company, http://www.mckinsey.com/insights/asia-pacific/indias_path_from_poverty_to_empowerment, accessed August 2015.

[xii] Richard Florida, "The World's Leading Startup Cities," The Atlantic City Lab on the Web, July 27, 2015, http://www.citylab.com/tech/2015/07/the-worlds-leading-startup-cities/399623/, accessed August 2015.

[xiii] Marten Carstens, "India's Entrepreneurial Landscape according to Ernst & Young," Ventureburn, January 22, 2014, http://ventureburn.com/2014/01/indias-entrepreneurial-landscape-according-to-ernst-young/, accessed August 2015.

[xiv] José Ernesto Amóros and Niels Bosma, "Global Entrepreneurship Monitor 2013 Global Report," 2014, http://www.gemconsortium.org/docs/3106/gem-2013-global-report, accessed August 2015.

[xv] Invest India, "Automobiles," http://www.makeinindia.com/sector/automobiles/, accessed August 2015.

Round Table

Commercial Academics: An Indian Story

Venkat Chandrasekhar and Vijay Chandru

Venkat Chandrasekhar is a Senior Consultant with PicoPeta Simputers and is a Partner at C&C Associates.He teaches popular courses on entrepreneurship at IISc and IIM-B. candc@vsnl.com

Vijay Chandru, Professor, Indian Institute of Science, National Institute of Advanced Studies, Jawaharlal Nehru Centre for Advanced Scientific Research, is a Co-Founder of Strand Genomics, PicoPeta Simputers, String Labs and MetaString. chandru@csa.iisc.ernet.in

Venture funding today is geared to entrepreneurial initiatives that are innovative, address new markets, highly scalable and thus growth oriented. On a time dimension, the sooner this is achieved the better. Given this scenario, successful technology entrepreneurs must possess good domain knowledge and deep-rooted technical expertise. The recent history of high-tech successes have shown us that in the above conditions, academic research based startups and entrepreneurs from academic environments have been highly successful with astounding contributions towards wealth creation.

These entrepreneurship activities are knowledge based and so is the current knowledge economy. In the era of technology start-ups, for a company to grow and achieve industry leadership, the fundamental requirement is technical superiority and development leadership. This calls for a great deal of depth in the domain knowledge. As Charles Geshke, Founder of Abode Systems, is reported to have said, " Technology travels with people. You cannot throw it over the wall and because it is such a good idea, expect another engineering group to simply pick it up and run with it". Therefore the universities have to build necessary bridges for entrepreneurs to travel with their technological innovations.

Kanter (1983) in the now famous article on innovation"when a thousand flowers bloom" notes that technological " innovation is facilitated by organizational complexity, diversity breadth of expertise, including experts who have a great deal of contact with other experts in other fields, openness to the environment, and integration across fields via intersecting territories, multiple communication links and interdisciplinary units". Very few organisational entities in India except universities seem to carry these characteristics.

In the current 'knowledge economy', by nature, academia will become the largest asset holders and it will be in their interest to exploit these assets for their own growth and entrepreneurship promotion gives the opportunity. However, universities need to change and adapt their objectives, structures and teaching programmes and note that:

- Knowledge is now the key determinant for competitive advantage in businesses and economies
- Universities are principal builders of knowledge

- Universities have a responsibility to apply that knowledge to the economy, businesses and the community

To this effect there is a need to conceptualise university activities in ways that will enhance its ability to interact while preserving the key and distinctive features responsible for it's intellectual vitality, independence, ensured quality and foresight.

Historically, the university involvement has been low. Our universities have produced great discoveries and inventions but very few new businesses. To a graduate from the university till a few years ago, entrepreneurship was not a relevant career option. However, the university was important in passively contributing to manpower, adding And enhancing skills, knowledge and capabilities. It has been a provider of facilities and a creative cauldron of intellectual property.

But the changing role requires that apart from teaching and research, the universities should also engage extensively in the application of its knowledge repository.

- Universities have a huge asset base of knowledge worth millions
- It is currently unexploited/under-exploited
- They have a responsibility to get value from these assets to support:

  existing university teaching and research
  economic performance and growth
  employment
  wealth creation

The entrepreneurship "nurture and creation" by universities and academic institutions has a benefit cycle. The entrepreneur's original research creates 'intellectual capital' -- case studies, textbooks, journal articles, award-winning dissertations -- and fuels constantly evolving innovations in course offerings. This in turn creates market driven, real world savvy human capital that gives entrepreneurs and entrepreneurial companies the knowledge workers they need for a sustainable competitive advantage.

It should also be noted that in this evolving knowledge economy organizations are realizing the crying need for innovation and entrepreneurship within organizations (Intrapreneurship). The traditional managers and executives roles are being redefined as we speak. Tomorrow's managerial elite will be those who are constantly in an innovative mode by zeroing in on customer needs and who show entrepreneurial orientation to deliver on those needs. This in itself will force some fundamental changes in university curriculums. Just as we focus on analytical ability today we may have to focus on innovation and entrepreneurial ability soon.

Faculty entrepreneurship is not new. Though there have been only a few cases in India there are many prominent examples abroad. Jim Clark of Stanford (Silicon Graphics, Netscape and Healtheon) , Amar Bose at MIT (Bose Systems), William A. Haseltine of Harvard (Human Genome Sciences), Charles Cantor of Berkeley (Sequenom). Faculty in universities, particularly in technology departments, are by nature innovators. They like to come up with new ideas and new ways of doing things. They also have a good network of people in their knowledge domain.They often carry a sense of duty to see their discoveries widely distributed. Often research labs and universities are not in a position to help them commercialise their innovations. In today's marketplace, unlike grant funding in research which is dwindling, venture funding for commercialization is available along with other managerial and market resources that a faculty entrepreneur might need.

If we accept the premise that India has great intellectual capital in several temporally relevant aspects of science and technology, then we must wonder why the premier research institutes and universities have been unable to drive an IP (intellectual property) based entrepreneurial sector.

Undoubtedly, there are many contributing factors to the emergence of the Silicon Valley around Stanford and Berkeley, Route 128 around MIT and Harvard, Cambridge around the legendary university, or most recently Tel Aviv around Tel Aviv University (*Newsweek* (November 2, 1998) deemed Tel Aviv one of the top ten capitals of the high technology industry worldwide. The nine other capitals include seven cities in the United States, *Bangalore* in India, and Cambridge in Great Britain. Tel Aviv is flourishing as a technological center mainly as a result of immigration from the former Soviet Union, a high percentage of which were trained in the computer science field. The magazine also states that the Israel Defense Forces provides talent to the industry with youngsters who obtain knowledge and experience during their military service. Some say that of all the cities in the world, Tel Aviv is the most serious competitor to the Silicon Valley.). The availability of capital, good incubation centers, local test markets for products and services are as important or more than the mere presence of universities which assures the availability of highly skilled manpower and a rich source of IP.

However, the single most important factor that has held India back on this score has been the extreme conservatism in university policies, sometimes referred to as the Saraswati-Lakshmi dichotomy. The trendsetter in progressive policies towards faculty entrepreneurship has certainly been the US. The seeds of commercial academics in the US were sown by the legislation known as the Bayh-Dole Act of 1980.

A difficult issue that has to be addressed is the issue of ownership of the venture and how much should the universities partake in the direct wealth of the company. Clearly one approach is that of Cambridge University, UK that takes the position that the university takes no equity at all (despite the fact that the entire IP for the venture may have been created in the university). The reasoning (as stated by Professor David King ScD FRS, Chief Scientific Adviser, Office of Science and Technology (UK), when he was in Bangalore a couple of months back) is that this laissez-faire approach by the university has a great benefit to society at large because of wealth and job creation.

If the university wishes to partake up to the point of noticeable liability (both reputational and financial), just under 10% equity is the right amount from the perspective of Company Law in India. Ideally the equity share should be reflective of the value of the IP being taken out of the university by the entrepreneur. This is sometimes a little hard to pin down and has to evolve from experience.

Academic Entrepreneurship in US and The Bayh-Dole Act, 1980

The primary intent of the Bayh-Dole Act of 1980 was to foster the growth of technology-based small businesses by allowing them to own the patents that arose out of federally sponsored research. Universities and other nonprofit recipients of federal funding were included in the definition of 'small entities' benefitting from the Bayh-Dole Act, largely as an afterthought. This law permitted universities two new mechanisms:

- It allowed universities to license patented technologies to industry, retain the royalties from such licensing with an explicit specification that the royalties would be shared as personal income to the inventors. By law, the university's share of the royalties must be ploughed back into its research and educational activities.

- A key aspect of the licensing of inventions under this new law was that the universities were now permitted to grant exclusivity. The justification of why a single company is given the advantage of intellectual property developed under public funding was that exclusivity is required as an inducement and reward for a company willing to step forward and take the risk of early-stage technology development.

The number of US patents granted to US universities in a year rose from 300 in 1980 to almost 2000 in 1995. More than 1,900 companies were formed through university licenses in the 1980-1996 period after the Bayh-Dole Act. The direct financial impact on the universities themselves has been relatively small contradicting the belief of many that royalties could compensate for declining state support of research. We always remember the few blockbusters that make some universities tens of millions but these are a precious few. In contrast, the impact of university technology transfer has had substantial impact on local and national economies. As a result the "town/gown" seems to be on the wane all over the US.

A culture of filing patents before publishing has gradually evolved in US universities thereby giving corporate sponsorship of research some confidence in partnering with universities. Contrary to the fear that patenting and commercialization would somehow shut out student access, the reality is that the students are motivated and their awareness of the potential commercial value of their research findings and inventions is on the increase.

Policy fiats or even attempts to categorize IP and formalize their handling are very likely to be doomed to have overly broad effects with harmful, unintended consequences. The answer seems to be to handle situations on a case-by-case basis, but with a continual dedication within the university to "do the right thing".

Extracted from Nelsen (1998).

The good news is that there are winds of change blowing across the Indian academic landscape. Incubation centres have come up on several campuses, alumni who have been successful entrepreneurs in the US and elsewhere are providing seed capital and networking for ventures emerging from their alma maters. In the following sections the reader will be subjected to a brief description of one such initiative at the Indian Institute of Science in Bangalore where four professors from the computer science faculty have been encouraged to spin off ventures to commercialise technologies developed in their laboratory at the institute. The institute has also worked out mechanisms to hold equity in the emerging ventures – a move that may help the university build on its endowment from holdings in successful ventures.

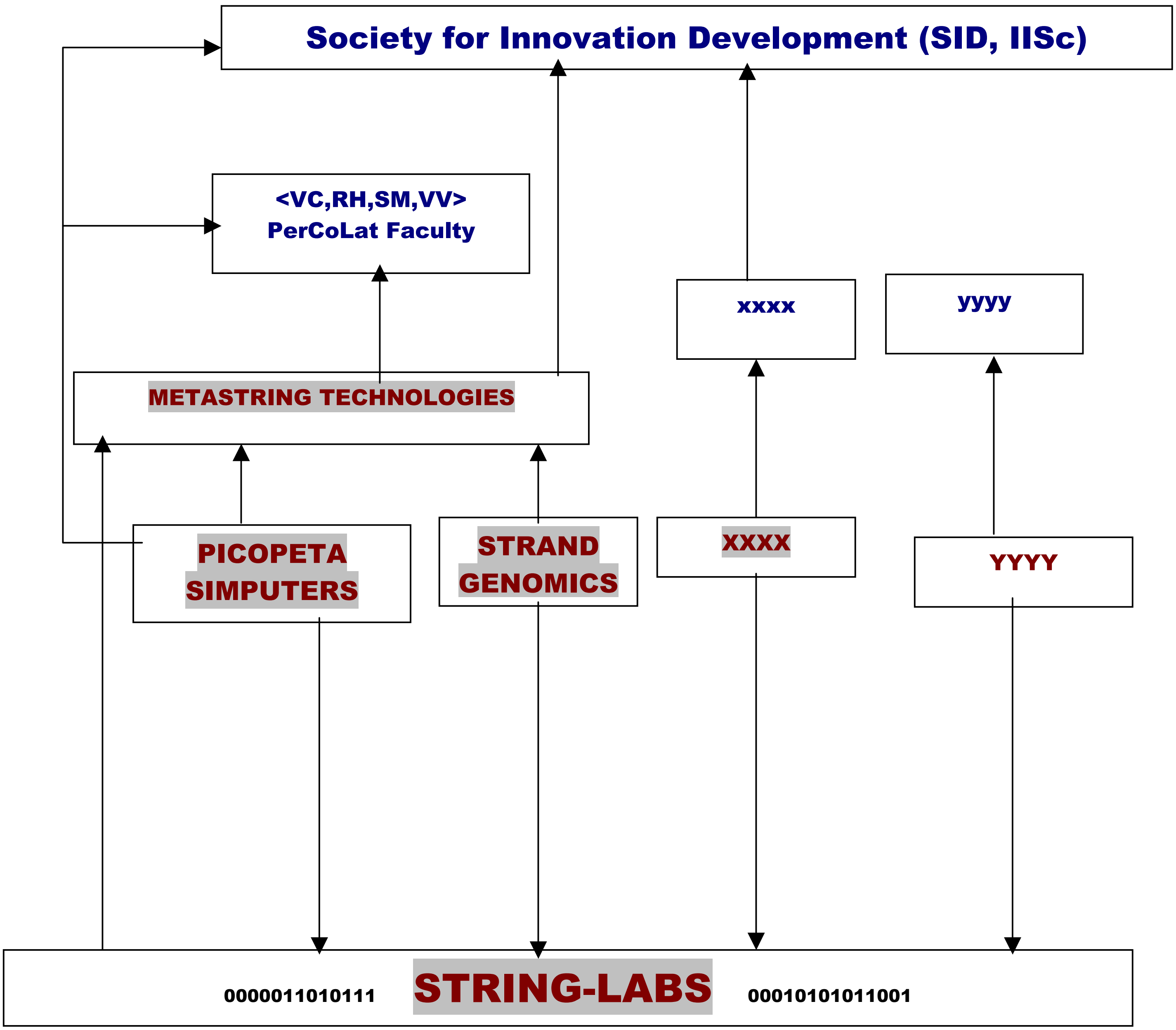
Faculty Entrepreneurs at Indian Institute of Science

In the autumn of 1999, four professors at the Indian Institute of Science in Bangalore responded to a call from the President of the Court, Mr. Ratan Tata, who in his inaugural address to the Court of IISc earlier that year had raised the very question that we posed above – i.e., why has an institution like IISc not been able to drive scientific and technology entrepreneurship in and around Bangalore. The professors: V Chandru, R Hariharan, S Manohar and V Vinay, all computer scientists, had been co-convenors of the Perceptual Computing Laboratory (PerCoLat) in the Department of Computer Science and Automation, IISc and had been collaborators in research and consultancy for about a decade. They proposed to the Council of IISc for permission to commercialise technologies developed in the laboratory through entrepreneurial ventures and requested a formulation of policies on this issue.

The Council and the Administration of IISc took a very positive view of this request and initiated a systematic enquiry into the issues that led to the creation of "The Office Of Technology Licensing and Entrepreneurship" (TOTLE) which would oversee the policy of faculty entrepreneurship and advise the Society for Innovation Development (SID) on its participation (as the representative of IISc) in the shareholding agreements with the concerned faculty in these ventures. These structures were in place in October of 2000 and PerCoLat spun off a holding company – MetaString Pvt Ltd – in partnership with SID soon after. By November 1, 2000 a complex structure (see the figure below) was in place. The media, both local and international (Business World, Business Today, Time Asia, Business Week, Technology Review), sniffed history in the making and heralded the developments.

PerCoLat had evolved from the "Virtual Prototyping Lab" which specialized in work on graphics, Geometric Modeling for CAD/CAM, and Virtual Reality experiments. PerCoLat continued excellent work in graphics and visualization, including some path breaking work on visualization of molecular dynamics and the creation of a voxel based 3D sculpting system called SIRPI-Lab. The Satyam group gave the seed funding for PerCoLat (1997-1999) to pursue a futuristic theme in natural language processing. The "Genomics Factory" software for Genomics Collaborative Inc., USA was created at PerCoLat (1999-2000). The ambitious Simputer project was initiated at PerCoLat in 1998 in collaboration with Ncore Technologies. This was clearly a ripe breeding ground for entrepreneurs in the making.

The structure for the commercialisation of the Perceptual Computing Laboratory (PerCoLat) of the Indian Institute of Science as proposed to The Office of Technology Licensing and Entrepreneurship (TOTLE), IISc.



Features:

- Creation of a Holding Company – **MetaString Technologies** (partners: Faculty of PerCoLat - V Chandru, R Hariharan, S Manohar and V Vinay - and SID representing IISc).

- Creation of **String Labs** - an incubator / software technology center devoted to hatching ventures of PerCoLat initially. String Labs began as a wholly owned subsidiary of Metastring. Future incubation of new ventures (XXXX,YYYY) at StringLabs - these ventures may be independent of Metastring and of the promoters of MetaString or for that matter of any entity related to IISc. In essence, StringLabs is a venture in its own right and could evolve into an incubation enterprise.

- Creation of **Strand Genomics**, a bioinformatics company – with a focus on developing advanced data-processing, annotation and visualisation software tools and services for the global bioinformatics industry.

- Creation of **PicoPeta Simputers** - a subsidiary of Metastring, PicoPeta is a solutions company that will build on the Simputer platform for handheld computing and communications.

- Start dates: November 1, 2000 for MetaString, StringLabs and Strand Genomics; May 31, 2001 for Picopeta.

The status of these ventures at the time of writing is that Strand Genomics is running with 45 scientific staff and a management infrastructure in place. First round venture funding has been obtained and a pipeline of high-end contract services and products is in place. PicoPeta Simputers has close to 30 employees and the first batch of Simputers have rolled out of the manufacturing plant this week. Some seed funding has been in place and a revenue pipeline is beginning to aid the cash flow and expansion. String Labs has had the experience of incubating these two companies and is getting ready to offer incubation services as a business proposition. These commercial academics are on their way.

References and Notes:

Rosabeth Moss Kanter "When a thousand flowers bloom: structural, collective and social conditions for innovation in organization," in *Entrepreneurship: The Social Science View*, edited by Richard Swedberg Oxford ; New York : Oxford University Press, 2000.

Lita Nelsen "Increase of intellectual property licensing at universities stems from changes in funding and legislation." *Science*, volume 279, number 5356 issue of March 1998, pp. 1460-1461.

COMMITTED TO IMPROVING THE STATE OF THE WORLD

World Economic Forum White Paper

Misaligned Stakeholders and Health Systems Underperformance

Industry Agenda Council: The Future of the Health Sector

June, 2016

Abstract

This document summarizes the findings of the World Economic Forum's Industry Agenda Council on the Future of the Health Sector's two-year inquiry into underperformance of health systems. The Council found that misalignments (i.e., situations involving conflicting incentives, behaviors, structures, or policies) among key stakeholders are likely to lead to significant waste, whether measured in terms of health for money or money for health. This paper defines three types of misalignments (those due to divergent objectives, power asymmetries, and cooperation failures), offers concrete examples of each type, and includes several cross-cutting examples from three particularly burdensome disease areas: cancer, diabetes, and mental health. The Council also examined situations in which stakeholders acted to correct misalignments and reduce waste. This paper details several case studies of achieving greater alignment and enumerates a set of lessons for stakeholders. It concludes by making recommendations for future research in this domain.

1. INTRODUCTION

Global health is a story of the good, the bad, and the ugly.[i]

The good. There have been extraordinary gains in longevity and health. Between 1990 and 2015, the global infant mortality rate fell by 42%, and the child mortality rate by 53%. Over the past six decades, global life expectancy has increased by more than two decades – with people living well into their 80s in many wealthy industrialized countries.

The bad. Millions of people continue to suffer and die from health conditions that can be easily and inexpensively prevented or treated with existing knowledge and tools. In 2014, nearly 6 million children died before celebrating their fifth birthday (the majority from prematurity and vaccine-preventable infectious diseases) and an estimated 642 million people will be living with diabetes by the year 2040, representing an increase of more than 50% from today's 415 million.

The ugly. Gross disparities are made apparent by juxtaposing our most impressive global health achievements with our most dismal failures. At the international level, there are drastic differences in life expectancy between high- and low-income countries, ranging from Japan at 83 years to Sierra Leone at 45. Massive disparities also occur at the national level – within the same health system. In the United States, the maternal mortality rate for black women is almost three times higher than for white women, although the prevalence of the most common birth complications is roughly similar. A recent review of health indicators (including child mortality, life expectancy, and obesity) among indigenous populations in over 20 countries found almost universally poor outcomes relative to corresponding non-indigenous populations. In Australia, Canada, Cameroon, and Kenya, the life expectancies of indigenous populations were at least 10 years lower than in the general population, and over 20 years in Cameroon.[ii]

The world's health deficits are especially disconcerting given indications of wasted resources within health systems in both rich and poor countries. In 2013, the United States spent nearly 50% more on healthcare as a percent of GDP than the next highest OECD spender, France, without superior health outcomes, such as life expectancy.[iii] That same year, Paraguay achieved similar health outcomes in terms of life expectancy and child mortality as El Salvador – despite spending 34% more per person on healthcare and having similar per capita income.[iv]

The world aspires to erase its glaring health deficits, as underscored by the 2015 Millennium Development Goals and the 2030 Sustainable Development Goals. The main focus has been, and continues to be, on increasing the scale and efficiency of resources devoted to (i) medical interventions (e.g., the provision of drugs, vaccines, and primary healthcare); (ii) non-medical health interventions (e.g., the expansion and strengthening of health delivery systems, the widening and deepening of human resources for health, and the more widespread provision of safe water and sanitation); and (iii) non-health interventions that can have a powerful impact on health outcomes (e.g., governance, education, housing, and the social conditions that influence daily life). Globally, many laudable initiatives are underway to extend potential longevity and enhance overall well-being, with technological, social, and institutional innovation featuring prominently in the effort.

Methodology

Against this backdrop, in March 2015 the World Economic Forum's Industry Agenda Council on the Future of the Health Sector decided to conduct in-depth research to gain further insights into health deficits and disparities throughout the world, and to identify ways to better use available resources to improve health outcomes and reduce waste. Research methods included desk research, extensive phone and in-person interviews, and face-to-face and virtual meetings.

Our working hypothesis is that global health underperformance is significantly driven by misalignments among health sector stakeholders. This focus on misalignments is underdeveloped in the literature, but it resonated with our Council members and nearly every professional interviewed. In all, 60 interviews were conducted across 16 countries, covering healthcare providers, payers, pharmaceutical companies, patient groups, employers, and governmental and inter-governmental organizations.

Principal findings

The misalignments perspective takes a step back and looks across the field of players to see how their delivery of healthcare is affected by the way they interact with each other, the barriers in their paths, and the size of the field. Reaching beyond the realities of each stakeholder community and looking at these disconnects with an open mind and creativity has shown us that: (i) these are issues of paramount importance to almost every stakeholder community, and (ii) there are compelling examples of alignment from which we can draw lessons to apply to a wide range of issues within the health sector.

The Council finds that, fortunately, where misalignments can be identified, they can potentially be corrected. Better-aligned systems may yield either of two positive results: (i) maximizing health outcomes and healthcare delivery for a given amount of spending, or (ii) minimizing spending while maintaining the same level of health. More specifically, we estimate that if used more effectively, current patterns of national health spending could potentially raise global life expectancy by over four years. Alternatively, current patterns of life expectancy could potentially be achieved with as little as one-third the current level of health spending.[1v] Whether viewed in terms of money for health, or health for money, these analyses are suggestive of considerable waste and inefficiency.[2]

We begin with the objectives of our research, before detailing our use of the term “misalignment” and outlining our taxonomy of the three most salient categories (divergent objectives, power asymmetries, and cooperation failures). We provide examples of each category, touching upon a wide range of geographical, developmental, and ecological contexts. We also examine several in-depth examples of misalignment in the cancer, diabetes, and mental health fields. Then we turn to our findings from case studies on alignment in the health sector and suggest steps that stakeholders could take to achieve better alignment in certain contexts. We conclude with ideas for future research and a brief summary of the Forum’s Value in Healthcare project, which seeks to improve outcomes and payment models in healthcare delivery while ensuring access to care and financial sustainability.

2. OBJECTIVES

The Council set out to: (i) better understand misalignment among stakeholders (such as patients, regulators, pharmaceutical and device manufacturers, providers, insurers, academia, policymakers, and investors); (ii) articulate a vision of a more fully aligned health systems and the

[1] All references to health spending in this white paper are based on the following definition of “health expenditure” from the World Bank: “Total health expenditure is the sum of public and private health expenditures as a ratio of total population. It covers the provision of health services (preventive and curative), family planning activities, nutrition activities, and emergency aid designated for health but does not include provision of water and sanitation.”

[2] For details, see Potential Magnitude of the Problem section below and the Methodological Appendix (available at www.weforum.org).

ways in which they will improve health outcomes; (iii) design a plan for how to achieve it; and (iv) estimate the potential health and economic gains associated with better alignment.

Following consultation with the leaders of other relevant councils within the Forum, the Council selected three specific disease areas within which to focus its attention so that practical recommendations could be generated within the resource and time constraints of the project. These disease areas – cancer, diabetes, and mental health – were chosen because of their significant and growing contribution to death and disability globally, and because Council members had a high degree of expertise and professional experience in these fields.

The focus of this exercise was global, and it sought to address the following questions for each of the three diseases:

- How do misalignments among stakeholders lead to increased incidence of disease or poor quality of care?
- What real-world examples demonstrate misalignment and its effects?
- What are real-world examples in which stakeholders have reasonably aligned incentives or structures – and how does this manifest itself in terms of health outcomes?
- What would it take for incentives, structures, and priorities for each disease area to be more systematically aligned at local, national, and global levels?

While our project is global in terms of relevance and scale, it is rooted in analyses at the national level. The aim is to identify instances in which systemic flaws lead to sub-optimal outcomes. It is worth noting that the Council did not consider cases of fraud or corruption for this study, though these represent a subject ripe for future research. While fraud and corruption constitute a significant class of misalignment in many countries, adequately addressing the complexity of the problem and its solutions would require a special focus beyond the scope of this white paper. For a thorough exploration of the conditions that lead to corruption and its effects, please see the work of the Forum's Partnering Against Corruption Initiative.[3]

3. PROBLEM OF MISALIGNMENTS

The Council conceptualizes misalignments as situations in which the incentives, structures, or policies that shape the behavior and interactions of different health sector stakeholders result in wasted resources or suboptimal health outcomes. Misalignments can often be corrected and brought to a state of alignment, but that requires a clear understanding of the nature of the misalignment and the reasons for it. In this vein, our ultimate goal is to understand the steps that may be taken to achieve better aligned health systems.

A key premise of our analysis is that the vast majority of health sector stakeholders desire better health, even if their actions may be driven in part, or even dominated by, alternative motivations. For example, governments may want their populations to be healthier so that they can have more productive and cohesive societies. At the same time, individual actors within governments may make decisions based on their self-interest, as in seeking re-election. In this case, misalignments may occur when governmental structures unnecessarily place political advancement at odds with the institution of sensible health policies.

[3] https://www.weforum.org/communities/partnering-against-corruption-initiative/

Similarly, while stakeholders who sell therapeutic services (e.g., medical care) or curative products (e.g., pharmaceuticals) operate to improve health, they are businesses that must weigh profit maximization and shareholder interests against optimizing health outcomes. This internal conflict in motivations is not a misalignment in and of itself. Businesses exist to sell goods and services for financial gain. In the case of for-profit businesses, the situation becomes a misalignment when incentives, or other forms of conflict, artificially divorce profit maximization from health optimization.

3.1. Taxonomy of misalignments

What are the most common types of misalignments? The Council's research shows that three categories of misalignments stand out: those due to (i) divergent objectives, (ii) power asymmetries, and (iii) cooperation failures. But, as in any complex system, many examples do not fit neatly into one category and may cross over to other types. Further, we do not consider these categories to be an exhaustive list but rather a synthesis of the most compelling ones that emerge from our research.

Divergent objectives. This category covers divides between the long- and short-term interests of stakeholders, myopic targets or payment mechanisms (such as fee-for-service), or inadequate data and tools for decision-making.

Example 1: Inadequate payment mechanisms result in inexpedient care.

Lack of reimbursement often deters clinicians from billing for over-the-phone or electronic consultation. As a result, most patients cannot contact their physician for a rapid, remote consultation even though this may be the most efficient way to receive timely medical advice. Patient health may deteriorate while they wait to see their provider in person, which can result in costlier treatments and worse health outcomes. Under these circumstances, restrictive payment mechanisms provide a disincentive for clinicians to act in the best interests of their patients and ultimately lead to excessive healthcare costs borne by public and private payers. This misalignment affects care across many health systems, and is particularly problematic in the U.S.

Example 2: Excessive and inappropriate use of antibiotics fuels drug resistance.

Prioritizing immediate comfort over long-term consequences has driven a number of undesirable phenomena when it comes to the overuse of antibiotics. These include the large-scale use of antibiotics in commercial farming to promote growth and control infections, the improper prescription of antibiotics for infections that will not respond to them,[4] the rampant misuse of over-the-counter antibiotics in many countries, and the lack of adherence to antibiotic regimens, even when properly prescribed. As a result, antimicrobial resistance is increasing globally, creating a truly transnational threat. Estimates show that drug resistant "superbugs" are likely to kill an additional 10 million people per year by 2050 – accounting for more deaths globally than all cancers combined.[vi] The perceived immediate benefits of antibiotics to individuals are being given preference over the social appropriateness of their application, causing everyone to face the stark long-term consequences of overuse.

Power asymmetries. This category covers imbalances of power (economic, legal, informational, or political) that arise among stakeholders and create waste – possibly because one stakeholder is

[4] For example, recent estimates indicate that one-third of all antibiotics prescribed in the United States are unnecessary: www.washingtonpost.com/news/to-your-health/wp/2016/05/03/1-in-3-antibiotics-prescribed-in-u-s-are-unnecessary-major-study-finds

able to impose his will in a way that severely restricts the ability of disadvantaged stakeholders to impact decision-making.

Example 1: The outsized power of the tobacco industry impedes some governments from implementing tobacco-control interventions.[5]

The main goal of most private companies is to earn a profit, and the tobacco industry claims no immunity to this ambition. While the tobacco industry's methods for generating revenue clearly have negative consequences for public health, this fact alone does not constitute a misalignment. Misalignment arises when the tobacco industry uses its economic power to threaten or instigate legal battles that governments do not have the finances to fight, preventing policymakers from successfully implementing tobacco-control laws, launching public health education campaigns, or utilizing other policy tools such as taxation. Without being able to employ legally and publicly sanctioned tools that are typically used to protect public health, governments are left to bear the full costs of tobacco consumption in the form of increasing medical costs and slowed economic growth due to premature death and disability. The Council acknowledges that tobacco consumption is driven by a complex array of forces, including addiction, cultural attitudes toward tobacco, and the diversity of economic and legal factors at play within unique national contexts. Nevertheless, power imbalances between the tobacco industry and some governments significantly contribute to the problem by preventing officials from spending resources in a manner that would be most effective for promoting health, and thereby creating waste.

Example 2: U.S. regulations on price negotiations for Medicare Part D cause higher drug prices for federal reimbursements.

Enrollees in Medicare (a social health insurance program administered by the U.S. government) can purchase prescription drug plans (PDPs) from private insurance companies and pharmacy benefit managers in "Medicare Part D." Federal law currently prohibits the government from negotiating prices with drug manufacturers on behalf of Medicare, with the intent of encouraging the private companies that administer PDPs to negotiate directly for lower prices.[vii] However, evidence suggests that these private companies fail to secure prices as low as those paid by federal programs that can negotiate (such as the Veteran's Health Administration and Medicaid). Recent estimates show that the government could save $15.2 to 16.0 billion annually if it obtained the same drug prices for Medicare as obtained by other federal health programs.[viii]

The legal regulations that restrict the government from negotiating on behalf of PDPs create an artificial power imbalance between the government and pharmaceutical companies. As a result, the government pays higher prices for drugs under Medicare Part D than through other federal health plans, thereby wasting taxpayer resources for the health outcomes achieved. It is worth noting that changes to this legislation have been included in the fiscal year 2017 budget from the office of the U.S. President and submitted to the U.S. Congress for revision and voting.

Cooperation failures. This category covers barriers (such as cultural, operational, and regulatory) that keep stakeholders from collaborating in their mutual best interests. These failures generally reflect disconnected budgets, lack of leadership, or barriers to data sharing.

[5] For this example, the Council chose to focus on countries with strong privately-controlled tobacco industries. Countries with state-controlled tobacco industries face a different set of challenges due to internal conflicts of interest. Namely, it can be difficult for the ministry of health to negotiate the implementation of anti-tobacco policies when tobacco production serves as a major source of revenue for the ministry of agriculture and other government bodies.

Example 1: Poor interoperability of electronic medical records systems leads to inefficient care and negative health outcomes.

In many countries, electronic medical records (EMR) systems have poor interoperability. Primary care providers, laboratories, specialists' offices, and hospitals may maintain separate EMRs for a single patient, without passing any information among them. This lack of sharing and connectivity contributes to redundant testing, medical errors, and disjointed care, all of which can result in poorer health outcomes for patients and higher costs across the system. Experts contend that the main obstacles to correcting this misalignment include the complexity of sharing information among hundreds of EMRs and the lack of incentives for manufacturers to collaborate on connecting their products. Vendors are actually incentivized not to share information by default so that they can charge data exchange fees when they do choose to connect to other EMR systems. While the stakeholders would stand to benefit as a whole from greater EMR interoperability, no single party will necessarily net a positive return if required to individually fund a project to increase compatibility. Unless cooperation among stakeholders can be marshalled by implementing a mechanism through which costs are shared in proportion to the benefits enjoyed, this issue is likely to remain unresolved.

Example 2: Poor collaboration among stakeholders leads to suboptimal uptake of technical innovations in diabetes care.

Cooperation failures prevent providers, insurers, and consumers from capitalizing on innovations that can improve patient health and quality of life. With diabetes, new technologies such as wearable monitors, tele-health interfaces, and predictive analytics to tailor treatments to each patient are all being underused. Medtech and software developers find it difficult to get their products implemented across health systems because doctors do not see it as their job to select and "prescribe" these kinds of innovations. Moreover, providers operate in fragmented systems and are most frequently compensated to provide drugs and treatments, but not preventive technologies. Plus difficulties sharing health data at every level of the system prevent effective and meaningful use of the information these devices might be able to provide. As a result of these barriers, technologies that hold tremendous potential for achieving health gains remain largely unused. Each stakeholder seems to be focused on immediate and incremental gains – unwilling to pull together to achieve a much bigger transformation. As with EMR interoperability, a lack of cooperation keeps stakeholders from making a collective investment in an innovation that would also yield collective benefits.

3.2. Cross-Cutting Examples from Cancer, Diabetes, and Mental Health

Besides identifying independent examples of misalignment that clearly illustrate the three categories outlined in the taxonomy, the Council closely examined interconnected instances of misalignment from three disease areas: cancer, diabetes, and mental health. Across all geographies and income levels, these three disease areas are responsible for a significant portion of the global burden of death and disability.

This section examines misalignments through a wider lens than in the previous section. Rather than looking at examples in isolation, we explore how specific misalignments (e.g., fee-for-service models) arise in different areas of the global health sector.

Divergent objectives

Across all three disease areas, the Council observed that *payment models* for physicians and providers typically reward the volume, rather than the value of care provided. The majority of the world's spending on healthcare is directed through care-based payment models such as fee-for-service. This means that financial incentives encourage hospitals and clinicians to engage patients with as many tests, prescriptions, appointments, and procedures as possible. Introducing new innovations that get diagnosis, treatment, and care right the first time, or that reduce the amount of treatment needed, can result in less income for the provider under such systems.

The United Kingdom provides a good illustration of how the focus on volume over value affects diabetic patients. Across medical jurisdictions, there is much variability in outcomes that are indicators of the quality of care, such as limb amputation and hospital readmissions. These discrepancies indicate that some providers are achieving higher quality care than others. However, in a fee-for-service model, the providers with greater frequency of readmission and amputation are rewarded with greater income. This not only incentivizes failure but also presents providers with a disincentive for investing in innovations that could keep patients healthier.

At their worst, volume-based payment models can undermine trust in the healthcare system completely. An extreme example can be seen in the recent trend of attacks on healthcare workers in China, where tensions between patients and providers have erupted into violence.[ix] Patients who are frequently pushed into poverty due to the high costs of medical care have lashed out at providers who are mostly paid on the basis of revenue; many physician contracts even include income-generation targets for the hospital. China provides an example of how fee-for-service payment schemes can incentivize unnecessary and lower quality healthcare at the expense of patient health.

Misaligned objectives can also come in less explicit forms, such as *disproportionate spending on treatment over prevention* and other solutions likely to produce a more lasting impact. This imbalance reveals disconnects between the short- and long-term interests of stakeholders**.** In the case of mental health, the WHO estimates that nearly 70% of funding globally goes to stand-alone mental health institutions, despite a long-standing consensus that most investment should go into rehabilitative care in the community, not "warehousing" in bedded facilities.[x] The preference for funding highly visible facilities reveals a desire on the part of policymakers to demonstrate their commitment to mental health programs without taking on the challenge of long-term, multi-stakeholder, community-based interventions, which may impact both the treatment and prevention of mental health conditions.

Similarly, while 52% of cervical cancer mortality could be prevented through effective HPV screening and vaccination,[xi] uptake has been slow, notably in low- and middle-income countries where 85% of cervical cancer deaths occur. Gavi, the Vaccine Alliance, has worked closely with governments to demonstrate the positive business case for investing now in a program that will only reap its full health and economic benefits decades into the future. In the interim, countries that choose not to provide the vaccine (and invest in prevention) will be forced to spend precious resources on cervical cancer treatment and absorb the negative social and economic impacts of an unnecessarily high cancer burden.

In developed countries, recent trends show the treatment-prevention divide getting worse. Across the OECD from 2009 to 2014, spending on prevention was the only area of healthcare apart from pharmaceuticals that saw cuts.[xii] This suggests that as health systems around the world face financial pressures, decision-makers invest in interventions with relatively short-term returns at the expense of those with long-term benefits, which may even be larger in magnitude.

For decades, intergovernmental bodies and ministries of health have called for reprioritizing prevention over treatment, but misalignments have deterred such investments. Performance targets and voter priorities still focus on what appears urgent and can be easily measured (e.g., waiting times and medical errors) rather than what is more beneficial in the long-term and harder to measure (e.g. curbing chronic disease trends and improving access to pre-natal care).

Power Asymmetries

Across the three disease areas, experts often touched on power imbalances lying at the heart of particular examples of misalignment.

Many pointed out *funding disparities among particular conditions* as evidence that healthcare resource allocation is skewed by power imbalances among particular groups. Funding for mental health services provides a compelling example. Globally, mental health services receive around 2% of healthcare spending, despite accounting for around 13% of the burden of disease, according to the WHO.[xiii] While disparities in health investment may occur for a variety of reasons, health experts interviewed suggested that the primary driver of low funding for mental health is stigma – or rather the lack of power held by patients and professionals in the mental health sector compared to more empowered and vocal groups focused on physical health. Less than half of low-income countries have any form of mental health civil society organization present.[xiv]

It is hard to definitively demonstrate that any particular funding allocation across diseases is "right." This is exacerbated by a lack of tools to rationally assess potential to benefit from healthcare spending and allocate resources to optimize effectiveness. Advocates of particular causes and political expediency exert great influence on decision-makers, which may cause inefficiencies and missed opportunities for greater health gain from existing health investments.

In several markets, interviewees reported perceptions of either *producers or purchasers wielding outsized power*, creating persistent under- or over-consumption of particular products, or unnecessary loss of money or human health. Even though tobacco products are a leading cause of death and disability worldwide, the number of smokers in low- and middle- income countries has been rising dramatically – more than offsetting the fall in smoking rates in high-income nations. Yet despite the WHO's recommending tobacco taxes as "the most effective policy to reduce tobacco use," many countries do not levy them, and only 10% of the world's population lives in a country with a tobacco tax rate that sufficiently compensates for its true and full costs.[xv] Government inaction can be partly explained by tobacco companies' disproportionate power, in terms of financial, marketing, and legal resources, relative to some governments' ability to push through evidence-based anti-smoking strategies. The persistence of this power imbalance contributes to the spread of cancer-causing lifestyles across low- and middle-income countries, potentiating high levels of preventable disease (along with their resulting economic burden on society).

Even when stakeholders are theoretically on the same side, power asymmetries can lead to misaligned investment and objectives. In low-income countries, *donor agencies exercise considerable sway over health system development*, with global funders providing around 25% of overall health spending.[xvi] Many interviewees noted that health aid flows often reflect donor priorities rather than those of the recipient, and that donors "neither set nor fund priorities in a rational way."[xvii] This misalignment between donor and recipient priorities can have a profound impact on the health services available to hundreds of millions of people.

One example is donors' preference for supporting vertical programs that target specific diseases rather than those that strengthen the broader health system.[xviii] This approach can cause redundancy, inefficiency, and neglect of certain health areas. In the case of non-communicable diseases (NCDs), programs are woefully underfunded in relation to their contribution to the global health burden. In 2009, NCDs accounted for 45% of the burden of disease, but only 1% of donor funding for health.[xix] Programs targeting NCDs require a commitment to long-term investment and difficult-to-measure results, neither of which is attractive to donor communities.

Cooperation failures

Across all three disease areas, the Council found misalignments that are not the result of any fundamental economic or political failure, but a more basic lack of cooperation. The constraining influence of organizational boundaries can narrow leaders' focus and lead them to miss opportunities for partnership that would create shared savings or improved health. Thinking in a more collaborative "systems-oriented" way can occasionally be impeded by regulation, but more often than not these misalignments result from stakeholders' failure to see beyond the boundaries of their own sectoral, institutional, or professional silos.

Diabetes experts raised the example of the fragmented and duplicative care that is often delivered to diabetic patients around the world. Several interviewees commented on a *misalignment between modern patterns of disease and the traditional structure of specialties* in which doctors are trained and care is delivered. In Scotland, a study of nearly 2 million patients found that 47% of people with diabetes had three or more additional long-term conditions, and only 14% suffered from diabetes alone.[xx] Multiple morbidity patients (such as those living with diabetes and hypertension) are increasingly common and particularly expensive to treat, yet most countries continue to treat each condition separately. This leads to not only wasteful duplication of tests and possible overmedication but also patient confusion over burdensome recommendations – which likely reflect each doctor's individual specialty rather than a patient's holistic health and well-being.

Health experts noted that *medical records and information systems are frequently not connected*, such as between hospitals and primary care providers. In the United States, the separation of physical and mental health records is required by the 1996 Health Insurance Portability and Accountability Act.[xxi] The segregated record-keeping created by this regulation means that only mental health professionals have access to a patient's mental health records – increasing the risk of fragmented, suboptimal treatment decisions and even medication errors. While patient privacy should be of paramount importance in managing access to patient records, this must be weighed against the benefits of making it easy for clinicians to cooperate and share information. Only mental health is singled out for this exceptional closed status in the United States, and no other country that we could find has such a regulation.

These experts also pointed out the *missed opportunities for shared savings through multisectoral solutions*. The costs of mental illness fall broadly and deeply across societies in the form of significantly higher rates of unemployment, incarceration, hospitalization, and lower productivity than countries with a lower prevalence of mental illness.[xxii] Australia has calculated the societal cost of mental illness as around 12% of GDP annually.[xxiii] Psychosocial interventions involving other elements of the health system (e.g., general practice physicians, paramedics, and geriatricians) and non-health community professionals (e.g., employers, educators, and police officers) can be highly effective at identifying and preventing mental illness and providing a broader base of social support to help people before they reach a point of crisis.[xxiv]

But multisectoral collaborations of this kind are rare. Often the gains that follow from an investment in one intervention may not directly benefit the budget of the organization making the investment. For example, the education sector might need to spend money that is saved by the criminal justice system, or primary care might need to spend money saved by the emergency department or services for the homeless. Thus, budget segregation can prevent rational programs that would increase overall social welfare.

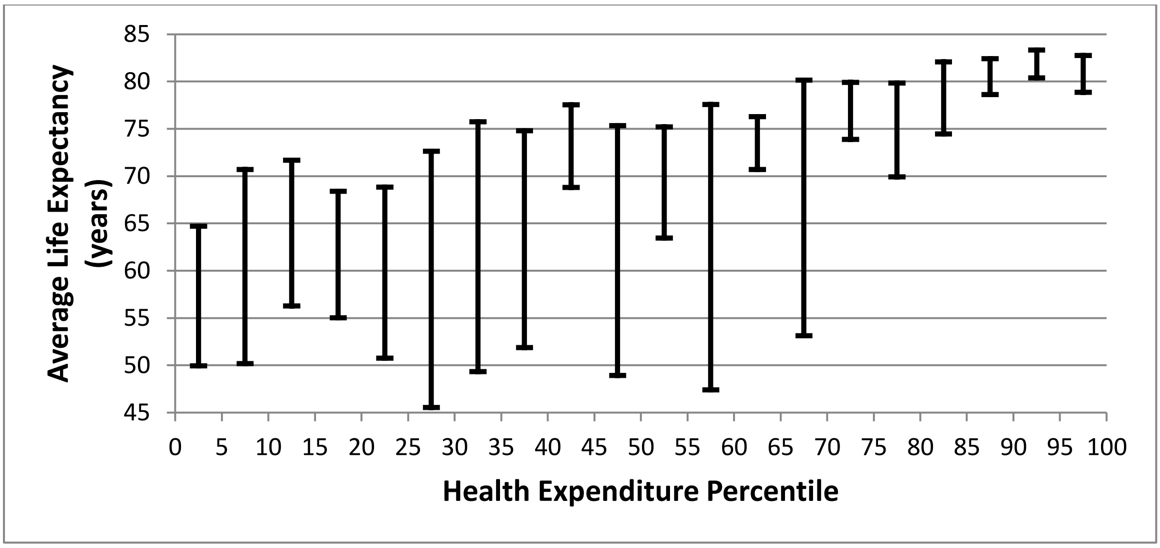
4. POTENTIAL MAGNITUDE OF THE PROBLEM

Waste may be understood either in terms of suboptimal health outcomes for a given allocation of resources, or in terms of excessive spending for a given set of health outcomes. The Council assessed the potential contribution of misalignments to waste by conducting a thought experiment based on two questions:

(1) If the world eliminated all inefficiencies, how much could a more effective use of economic resources improve global health? Our analysis reveals that:

Global life expectancy could be increased by over four years at current health expenditure levels if each country matched the health outcomes of the best country in its spending class. We reached this estimation by dividing all countries into 20 groups based on their health expenditure per capita and examining life expectancy within those 20 groups (see Figure 1). If one assumes that shortfalls in the maximum life expectancy in every group are due to misalignments that could be corrected, global life expectancy could be increased over 4 years at the same cost (representing about 15 years of normal improvement).[6]

Figure 1: ***Big variations in health outcomes among countries with similar health spending per capita.***



Source: Author calculations based on World Bank data for the year 2013. See Methodological Appendix for more details.

(2) If the world eliminated all inefficiencies, how much could be saved in healthcare spending? Our analysis reveals that:

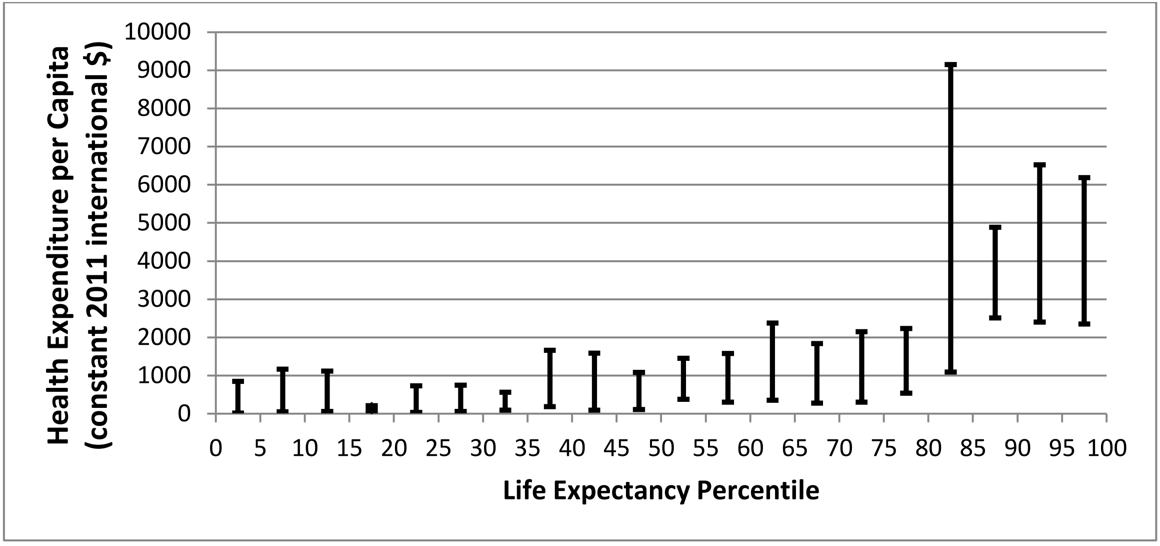
Current global life expectancy could be sustained at one-third of current healthcare expenditure. We reached this estimation by dividing all countries into 20 groups based on their life expectancy and examining health expenditure per capita within those groups (see Figure 2). Assuming health expenditures above the minimum needed to achieve that life expectancy are due to misalignments, correcting those misalignments could yield the same

[6] If the analysis is conducted using the second-highest life expectancy from each segment (to moderate the effect of outliers), the results suggest a more-than-three year increase in global life expectancy.

life expectancy at roughly one-third the health expenditure.[7][8]

[7] If the analysis is conducted using the second-lowest level of health expenditure from each segment, the results suggest that global health expenditure could be reduced to roughly one-half its current level.
[8] Alternatively, if the United States is removed from this analysis, the results suggest that health expenditure for all other countries in the sample could be reduced to roughly one-half its current level (for those countries).

Figure 2: ***Big variations in health spending per capita among countries with similar health outcomes.***



Source: Author calculations based on World Bank data for the year 2013. See Methodological Appendix for more details.

These calculations are, at most, suggestive of the potential magnitude of gains to be had from achieving greater alignment across the spectrum of economic development. The Council acknowledges that in addition to misalignments, cultural and social forces may contribute to health systems underperformance. Nonetheless, figures derived from these calculations suggest that the scale of waste due to misalignments and the gains that could be realized by correcting them are potentially quite larger. Indeed, by maximizing the effectiveness of available health resources within countries, the global community could have a substantial impact on overall well-being either by effecting massive improvements in health or by substantially reducing health expenditures – thereby freeing up funds for investing in other key areas, such as education, economic development, early childhood development, and provision of water and sanitation. (The methodology, data sources, and assumptions and limitations of these estimates are described in detail in the Methodological Appendix available at www.weforum.org.)

5. CORRECTING MISALIGNMENTS

Given that misalignments likely contribute to numerous instances of health sector underperformance, the Council focused considerable effort on identifying examples of health systems and their subcomponents that feature well-aligned health stakeholders. Dozens of examples of resolutions to misalignments at the local, national, and international level were collected. While misalignments lie at the heart of some of the most significant inefficiencies and threats in global health, they are neither static nor irresolvable. And although there is no universal solution for achieving alignment, the Council found a variety of strategies that may be applied in broadly similar contexts.

Are some systems better “aligned” than others?

At a national level, variations in health outcomes per dollar invested suggest that certain countries’ systems may reflect better stakeholder alignment than others. Rankings of health systems have for some time highlighted these disparities. For example, the Bloomberg “Most Efficient Healthcare” Index has recognized Singapore, Hong Kong, and Italy for achieving particularly long and healthy lives for their populations despite comparably low levels of spending.[xxv] The Economist's 2014 Outcomes Index tells a similar story, with Japan, Singapore, and Switzerland occupying top rankings.[xxvi] Though these rankings rely on a small number of very high-level indicators, two key lessons from the high performing health systems emerge: (i) cost management efforts are best led by powerful entities dedicated to this task, and (ii) strong systems of public health and primary care must underpin the provision of health services overall. Additional factors outside of the health system that promote the efficiency seen in well-aligned systems include dietary norms, climate, and labor costs.

A far more detailed attempt to rank the performance of health systems was conducted by the WHO in 2000, with France and Italy as the top two.[xxvii] The exercise was contentious, with some questioning the academic basis of the rankings[xxviii] and others questioning the integrity of the underlying data.[xxix,xxx] Nevertheless, the exercise highlighted some of the common attributes of top-performing countries. In particular, the rankings revealed that high-functioning systems all effectively articulated clear long-term goals focused on tackling health inequalities, promoted shared stewardship of resources, and had effective systems of regulation.

But the Council's closer analysis of even these top health systems reveals a much more complex picture. Model examples of alignment often coexist with significant misalignments, conflicts, and inefficiencies. Similarly, low-ranking countries will often display innovative resolutions to misalignments that confound countries higher up the index.

Health systems are generally too complex to have their degree of alignment captured in a single index, and there is certainly no one example of a perfectly aligned health system. But on some of the most fundamental misalignments, there are models in numerous countries that show how aligning particular aspects of a health system might be possible. For example, commentators have pointed to the primary care system of Israel, health promotion in Sweden, and aged care in Japan as demonstrating elements of "perfect" alignment.[xxxi] The same could equally be said for certain low- and middle-income countries, where the impressive health gains made by nations such as Costa Rica, Rwanda, and Cuba show that striving for alignment over particular priorities can lead to outcomes that surpass those in some wealthier countries.

It is also important to remember that misalignments and alignments do not exist only at the national level. Highly regarded local systems of care will often resolve misalignments at the national level – for example, the well-documented successes in value-based payment models of the Kaiser Permanente and the Geisinger health delivery systems in the United States.[xxxii]

Achieving alignment case studies

The Council identified several specific cases in which strategies were successfully designed and implemented to achieve alignment. Mechanisms employed to remove obstacles to alignment and facilitate efficient behaviors include established instruments, such as taxes, subsidies, regulations, and public health education. They also include technical and institutional innovations, such as new modalities for high-quality digital health consultations and harmonized electronic medical records. In addition, they include new models of value-based care, payment mechanisms that are tied to health outputs rather than inputs, and pricing schemes for health and life insurance that incentivize consumers to engage in healthy behaviors. And they include situations in which stakeholders find new ways to cooperate, reducing waste and benefiting all parties; the introduction of strong leadership is often the underlying mechanism.

Providing a new mechanism for expedient care – HealthTap, USA

What can be done about the difficulty that patients often have in accessing expedient health advice, due to clinicians lacking billing mechanisms for remote consultation? HealthTap, which was founded in 2010, uses technological innovation to address this misalignment, which stems from divergent objectives. In doing so, it increases opportunities for a globally diverse clientele to conveniently access credible, personal health advice in a timely fashion, and at a low cost.

HealthTap is an online platform that allows members from anywhere in the world to obtain virtual access to over 100,000 U.S.-licensed doctors with expertise in over 140 specialties and multiple languages. As of May 2016, its providers had answered over 4.5 billion queries and served over 250 million people from 175 countries, mostly from the United States. The most basic subscription is free of charge and allows members to post anonymous questions – with a 150-character limit – to the HealthTap medical community. Brief informational answers are provided by at least one doctor, typically within 24 hours, with provision for peer review. Members can also access health tips, medical news, mobile app recommendations, checklists, and millions of other answers posted by HealthTap providers. Paid subscribers have instantaneous access to real-time consultations for minor conditions with a medical provider of their choice via video, voice, or text-based chat.

By facilitating remote consultation, HealthTap makes it more likely for patients to obtain the care they need in an expedient fashion. This has the potential to prevent troublesome illnesses and injuries from progressing unnecessarily, which can lead to inflated healthcare costs and suboptimal health outcomes.

Encouraging insurers to cover a new heart drug by minimizing risk through pay-for-performance – Novartis, Cigna, Aetna

A recent pricing accord between a drug manufacturer and two insurance companies shows how a misalignment due to divergent objectives can be corrected by ensuring that risks and rewards are shared more equitably by stakeholders. In February 2016, the Swiss pharmaceutical company Novartis AG reached an agreement with U.S. insurers Cigna Corp. and Aetna Inc. that specifies performance-based pricing for its new heart drug Entresto. This drug has shown promising results in clinical trials, outperforming alternative drugs for heart failure. Significantly, it has also been linked to reduced hospitalization rates.[xxxiii] Thus besides being a boon for patient health, it could help reduce long-term costs for both patients and insurers. However, a high price point could slow Entresto's uptake. Priced at $12.50 per day (or $4,560 per year) under traditional models, it stands to cost significantly more than the drugs it is positioned to replace (such as captopril, lisinopril, enalapril, and losartan).

The pay-for-performance structure might allay insurers' concerns about Entresto's cost. David Epstein, the Division Head and CEO of Novartis Pharmaceuticals, has indicated that payment will be founded on a "fairly modest" base rebate price, which is then adjusted depending on whether the drug's administration results in sufficiently reduced patient hospitalization rates.[xxxiv] Essentially, the closer Entresto gets – in the real world – to achieving the same results seen in clinical trials, the greater the payout will be for Novartis. Although insurers will have to pay more if the drug is more effective, they stand to benefit overall from reduced hospitalization costs. It is worth noting that the potential complexity of tracking health outcomes may pose a challenge when implementing this type of contract. Nevertheless, the pay-for-performance model has the potential to align manufacturer and insurer incentives, encouraging investment in solutions with great potential to improve patient outcomes and reduce long-term costs.

Deterring the consumption of unhealthy products – Mexico's soda tax

A growing obesity epidemic – driven by overconsumption of unhealthy foods and beverages (such as sugared beverages) – has contributed to a rising incidence of NCDs (such as diabetes and heart disease) in many countries. The increasing costs of treating and managing NCDs are often borne by governments that lack the financial means to accommodate the growing demand for care. But despite the social costs that this excessive consumption entails, the purchase price to consumers does not reflect these costs. In this power asymmetry, consumers have the full autonomy to consume these legal products as they wish; however, the consequences of their choices are partially borne by society.

In Mexico, which has the second highest rates of obesity and diabetes in the world,[xxxv] the government has implemented a one-peso-per-liter tax on sugary soft drinks as part of broader anti-obesity reforms aimed at reducing consumption and collecting revenue for health promotion programs. In addition to the tax, the government implemented reforms to nutritional labeling on school lunches and banned advertisements that target children for the promotion of unhealthy products.[xxxvi] Recent research has shown that sales of taxed beverages fell by 6% from 2012-2014, while sales of nontaxed beverages rose by an average 4%.[xxxvii] If this trend persists, it would prevent an estimated 400,000 to 630,000 cases of diabetes by 2030 – a significant bend in the diabetes epidemic curve.[xxxviii] One recent economic analysis of the tax's effects found that it could result in a two- to four-pound reduction in Mexico's mean weight, which would be enough to produce meaningful health gains.[xxxix] The early success of Mexico's soda tax underscores the potential for policy instruments to play an influential role in correcting misalignments.

Sharing the savings from prevention – Discovery Vitality, South Africa

Under traditional life insurance models, premiums are determined based on the policyholder's demographics, medical condition, and lifestyle choices at the time the policy contract is signed. Once established, the premium does not change, regardless of whether the policyholder's risk of death increases or diminishes. While it is naturally in an individual's self-interest to take better care of him or herself, many people do not fully avail themselves of preventive care and optimal health behaviors. Although external motivation could further encourage adoption of these practices, most life insurance policies do not offer incentives to get policyholders to improve their health, even though these improvements benefit both parties. All of the insurance-related financial benefits of decreased risk accrue to the insurer; none are returned to the policyholder.

The South African company Discovery addresses this cooperation failure by packaging life insurance, health insurance, and other health-service benefits for its customers through its Vitality program. This allows Discovery to return some of the financial returns of better health to its policyholders. By meeting healthy living goals, members of its Vitality plan receive rewards such as reduced insurance premiums, discounted gym membership and wearables, cash back on healthy food purchases, or vouchers for travel, shopping, and entertainment.

Sharing the savings of healthy behaviors with policy holders incentivizes healthy behaviors at an individual level, which decreases the need for healthcare. The effects can be wide-ranging: a research study of 300,000 people across 5 years showed that the cost of healthcare for people who were engaged with Vitality was reduced compared to that for non-engaged clients.[xl] Moreover, the likelihood of being admitted to the hospital and the length of average stay in the hospital were also reduced in patients who were engaged with Vitality. Additionally, Vitality plans offered as part of employee benefit packages have substantially reduced healthcare costs. Discovery has designed life and health insurance contract that uses incentives to create health-related financial benefits for themselves and their policyholders.

Empowering caregivers – Noora Health, India

At the regional level, the innovative programming from Noora Health in India serves as an example of correcting a cooperation failure in developing country health systems. This non-profit recognized that – by working separately – patients, providers, and families were foregoing an opportunity to achieve better health outcomes. To remedy this situation, they designed a program to better harness the time and energy of caregivers attending to loved ones in the hospital. When a patient is admitted, their caregiver undergoes interactive group training, followed by hands-on training, certification, and supervision, while working with the patient. The aim is to give basic but vital training on post-surgical recovery, physical therapy, prevention of infections, and recognition of early warning signs of further health problems.

Over 45,000 caregivers have now been trained, with the program rolled out across more than 25 hospitals (both private and public) and looking to expand across India and beyond. The program has led to lower rates of preventable complications and unplanned readmissions, lower lengths of stay, improved quality of care, and greater patient satisfaction. By fostering cooperation between hospital staff and family caretakers, Noora Health helps patients achieve better outcomes without requiring significant additional resources.

Enabling targeted patient care and generating evidence for alternative pricing: COTA, USA

A hospital-level example of correcting cooperation failures – in this case through technical innovation – comes from the Clinical Outcome Tracking Analysis (COTA) program at the John Theurer Cancer Center in New Jersey. COTA uses data from patients' electronic medical records to assign each patient an evolving identification number that reflects the specifics of their cancer (type, stage, hormone receptors, etc.), courses of treatment (dose intensity delivered for each treatment, toxicity, quality of life, and overall costs for each episode of care), comorbidities, and progression of the disease.

By creating a common system for making apples-to-apples comparisons across patients and treatments, COTA enables providers to generate more reliable real-world outcome benchmarks, empowers oncologists with actionable insights and feedback, enables alternative payment models (bundles), and empowers patients to make better decisions about their care. It thereby helps improve patient outcomes while reducing costs and provides a robust tool for research (on topics such as navigating clinical trials and accelerating drug discovery and approval processes). And because COTA operates at the hospital level, its costs and benefits can be easily shared. The challenge is to encourage implementation of similar technology in situations in which multiple stakeholders will have to share costs, but will also share much greater benefits over the long term.

Bridging the global pain gap – American Cancer Society's "Treat the Pain"

Despite being both affordable and included on the WHO's Essential Medicines List, cooperation failures restrict access to opioid pain medications in many low- and middle-income countries. Globally, about 7 million cancer or HIV sufferers die annually in moderate or severe pain – with 3.6 million having no access to palliative-care opioids, such as morphine. In countries such as Nigeria, 99 percent of the 162,000 people that die in pain each year will not receive opioids.[xli] Notwithstanding the current epidemic of opioid addictions in some high-income countries, the inability of suffering patients to access proper treatment represents a significant failure of health systems in many places around the world.

The American Cancer Society's program "Treat the Pain" tackles the problem by applying a framework – known by the acronym "MORPHINE" – to identify and address the needs of key stakeholders. The framework, which has eight stages, starts with (i) influencing the **M**indset of policymakers, (ii) **O**rganizing to identify the barriers to access, (iii) updating **R**egulations, and (iv) fixing **P**rocurement issues (which often results in establishing local manufacturing facilities). They then (v) offer **H**ealth worker education (many health workers have not been previously trained in pain relief), (vi) **I**nitiate the program in pilot hospitals, (vii) **N**ationalize the effort, and (viii) work to **E**mpower local stakeholders to maintain progress. The whole model is built on mobilizing partners at every level from the police to the ministry of health, including health workers and patients. The MORPHINE framework achieves alignment by sensibly and productively engaging all necessary stakeholders in a way that addresses their individual and collective interests.

This alignment case study provides an excellent example of how overcoming cooperation failure can lead to substantial gains in health outcomes (in the form of enhanced quality of life for patients and lessened stress and anxiety for caregivers) without requiring a significant addition of resources. Although administering opioids to suffering patients requires a small net input of money without necessarily prolonging life, the Council contends that the MORPHINE framework nevertheless offers enhanced value to patients and health systems.

6. LESSONS ON ACHIEVING ALIGNMENTS

Our analysis of the foregoing examples reveals a number of generalizable lessons about how systems can successfully move from misalignment to alignment.

First, alignment does not occur by happenstance. Rather, it is almost exclusively the result of leadership and a commitment to improving quality, affordability, equity, and efficiency of care. Most of the alignment examples rely on one stakeholder taking a strong system leadership role; however, there was no discernible pattern with regard to which stakeholder. This suggests that one should not saddle any particular group (such as the government) with the lead responsibility for solving misalignments – progress is as likely to be driven by a payer, provider, supplier, or community group. The key is to create a case and space for system-thinking to flourish, and to challenge institutional intransigence and short-term self-interest (or at least, show how alignment would better meet that self-interest).

Second, stakeholders should work together to highlight and overcome the misalignment. Best practices show that stakeholders should: (i) quantify and highlight a particular area of inefficiency and the value to be unlocked from a different way of working together; (ii) agree on a vision for how health care could be improved; (iii) determine how the benefits of that improvement will be shared; (iv) create new – or alter existing – targets and incentives to promote that alignment; and (v) introduce and standardize new care processes across the system.

Depending on the formality of the new targets and incentives, stakeholders may also implement a mechanism to measure and track the impact of change. As in most public and private domains, keys to success in overcoming misalignments include a willingness to approach problem solving in a cooperative manner – coupled with hefty doses of accountability and transparency, and sensitivity to potential unintended consequences.

Third, occasionally, misalignments are overcome by a much more disruptive and unpredictable change. Often a new entrant or technology is introduced that, rather than solving the underlying misalignment, *sidesteps* it by creating a totally new business or care model – as exemplified by HealthTap, which can be seen as a "disruptive alignment."

Fourth, particular alignment approaches tend to pair with different types of misalignment.

- In addressing ***divergent objectives***, it is critical to identify the value being lost to the system (either in health or resources) – which creates the case for finding a new approach that would be better for all or most stakeholders and redesigning incentives to promote that. Possible instruments include shifting business models from the sale of pills and procedures to the sale of health outcomes, and in some cases, the dissolution of unnecessary organizational boundaries through mergers (as with the creation of Accountable Care Organizations).

- In correcting ***power asymmetries***, it is generally not the disempowered group that drives alignment forward but another, more powerful stakeholder. This could be the overly dominant player who comes to recognize the value lost through disempowerment (which is reversed when providers better engage their patients), or a separate organization that acts on behalf of the less powerful party (such as a government or regulator). Common instruments include new taxes or subsidies, or empowering weaker groups with money, information, or advice.

- In resolving ***cooperation failures***, a key step is to create a situation in which the benefits of alignment are shared across everyone who needs to act. Mechanisms to do this include direct instruments (such as shared savings schemes and joint ventures) and indirect means of fostering collaboration (such as forums, industry associations, and data sharing agreements). Some cooperation alignments are also spurred by greater media attention to a particular societal problem.

These principles can be used to guide the achievement of better alignment. Though misalignments are common, pervasive, and deeply rooted, motivated and engaged stakeholders can do much to correct them and deliver healthcare that improves both health outcomes and financial bottom lines.

CONCLUSIONS AND RECOMMENDATIONS

At the outset of this project, the Council aimed to better understand misalignments, articulate a vision of a more aligned system, design a plan for achieving alignment, and estimate the potential health and economic dividends that alignment may offer. By outlining a typology of misalignments based on real-world examples, calculating the possible magnitude of the problem, recounting scenarios in which alignment has been achieved, and illustrating key lessons to be drawn from these examples of alignment, our white paper offers a framework for future research into misalignments and their consequences.

The next phase of research and dialogue should focus on developing methods for applying the lessons that we have learned so far – ideally including direct multi-stakeholder involvement. One goal is to undertake a mock or real-life alignment exercise with representatives from key stakeholder groups, which could be done as part of the Forum's work in healthcare. Another goal is to create a toolkit, including diagnostic software that would enable researchers and stakeholders to catalogue and classify misalignments. Such a toolkit would offer proven approaches for alignment, drawn from the relevant scientific literature.

As for recommendations for future research on the nature of misalignments, the Council suggests: (i) investigating how misalignments impact health equity, (ii) identifying examples of system-level alignments in health systems in which universal healthcare is absent, and (iii) determining what barriers currently restrict the use of data to assess and correct misalignments.

This white paper also helps lay the foundation for a new project launched by the Forum and its partners called *Value in Healthcare*. Recognizing the inefficiencies and drawbacks of volume-based compensation models (such as fee-for-service), the Global Health and Healthcare team under the recommendation of the Health and Healthcare Governor’s Community is embarking upon a multi-year impact initiative of how to accelerate the adoption and implementation of strategies that deliver the best possible value to the patient at a sustainable cost. The project will focus on three objectives:

- To increase understanding of what value-based care truly means and the implications for each stakeholder in the healthcare ecosystem.
- To identify concrete areas for extracting value for stakeholders, either by reducing costs or improving health outcomes.
- To accelerate shifting the paradigm for healthcare delivery from volume fee-for-service to outcome-based approaches that maximize the value for patients.

By advancing knowledge on how to achieve a double bottom line of better patient outcomes and cost efficiency, this project hopes to catalyze new approaches to healthcare delivery.

Acknowledgements

The Industry Agenda Council would like to acknowledge and express its gratitude to KPMG – especially Mark Britnell and Jonty Roland – for their collaborative, diligent, and creative work. Special thanks also go to Jessica Sullivan and Daniel Cadarette (both at the Harvard T.H. Chan School of Public Health) for their significant and thoughtful contributions to the drafting of this white paper. The Council is also grateful to Laura Wallace for her valuable comments and skillful editing of the document. The writing team for this white paper consisted of David Bloom, Daniel Cadarette, Jonty Roland, and Jessica Sullivan.

Key Events:

- World Economic Forum Summit on the Global Agenda; Dubai, UAE; October 2014
- World Economic Forum Annual Meeting; Davos, Switzerland; January 2015
- World Economic Forum Meeting of the Technology Pioneers; San Francisco, USA; June 2015
- World Economic Forum Annual Meeting of the New Champions; Dalian, China; September 2015
- Mental Health Roundtable; Luxembourg; October 2015
- World Economic Forum Summit on the Global Agenda; Abu Dhabi, UAE; October 2015
- World Economic Forum Annual Meeting; Davos, Switzerland; January 2016
- World Economic Forum Annual Meeting of the New Champions; Tianjin, China; June 2016

Key Contributors and Supporters:

IAC Council Members: David Agus, Peter Bach, Raymond Baxter (vice-chair), David E. Bloom (chair), Mark Britnell, Vijay Chandru, Francis Collins, Victor Dzau, Geoff Ginsburg, Adrian Gore, Andre Goy, Bernard Hoet, Stefan Kapferer (vice-chair), Suresh Kumar, Petra Laux, Javier Lozano, Bobby Prasad, Peer Schatz, Alfred Sommer, Alan Tennenberg, Mary Lou Valdez, Andrey Zarur, and former Council members John Glaser, Margaret Hamburg, and Pascale Witz

KPMG research team: Jonty Roland (team lead), Chris Hardesty, Randall Baran-Chong, Dhara Desai, Tommy Hana, Chris Leiter, and Aman Kumar

Harvard research team: Jessica Sullivan and Daniel Cadarette

WEF: Arnaud Bernaert, Vanessa Candeias, Olivier Oullier, Roderick Weller, and former WEF team members Shubhra Saxena Kabra, Daniel Akinmade Emejulu, and Susanne Weissbaecker

Interviewees: 60 interviewees: thank you for sharing your time and insights

Others: Farah Al Besher, Aneesh Chopra, Nicolaus Henke, Linda Fried, Hu Bo, Sangita Reddy, Edith Schippers, Derek Yach, Ruilin Zhao, Ron Gutman, and participants at the events in San Francisco, Dalian, and Luxembourg

[i] Bloom, D.E. “The Shape of Global Health”. *Finance and Development.* 2014, http://www.imf.org/external/pubs/ft/fandd/2014/12/bloom.htm
[ii] Anderson, I. et al. “Indigenous and tribal peoples' health (*The* Lancet-Lowitja Institute Global Collaboration): a population study”. *The Lancet.* 2016, http://www.sciencedirect.com/science/article/pii/S0140673616003457
[iii] Squires, D. and Anderson, C. “U.S. Health Care from a Global Perspective: Spending, Use of Services, Prices, and Health in 13 Countries”. The Commonwealth Fund. 2015, http://www.commonwealthfund.org/publications/ issue-briefs/2015/oct/us-health-care-from-a-global-perspective
[iv] The World Bank. “World Development Indicators”. 2015, http://data.worldbank.org/data-catalog/world-development-indicators
[v] The World Bank. “Health Expenditure, Total (% of GDP)”. 2016, http://data.worldbank.org/indicator/SH.XPD.TOTL.ZS
[vi] Walsh, F. “Superbugs to kill 'more than cancer' by 2050”. *BBC.* 2014, http://www.bbc.com/news/health-30416844
[vii] Shih, C., Schwartz, J., and Coukell, A. “How Would Government Negotiation of Medicare Part D Drug Prices Work?” *Health Affairs.* 2016, http://healthaffairs.org/blog/2016/02/01/how-would-government-negotiation-of-medicare-part-d-drug-prices-work/
[viii] Gagnon, M. and Wolfe, S. “Mirror, Mirror on the Wall: Medicare Part D pays needlessly high brand-name drug prices compared with other OECD countries and with U.S. government programs.” Carleton University. 2015, http://carleton.ca/sppa/wp-content/uploads/Mirror-Mirror-Medicare-Part-D-Released.pdf
[ix] Rauhala, E. “Why China's Doctors Are Getting Beat Up”. *TIME.* 2014, http://time.com/15185/chinas-doctors-overworked-underpaid-attacked/
[x] World Health Organization. “Mental Health Action Plan 2013 – 2020”. 2013.
[l]http://www.who.int/mediacentre/factsheets/fs380/en/
[xii] OECD. “Growth rates of health spending for selected functions per capita”. 2015.
[xiii] World Health Organization. “Mental Health Action Plan 2013 – 2020”. 2013.
[xiv] World Health Organization. “Mental Health Action Plan 2013 – 2020”. 2013.2013)
[xv] World Health Organization. *WHO Report on the Global Tobacco Epidemic, 2015: Raising taxes on tobacco.* 2015, http://apps.who.int/iris/bitstream/10665/178574/1/9789240694606_eng.pdf
[xvi]Moon, S. and Omole, O. *Development Assistance for Health: Critiques and Proposals for Change,* London: Chatham House. 2013, https://www.chathamhouse.org/sites/files/chathamhouse/public/Research/Global%20Health/0413_devtassistancehealth.pdf
[xvii] Sridhar, D. and Woods, N. “Are there simple conclusions on how to channel health funding?” *The Lancet.* 2010, http://www.thelancet.com/journals/lancet/article/PIIS0140-6736(10)60486-2/fulltext
[xviii] Sridhar, D. and Tamashiro, T. “Vertical Funds in the Health Sector Lessons for Education from the Global Fund and GAVI”. UNESCO. 2009, http://unesdoc.unesco.org/images/0018/001865/186565e.pdf
[xix] Murray, C., Hanlon, M., and Leach-Kemon, K. *Financing Global Health 2011: Continued Growth as MDG Deadline Approaches.* Institute for health Metrics and Evaluation, University of Washington. 2011, http://www.healthdata.org/policy-report/financing-global-health-2011-continued-growth-mdg-deadline-approaches
[xx] Barnett, K. et al. Supplementary appendix for “Epidemiology of multimorbidity and implications for health care, research, and medical education: a cross-sectional study”. *The Lancet.* 2012, http://www.thelancet.com/cms/attachment/2009245825/2032056168/mmc1.pdf
[xxi] U.S. Department of Health and Human Services. “Summary of the HIPAA Privacy Rule”. http://www.hhs.gov/hipaa/for-professionals/privacy/laws-regulations/
[xxii] NHS Confederation, Key facts and trends in mental health, (NHS Confederation, 2011)
[xxiii] Lancy, A. and Gruen, N. “Constructing the Herald/Age – Lateral Economics Index of Australia's Wellbeing”. *The Australian Economic Review.* 2013, http://onlinelibrary.wiley.com/doi/10.1111/j.1467-8462.2013.12000.x/abstract
[xxiv] World Health Organization, Mental Health Action Plan 2013 – 2020, (WHO, 2013)
[xxv] Bloomberg. “Most Efficient Health Care 2014: Countries”. 2014, http://www.bloomberg.com/visual-data/best-and-worst//most-efficient-health-care-2014-countries

[xxvi] The Economist Intelligence Unit. “Health outcomes and cost: a 166-country comparison”. 2014, http://www.eiu.com/Handlers/WhitepaperHandler.ashx?fi=Healthcare-outcomes-index-2014.pdf&mode=wp&campaignid=Healthoutcome2014
[xxvii] World Health Organization. *The World Health Report 2000 – Health Systems: Improving Performance.* 2000, http://www.who.int/whr/2000/en/
[xxviii] Jamison, D. and Sandbu, M. “WHO Ranking of Health System Performance”. *Science.* 2001, http://science.sciencemag.org.ezp-prod1.hul.harvard.edu/content/293/5535/1595.full
[xxix] Musgrove, P. "Judging Health Systems: Reflections on WHO’s Methods." *The Lancet*. 2003, http://www.sciencedirect.com/science/article/pii/S0140673603134083
[xxx] Brundtland, G., Frenk, J., and Murray, C. “WHO assessment of health systems performance." *The Lancet.* 2003, http://www.sciencedirect.com.ezp-prod1.hul.harvard.edu/science/article/pii/S0140673603137026
[xxxi] Britnell, M. *In Search of the Perfect Health System.* Palgrave Macmillan. October 2015, https://he.palgrave.com/page/detail/?sf1=barcode&st1=9781137496614
[xxxii] Shortell, S. et al. “Accountable care organisations in the United States and England: Testing, evaluating and learning what works”. The Kings’ Fund. 2014, http://www.kingsfund.org.uk/sites/files/kf/field/field_publication_file/accountable-care-organisations-united-states-england-shortell-mar14.pdf
[xxxiii] Pollack, A. “F.D.A. Approves Heart Drug Entresto Said to Cut Death Risk by 20%”. The New York Times. 2015, http://www.nytimes.com/2015/07/08/business/international/fda-approves-heart-drug-entresto-after-promising-trial-results.html
[xxxiv] Comstock, J. “Novartis signs Aetna, Cigna for pay-or-performance drug deal, but no remote monitoring yet”. Mobi Health News. 2016, http://mobihealthnews.com/content/novartis-signs-aetna-cigna-pay-performance-drug-deal-no-remote-monitoring-yet
[xxxv] World Health Organization. “Putting taxes into the diet equation”. Bulletin of the World Health Organization. 2016, http://www.who.int/bulletin/volumes/94/4/16-020416/en/
[xxxvi] O’Connor, A. “Mexican Soda tax Followed by Drop in Sugary Drink Sales”. *The New York Times.* 2016, http://well.blogs.nytimes.com/2016/01/06/mexican-soda-tax-followed-by-drop-in-sugary-drink-sales/
[xxxvii] Colchero, M. et al. “Beverage purchases from stores in Mexico under the excise tax on sugar sweetened beverages: observational study”. *BMJ.* 2016, http://www.bmj.com/content/352/bmj.h6704
[xxxviii] The Update team. “Mexican diabetes devastation. It is essential to tax soda”. *World Nutrition.* 2015, http://wphna.org/wp-content/uploads/2015/06/WN-2015-06-03-137-139-Update-Mexico-diabetes-soda-tax.pdf
[xxxix] Grogger, J. “Soda Taxes and the Prices of Sodas and Other Drinks: Evidence from Mexico”. National Bureau of Economic Research. 2015, http://www.nber.org/papers/w21197.pdf
[xl] Patel, D. et al. “Participation in fitness-related activities of an incentive-based health promotion program and hospital costs: a retrospective longitudinal study”. *Am J Health Promot*. 2011, http://www.ncbi.nlm.nih.gov/pubmed/21534837

[xli] Hughes-Hallett T. et al. “Dying Healed: Transforming End-of-Life Care through Innovation”. World Innovation Summit for Health. 2014, http://www.wish-qatar.org/app/media/386

THE HINDU

OPINION » LEAD

Published: August 3, 2015 02:38 IST | Updated: August 3, 2015 07:17 IST August 3, 2015

Revamping India's scientific ecosystem

- Vijay Chandru

Vijay Chandru.

There is a massive wave of entrepreneurial energy coursing through India's arteries. The need is to connect this enthusiasm with excellence in basic and applied research in our higher educational institutions.

A newspaper headline, "The Lament and the Lash", made waves recently. Interestingly, it was a report on how Chairman Emeritus, Infosys Ltd., N.R. Narayana Murthy had lamented, while delivering the convocation address at the Indian Institute of Science, on July 15, 2015, that India has not produced a single invention that became a global household name. "... let us pause and ask what the contributions of Indian institutions of higher learning particularly IISc and IITs [Indian Institutes of Technology], have been over the last 60-plus years to make our society and the world a better place. Is there one invention from India that has become a household name in the globe? Is there one technology that has transformed the productivity of global corporations? Is there one idea that has led to an earth-shaking invention to delight global citizens? Folks, the reality is that there is no such contribution from India in the last 60 years....," were his words.

Mr. Murthy then went on to contrast local achievements with some by MIT (the one in Massachusetts, and not Madras or Manipal), recalled the contributions made by Indian scientists in the 1950s and 1960s and then challenged IISc's finest to revisit that passion. This is a bit of an unfair comparison since most global universities, the West included, would fare poorly in comparison with MIT. According to the Kauffman Foundation report in 2009, MIT alumni founded companies that generated over $2 trillion in revenue a year (the 11th largest economy, when viewed as the GDP of a nation) and provided gainful employment to over 3.3 million people.

'Translational research'

However, Mr. Murthy's focus on the value that could be created for society by the outgoing class of 2015 is not misplaced. IISc's founders, J.N. Tata and the Mysore Maharaja, Krishnaraja Wodeyar IV, had hoped that the institute would contribute to the "material benefit" of society. In turn, Mr. Murthy asked the students to realise the ideals of the founders. However, these ideals have evolved based on interpretations by the then leadership; therefore, today, some realignments may be needed.

For example, there is a massive wave of entrepreneurial energy coursing through the nation's arteries. If we could connect this enthusiasm with excellence in basic and applied research in our higher educational institutions, the possibility of a new growth engine that has more enduring value seems within reach. This synergy needs a new alignment around the theme of "translational research" — a concept easy to describe but hard to execute as it disturbs the order.

At SelectUSA 2015, Google executive chairman Eric Schmidt reasoned why the United States continues to hold the innovation advantage — its outstanding centres of higher education and the clusters of enterprises around them. He talked about Vannevar Bush, the engineer, inventor, founder of Raytheon and science administrator par excellence, who had enabled this in the post-World War II era. President Roosevelt had challenged Bush, as Director of the Office of Scientific Research and Development, to make a plan for the post-war period. Bush then penned the now classical declaration, "Science the endless frontier" in July 1945. He was not advocating "applied" over "basic" research. It must be noted that basic research leads to new knowledge and provides scientific capital. New products and processes do not appear in final form as if by magic, but are painstakingly developed by scientific enquiry.

Bush presented the blueprint for the National Science Foundation. For historians, Bush is to be noted for having "insisted upon the principle of Federal patronage for the advancement of knowledge in the United States, a departure that came to govern Federal science policy after World War II."

Science, the endless frontier

In post-Independent India too, we have seen similar tasks being presented to Homi Bhabha, Vikram Sarabhai, Meghnad Saha and C.V.

Raman by Jawaharlal Nehru. For example, in 1947, C.V. Raman proclaimed that "vast powers are placed in the hands of man by successful research which opens up a vista of possibilities for its beneficient application in the relief of the fundamental ills of humanity, namely hunger, poverty and disease." For perhaps the first three decades after 1947, India did experience a great period of development driven by the passion to create and innovate for the "material advancement" of the nation. This was manifested in the public sector undertakings as there was no real capital available from the private sector for this kind of institution building. Committed leadership was present through those years. The image of IISc created by scientist-engineer Satish Dhawan to drive important initiatives in space, electronics, machine tools, aviation and rural technologies remains its legacy.

So early on, India's science and technology strategy was in step with those of the most progressive nations. Scientists dedicated themselves to institution and capacity building. Despite inheriting an economy ravaged by colonialism, the Indian leadership seemed to have found the resources to carve a special role for science in the country's development.

Incidentally, the Indian IT industry owes a great deal to visionaries in the government in the 1960s and 1970s. The Department of Electronics identified "software led exports" as a segué for Indian export promotion in 1972 and provided resources to buy computers and get the private sector up to speed. Software technology parks in the 1980s and 1990s enabled industry to import equipment at competitive prices, avail tax holidays and access free connectivity to the Internet. The leg up that industry received in these two decades is often overlooked in the facetious praise of "benign neglect by the government" as the reason for the IT sector's success. The truth is that the government provided the right help at the right time and did not over-regulate the sector — a fine example of directed public policy in technology.

"Innovations" such as the "24-hour work day and global delivery" were borrowed from the supply chain and hospitality industry as the IT industry needed to maintain a fantastic growth rate triggered by the fortuitous service opportunities that the Y2K (millennium) crisis provided. The sector has continued to maintain scale at a breakneck speed.

A contrast in cultures

In the late 1970s and 1980s, science and higher education in the U.S. was going through a major realignment. The bets were on translation and commercialisation of federally funded research output. The Bayh-Dole Act (1980) allowed individual researchers and universities "to leverage funded research output as intellectual property that could be taken forward, licensed or commercially exploited to create new products, solutions and private companies". Overnight, professors transformed themselves from "geeks into suits", learned to hire lawyers, negotiate with universities, attract investors, build companies and hand off to professionals to scale and build value. For example, Boston and the Bay Area in the U.S. began to bustle with high technology companies that sprouted at regular intervals. Clusters formed around the hotspots of scientific research and "regional advantage" in Saxenian terms took root.

In contrast, India chose a new alignment which might be considered a conservative and almost retrograde pivot. There was a culture of "singular focus on excellence in research". Institutes of higher education were bound tightly to purist academic metrics — "publish in high quality journals, place your postgraduate students in centres of excellence and ensure that the most tutored young scientists entered the campus". We were promoted, feted and given awards based on these metrics.

Some correction in the quality of research may have been warranted. The National Centre for Biological Sciences and the Jawaharlal Nehru Centre for Advanced Scientific Research in Bangalore were exemplary new additions to the "Science City" and within proximity to IISc and the Raman Research Institute. Over the last three decades, many new IITs, Indian Institutes of Science Education and Research and Indian Institutes of Management have come up. Publications seem to be showing a hockey stick growth in numbers and many young and senior doctoral students in the diaspora are returning home as universities improve.

Yet, engineering and applied scientific research have not shown relevance to practice. We have clearly lost the plot as far as the accountability of science to society is concerned and need to make some course corrections. Self and peer review seem to have been ineffective in auditing and nudging institutions and faculty to be more accountable to society. One solution is to introduce frequent third party external reviews with mechanisms for actionability. We can be sure that such a move will meet resistance.

Entrepreneurship and innovation

The drumbeat on innovation and entrepreneurship has been rolling for over a decade in India. The virtues of innovation are being written and spoken about, with grass roots and business innovators, incubators, accelerators and entrepreneurs coming up in every corner. There is a rapid conversion of demographic dividend into a "hungama" of entrepreneurship. India now has the world's second largest number of entrepreneurs, as a Google search will show.

With incubators and seed capital becoming accessible, a handful of start-ups attempting to translate research are visible. Localised versions of the Bayh-Dole legislation have been doing the rounds of Parliament and institutes of higher education. While the trends are right, the volume and velocity of impact is still quite disappointing.

Developing on this, we need to blend the emerging entrepreneurial ecosystem with centres of excellence in higher education, and soon we would be a nation transformed. This would require great focus on translation of research while extending support for excellence in research in the basic and applied sciences.

I come back to the point made in the first paragraph, that of Mr. Murthy's "lament and lash" which could be turned into "appreciation and applause". For this to happen, we need to pay close attention to the realignment needed for translation to be elevated to a high priority and processes to be in place to enable a scaled effect. Incubation centres, patent cells and technology transfer offices are visible in many institutions but as satellite activities and translation is not core to the academic mission. Vannevar Bush also said: 'Science can be effective in the national welfare only as a member of a team, whether the conditions be peace or war. But without scientific progress, no amount of achievement in other directions can insure our health, prosperity, and security as a nation in the modern world."

A prescriptive plan of realignments is being formulated and will be made public soon. But the execution of the plan will need commitment from institutions of higher education, science administrators, the state, and, most importantly, by scientists.

(Vijay Chandru is an engineer-scientist, a former professor of IISc and an entrepreneur. He is a member of the expert committee on innovation and entrepreneurship at Niti Aayog. The views expressed are personal.)

Printable version | Sep 16, 2015 9:38:39 AM | http://www.thehindu.com/opinion/lead/revamping-indias-scientific-ecosystem/article7492142.ece

Genet. Res., Camb. (2014), vol. 96, e009. © Cambridge University Press 2014
doi:10.1017/S0016672314000111

Organization for rare diseases India (ORDI) – addressing the challenges and opportunities for the Indian rare diseases' community

HARSHA KARUR RAJASIMHA[1]*, PRASANNAKUMAR BASAYYA SHIROL[1], PREVEEN RAMAMOORTHY[1], MADHURI HEGDE[1], SANGEETA BARDE[1], VIJAY CHANDRU[1], M. E. RAVINANDAN[1], RAMANI RAMCHANDRAN[2], KASTURI HALDAR[3], JIMMY C. LIN[4], IMRAN A. BABAR[4], KATTA M. GIRISHA[5], SUDHA SRINIVASAN[6], DURAISWAMY NAVANEETHAM[7], RAJANI BATTU[8], RAJASHREE DEVARAKONDA[9], USHA KINI[10], KINNIMULKI VIJAYACHANDRA[11] AND ISHWAR C. VERMA[12]

[1]*Organization for Rare Diseases India, No 10, 4th Main Road, # 2 (Ground Floor), Jaladarshini Layout, Bangalore – 560054 India*
[2]*Developmental Vascular Biology Program and Zebrafish Drug Screening Core, The Medical College of Wisconsin, 8915 W. Connell Ave, Milwaukee, WI 53226, USA*
[3]*Center for Rare and Neglected Diseases, University of Notre Dame, Notre Dame, IN 46556, USA*
[4]*Rare Genomics Institute, 4100 Forest Park Ave, Suite 204, St. Louis, MO, 63108, USA*
[5]*Department of Medical Genetics, Kasturba Medical College, Manipal University, Manipal, India*
[6]*Centre for Human Genetics, Biotech Park, Electronics City (Phase I), Bangalore 560 100, India*
[7]*Foundation for Research on Rare Diseases and Disorders, C252, Kandasamy Salai, Periyar Nagar, Chennai 600082, India*
[8]*Narayana Nethralaya, #121/C, Chord Road, 1st 'R' Block, Rajajinagar, Bangalore – 560 010, India*
[9]*Voisin Consulting Private Limited, #1, Hara House, 2nd Floor, Gutte Anjaneya Temple Road, Hosur Main Road, Wilson Garden, Bangalore – 560027, India*
[10]*Department of Pathology, St. John's Medical College and Hospital, Bangalore – 560034 India*
[11]*Ixora Biosciences Pvt Ltd, Institute of Bioinformatics and Applied Biotechnology, Biotech Park, Electronic City Phase-1, Bangalore – 560100, India*
[12]*Sir Ganga Ram Hospital, Rajinder Nagar, New Delhi – 110060 India*

(*Received 19 June 2014; accepted 25 June 2014*)

Summary

In order to address the unmet needs and create opportunities that benefit patients with rare disease in India, a group of volunteers created a not-for-profit organization named Organization for Rare Diseases India (ORDI; www.ordindia.org). ORDI plans to represent the collective voice and advocate the needs of patients with rare diseases and other stakeholders in India. The ORDI team members come from diverse backgrounds such as genetics, molecular diagnostics, drug development, bioinformatics, communications, information technology, patient advocacy and public service. ORDI builds on the lessons learned from numerous similar organizations in the USA, European Union and disease-specific rare disease foundations in India. In this review, we provide a background on the landscape of rare diseases and the organizations that are active in this area globally and in India. We discuss the unique challenges in tackling rare diseases in India, and highlight the unmet needs of the key stakeholders of rare diseases. Finally, we define the vision, mission, goals and objectives of ORDI, identify the key developments in the health care context in India and welcome community feedback and comments on our approach.

Introduction to rare diseases

By definition, a rare disease occurs infrequently in a population, but there is no universal definition. There are three elements to the definition as used in various countries – the total number of persons having the disease, its prevalence and non-availability of treatment for the disorder. This definition was introduced to identify disorders that are neglected by health professionals. A formal definition helps a nation to identify diseases that require financial incentives for discovery and development of drugs and biologics, so as to encourage product development as well as funding for basic and clinical research on those diseases. Many countries define 'rare' or 'orphan' diseases as those affecting less than a specific number of persons in the populations. For example, in the USA, it is defined strictly according to its prevalence, specifically

* Corresponding author: E-mail: harsha@ordindia.org

'any disease or condition that affects less than 200,000 persons' (Shire Human Genetic Technologies, 2013). In Japan, the number is 50,000 persons, in Korea 20,000, in Taiwan 10,000, and in Australia 2000 (Lavandeira, 2002; Tang, 2013). The World Health Organization (WHO) has suggested that a rare disease should be defined as one with a frequency of less than 6·5–10 per 10,000 persons (Song, 2012), although, some experts feel this is rather high (Aronson, 2006). Stated as the prevalence per 10,000, the number used in the USA is 7·5, in Europe 5, in Japan 4, in South Korea 4, Australia 1·1 and Taiwan 1·0 (Song, 2012). In China, a rare disorder is defined as one that affects less than 1/500,000 persons, or one that has a neonatal morbidity of less than 1/10,000 (Song, 2012; Ma *et al.*, 2011). Thus, a country should define a rare disease in the context of its own population, health care system and resources. India, like many developing countries, currently has no standard definition. Considering the large population of India, we suggest the threshold for a disease to be defined as rare to be 1 in 5000. This would include diseases that have a higher prevalence, but do not have definitive therapy. Although this definition may suggest that the number of affected patients is small, it is important to note that when taken together, the number of patients living with a rare disease in India is over 70 million (Verma, 2000). By contrast, about 30 million Americans (Shire Human Genetic Technologies, 2013), and about 29 million persons in the EU are affected with rare disorders (Nogales, 2004).

The exact number of rare diseases is not known, but is estimated to be around 7000–8000 worldwide (Global Genes. RARE Facts and Statistics). With the rapid advances in genomic technologies in the last decade, the number is increasing steadily each year with new diseases and associated genes being discovered. About 80% of rare diseases are genetic in origin, many of which are thought to be monogenic (Global Genes. RARE Facts and Statistics). Rare diseases also include rare inherited cancers, autoimmune diseases, congenital malformations and infectious diseases amongst others. All rare diseases taken together affect about 6–8% of the world's population. About half of the rare diseases affect children causing significant social and economic burden, while the other half manifest in adulthood. Some examples of rare diseases include hemangiomas (Haggstrom, 2006), Hirschsprung disease (Butler Tjaden & Trainor, 2013), Gaucher disease (Rosenbloom & Weinreb, 2013), cystic fibrosis (Ehre, 2014), muscular dystrophies (Mercuri & Muntoni, 2013) and Pompe disease (Ausems *et al.*, 1999).

Treatments for rare diseases

According to a *Thomson Reuters* report (http://thomsonreuters.com/business-unit/science/subsector/pdf/the-economic-power-of-orphan-drugs.pdf), the global market for 'orphan drugs' (drugs that are meant to treat rare medical conditions) accounted for more than $50 billion in 2011. A majority of these diseases lack proper treatment options. A key challenge associated with rare diseases globally is the inability of the medical system to properly diagnose these diseases in a timely manner, leading to a delay in therapy. Early diagnosis is essential for proper disease management. The newborn screening program in the USA covers about 31 metabolic disorders, which, when detected in the neonatal period, can be treated to prevent disability. An example of this is phenylketonuria (PKU), which can be managed nutritionally to save the child from the devastating effects of PKU. Recently, a drug called 'Kuvan' was launched for the BH4 responsive version of PKU (BIOMARIN Pharmaceuticals; http://www.bmrn.com), which precludes the need for expensive dietary therapy. The average time to diagnose most rare diseases in the USA is about 7 years (Shire Human Genetic Technologies, 2013), causing significant anxiety and financial hardship to the families let alone increasing the morbidity in patients. In developing countries, the time to diagnosis is even longer. Even after proper diagnosis, there is little hope for cure. Only around 400 FDA approved 'orphan drugs' are available on the US market, and ~100 drugs approved by the European Medicines Agency (EMA) are available in the EU (Orphanet, 1997). Together, these approved drugs cover only about 11 million patients suffering from rare diseases leaving a majority of patients with no treatment options. Even where treatment is available, the cost is often prohibitive due to high development costs, fewer patients and lack of competition (Nogales, 2004). This is true for the enzyme replacement therapies (ERT) that have been approved for mucopolysaccharidosis types I, II, IV-A and VI, Gaucher disease, Fabry disease and others. The prohibitive costs limit their use to very few patients in India and other low resource countries. Charitable programs started for lysosomal storage disorders by companies like Genzyme, and on a smaller scale by Shire Human Genetic Technologies, are praiseworthy, and provide hope to some patients. These programs have also helped raise awareness among physicians, and stimulated them to make early and precise diagnosis. In India, enzyme therapies are provided either by the Pharma companies under their charitable programs, or by employers in India who are committed to giving 'free' health care to their employees and their dependents. A few associations of families of patients with rare disorders are also trying to persuade the government to cover the cost of therapies. Among the success stories in India is the case of providing Factor VIII to patients with hemophilia A and chelating agents to patients with thalassemia major. In an interesting development a plea was filed in the

Delhi High Court by the father of a 7 year old child suffering from Gaucher disease after being denied treatment by the All India Institute of Medical Sciences (AIIMS), New Delhi, for want of funds. He had lost four children with Gaucher disease. Justice Manmohan, remarking that ‘health is not a luxury’, and ‘should not be the sole possession of a privileged few’, asked the Delhi government to discharge its constitutional obligation and provide the child with ERT at AIIMS, free of cost, when required (Provide free treatment to Gaucher disease patient: High Court. News story at: http://www.newkerala.com/news/2014/fullnews-40628.html#.U22oX2xZrVI).

Global organizations devoted to rare diseases

Numerous organizations across the globe are tackling the challenge of rare diseases head on. The names of some such organizations, with URL addresses, are given below in alphabetical order:

- CORD: Canadian Organization for Rare Disorders (http://www.raredisorders.ca)
- EURORDIS: European Organization of Rare Diseases (http://EURORDIS.org)
- GARD: Genetic and Rare Diseases Information Center (https://rarediseases.info.nih.gov/GARD/)
- HMDSN: Hirschsprung’s and Motility Disorders Support Network (http://www.hirschsprungs.info)
- INOD: In Need Of Diagnosis (http://www.inod.org)
- IRDiRC: International Rare Disease Research Consortium (www.irdirc.org)
- Jain Foundation (http://www.jain-foundation.org/)
- Madisons Foundation (http://www.madisonsfoundation.org/)
- NORD: National Organization for Rare Disorders (http://rarediseases.org)
- ORDR: Office of Rare Diseases Research (http://rarediseases.info.nih.gov)
- Orphanet (http://www.orpha.net)
- RARE – Rare disease, Advocacy, Research, Education (http://globalgenes.org/leadership)
- Rare Genomics Institute (RGI, USA) (http://raregenomics.org)
- Rare Health Exchange (http://rarehealthexchange.org)
- SWAN: Syndromes Without a Name (http://www.undiagnosed-usa.org)
- Vascular Birthmarks Foundation (http://birthmark.org)

ORDI will extend the work of these existing organizations in rare diseases, and collaborate at the international level, to advance the common objective of finding solutions to the problems of rare diseases and advocate for these patients. ORDI has partnered with RGI USA to institute a process for recruiting patients and their families with undiagnosed diseases (suspected to be familial) into exome-sequencing programs to identify potential causal mutations. ORDI is in discussion with other prominent international organizations to explore opportunities for collaboration and is already mutually cross-referenced with several of them.

Indian organizations devoted to rare diseases

Verma reviewed the burden of rare genetic diseases in India in 2000 and subsequently in 2002 and 2004 (Verma & Bijarnia, 2002; Verma, 2000, 2004). Although there has been improvement in the ability to reduce this burden over the years, still the services are inadequate and much remains to be done. Some of the organizations and resources for patients with rare diseases in India are listed below:

- ARDSI – Alzheimers and Related Disorders Society Of India (http://www.alzheimer.org.in)
- Birth Defects Registry of India (http://www.fcrf.org.in/bdri_abus.asp)
- Down Syndrome Federation India (http://downsyndrome.in/)
- Fragile X Society – India (www.fragilex.org)
- Genetic Alliance (http://www.geneticalliance.org)
- Hemophilia Federation (http://www.hemophilia.in/)
- Indian RETT Syndrome Foundation (www.rettsyndrome.in)
- Indian Association of Muscular Dystrophy (www.iamd.in)
- Indian Prader-Willi Syndrome Association (http://pwsindia.hpage.com)
- IPSPI – Indian Patients Society for Primary Immunodeficiency (www.ipspiindia.org)
- LSDSS – Lysosomal Storage Disorders Support Society (www.lsdss.org)
- MERD – Metabolic Errors and Rare Diseases (http://merdindia.com)
- Muscular Dystrophy Association India (http://mdindia.org/)
- Muscular Dystrophy Foundation India (http://www.mdfindia.org)
- Muskaan (intellectually disabled) (http://muskaan-delhi.com/)
- National Thalassemia Welfare Society (http://www.thalassemiaindia.org/)
- Pompe Foundation (http://pompeindia.org/)
- Rare Diseases India (http://www.rarediseasesindia.org)
- Retina India (http://www.retinaindia.org)
- Sjogren’s India (http://www.sjogrensindia.org)
- Thalassemics India (www.thalassemicsindia.org)

These organizations render yeomen services to the patient community. Internationally, umbrella organizations such as NORD, EuroRDIS, Genetic Alliance and Global Genes RARE have played key roles in engaging with stakeholders primarily in the USA and Europe. In India, an organization that can unite all rare disease stakeholders under a single umbrella, and speak in a single voice for them does not exist. The lack of such an umbrella organization has reduced the effectiveness of the above mentioned organizations as much of their resources are directed towards common causes such as raising general public awareness about rare diseases. As a result, progress in assisting patients with rare diseases is slow, leaving many patients hapless. The need for an umbrella organization that can provide a common framework for these disease-specific organizations to function effectively and to focus on their mission is clear. Such an organization could provide generic patient registries compliant with regulatory requirements in India, develop and maintain a comprehensive information portal about rare diseases, create and maintain a sample biorepository for use by Indian rare disease researchers in approved translational research studies, interface with international resources, broadcast best practices, raise public awareness about rare diseases, host national and international conferences and other events to engage key stakeholders, and create an ecosystem of incentives to accelerate research, development and delivery of affordable diagnostics and treatment options for patients with rare diseases. Bringing the various associations under one organization will give it greater leeway to lobby with the government, international agencies and philanthropists for help and support. It is ORDI's objective to fill these gaps. It must be stated that getting these different organizations in India to work under one umbrella organization would not be an easy task. In this context, the response from Indian organizations, since the launch of ORDI on 18 February 2014 in Delhi, has been most encouraging. The ORDI's nationwide rare disease telephone helpline receives on average, 3–4 enquiries every day. A total of 13 Indian organizations have already joined, as they are convinced that joint effort will be more rewarding than their individual efforts.

History of rare diseases in India

Extensive haplotyping studies have predicted that most present day Indian populations are descendants of ancestors who migrated out of Africa to the Indian continent about 65,000 years ago (Tamang *et al.*, 2012) Evidence suggests that today's Indian population is an admixture of two genetically divergent ancient populations, referred to as the Ancestral North Indians (ANI), who are genetically close to Middle East, Central Asian and European populations, and the Ancestral South Indians (ASI), who are less so (Reich *et al.*, 2009). This study estimated that ANI ancestry ranges from 39–71% among the various current Indian populations, with pure ASI groups represented by the indigenous Andaman Islanders (Reich *et al.*, 2009). Consanguineous marriages take place preferentially in many communities, while in other ethnic groups, endogamous marriages have occurred over long period of time. As a result, the frequencies of founder and common mutations are likely to be relatively higher in the Indian subpopulations. Rough estimates show that more than 56 million individuals in India are likely to be affected by single gene disorders (monogenic disorders) (Global Genes. RARE Facts and Statistics). With the lack of awareness in the general population about genetic disorders, and scarcity of specialized medical professionals and affordable genetic tests, the burden from these disorders is growing rapidly. This is, in part, due to the absence of a properly functioning social health care system in India, where the health professionals, including doctors and nurses, are not given enough exposure to medical genetics, molecular biology and rare disorders in their curriculum. There is also insufficient encouragement by the government for individual health insurance. Consequently, most of the population, especially in rural areas, does not opt for prenatal testing, predictive genetic diagnosis or timely genetic counseling with some families having more than one affected individual. The WHO has stressed the need for prevention, early diagnosis and management of genetic disorders in developing countries, and has issued detailed guidelines (http://www.doh.gov.za/docs/policy/humangenetics.pdf).

Over a decade since the completion of the Human Genome Project, awareness about genetic disorders among physicians as well as the general population of India is still lacking. As a result, early and affordable diagnostic tests, even where available, are not widely prescribed. To remedy this, a few accredited and reputed educational institutions, hospitals and laboratories across the country have initiated genetic diagnostic services covering selected disorders. A directory of accredited genetic testing service centers in India compiled in 2007 (Singh *et al.*, 2010) showed that there were 47 such centers offering genetic services, including cytogenetic (40 centers), biochemical (26 centers) and molecular diagnosis (26 centers), along with genetic counseling. A current directory of genetic centers and services is now available online http://www.geneticsindia.org. This is supported by the Indian Council of Medical Research (ICMR). It has listed 649 disorders, 66 genetic centers and 35 prenatal diagnostic centers. Another ten genetic counseling centers are known to the authors that are not yet listed on the website. It has also started listing the parent support groups for various diseases in India. The number and distribution of these centers are abysmally small to serve the massive Indian rare disease community. Many of the centers need to upgrade capabilities to include recent advances such as next-generation sequencing (NGS)-based molecular diagnosis of rare diseases. Most of the genetic centers offer targeted tests that involve screening for common mutations, although in recent years, many centers provide sequencing of entire disease-associated genes. A number of private companies have set up NGS technologies, harnessing the excellent bioinformatics resources available in India. They are providing molecular diagnosis based on NGS of disease gene panels at a fraction of the cost of these tests abroad, making them affordable to the Indian population. However, some samples are still sent abroad for genetic testing. International laboratories (Quest, Centogene, BGI China and Core Diagnostics) have also set up offices in India. Some Indian national laboratories such as SRL Labs and Lal Path Labs also offer advanced medical and genetic tests. Therefore, an extensive network of collection centers has spread all over India. This enables every person, even in small remote towns, to provide samples for advanced genetic and biochemical tests. This has tremendously improved the diagnostic abilities for genetic tests in the country.

Education and genetic counseling (GC) are critical necessities to help patients and physicians deal with rare diseases. GC is needed at various levels such as prior to genetic testing, post-testing, prenatal diagnosis and family planning particularly in consanguineous marriages. GC needs to be made an integral part of all genetic testing centers in India. This has been realized by the private genetic laboratories and they have appointed genetic counselors on their staff. There are at least three institutions offering courses in GC for non-medical personnel. There is a move by one of the authors of this paper to initiate a GC course under the Indira Gandhi Open University. ORDI will provide expertise for this venture by enlisting experts from the USA to assist the faculty in India. This will, to some extent, fill the gap of qualified genetic counselors in India.

The burden of providing care for an individual affected by rare disease is not easy to meet in India due to lack of infrastructure. Therefore, determining carrier status for genetic diseases can help individuals make decisions about prenatal testing when both partners are carriers of a particular disease. In urban areas at least, most obstetricians screen for common infections and thalassemia. After that, the family history guides the screening strategy. Carrier testing for genetic diseases using NGS of a panel of genes is being offered by one company now, while others are in the process of validating such tests. A number of institutions, both in the public and private domains, provide GC to patients.

The National Board of Examinations, with support from the Department of Biotechnology, Government of India, is due to start a national postgraduate course in Medical Genetics and Genomics. The Medical Council of India (MCI) has expanded the curriculum for medical students to include genetics and molecular biology (Vision Document 2015: http://www.mciindia.org/tools/announcement/MCI_booklet.pdf). What is still woefully inadequate is the number of medical genetics departments in the country. To meet the shortage of medical doctors and to ensure that rural populations are better served, the Ministry of Health and Family Welfare, Government of India, pushed for a 3 year B. Sc. course in Community Medicine to be imparted to doctors in Ayurveda and other indigenous systems of medicine. The course has been approved by the MCI (Sinha, 2012). The number of seats in medical courses, at both the graduate as well as the postgraduate level has increased by 1·5 to 2 times in a short time span. Kapoor *et al.* (2013) reviewed the challenges and opportunities for newborn screening (NBS) in India and recommended widespread implementation of the NBS program across the country starting with the metro cities. At least six private laboratories offer newborn screening through tandem mass spectrometry and gas chromatography mass spectrometry at a nominal fee. Many private hospitals have instituted NBS programs. Some states like Gujarat, Chandigarh, Delhi, Maharashtra, Kerala, Goa and Tamil Nadu, have started pilot NBS programs, and it is expected that these will be further extended to cover larger populations in the near future.

India-specific challenges in the management of rare diseases

India faces numerous challenges in awareness, public perceptions, diagnosis, treatment and public policy on rare diseases. Some of the most significant ones are listed below.

Lack of awareness

There is significant lack of awareness about rare diseases among the lay public, and unfortunately even among physicians in India. The health care training and education system in India are more focused on training physicians and health care personnel to treat common diseases such as infectious diseases, diabetes mellitus, cardiovascular disease and common cancers. This has led to a pronounced dearth of trained physicians and health care personnel to care for patients with rare diseases.

Lack of infrastructure

Inadequate training and facilities to properly diagnose rare diseases in a timely manner is another disadvantage. If it takes an average of 7 years to diagnose a rare disease in developed nations like the USA (Shire Human Genetic Technologies, 2013) the average time to diagnose is likely to be greater in developing countries (Christianson & Modell, 2004). It is unclear as to the average time that it may take to diagnose a rare disease in India. There is lack of adequate statistics and data on incidence as well as prevalence of rare diseases in India. A systematic catalogue of the different rare diseases in India does not exist. The availability of such basic information is essential for use by policy makers, physicians, scientists, drug or device manufacturers, patients and the community at large.

Prohibitive costs

A majority of patients with rare diseases in India cannot afford the high costs of treatments even when available. Most orphan drugs are not curative but palliative, and need to be administered regularly. These drugs are highly expensive and inaccessible to the majority of the Indian population affected by rare diseases.

Cultural influences

The incidence of rare diseases is believed to be higher in India compared to western countries due to practice of consanguineous marriages in many communities. Social stigmas for disabled individuals and patients with rare diseases continue to be a societal challenge that can only be addressed with education and awareness.

Government initiatives

There seems to be no official policy on rare diseases in India. There is no specific push for research and development in the field of rare diseases in India, although the programs by funding agencies like the ICMR, Department of Science and Technology (DST), Council of Scientific and Industrial Research (CSIR) as well as the Department of Biotechnology (DBT) are noteworthy. The Government should encourage and fund Indian academic research laboratories as well as pharmaceutical and biotechnology industries to take up scientific research work leading to the development of diagnostics and drugs for rare diseases. ORDI would persuade and assist the government to (i) enact an act similar to the Orphan Drugs Act (ODA) of USA, and (ii) create a fund to support work related to rare disease research, education and treatment by institutions that are prepared to take up this work. The Government has already started funding such work in a small way but we believe that a lot more needs to be done. A national plan for rare disease research needs to be developed.

Funding

The lack of significant public funding for rare disease research is another critical impediment in creating the much needed momentum for education, diagnosis and treatment of rare diseases in India.

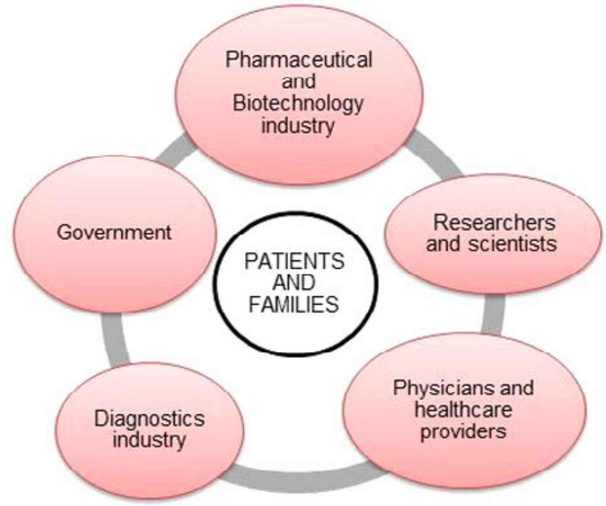


Fig. 1. Key stakeholders of rare diseases in India.

The rare disease stakeholders in India

Fig. 1 depicts the key stakeholders of rare diseases in India. The patients and their families are the most important stakeholders as they experience the disease first hand. The other important stakeholders such as physicians, health care providers, researchers, diagnostic laboratories, biotechnology and pharmaceutical companies, government, regulatory bodies, non-profit patient organizations and health insurance companies have a significant impact on the rare disease patients and their families.

Patients and families

Patients with rare diseases and their families need quick and easy access to information about affordable diagnosis and treatment options. A single portal for such information would serve the purpose and is urgently needed. The ORDI has already provided such a portal and receives many enquiries every day. The existing parent support groups also play an important role in providing such information, and in providing a forum for parents of children with specific diseases to exchange experiences.

Physicians and health care providers

Indian doctors need access to more clinical training, research and education on rare diseases. In addition, they need access to state-of-the-art diagnostic tests, information on clinical trials and currently available drugs for rare diseases in India. ORDI could help organize workshops to meet this need. The ratio of doctors to patients in India continues to be significantly lower compared to those in developed countries and compared to the ratio recommended by WHO. Training in medical genetics needs to be scaled up urgently across the country.

Scientists and researchers

Indian scientists need access to rare disease patient registries and biorepositories to advance science and understanding of these diseases in order to translate research findings into diagnostic tests and therapeutics. Biorepositories that house patient samples and associated clinical, genomic, transcriptomic, proteomic and metabolomic data will be the main catalysts of therapeutic and diagnostic solutions for rare diseases. Defective genetic pathways contribute to a great extent to the pathogenesis of rare diseases (Global Genes. RARE Facts and Statistics). Molecular diagnostic technologies such as NGS offer great opportunities to sequence the genome or targeted regions of a patient's DNA at an affordable cost to identify mutations that may be associated with a disease. There is a need and immediate opportunity to accelerate the identification of as yet unknown critical genes involved in rare diseases. To establish a strong association between a mutation and disease, a large number of patient samples are often required. Dalal *et al.* (2012) report on an Indian cohort study involving 35 progressive pseudorheumatoid dysplasia patients harboring mutations in the *WISP3* gene. Such studies will clearly benefit the populations in the Indian subcontinent. Inherited disease-causing mutations could be detected by comparing DNA sequencing data from the trio (patient and parents). Being able to access patient samples could help identify associated mutations that could lead to drug targets and thereby therapies. Moreover, identification of rare disease genes and mutations can advance our overall understanding of underlying biological mechanisms of more common human diseases. For example, understanding the involution process in vascular tumors such as hemangiomas (Haggstrom *et al.*, 2006), a rare disease, will benefit the understanding of human solid tumor regression biology.

Pharmaceutical and biotechnology industries

The lack of an Orphan Drug Act type of legislation in India is hampering the development of indigenous development of drugs for rare diseases. While the opportunity for growth in the rare disease market is high from patients/providers' perspectives, industry faces mixed benefits and barriers. Orphan drugs are relatively non-profitable as rare diseases affect a small patient population, and usually are unaffordable to Indian patients as the drug cost per patient far exceeds the per capita income. Hence, the Indian pharma/biotech industry needs incentives for investing in research and development of orphan drugs. A stable and reliable regulatory environment conducive to investments in long-term R&D is also desired. If the Indian pharmaceutical companies take up manufacturing of products for rare disorders this would bring the cost down not only for the Indian market, but also for the whole world. An example is the manufacture and supply of anti-retroviral drugs by CIPLA, a pharmaceutical company in India that helped tremendously in controlling AIDS across the world. Some global pharma companies, such as Genzyme, have developed charitable access programs for orphan drugs in developing countries including India. The Gaucher initiative to provide Genzyme's first approved orphan drug (1999), called Cerezyme, is one such example. Such initiatives work with humanitarian organizations but a commitment from the Government of India to support the cause of rare diseases in the form of an ODA will clearly improve accessibility to orphan drugs. The policy to import orphan drugs into India also needs a fresh re-evaluation, as companies should be permitted to import free of duty.

Government of India

The support of the Indian government is critical to the success of health initiatives such as the one we are proposing for rare diseases. The government should play an active role in addressing the enormous health care challenges posed by rare diseases through various mechanisms such as public funding for research in orphan diseases, creating business friendly policies for pharmaceutical and biotech companies and developing a balanced regulatory framework that catalyses innovation and protects the safety of patients.

ORDI vision

Our vision is to make rare diseases as easily diagnosed and treated as common diseases are in India. Collection of epidemiologic data, catalysing research and facilitating creation of registries and biorepositories would be high on our agenda.

ORDI mission

We at ORDI aim to be the umbrella organization, uniting and providing a common forum for all individual disease-specific organizations in India. It will have branches in all the state capitals. It will collaborate with other parent support groups and help to initiate new parent groups for disorders that currently do not have one. It will obtain funds from corporate houses, pharmaceutical companies, private genetic laboratories and foundations both in India and abroad. It will formulate plans of action on various topics involving rare disorders, such as epidemiology, natural history, mechanisms of disease and treatments. It will seek support for these plans from the Government of India – DST, ICMR, DBT, CSIR, other Government agencies, and philanthropists. Memberships will be available for parent support organizations, individual patients, various categories of health professionals, hospitals (both public and private), corporate houses and pharmaceutical companies for the following mission:

- Create awareness for rare diseases all over India using mass media, newspapers, television, social media, pamphlets and posters.
- Set up patient registries for the more prevalent 'rare disorders'.
- Represent the collective voice and concerns of over 70 million patients with rare diseases in India.
- Obtain concessions from the local governments (as health is a state subject in India) for travel, treatments and jobs.
- Work with the government of India to create an optimal business and regulatory environment for the diagnostics and drug development industry (pharmaceutical and biotechnology).
- Catalyse rapid development and delivery of affordable diagnostics and treatments for rare diseases in India through innovative collaborations and partnerships among stakeholders.
- Advocate investments in rare disease research, diagnostics and drug development.
- Work with the insurance regulatory agency to ensure non-discrimination in health insurance based on the genetic constitution of an individual.
- Organize national and international conferences in India on rare disorders to create awareness and promote research and development of therapies.
- Provide assistance for rare disease patients to the largest possible extent.
- Service a 24/7 rare disease helpline that has been launched to provide assistance to patients. A proposal by ORDI to develop this helpline into a comprehensive rare disease care coordination center is currently being reviewed by pharmaceutical companies in India.

Future perspectives

Although the challenges for rare diseases appear to be of Himalayan (pun intended) proportions, the advent of precision medicine, otherwise referred to as personalized medicine, offers hope in the diagnosis and treatment of rare diseases. Precision medicine is being applied in relatively common disorders such as cancer, immune diseases and infectious diseases. It is based on the premise that by stratifying patients with similar genomic, transcriptomic, proteomic and metabolic biomarker profiles, it is possible to direct specific therapies to achieve higher levels of efficacy and reduce drug toxicities. The 'blockbuster model' of pharmaceutical companies is no longer tenable in this scenario and hence they are already developing therapies for smaller patient populations with relatively common disorders such as cancer, immune diseases and infectious diseases. All of the recent epochal advances in drug development, diagnostics and regulatory paradigms are likely to have a beneficial impact in driving precision medicine into rare disease clinics across the world.

India has a high potential to cost-effectively develop and manufacture small molecule drugs, biologics and vaccines for rare diseases due to its inherent capabilities in drug and vaccine development and manufacturing (Smita, 2006; Chakma *et al.*, 2011). India is already considered as a global hub for vaccines as it supplies close to half of the world's childhood vaccines (Virk, 2010). In addition, the small molecule drug development and manufacturing capabilities of Indian pharmaceutical companies are well recognized, especially in the generic market segment (Genetic Engineering and Biotechnology News, 2006; Kale, 2012). The biosimilars (generic biologic drugs that are produced using recombinant DNA technology such as monoclonal antibodies, hormones and cytokine therapeutics) segment is likely to experience significant growth in India in the coming years with numerous biologics likely to go off patent adding to India's capabilities in the biologic manufacturing arena (Mukherjee, 2010).

The rapid growth and innovations in molecular diagnostics and bioinformatics globally are also likely to have a beneficial impact on precision diagnostics for rare diseases. The advent of NGS in the clinic has led to the application of multi-gene, exome and whole genome-based diagnostic tests. India's strength in information technology and bioinformatics will be highly beneficial in the development and democratization of precision diagnostics for rare diseases.

The government of India has a crucial leadership role to play in advancing progress in this area. We persuade the government agencies to sponsor various activities beginning with a national level assessment of the needs of various stakeholders in the rare diseases community. India can draw upon the model that the US government developed to support rare diseases. The US government funded rare disease community needs assessments on three occasions: first, in the 1970s, approximately 5 years prior to the enactment of ODA; second, in the 1980s about 5 years after the ODA and then as part of a Special Emphasis Panel on the Coordination of Rare Diseases Research in 1998. The surveys completed examined the needs and priorities of the patients/families, physicians and medical specialists, research investigators, voluntary health organizations (patient advocacy groups), the pharmaceutical industry, and philanthropic foundations, regulatory agency (FDA), biomedical research agency (National Institutes of Health (NIH)) and other government agencies such as Health Resources and Services Administration (HRSA) and the Centers For Disease Control And Prevention (CDC). According to the office of rare disease research (ORDR) at the NIH, these surveys helped identify the needs and opportunities to implement specific recommendations provided by the stakeholders. In many cases, these recommendations even became part of the legislative initiatives such as the ODA in 1983 and then the Rare Disease Act of 2002. Innovative government funded programs such as the Rare Disease Clinical Research Network (RDCRN; http://rarediseasesnetwork.epi.usf.edu), have made significant positive impact to the overall cause of rare disease patients in the USA. We urge the government of India to similarly take a lead in initiating and funding such needs assessments formally, to pave the way towards developing a roadmap for tackling rare diseases in India.

Developments in India

In recent years, some significant developments have taken place that will change the future of health care in India. First, the launch of the Rasthtriya Bal Swasthya Karyakram (National Health Program for Children), on 6th February 2013 (Rashtriya Bal Swasthya Karyakram, 2013). It covers 270 million children starting from birth to 18 years of age, in a phased manner and moving towards the goal of Universal Health Coverage. This program screens for 30 health conditions among children including defects at birth, deficiencies and diseases, development disabilities and also helps manage these conditions. The following conditions have a significant genetic component and will be screened and managed: impairment of vision, hearing, neuro-motor system, delay of motor functions, cognition and language, autism and attention deficit hyperactivity disorder. Birth defects such as neural tube defects, Down syndrome, cleft lip and palate, club foot, developmental dysplasia of hip, congenital heart disease, congenital deafness and congenital cataracts will also be included. Second, is the significant lowering of the infant mortality rate (IMR) across India due to government initiatives in health care. The IMR is currently 42 per 1000 at the national level, 46 in the rural areas and 28 in the urban areas (Mukherjee, 2010). Many states have an IMR of less than 20. The WHO has recommended that genetic services should be established without fail in countries with an IMR of less than 50 (Sample Registration System Bulletin, 2013). Third, is the changing pattern of diseases in India from communicable and nutritional disorders to predominance of non-communicable disorders. Finally, an enlarging and expanding private health sector and genetic laboratory services are driving health care and public awareness for genetic disorders. These events are likely to effect a remarkable change in the care and cure of rare disorders in India. ORDI has its work cut out to play a major role in this transformation.

Contact us

Visit us online at www.ordindia.org. Email your comments and suggestions to contactus@ordindia.org. We have launched the first rare disease telephone helpline in India to guide patients with rare diseases through their journey by connecting them with experts and parent support groups: +91 8892 555 000.

Funding

No specific funding was available to the authors to carry out this work.

On 16 May 2013, the first broader brainstorming session was organized to gather feedback from various stakeholder groups in India. We received valuable suggestions from the diverse participants at this event hosted by Strand Life Sciences, Bengaluru, India, as well as offline from thought leaders in India and abroad. Individual team members have gathered feedback from a large number of relevant people via informal discussions, e-mails, the Facebook page and LinkedIn group discussions. Much of this feedback has defined the core principles of ORDI as a neutral, non-profit organization. The authors wish to thank all organizations and individuals for volunteering their time and sharing their perspectives during informal meetings, discussions, brainstorming sessions, and teleconference calls. Many of these organizations are listed in the body of this manuscript. In particular, we thank Dr Stephen C. Groft and Dr Rashmi Gopal-Srivastava at the NIH office of rare disease research for their guidance, support and contribution, based on 30+ years of experience in rare diseases in the USA and worldwide. Dr Rajat Agrawal from Retina India and Dr Linda Shannon-Rozell from vascular birthmarks foundation are among other organizations that offered detailed discussion and suggestions. We thank everyone who has contributed directly or indirectly to the formation of this important initiative.

References

Aronson, J. K. (2006). Editor's view – rare diseases and orphan drugs. *British Journal of Clinical Pharmacology* **61**, 243–245.

Ausems, M. G., Verbiest, J., Hermans, M. P., Kroos, M. A., Beemer, F. A., Wokke, J. H., Sandkuijl, L. A., Reuser, A. J. & van der Ploeg, A. T. (1999). Frequency of glycogen storage disease type II in the Netherlands: implications for diagnosis and genetic counseling. *European Journal Human Genetics* **7**, 713–716.

Butler Tjaden, N. E. & Trainor, P. A. (2013). The developmental etiology and pathogenesis of Hirschsprung disease. *Translational Research* **162**, 1–15.

Chakma, J., Masum, H., Perampaladas, K., Heys, J. & Singer, P. A. (2011) Indian Vaccine Innovation: the case of Shantha Biotechnics. *Globalization and Health* **7**, 9.

Christianson, A. & Modell, B. (2004). Medical genetics in developing countries. *Annual Review of Genomics and Human Genetics* **5**, 219–265.

Dalal, A., Bhavani, G. S. L., Togarrati, P. P., Bierhals, T., Madhusudan, R., Nandineni, , Danda, S., Danda, D., Shah, H., Vijayan, S., Gowrishankar, K., Phadke, S. R., Bidchol, A. M., Rao, A. P., Nampoothiri, S., Kutsche, K. & Girisha, K. M. (2012). Analysis of the *WISP3* gene in Indian families with progressive pseudorheumatoid dysplasia. *American Journal of Medical Genetics Part A* **158A**, 2820–2828.

Ehre, C., Ridley, C. & Thornton, D. J. (2014). Cystic fibrosis: an inherited disease affecting mucin-producing organs. *International Journal of Biochemical Cell Biology* **2014**, pii: S1357–2725(14)00086–7. doi:10.1016/j.biocel.2014.03.011.

Genetic Engineering and Biotechnology News. (2006). Drug discovery and development in India: an untapped seam of outsourcing opportunities. Vol. 26, No. 3, 1 February (http://www.genengnews.com/gen-articles/drug-discovery-and-development-in-india/1314/). Accessed July 2014.

Global Genes. RARE Facts and Statistics (http://globalgenes.org/rarefacts). Accessed March 2013.

Haggstrom, A. N., Drolet, B. A., Baselga, E., Chamlin, S. L., Garzon, M. C., Horii, K. A., Lucky, A. W., Mancini, A. J., Metry, D. W., Newell, B., Nopper, A. J. & Frieden, I. J. (2006). Prospective study of infantile hemangiomas: clinical characteristics predicting complications and treatment. *Pediatrics* **118**, 882–887.

Kale, D. (2012). Innovative capability development in the Indian pharmaceutical industry. *International Journal of Innovation and Technology Management* **9**, 1250013.

Kapoor, S., Gupta, N. & Kabra, M. (2013). National newborn screening program – still a hype or a hope now? *Indian Pediatrics* **50**, 639–643.

Lavandeira, A. (2002). Orphan drugs: legal aspects, current situation. *Haemophilia* **8**, 194–198.

Ma, D., Li, D. G., Zhang, X. & He, L. (2011). The prevention and treatment on rare diseases in China: opportunities and challenges. *Chinese Journal of Evidence Based Pediatrics* **6**, 81–82. (in Chinese).

Mercuri, E. & Muntoni, F. (2013) Muscular dystrophy: new challenges and review of the current clinical trials. *Current Opinions in Pediatrics* **25**, 701–707.

Mukherjee, S. (2010). Pharmaceutical R&D: India at the crossroads. *Financial Chronicle* December 2010.

Nogales, E. A. (2004). Rare diseases: a new chapter in medicine. *Anales de la Real Academia Nacional de Medicina* **121**, 139–151; discussion 151–155. (in Spanish; Castilian).

Orphanet: an online database of rare diseases and orphan drugs. Copyright, INSERM 1997 (http://www.orpha.net). Accessed May 2014.

Rasthtriya Bal Swasthya Karyakram (2013). Child health screening and early intervention services under NRHM. Ministry of Health and Family Welfare, Government of India, February 2013 (http://www.unicef.org/india/7). Accessed July 2014.

Reich, D., Thangaraj, K., Patterson, N., Price, A. L. & Singh, L. (2009). Reconstructing Indian population history. *Nature* **461**, 489–494.

Rosenbloom, B. E. & Weinreb, N. J. (2013). Gaucher disease: a comprehensive review. *Critical Reviews in Oncogenesis* **18**, 163–175.

Sample Registration System Bulletin. (2013). Registrar General India, New Delhi. September **48**, 2 (http://tripuranrhm.gov.in/pdf/SRS_BULLETIN_SEPETEMBER_2013.pdf). Accessed July 2014.

Shire Human Genetic Technologies (2013). Rare disease impact report (http://www.rarediseaseimpact.com). Accessed April 2014.

Singh, J. R., Singh, A. R. & Singh, A. R. (2007). Directory of Human Genetic Services in India 2007. *International Journal of Human Genetics* **10**, 187–192.

Sinha, K. (2012). Medical Council of India approves 3·5 years medical course. The Times of India, 24 September. (http://timesofindia.indiatimes.com/home/education/news/Medical-Council-of-India-approves-3-and-a-half-year-medical-course/articleshow/16523446.cms). Accessed July 2014.

Smita, S. (2006). Industrial development and innovation: some lessons from vaccine procurement. *World Development* **34**, 1742–1764.

Song, P., Gao, J., Inagaki, Y., Kokudo, N. & Tang, W. (2012). Rare diseases, orphan drugs, and their regulation in Asia: current status and future perspectives *Intractable and Rare Diseases Research* **1**, 3–9.

Tamang, R., Singh, L. & Thangaraj, K. (2012). Complex genetic origin of Indian populations and its implications. *Journal Bioscience* **37**, 911–919.

Tang, W. (2013) Editorial. Hopes for intractable and rare diseases research. *Intractable and Rare Diseases Research* **2**, 1–2.

Verma, I. C. & Bijarnia, S. (2002). The burden of genetic disorders in India, and a framework for community control. *Community Genetics* **5**, 192–196.

Verma, I. C. (2000). Burden of genetic disorders in India. *Indian Journal of Pediatrics* **67**, 893–898. Erratum in: *Indian Journal of Pediatrics* (2001). 68, 25.

Verma, I. C. (2004). Genetic disorders and medical genetics in India. In *Genetic Disorders of the Indian Subcontinent* (ed. D. Kumar), pp. 50–518. Dordrecht: Kluwer Academic Publishers.

Virk, PK (2010). India's future as a biologics manufacturing hub. *BioPharm International* **23**, 1.

Boldness has genius, power, and magic in it.

BITS Pilani 2016 Convocation Address, July 24th 2016

Vijay Chandru

Thank you Chancellor Shri K M Birla, Vice-Chancellor Professor Souvik Bhattacharya, BITS Board of Governors and the BITS community for according me the great honour of addressing the class of 2016. I recall the last time I was up on this stage at my alma mater BITS Pilani. This was as a student 41 years ago in 1975 as an amateur actor in a play staged by the drama club. My hearty congratulations to the graduating students, the distinguished faculty and the proud family members who have supported their loved ones in achieving excellence in education from a young age to successful graduation from Pilani which stands high among the list of India's finest universities.

It is the vision of the Birla family starting with GD Babu that has guided this University from its founding, on the fringes of the Rajasthani desert, more than five decades ago to being placed ninth in the list of India's leading universities for the year 2016. This achievement is singular because BITS Pilani is the only private institution amongst the top ten universities in a ranking framework created by a public agency, i.e., the National Institutional Ranking Framework of the MHRD, Govt. of India. This achievement is the cumulative effort of dozens of university administrators, hundreds of teachers and thousands of students across your three national campuses at Pilani, Goa and Hyderabad as well as your international campus at Dubai.

There is a massive wave of entrepreneurial energy coursing through the nation's arteries. If we could connect this enthusiasm with the excellence in basic and applied research at our higher educational institutions, the possibility of a new growth engine that has more enduring value to society seems within reach. This needs a new alignment around the theme of "translational research" – a concept that is easy to describe but hard to execute as it disturbs the natural order. In my mind, BITS should be projected as an example by agencies such as the NITI Aayog in transforming India through translational impact. Industry engagement is one of the cornerstones of the educational philosophy of BITS Pilani exemplified by the *Practice School*, and the *Work Integrated Learning Programmes* which are outstanding models of industry-academia cooperation.

As I look at you, the graduating class of 2016 of BITS-Pilani, I am both awestruck and nervous. I am awestruck because you represent such an awesome force of nature, a large auditorium full of exceedingly bright, well-educated and highly talented young scientists and engineers. Many of you will become leaders and will also have a great impact through your chosen careers in the industry as well as on society in all corners of the country and across the world. You will doubtless make your families and your alma mater proud with your creativity and success. I have only one simple piece of advice.

"Whatever you can do, or dream you can do, begin it. Boldness has genius, power, and magic in it. Begin it now." These lines are from the epic play "Faust" written by the German writer von Goethe who goes on to say that the moment one definitely commits oneself, then Providence moves too - भाग्य बहादुर का साथ देता है| (Fortune favours the brave).

If I have had some success in my academic and professional endeavours it has been because of the power of commitment and perseverance.

As a student at BITS- from 1970-1975, I had not only the good fortune of the tutelage of the inspiring faculty of BITS , but was also exposed to great intellectual minds from across the world. In our final year, the legendary director of BITS, Dr Chittaranjan Mitra, had invited Dr Stafford Beer to visit and give a few lectures about Operational Research and his adventures in Chile as a technical advisor to Salvador Allende. I was enthralled by Beer's descriptions of the field of "cybernetics" and its application to real world systems. I had found my academic passion - my segue into a career in research.

After getting a master's degree in engineering systems at UCLA, I opted to do a PhD in Operations Research at MIT. My guru and friend Professor Jeremy F Shapiro of MIT was a special man. Jerry taught me more than anyone else how to be courageous and bold in research. He was never one to be unduly impressed by "reigning paradigms" - mutual admiration clubs trapped in cul-de-sacs of specialization. If my career has taken on such a variety of experiences, it is because of Jerry's encouragement to seek them.

In years to come, many of you will also realize and value how much you have learned and benefitted from your Gurus at BITS. In 2009, my batch of 1975 decided to honor our faculty by creating the BITS 75 Charitable Trust. From this endowment, we have honoured over 27 of our BITS gurus through a Guru Dakshina program. As recently as this January, Professors Kundu, Arora, Chaturvedi and Radhakrishnan were honoured here on campus by our batch. I do hope the graduating class of 2016 will keep such traditions going.

I began by saying that I am also nervous. Why am I nervous? I am nervous because you have such enormous responsibilities in the days ahead. India is blessed with what we have been calling a "demographic dividend" – a large pool of young people entering the workforce relative to much of the developed world which is ageing rapidly. This also represents an urgent need for disruptive change; that bold reforms are the order of the day. India needs to generate 115 million non-farming jobs over the next decade, to gainfully employ its workforce and reap its "demographic dividend".

Given this context, encouraging and promoting self-employment as a career option for young people will be of highest importance. The culture of entrepreneurship should be inclusive and focus on a variety of enterprises, such as young and innovative technology firms, upcoming manufacturing businesses and rural innovator companies. Entrepreneurs should also be encouraged to help solve pressing socio-economic problems. As Peter Drucker says, entrepreneurs create something new, something different; they change or transmute values. This scale of employment can only be created by entrepreneurship.

The drumbeat on innovation and entrepreneurship has been rolling for over a decade in India. There are many gurus writing, speaking and extolling the virtues of innovation. Grassroots innovators, business innovators, incubators, accelerators and entrepreneurs are now at every corner-in both urban and rural India. Our country now officially has the world's second largest collection of entrepreneurs. A google search on "entrepreneur, founder" in a 5km radius around Koramangala in Bangalore yields a hit of 9,662 LinkedIn sites. While etymologists may argue that the word "entrepreneur" strictly has French and Latin origins, I am tempted to believe that it may have derived from the Sanskrit "Anthah-Prerna" or "self-motivated".

In a small way, our batch of 1975 has also tried to seed and mentor entrepreneurship in our BITS campuses. Two startups that have benefitted and are flourishing are Phyzok in e-learning and Sattva in the medical device space. Phyzok is already profitable and has won many awards, and Sattva was selected by PMO to make a presentation in the presence of the PM in Silicon Valley early this year.

There is a dire need to find financially viable solutions for the challenges of the disenfranchised. This will be of paramount importance, so as to ensure long-term social and economic stability. India needs a model that pulls along the 350-400 million people that currently reside outside mainstream society. Social inclusion not only fulfils higher altruistic purposes, but can also be financially viable. Furthermore, measures taken to enhance social inclusion would inevitably result in the opening up of a significant new market. We are at an inflection point in history where Science and Technology can play an extraordinary role in transforming our society and our country. It is not very different from what some nations including the US faced at the end of World War II just as India became independent. Let us take a look at the US at that stage.

President Roosevelt had challenged Professor Vannevar Bush, the director of the office of scientific research and development to make a plan for the post war period and the latter did just that based on the classical declaration “Science the endless frontier” that he penned in July 1945. “Scientific progress is one essential key to our security as a nation, to our better health, to more jobs, to a higher standard of living, and to our cultural progress. Science can be effective in the national welfare only as a member of a team, whether the conditions be peace or war. But without scientific progress no amount of achievement in other directions can insure our health, prosperity, and security as a nation in the modern world."

India, impoverished by colonial rule for over two centuries also made deliberate choices as a free nation to leverage science and technology to regain its lost glory. And naturally it has succeeded in some quarters and not in others. After close to seven decades we have witnessed a few examples of science driven innovation that have been quite noteworthy. The list below is representative:

- In agriculture, we have moved from a status of massive famines to self sufficiency in gross output albeit with challenges in storage and equitable distribution of supplies.
- In automobile manufacturing we moved from archaic hand-me-down models from Europe to the ability to design, manufacture and export electric sedans, MUVs, mini-Vans, Farm Tractors, …
- Design and manufacture electronic voting machines and successfully scale to run the world’s largest democratic elections electronically.
- Formulate and manufacture in the world’s fourth largest (by volume) production facilities of pharmaceutical therapeutics.
- We are the world’s largest manufacturer of vaccines with approximately 33% market share.
- A software development and software enabled services – scaled to a 150B$ industry
- Demonstrated ability to develop indigenous missile and rocket science, launch satellite and space missions.

After WWII, Professor Vannevar Bush returned to Cambridge, Massachusetts to guide both MIT and Raytheon for a couple of decades. According to the Kauffman Foundation report in 2009, MIT alumni founded companies generated over two trillion US dollars in revenue a year (the 11th largest economy, when viewed as the GDP of a nation) and provided gainful employment to over 3.3 million people. We should similarly assess the impact of enterprise driven by BITS alumni and while the numbers may be modest in comparison to begin with, it would be a valuable metric to track as a measure of social impact.

I circle back to the issue of leadership and while congratulating you once again for your brilliant performances as young graduates of this great campus, exhort you to become someone whose actions inspire others to dream more, learn more, do more and become more. And remember whatever you can do, or dream you can do, begin it now. Boldness has genius, power, and magic in it.

Thank you.

Professor Vijay Chandru
chandru@alum.mit.edu

Vijay Chandru (PhD, MIT '82) is an academic turned entrepreneur. His academic career in data science spanned over two decades as a professor of computational mathematics first at Purdue University and then Indian Institute of Science. He is a fellow of the national academies of science and engineering in India and the winner of the Presidents Medal of INFORMS in 2006.

A Technology Pioneer of the World Economic Forum since 2006, Professor Chandru has been on the global advisory council for the future of healthcare at WEF for 2014-2016. Professor Chandru is a founder and former Honorary President of ABLE (the apex body for the biotechnology industry in India) from 2009-2012 and continues as a special invitee to the executive council. He serves on the vision groups for biotechnology, Science & Technology and as a member of the Karnataka State Council for Science and Technology

As an entrepreneur, Vijay leads the new generation healthcare company Strand Life Sciences. Since 2007, Strand has been a global leader in bioinformatics products licensed to over 2000 research labs and hospitals worldwide. With a team of over 200 high calibre scientists, Strand is now leading the charge in precision medicine in India with over 300 referring hospitals and clinics. Strand has the only CAP and NABL accredited labs for clinical genomics using NGS technologies in South Asia helping physicians diagnose germline risks of familial disease and decision support for clinical delivery of individualized medicine.

To appear in “BITSian Achievers 50@50” January 2014

The Age of Irresponsibility

Vijay Chandru
chandru@alum.mit.edu

I recently turned 60 and friends sent the usual “Shastipoorti” messages. One older friend congratulated me on entering the age of irresponsibility. After all at the age of 60, one is usually done with householder duties and in some situations with professional careers. I tried to think back to when the “responsible” phase had begun and that took me back to 1970 when I set foot in BITS Pilani as a fresher.

I was born into a prominent family in Bangalore in 1953, the year that Mount Everest was officially scaled and the double-helical structure of DNA was officially solved. My father, Dr.H.G.V. Reddy was a lecturer in zoology at Bangalore University who had switched to the national civil services (IAS) out of a sense of duty to participate in nation building. He hailed from a family of lawyers dedicated to politics and public service. My mother was the only child of B.N. Reddi, a pioneer producer-director of Telugu films in Madras (now Chennai). Due to a combination of circumstances, my sisters and I were raised by my maternal grandparents in Chennai with my father playing a larger role after I had reached the age of eight.

I was a sickly child who had to be home schooled in 3rd grade because of ill health. My health improved considerably after my father, a skilled sportsman, moved to Chennai and took charge of my lungs and limbs. I swam and played cricket and tennis almost every day and my lungs settled. My father's interest in science was infectious although I was also fascinated by films and saw an average of 3 feature films a week in our private studios. I was gifted in mathematics and Don Bosco's Matriculation School in Egmore had fabulous math instructors who gave me a lot of challenge problems to solve beyond the regular curriculum. My mind was made up to be an engineer when I interacted with my Dad's brother who was a chemical engineering PhD trained at the Imperial College in London. He had returned to be a technical director of the Shell Petroleum Company. I had an engineering drafting set and a slide rule by the time I hit my teens!

A couple of my older cousins who had grown up in Delhi went to study mechanical engineering in BITS Pilani in the late 60s (BITS had been established in 1964) and I was keen to follow their lead. So in 1970 I made the transition from the incredibly nurturing nest of my grandparents' home to the somewhat rough and tough dormitory life in the magnificent oasis of Jhunjhunu District. Pilani gave us an outstanding 5 year undergraduate education in engineering sciences and I ended up enjoying every aspect of campus life to the extreme. I captained the cricket team, acted and directed in plays, played water polo and student politics. My academic performance was above average and I learnt to get by with a regular habit of a couple of hours in the library in the evenings after cricket practice. My abilities to multi-task and manage my time to fit in a large number of activities at BITS would serve me well all through my professional life.

My branch of choice was Electrical Engineering and I enjoyed working with hardware - I was into DIY electronics and had built my own audio amplifiers and a small medium wave radio transmitter in my room in Ashok Bhavan. I would transmit a “castor” music session late on Saturday nights – Doors, Pink Floyd, Supertramp, Deep Purple, etc. While building gadgets was more like a hobby, what really fascinated me intellectually was the mathematical complexity of signals, systems and control. In our final year, the director of BITS, the enigmatic Dr CR Mitra, had invited Dr Stafford Beer to visit him in Pilani and give a few lectures about Operational Research and his adventures in Chile as a technical advisor to Salvador Allende. I was enthralled by Beer's descriptions of the field of "cybernetics" and its application to real world systems. I had found my academic passion - my segue into a career in research.

My training in systems science began in 1975 when I was a Masters student in Engineering at UCLA. I moved to MIT for my PhD and my big influences there were Jerry Shapiro, Tom Magnanti, Ravi Kannan, Sanjoy Mitter, Richard Larson, Dmitri Bertsekas, Christos Papadimitriou and Jim Orlin. I wrote a thesis on computational complexity of super group methods in integer programming bringing together the exposure to theoretical computer science (Kannan) and Gomory constructions for integer programs (Shapiro). I could not have asked for better teachers. My Guru and friend Jeremy F Shapiro, who sadly passed away in 2012, was a special man. Jerry taught me more than anyone else how to be courageous in research. He was never one to be unduly impressed by "reigning paradigms" - mutual admiration clubs trapped in cul-de-sacs of specialization. If my career has taken on such a variety of experiences, it is because of Jerry's encouragement to seek them.

As a professor of engineering in Purdue University, I started looking at computational geometry in the mid-80s and together with a few graduate students and colleagues we worked on many aspects of optimization problems posed on planar and 3D geometries. I also became a part-time student again and studied algebraic geometry with (late) Shreeram Abhyankar - a colleague and a great mathematician at Purdue. I had a lot of fun working in geometry because it brought back many high school math skills in ruler and compass constructions, conic sections and high school algebra. Two of my students from these geometry days stood out - Debasish Dutta who is currently a chaired professor and dean of graduate school at University of Illinois, Urbana Champaign and Ravi Venkatesan who just stepped down as the Chairman of Microsoft in India.

In the mid-80s another important influence in my research ideas came from the late Bob Jeroslow at the ARIDAM workshops at Rutgers. Bob asked integer programmers to look more closely at the interface of integer programming and logic. John Hooker of CMU and I responded to his call. My book with Hooker called "Optimization Methods for Logical Inference" published by Wiley Interscience in 1998 was the culmination of a decade of joint work that the remarks of Jeroslow had triggered.

Between 1989 and 1992, I tested the idea of returning to India to teach at IISc, Bangalore. After almost two decades in the US, this was clearly an adventurous move. I was at the top of my academic career, a high flying tenured professor of engineering at Purdue University. But I was

36 years old and knew deep down that it was time to go home and make a difference. When I landed in Bangalore, a stanza from a poem by Sir Walter Scott, from childhood memories kept coming to me

Breathes there the man with soul so dead
Who never to himself hath said,
This is my own, my native land!
Whose heart hath ne'er within him burned,
As home his footsteps he hath turned

Ensconced at the Indian Institute of Science (IISc), I was on the prowl again for a new research experience. The students from Molecular Biophysics Unit came over to ask me to teach them combinatorial computational geometry to use in exploring functional structural properties of proteins in biology. I got excited with the optimization models associated with minimum energy conformations also called the protein folding problem - the so called *holy grail* of structural biology. The more I looked at biology, the more excited I became about possibilities for data science and decision sciences in the life and health sciences. We began with a small academic effort in computational biology seminars and workshops at IISc but after a sabbatical at University of Pennsylvania in 1999-2000, I was ready to plunge headlong into computational biology.

By the mid-90's, I was increasingly concerned that my work at IISc continued to be quite abstract and unrelated to the society I lived in. I may as well have done all this in the US. I needed to look beyond the usual academic conquests (by 1996 I was a full professor and a fellow of the academy of sciences). It was my great fortune that I discovered a few younger colleagues who felt likewise and were willing to think outside the ivory tower. So Swami Manohar, Vinay Vishwanathan, Ramesh Hariharan and I began our journey by first convening an applications lab at IISc, we called it the "Perceptual Computing Laboratory" (*PerCoLat*) and began building relevant technologies for pursuing interests in ICT for development and some practical aspects of bioinformatics. In 1996, PerCoLat started off with several funded consulting projects with Indian IT companies and international clients.

In 1998, as a consequence of our focus on ICT for development, we had the concept design of the "*Simputer*", a unique handheld computer that would put India on the global technology map for innovation. Instead of getting into a lot of details, let me point the interested reader to www.simputer.org and the citation we received in 2001 from the New York Times as the ICT innovation of the year http://nyti.ms/1e5f0pP . The Simputer project rolled into a dynamic startup called *PicoPeta Simputers* which brought out a commercial Amida Simputer in 2004. Picopeta was eventually merged into Geodesic, a public company based out of Mumbai and the Simputer has now morphed into the Geo Amida.

The transition from convening an applied lab PerCoLat to starting companies was a big step for a group of professors at IISc. It had never been done before and so we were pioneers in the launch of academic entrepreneurship of this kind in India. The dream that drove us was to create world beating technology innovation companies from India. We felt that while the ICT sector in India had made great strides in services business, there would be an important niche for innovators who could help the sector add value through research and translation of intellectual property. If this group of smart technologists from IISc could show the way, our dream would be realized.

The other startup *Strand Life Sciences* from PerCoLat was born in late 2000. Strand has grown into a large and well-respected bioinformatics company. It was not an easy journey to success with a software product line from India and a global market in research biology. The frustrations of working in a niche, under-appreciated scientific field with the market size being a small one almost overwhelmed us at the end of 2003 when we had to retrench and refocus the company. Most bioinformatics companies that started out with the completion of the human genome project in 2000 have received a quiet burial.

The bioinformatics products from Strand for Array, Mass Spec and Next Generation Sequencing (NGS) analysis are used by over 2000 laboratories worldwide (about 30% global market share) with citations in thousands of publications. Strand has received many accolades for innovation and the most recent one was for biotech innovation of the year 2012 for the health sciences awarded by DBT/BIRAC. This award was in recognition of the extraordinary work by Strand scientists in quantitative computer modeling of the biochemistry of the liver to be able to predict and profile the mechanisms of toxicity of chemicals (including drugs) in the liver.

Strand's new clinical genomics product provides deep interpretation and reporting pipelines driven by real clinical experiences, computational expertise and the large-scale aggregation of genomic knowledge. Strand has the potential to bring genomics based personalized medicine at prices affordable for the Indian healthcare needs. The next few years will tell if the 200 brilliant scientists and technologists at Strand can deliver the goods. We are racing forward to be the first at the finishing line with an end-to-end platform for sequencing based diagnostics. We expect to reach there by the summer of 2014.

The team at Strand is perhaps a unique experiment in India of gathering the best and the brightest talent in a multi-disciplinary, gender balanced, collegial, and highly performance driven environment in an indigenous private enterprise. I arrive at work every morning knowing this is a team that can deliver almost the impossible. So Ramesh Hariharan and I demand the impossible of ourselves and the team. The work life balance does get challenged at times and we have to make some conscious choices about travel, vacations and work hours to keep some sanity in our personal lives.

On December 28th 2005, I had faced a real test. In a terrorist attack at the Indian Institute of Science, I was shot repeatedly at close range with an AK56 and gravely injured. Both my arms were shattered and a bullet to the chest had taken out my left shoulder musculature as well. I survived that night and, in the eight months that followed my courageous wife Uma and daughter Maya brought me back to normal. Of course there were some great surgeons and physiotherapists who pulled off some miracles as well. My second chance in life has given me an even greater sense of responsibility than I started out with in 1970 when I landed at BITS, Pilani. I will accept the option of claiming *the age of irresponsibility*, but I can wait a few more years before I play that card.
